

Chapter5

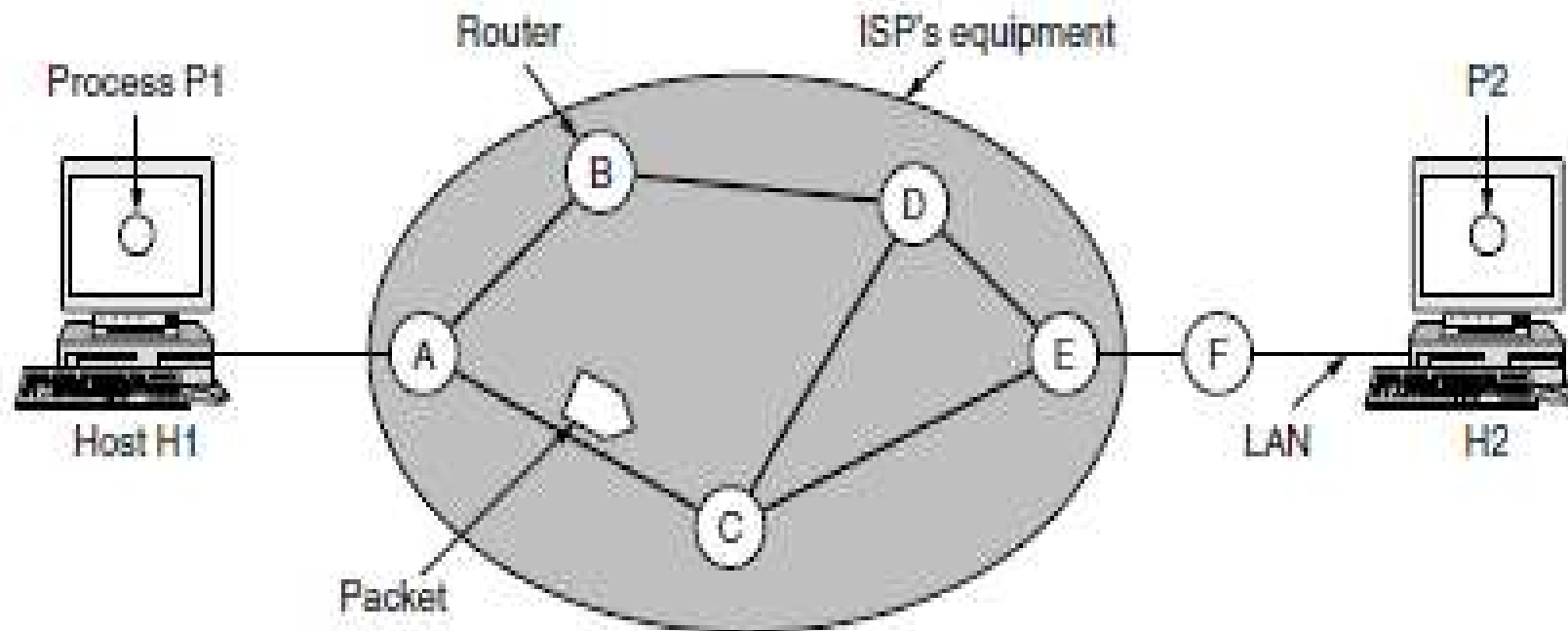
The Network Layer

(around 14 hours)

Outline

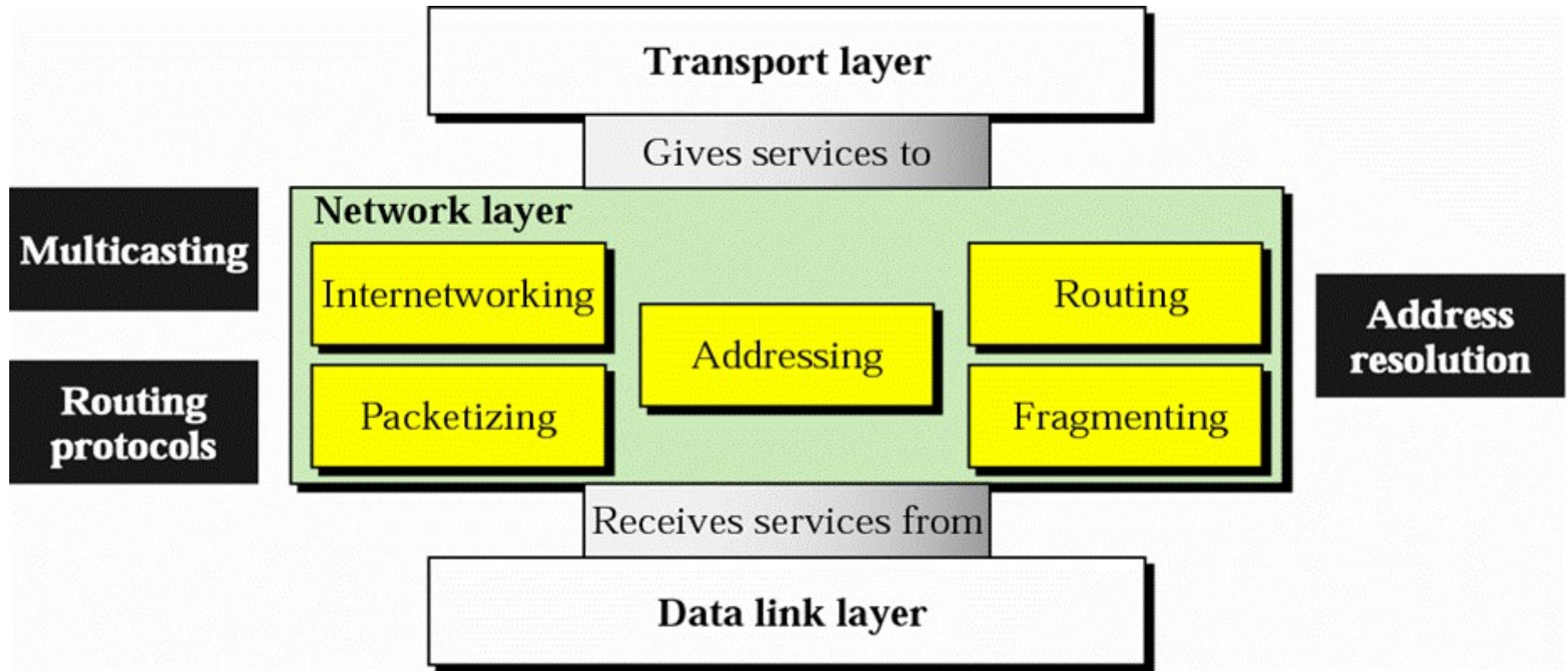
- Introduction to Network Layer
 - ◆ Position and functions of network layer.
 - ◆ Virtual Circuit Switching and Datagram Routing.
- Routing Algorithms
 - ◆ Distance Vector Routing (DVR)
 - ◆ Link State Routing (LSR)
 - ◆ Hierarchical Routing
- Congestion Control and Internetworking
- The network layer in Internet
 - ◆ IP Addressing
 - ◆ Packet format
 - ◆ IPv6
 - ◆ ICMP、 ARP and DHCP
 - ◆ Routing Protocols: RIP, OSPF and BGP
 - ◆ IP Multicast
 - ◆ Mobile IP

Environment of Network Layer Protocols



Store-and-Forward Packet Switching (存储转发分组交换)

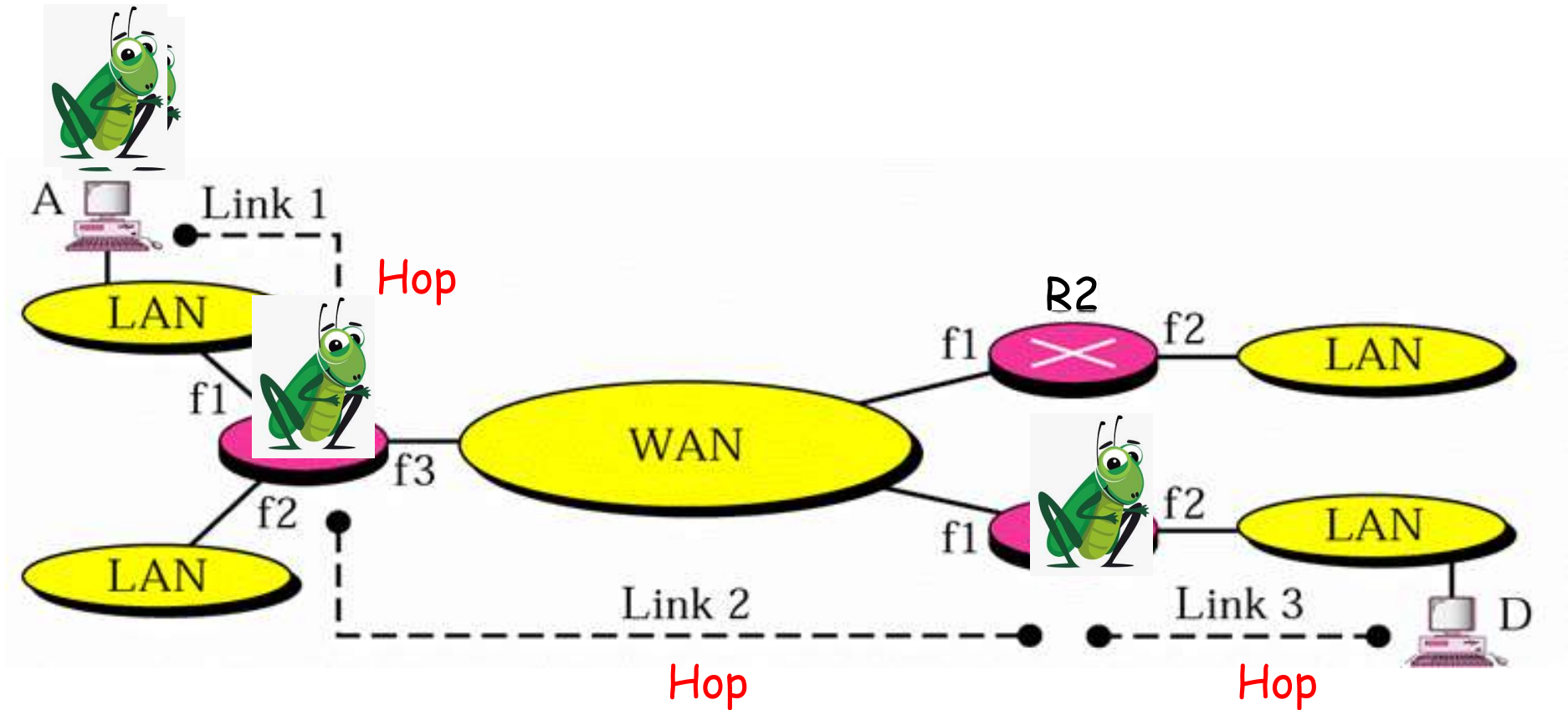
Positions and Functions of Network Layer



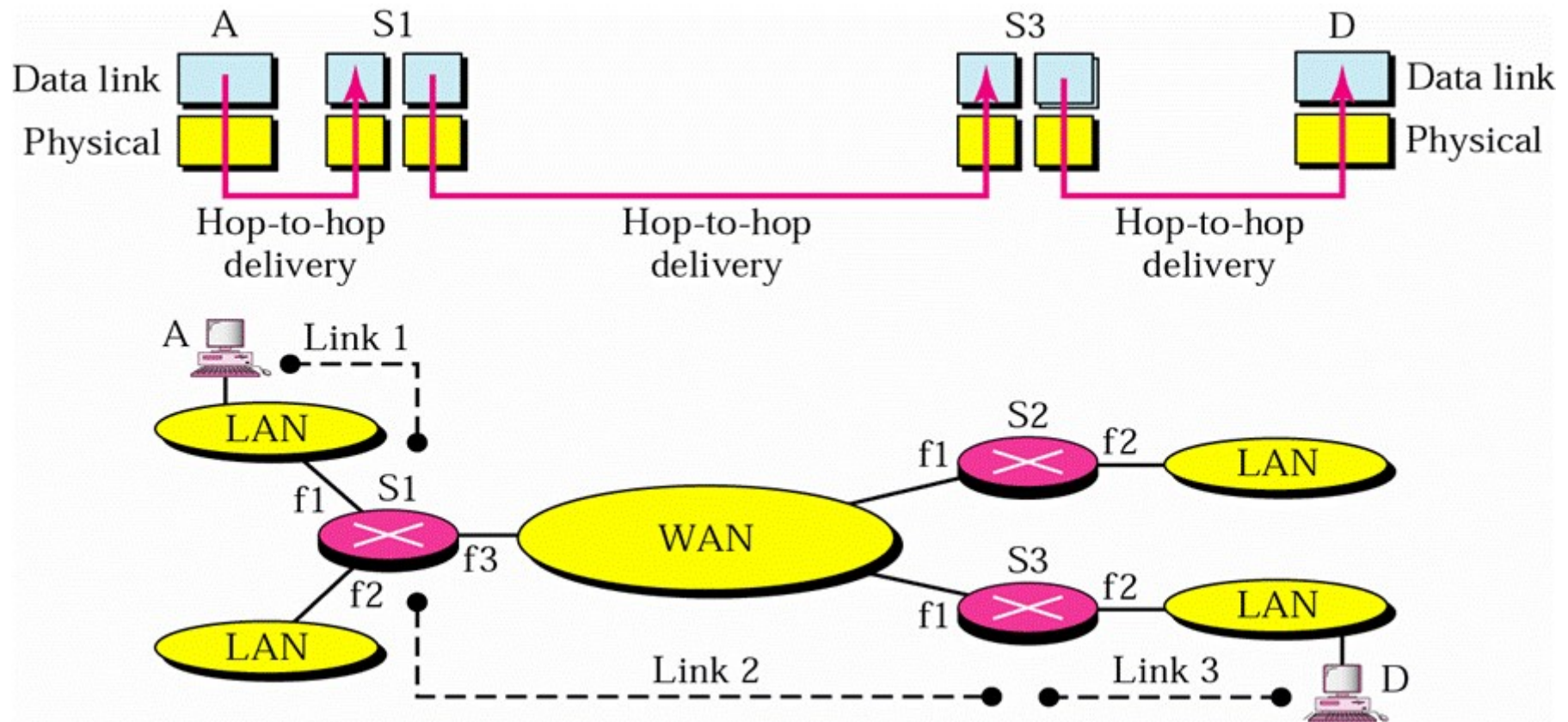
Network layer is the lowest layer that deals with **end-to-end (host-to-host)** transmission (端到端传输)



The concept of Hop(跳)

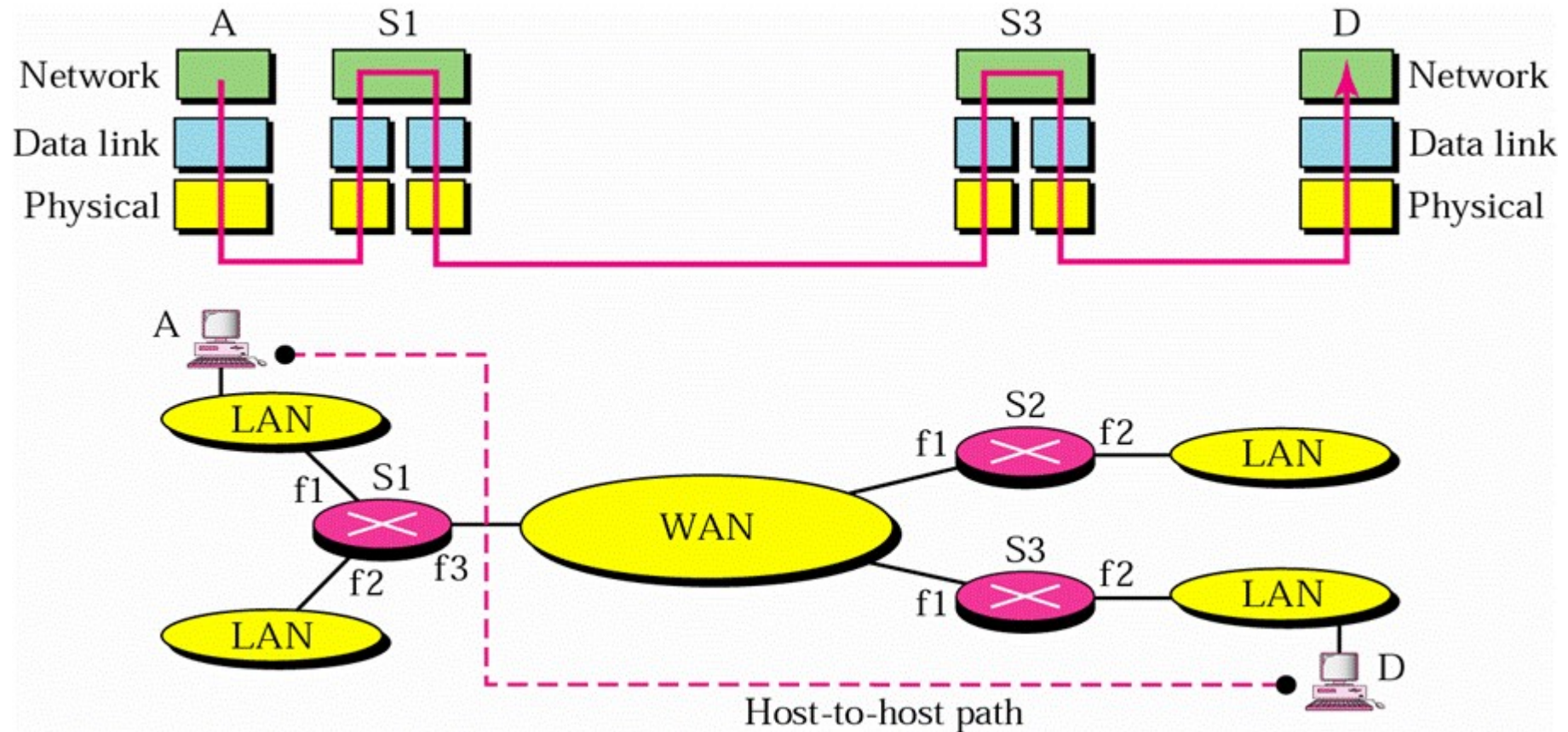


Data Link Layer in an Internetwork: Hop-to-Hop Delivery(逐跳传输)



[Forouzan]

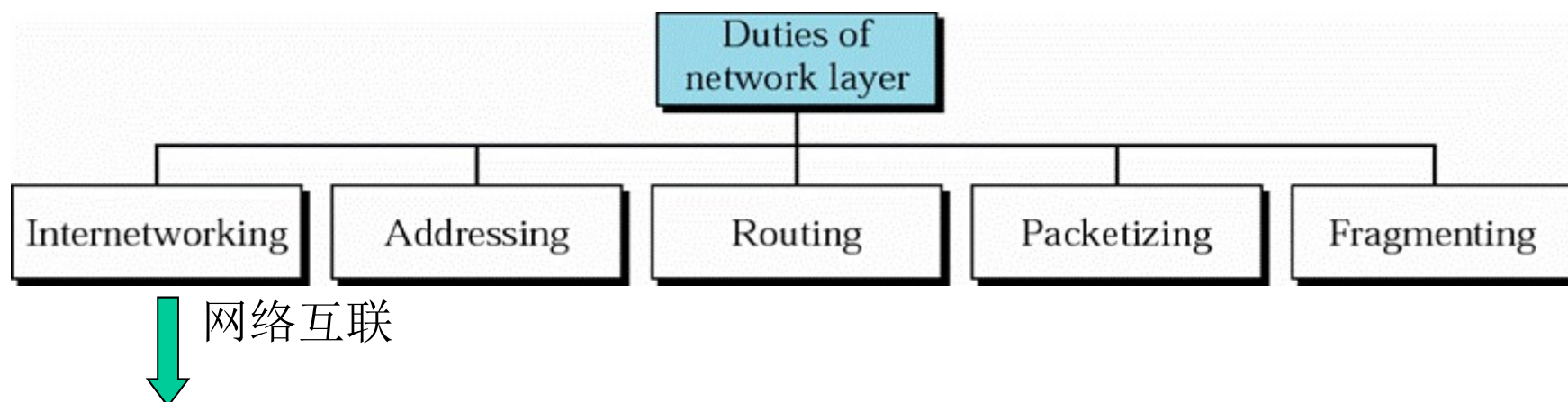
Network Layer in an Internetwork: Host-to-Host Delivery(主机-主机传输)



主机-主机（端-端）

[Forouzan]

Network Duties(1)



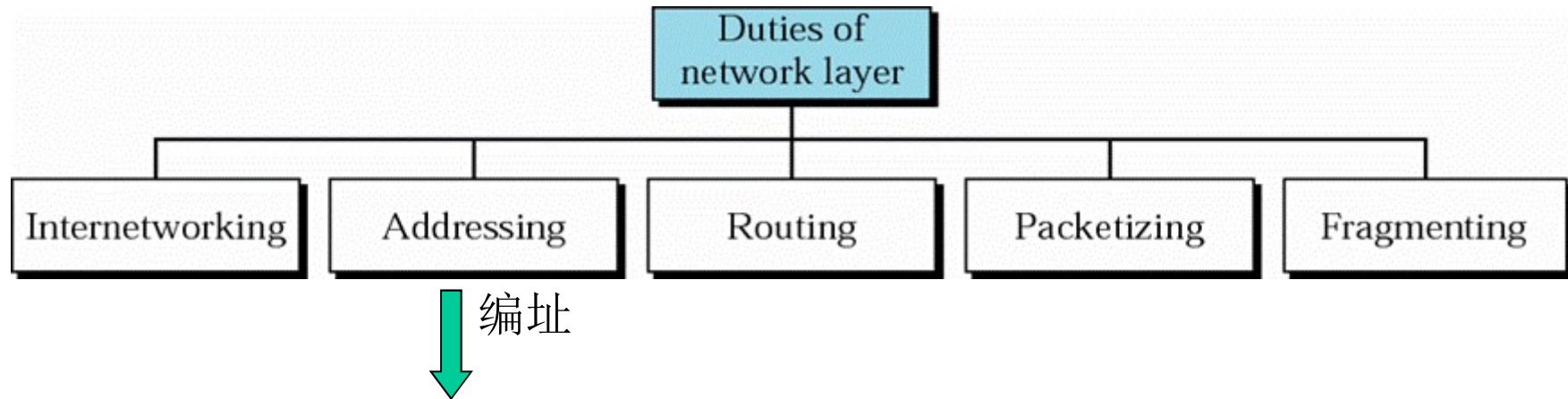
Interconnecting **different** networks and making them look the **same** to the transport layer.

不同的网络：网络地址不同、实现技术可能不同、包格式可能不同

The transport layer should not be worried about the underlying physical network! （网络的个数及其具体技术对于传输层透明）

[Forouzan]

Network Duties(2)

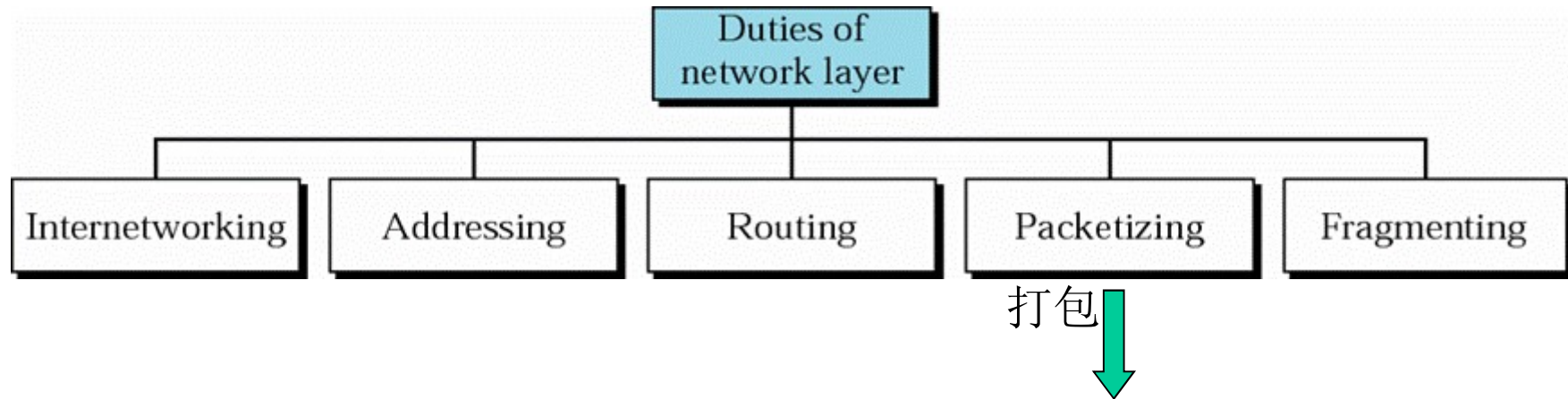


The addresses of a host/router must be **uniquely** and **universally**. (统一的地址格式)

Two devices on the internet can never have the same address. (Address per interface to network, 接口地址唯一)

[Forouzan]

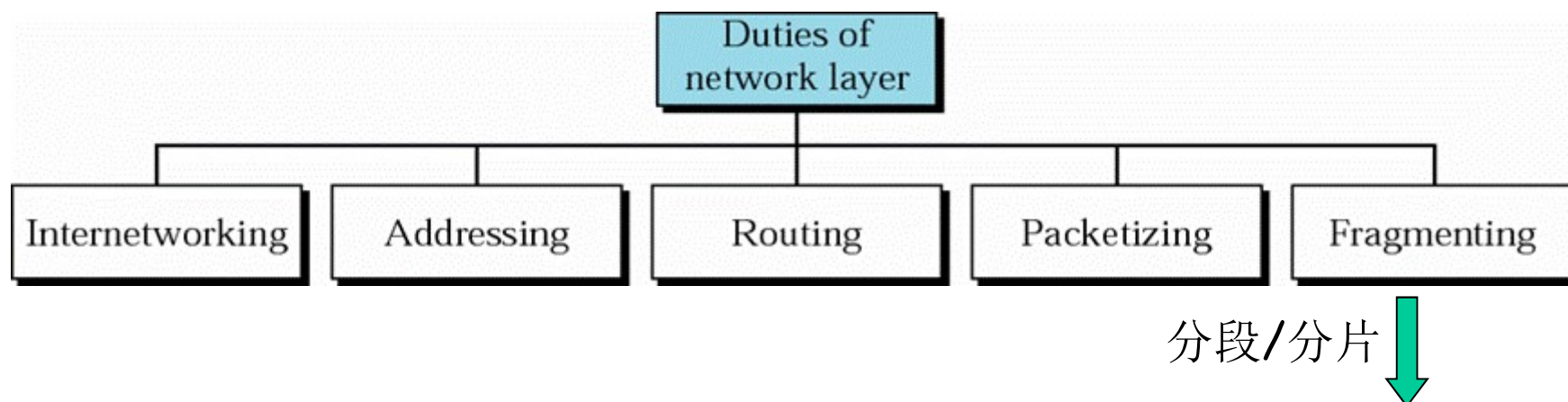
Network Duties(3)



Encapsulated message received from upper layer and forming **packets**, a task common to all layers. In Internet, packetizing is done by **IP**.

[Forouzan]

Network Duties(4)



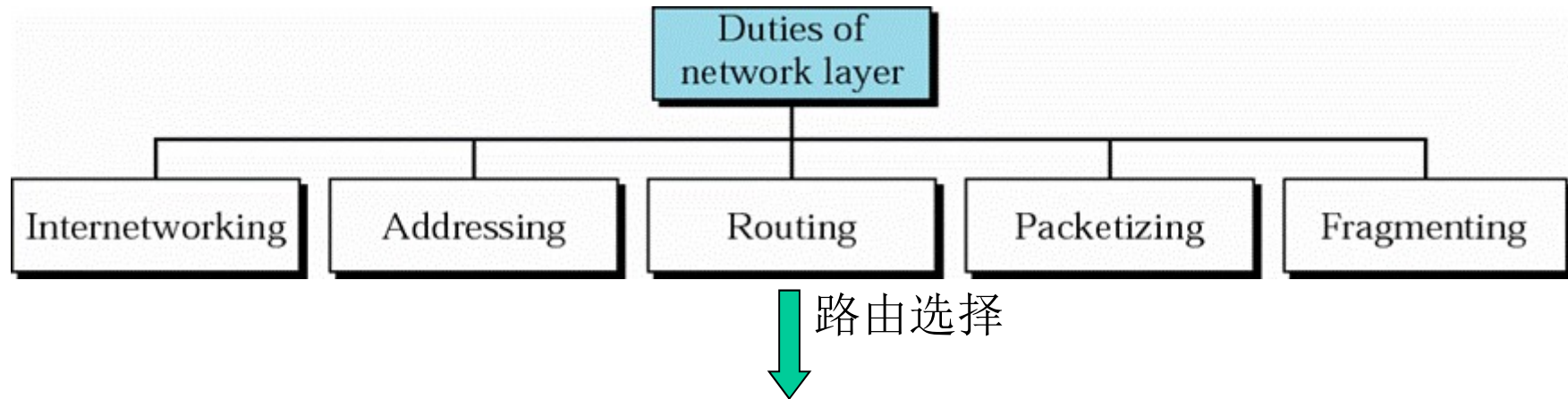
The network layer must be able to operate on top of **any** data-link layer technology.

All these technologies can handle a different packet length. (不同的数据链路层网络支持的最大数据长度不同)

The network layer must be able to **fragment** transport layer PDUs into smaller units so that they can be transferred over various data-link layer technologies.

[Forouzan]

Network Duties(5)

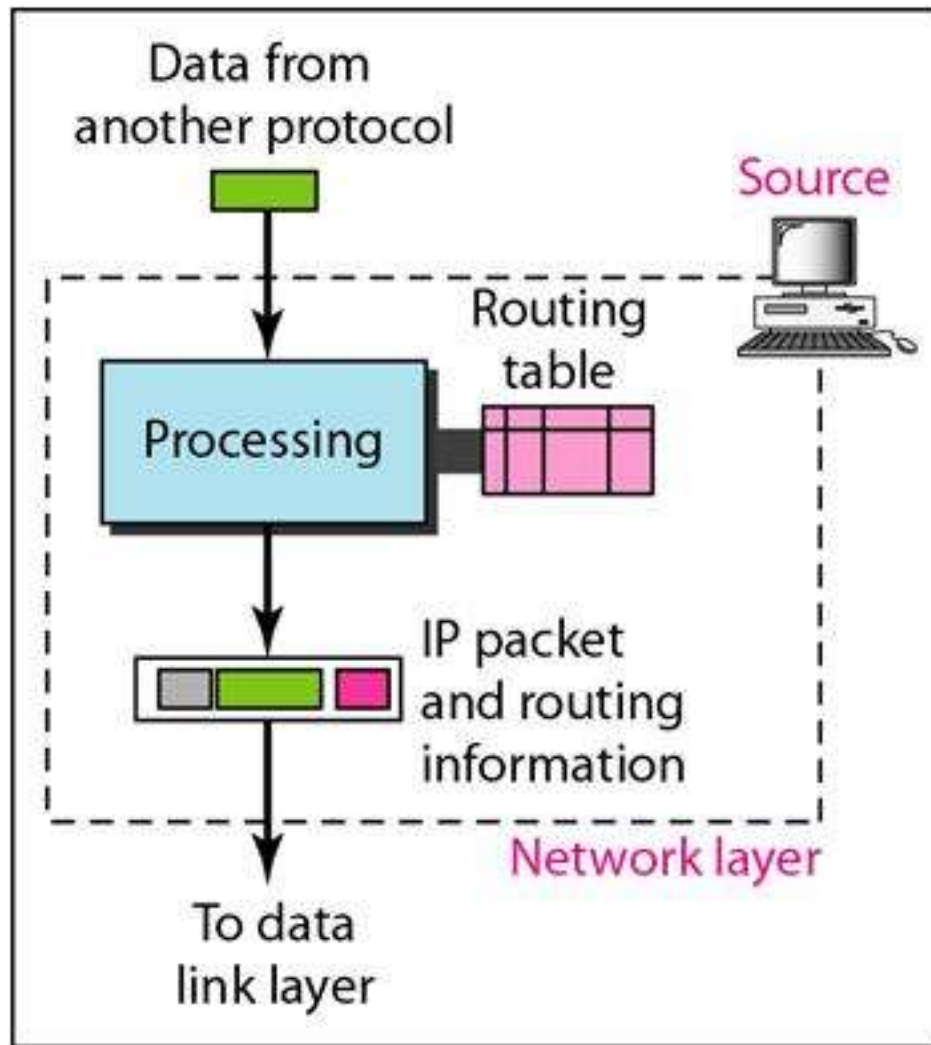


To send the packet to destination, who is the **next hop** (下一跳) ?

Packet cannot choose the route; the routers connecting the LANs/WANs makes this decision.

[Forouzan]

Network Layer at the Source



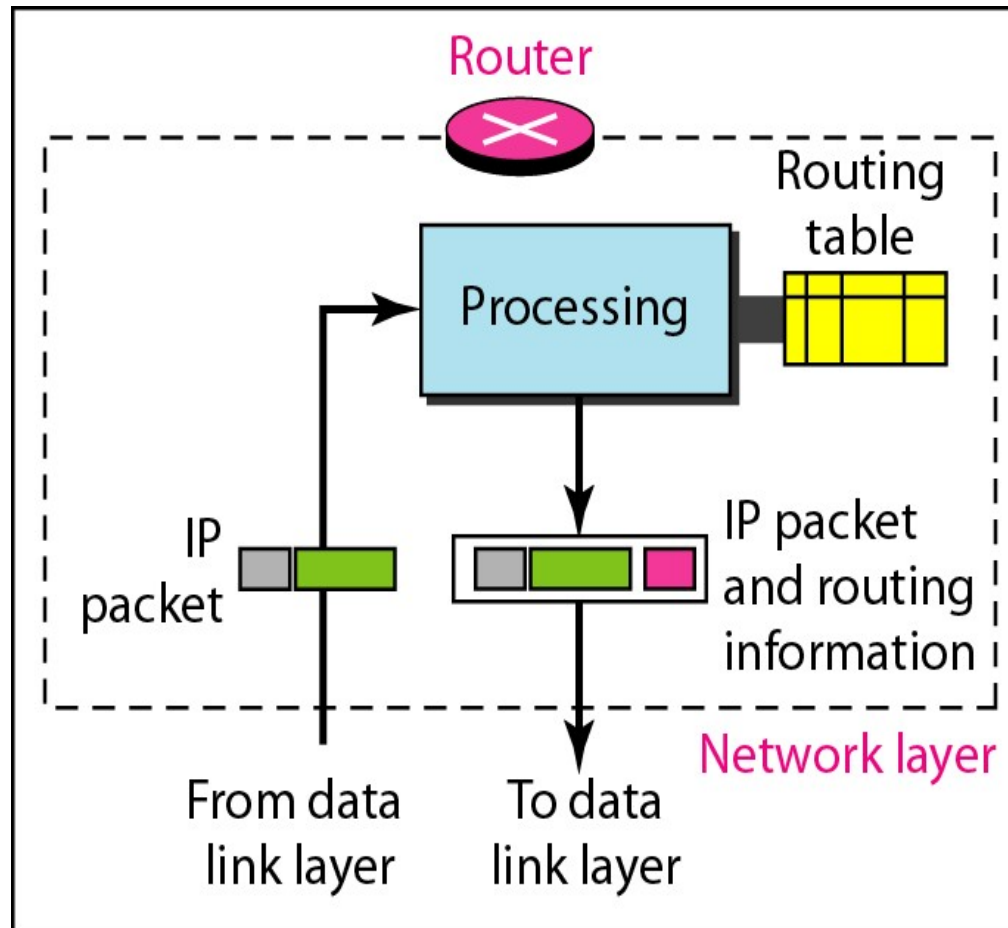
The source network layer **create a packet** carrying addresses of source host and destination host.

The source also checks routing table for find the **routing information** (e.g. outgoing interface, 输出接口, such as Ethernet or wifi).

[Forouzan]

Network Layer at the Router

Responsible for **routing** the packet



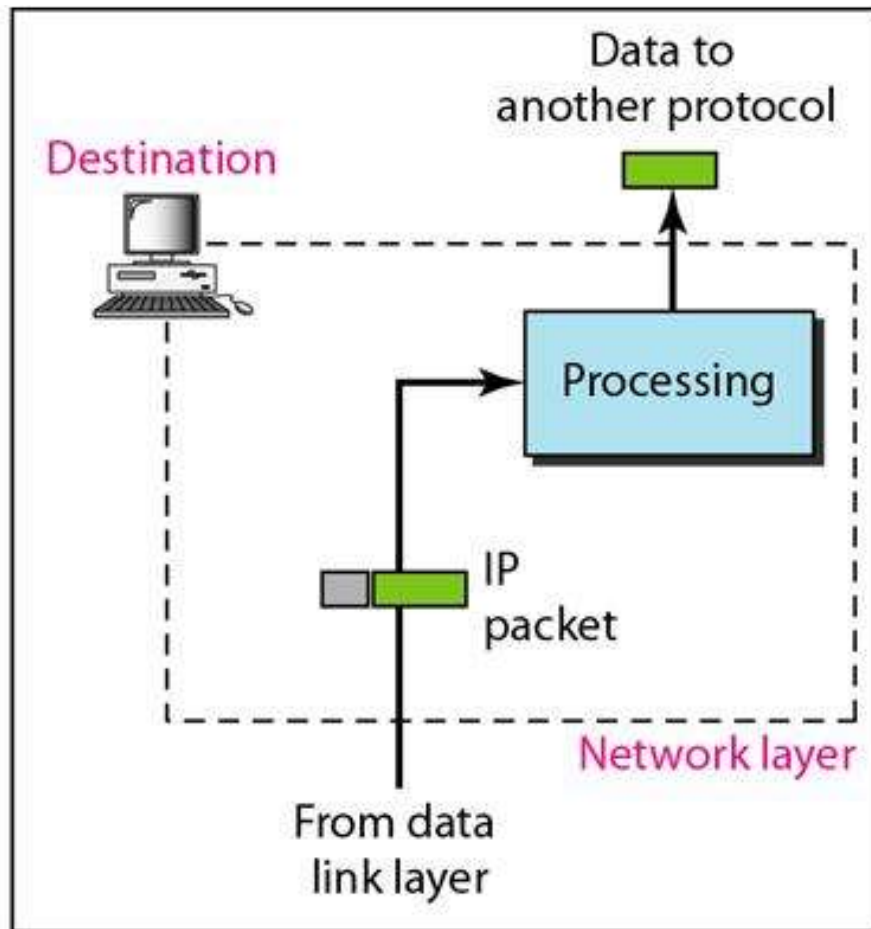
When a packet arrives, the router checks its routing table and finds the **interface** from which the packet must be sent using **routing table**.

c. Network layer at a router

[Forouzan]

Network Layer at the Destination

Responsible for **address verification**



b. Network layer at destination

The destination makes sure that the **destination address** on the packet is the same as the address of the host.

[Forouzan]

Services Provided to Transport Layer

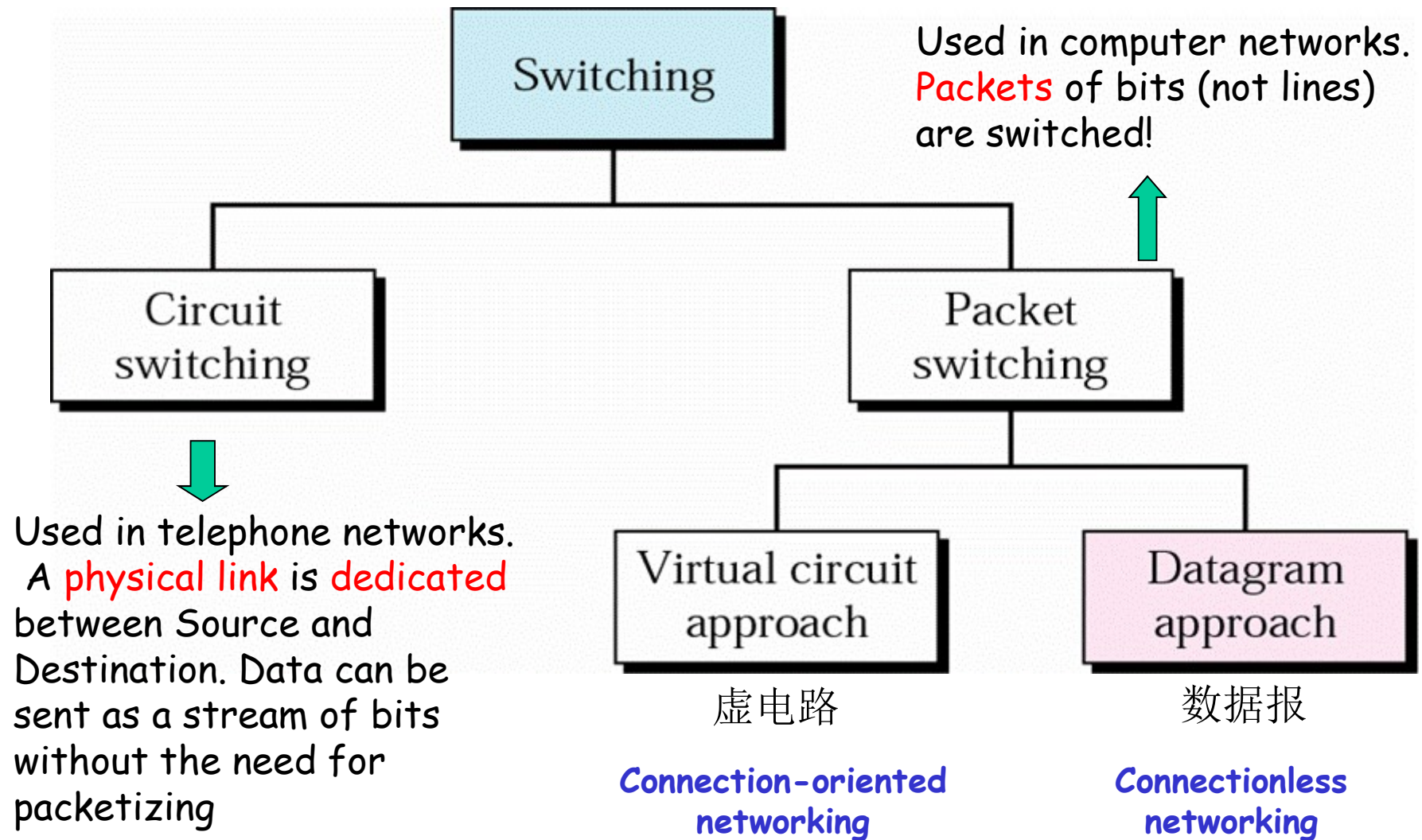
■ Design Goal

- ◆ The services should be independent of the router technology.
- ◆ The transport layer should be shielded from the number, type, and topology of the routers present
- ◆ The network addresses made available to the transport layer should use a **uniform** numbering plan, even across LANs and WANs

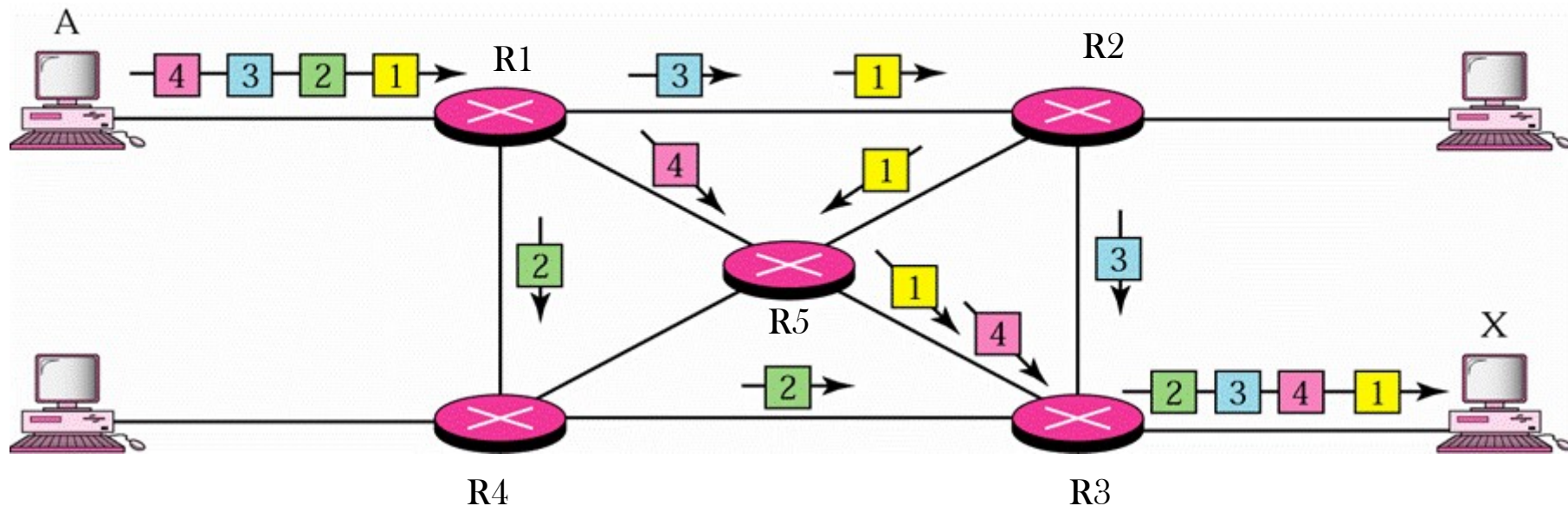
■ Two classes of service

- ◆ Connection-Oriented Service
- ◆ Connectionless Service

Switching



Implementation of Connectionless Service: Datagram(数据报) Network



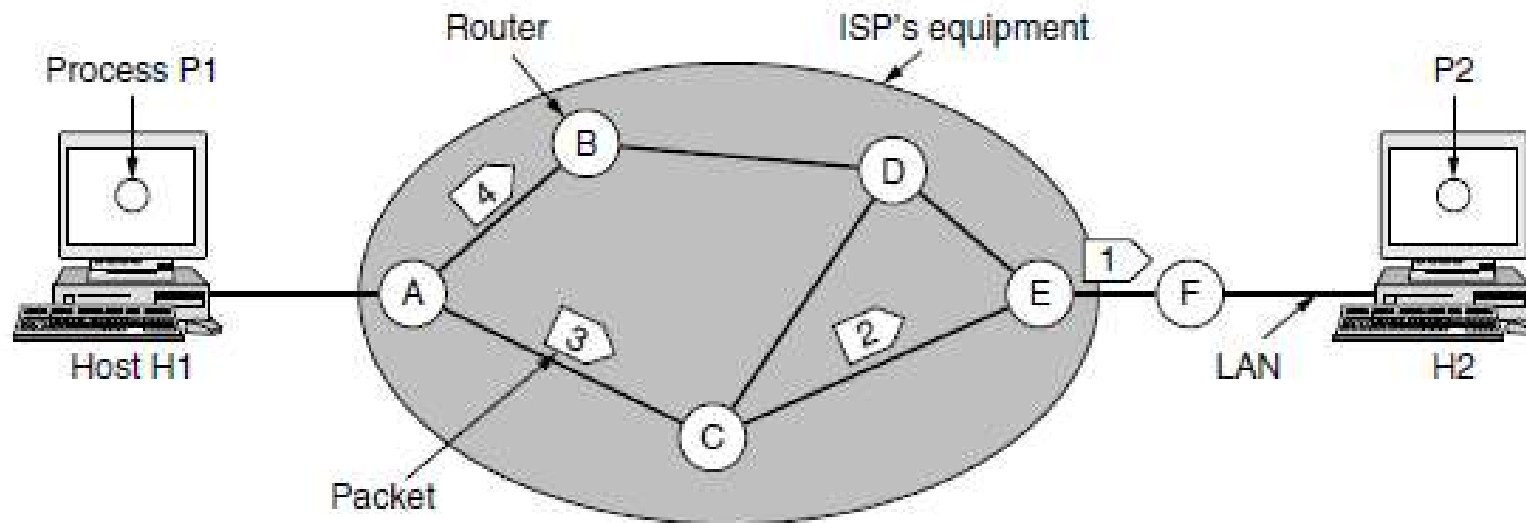
Routing: where to
forward the packet?

Routing Table of R1

Dest.	Line(Next Hop, 下一跳)
R1	---
R2	R2
R3	R5
R4	R4
R5	R5

[Forouzan]

Routing within a Datagram Subnet



A's table (initially)

A	-
B	B
C	C
D	B
E	C
F	C

Dest. Line

A's table (later)

A	-
B	B
C	C
D	B
E	B
F	B

C's table

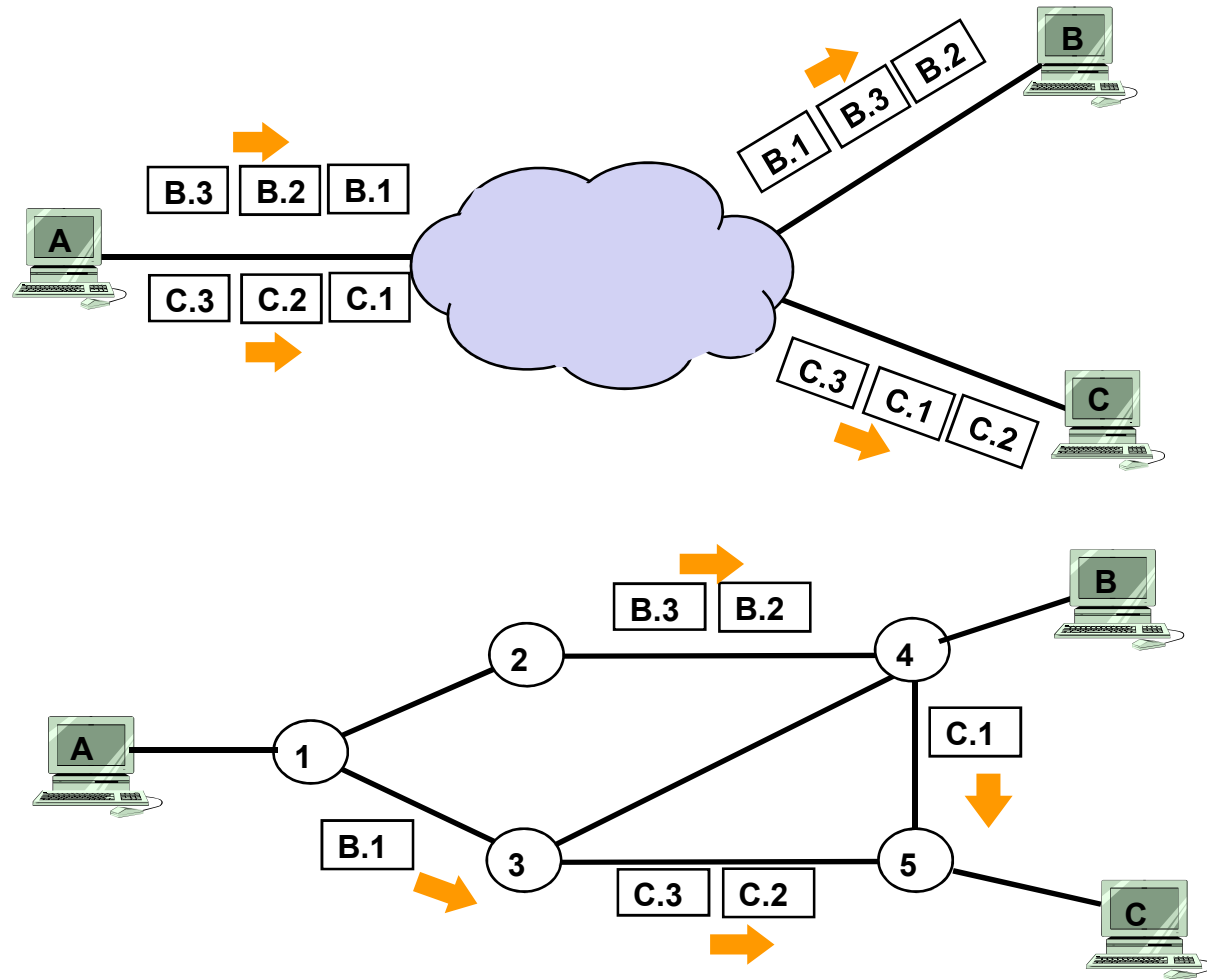
A	A
B	A
C	-
D	E
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	-
F	F

The routing tables are **dynamic** and are **updated** periodically.

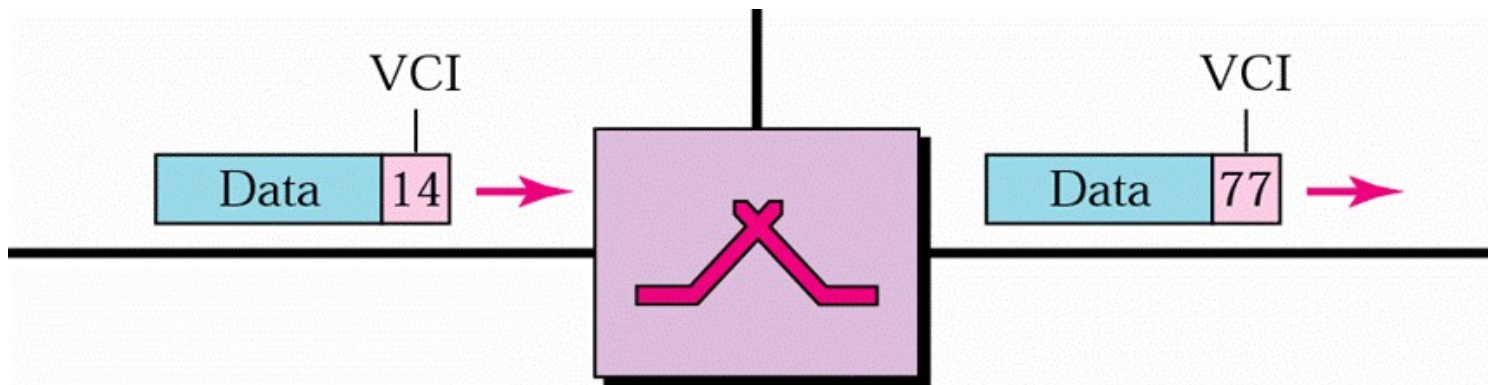
Routing within a Datagram Network(2)



包中需要携带完整地址；包到达可能乱序

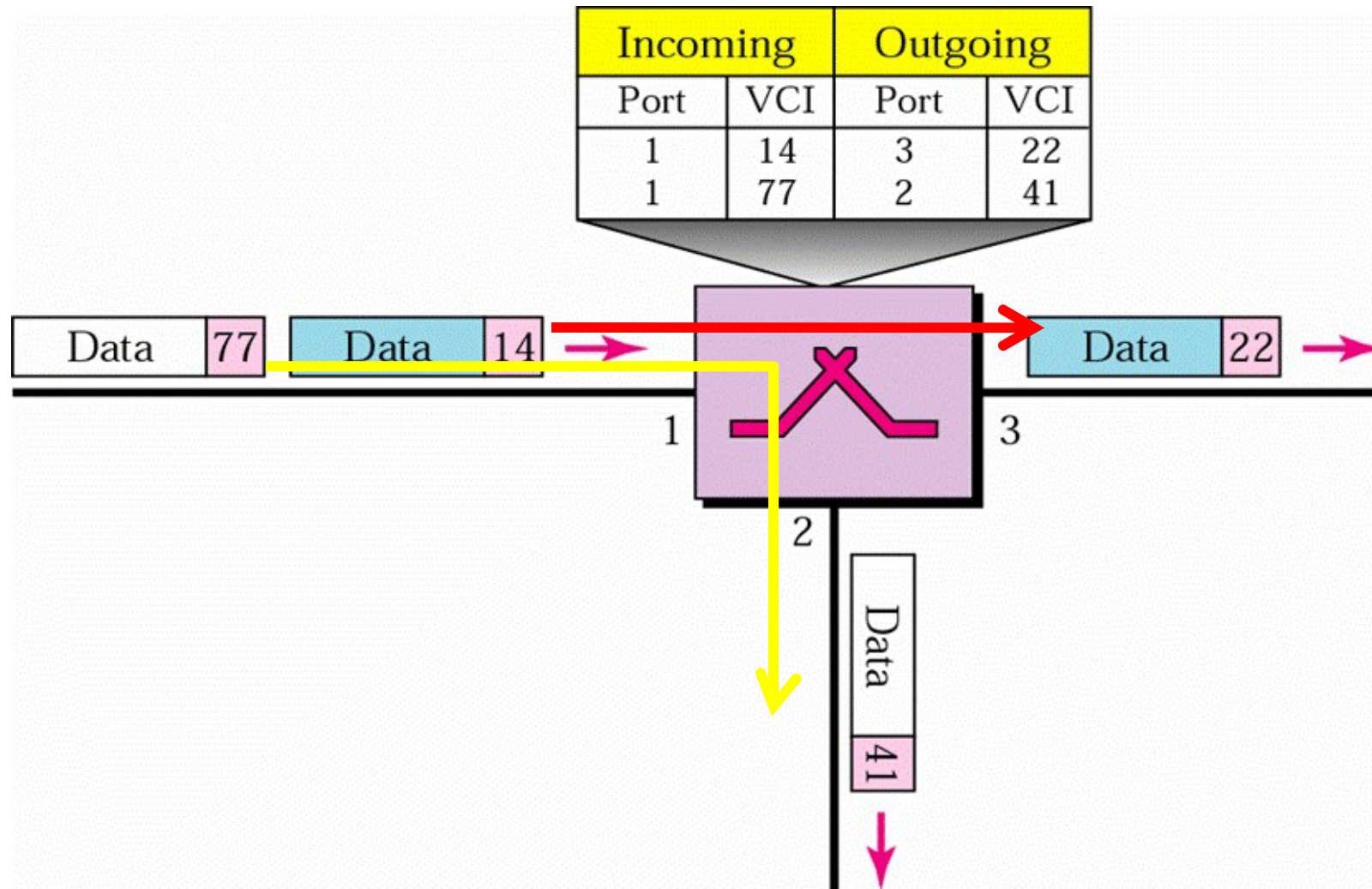
Implementation of Connection-oriented Service: Virtual Circuit(虚电路) Network

- A cross between a circuit-switched network and a datagram network.
- **Virtual circuit** is established and end-to-end path is decided before data transmission
- Connection-oriented service
- Virtual Circuit Identifier(**VCI**): local significant
 - ◆ Much shorter than address 虚电路 编号



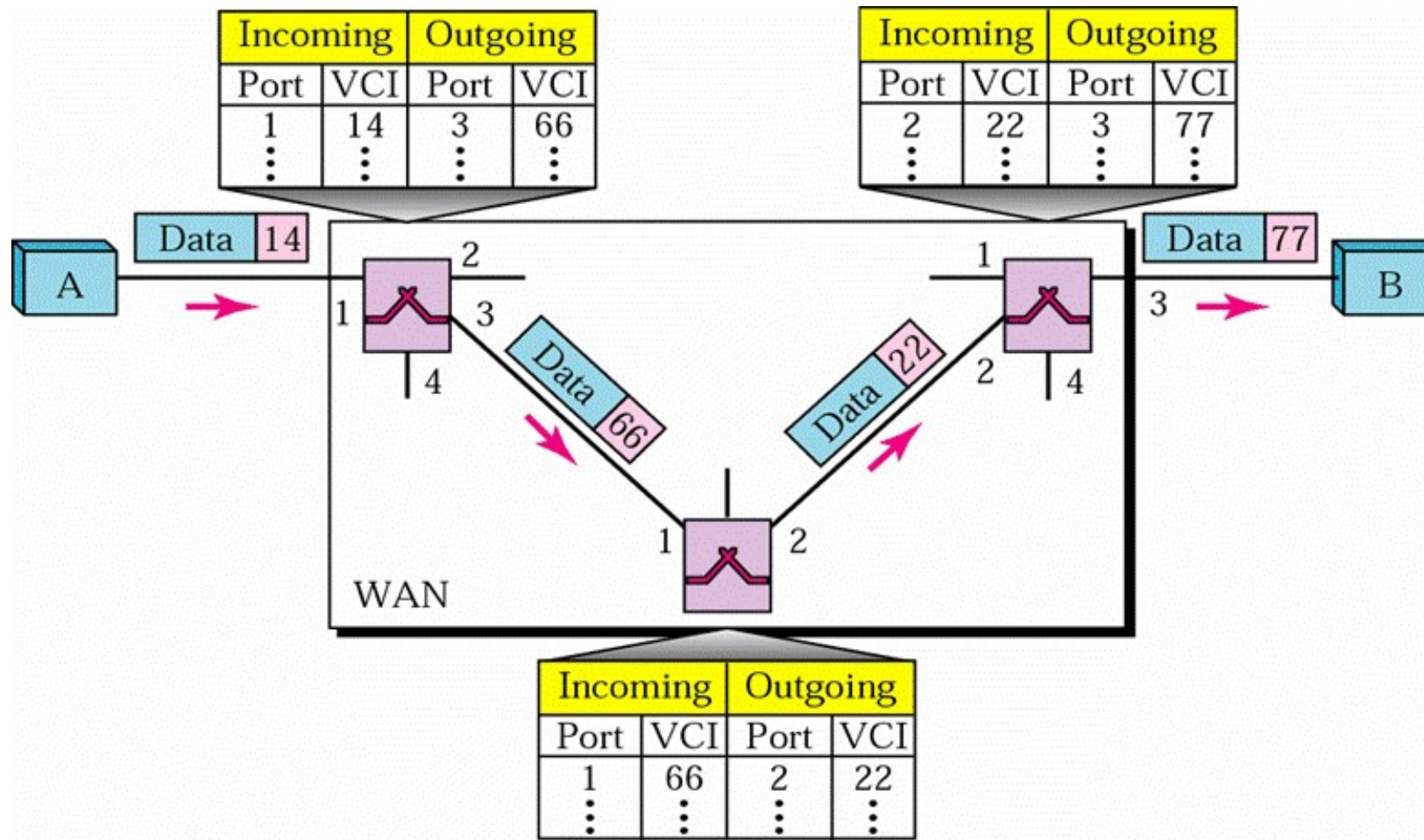
[Forouzan]

Virtual Circuit Approach: Routing Table



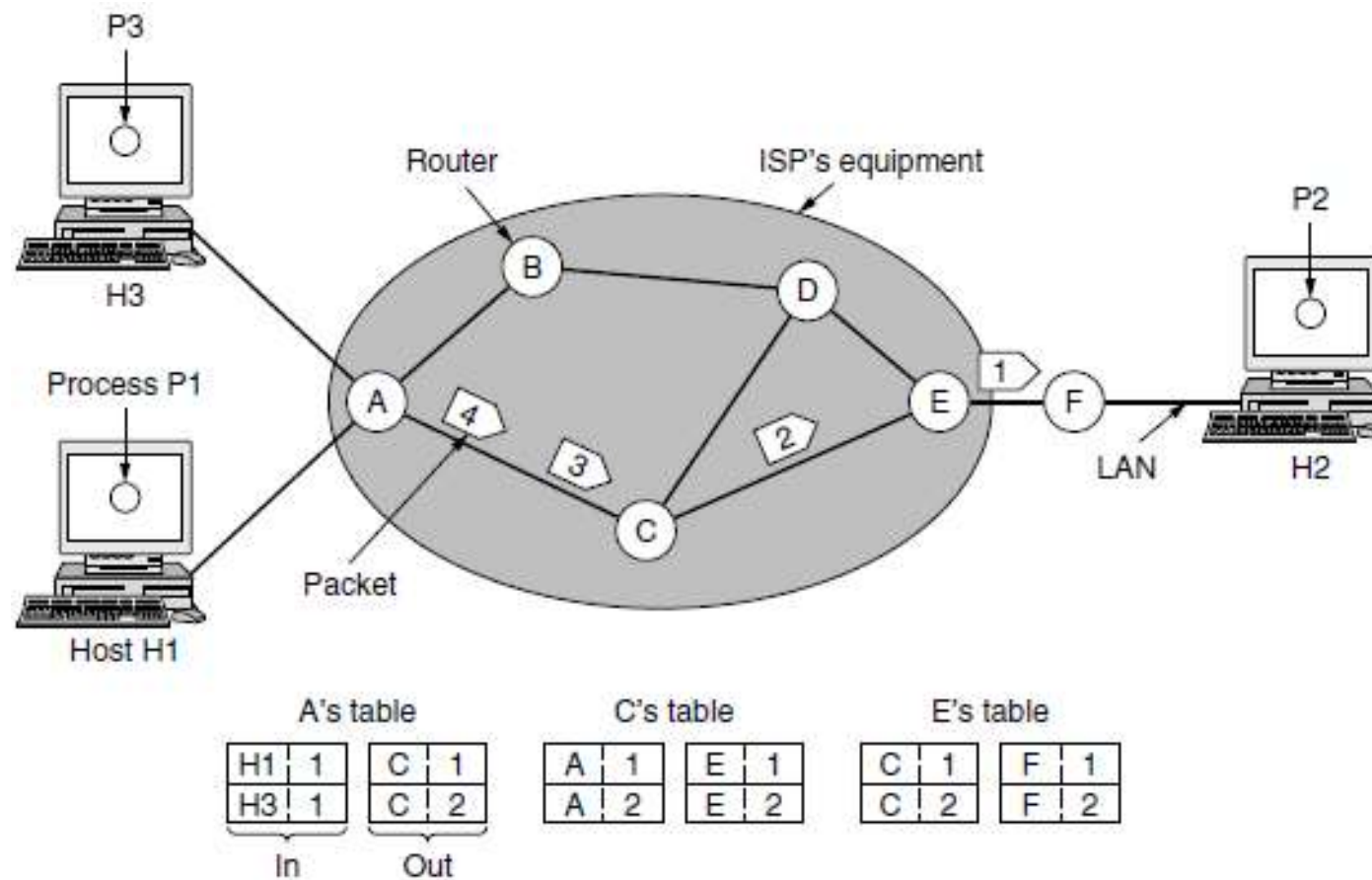
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Virtual Circuit Approach: Routing



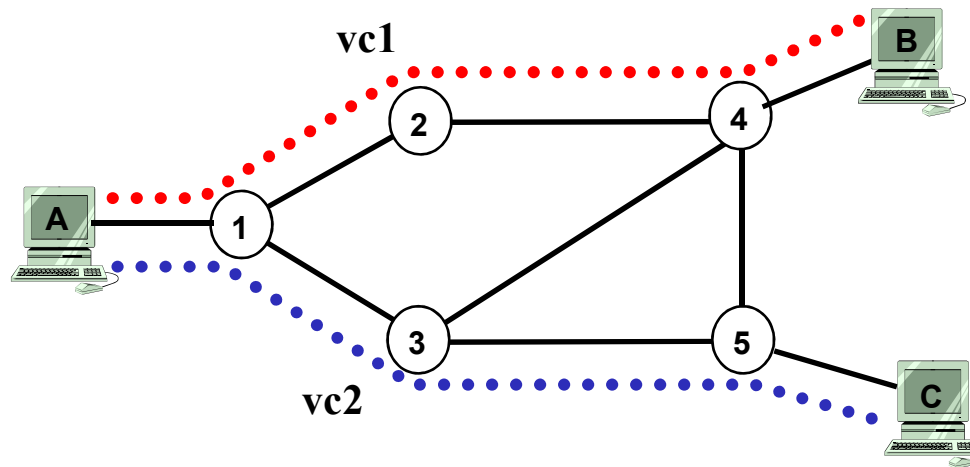
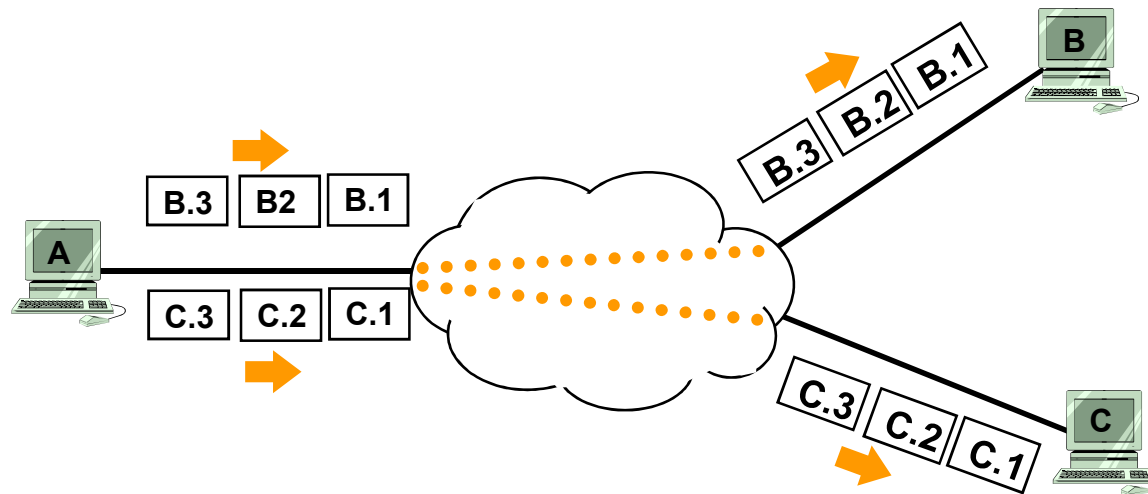
[Forouzan]

Routing within a Virtual-circuit Subnet



VC1: H1-A-C-E-F-H2
VC2: H3-A-C-E-F-H2

Routing within a Virtual-circuit Subnet(2)



● vc1: A--1--2--4--B

● vc2: A--1--3--5--C

Virtual Circuit vs. Datagram

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service 服务质量	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control 拥塞控制	Difficult	Easy if enough resources can be allocated in advance for each VC

Outline

- Introduction to Network Layer
- Routing Algorithms
 - ◆ *Concept of Routing and Routing Table*
 - ◆ Optimality principle
 - ◆ Flooding
 - ◆ Dijkstra's "Shortest Path" Algorithm
 - ◆ Distance Vector Routing (DVR)
 - ◆ Link State Routing (LSR)
 - ◆ Hierarchical Routing
- Congestion Control and Internetworking
- The network layer in Internet

Routing: Overview(1)

- An **internet(互联网)** is a combination of networks connected by routers.
- When a packet goes from a source to a destination, it may pass through many **routers**.
- A router consults a **routing table** when a packet is ready to be forwarded. The routing table specifies the **optimum path (最佳路径)** for the packet.
- To make the lookup efficiency, need to make the size of table manageable and handles issues such security at the same time.
 - ◆ Key question is **how to design the routing table**.

Routing: Overview(2)

- Routing table can be Static or Dynamic.
 - ◆ **Static table** does not change frequently.

A **static routing table** contains information entered manually
Usually remained unchanged.

- ◆ **Dynamic table** is updated automatically when there is a change somewhere in the network.

A **dynamic routing table** is updated periodically or whenever necessarily using one of the **dynamic routing protocols** such as RIP, OSPF, or BGP.

Routing: Overview(3)

- **Routing Algorithms:** algorithms used to construct and update routing tables
 - ◆ DVR, LSR, PVR
- **Routing Protocols:** a combination of rules/procedures that lets routers inform one another when changes occur; mostly based on sharing/combining information between routers at different networks.
 - ◆ RIP, OSPF, BGP

Routing table: Next-hop(下一跳) routing

Routing table for host A

Destination	Route
Host B	R1, R2, Host B

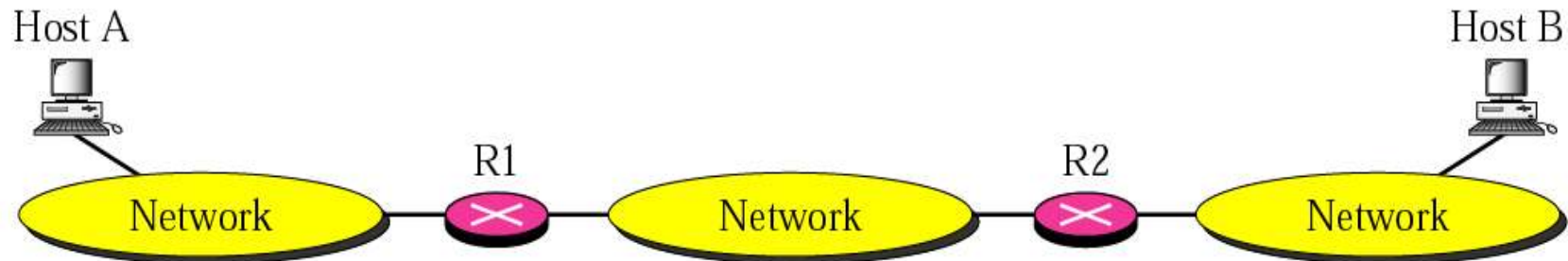
Routing table for R1

Destination	Route
Host B	R2, Host B

Routing table for R2

Destination	Route
Host B	Host B

a. Routing tables based on route (完整路径)



Routing table for host A

Destination	Next Hop
Host B	R1

Routing table for R1

Destination	Next Hop
Host B	R2

Routing table for R2

Destination	Next Hop
Host B	—

b. Routing tables based on next hop

Routing Algorithms

- Concept of Routing and Routing Table
- *Optimality principle*
- Flooding
- Dijkstra's "Shortest Path" Algorithm
- Distance Vector Routing (DVR)
- Link State Routing (LSR)
- Hierarchical Routing

How to evaluate routing algorithms?

■ Goals

- ◆ Correctness
- ◆ Simplicity
- ◆ Robustness
- ◆ Stability
- ◆ Fairness
- ◆ Optimality
 - Minimizing mean packet delay
 - Maximizing total network throughput

■ Performance Criteria

- ◆ Least cost, such as minimum hops

Responsibility of Routing Algorithms

- Responsible for **deciding** which **output line** an incoming packet should be transmitted on
 - ◆ If the subnet uses **datagrams** internally, this decision must be made for **every** arriving data packet 数据报网络对每个包选路
 - ◆ If the subnet uses **virtual circuits** internally, routing decisions are made only when a new virtual circuit is being **set up**. Thereafter, data packets just follow the previously-established route 虚电路网络，只在建立虚电路时选路
- **Routing(选路) & Forwarding(转发)**

Classification of Routing Algorithms

■ *Nonadaptive algorithms (Static Routing)*

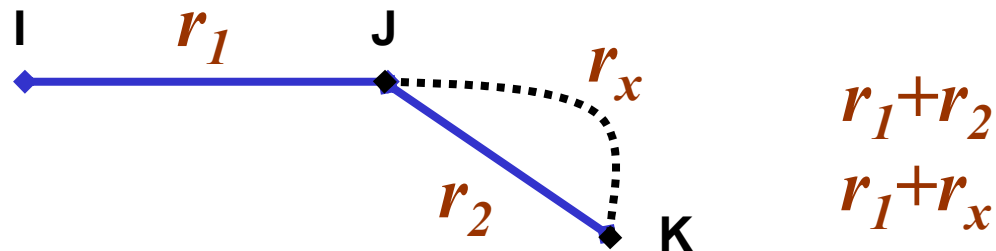
- ◆ The choice of the route to use is computed in advance, off-line, and downloaded to the routers when the network is booted
- ◆ The routing decisions are not based on measurements or estimates of the current traffic and topology

■ *Adaptive(自适应) algorithms*

- ◆ Changing their routing decisions to reflect changes in the topology, and usually the traffic as well 适应网络拓扑和业务量的变化

The Optimality Principle(最优化原则)

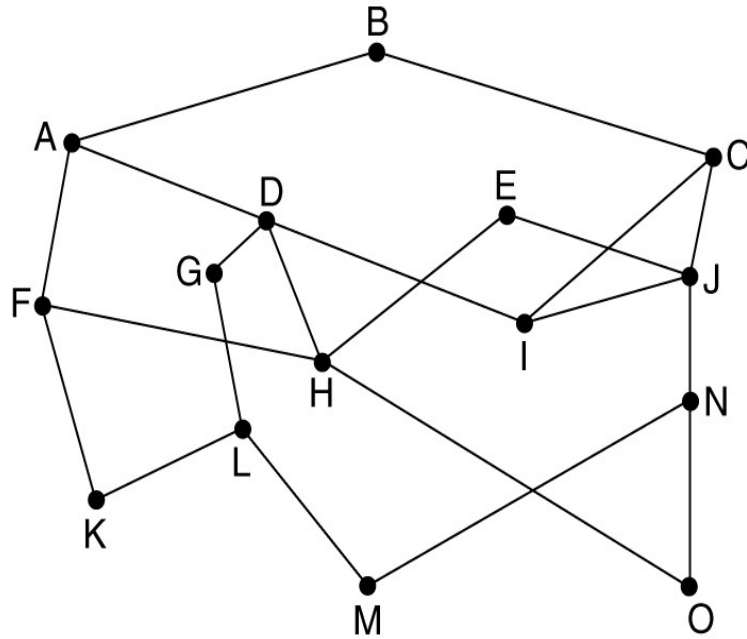
- If router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route
 - ◆ If a route r_x better than r_2 existed from J to K, it could be concatenated with r_1 to improve the route from I to K
 - ◆ Contradicting our statement that $r_1 r_2$ is optimal



Sink Tree(宿树/汇集树)

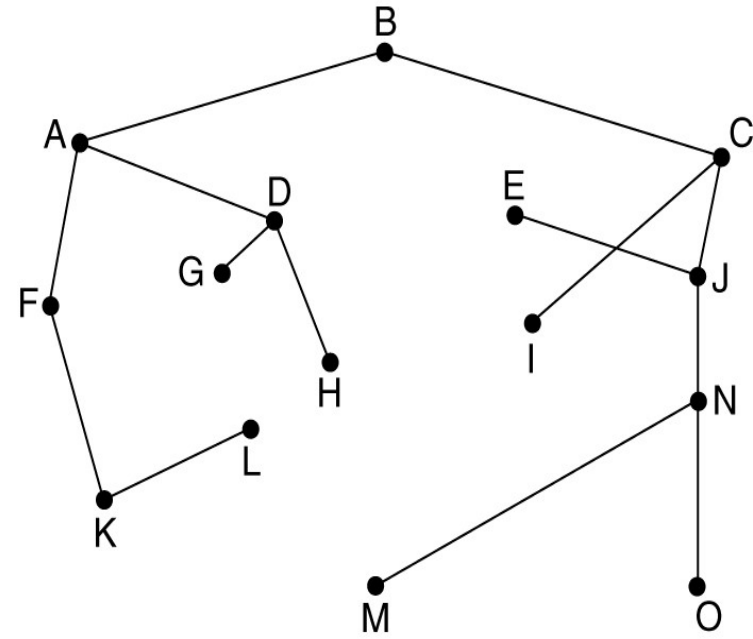
- **Optimal routes** from all sources to a destination, form a tree rooted at the destination 以目的节点为根，由所有源节点到目的节点的最优路由构成
- Does not contain any loops, so each packet will be delivered within a finite number of hops
- A **benchmark** to measure routing algorithms

Example of Sink Tree



(a)

(a) A subnet



(b)

(b) A sink tree for router B

Major Routing Strategies

- Fixed Routing(固定查表选路)
 - ◆ Single permanent route for each source to destination pair
- Flooding(洪泛/泛洪)
 - ◆ Packet sent to every neighbor
 - ◆ Incoming packets forwarded to every link except incoming link
- Random Routing(随机选路)
 - ◆ Selects one outgoing path to forward incoming packet
 - ◆ Selection can be random or round robin
- Adaptive Routing(自适应选路)
 - ◆ Routing decisions change as conditions on the network change
 - ◆ Requires information about network
 - ◆ Used by almost all packet switching networks

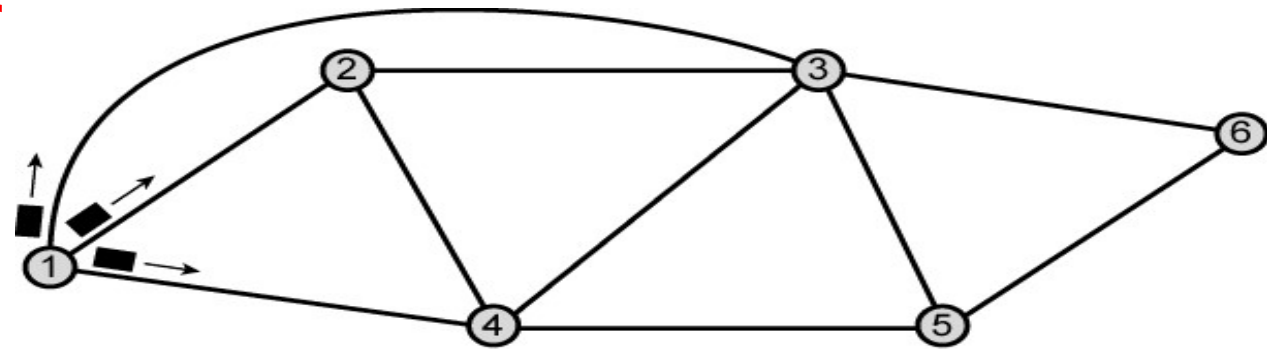
Routing Algorithms

- Concept of Routing and Routing Table
- Optimality principle
- *Flooding*
- Dijkstra's "Shortest Path" Algorithm
- Distance Vector Routing (DVR)
- Link State Routing (LSR)
- Hierarchical Routing

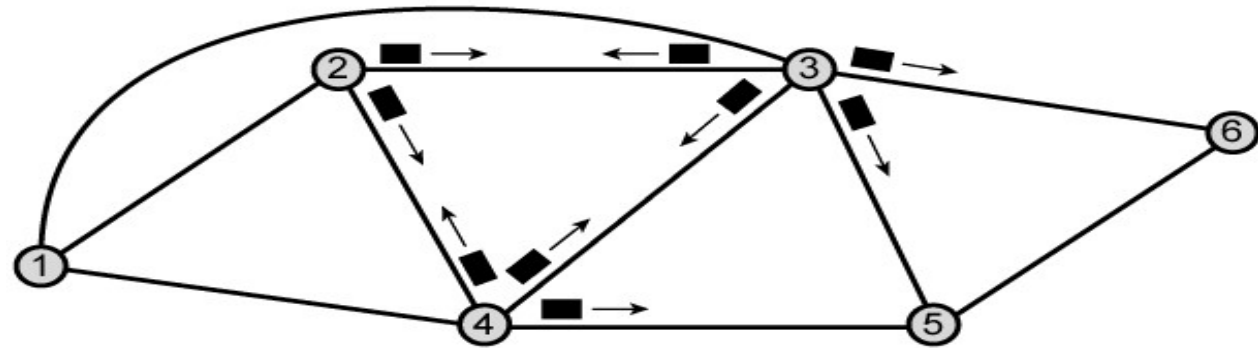
Flooding(洪泛法): Ideas

- 无需路由表
- No network information required
- Incoming packets retransmitted on every link except incoming link
- Eventually a number of copies will arrive at destination

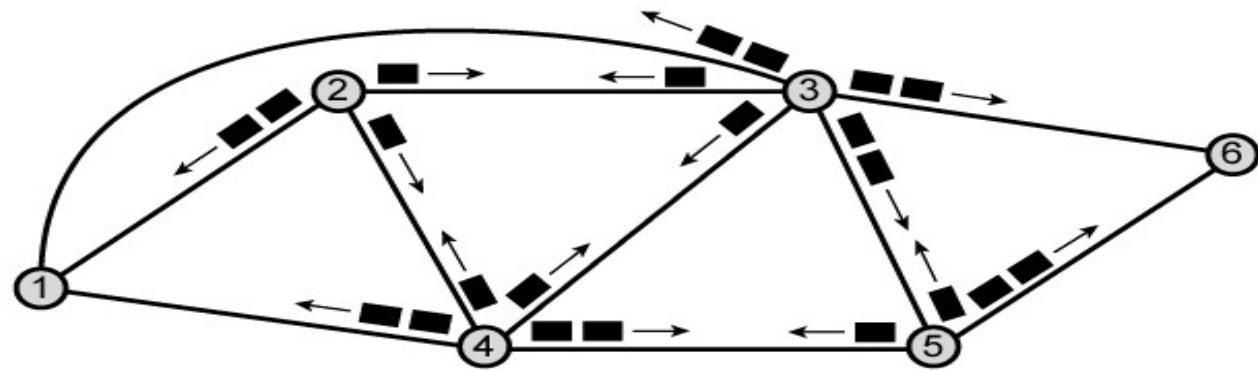
Example of Flooding



(a) First hop



(b) Second hop



(c) Third hop

Flooding: Problems and Solutions

■ Problem

- ◆ Generates vast numbers of duplicate packets

■ Solutions

- ◆ The header of each packet contains a **hop counter**(跳计数器)
 - Counter is decremented at each hop
 - When the counter reaches zero, discard the packet
 - Sender initializes the hop counter as the length of the path from source to destination, or the subnet diameter
- ◆ Source puts a **sequence number** in each packet (序列号)
 - Each router records maximum seq per source telling which seq have already been seen

Properties of Flooding

- All possible routes are tried
 - ◆ Very robust 健壯
- At least one packet will have taken minimum hop count route
- All nodes are visited
 - ◆ Useful to distribute routing information

Routing Algorithms

- Concept of Routing and Routing Table
- Optimality principle
- Flooding
- *Dijkstra's "Shortest Path" Algorithm*
- Distance Vector Routing (DVR)
- Link State Routing (LSR)
- Hierarchical Routing

Shortest Path Routing(最短路径选路)

- Build a weighted and directed graph of subnet(有向图)
 - ◆ Nodes represent routers
 - ◆ Arcs represents communication lines
- Weight
 - ◆ Hops
 - ◆ Geographic distance in kilometers
 - ◆ Mean queuing and transmission delay
 - ◆ A function of the distance, bandwidth, average traffic, communication cost, mean queue length, measured delay, and other factors
- Dijkstra Algorithm
 - ◆ Finds the shortest path between a given pair of routers

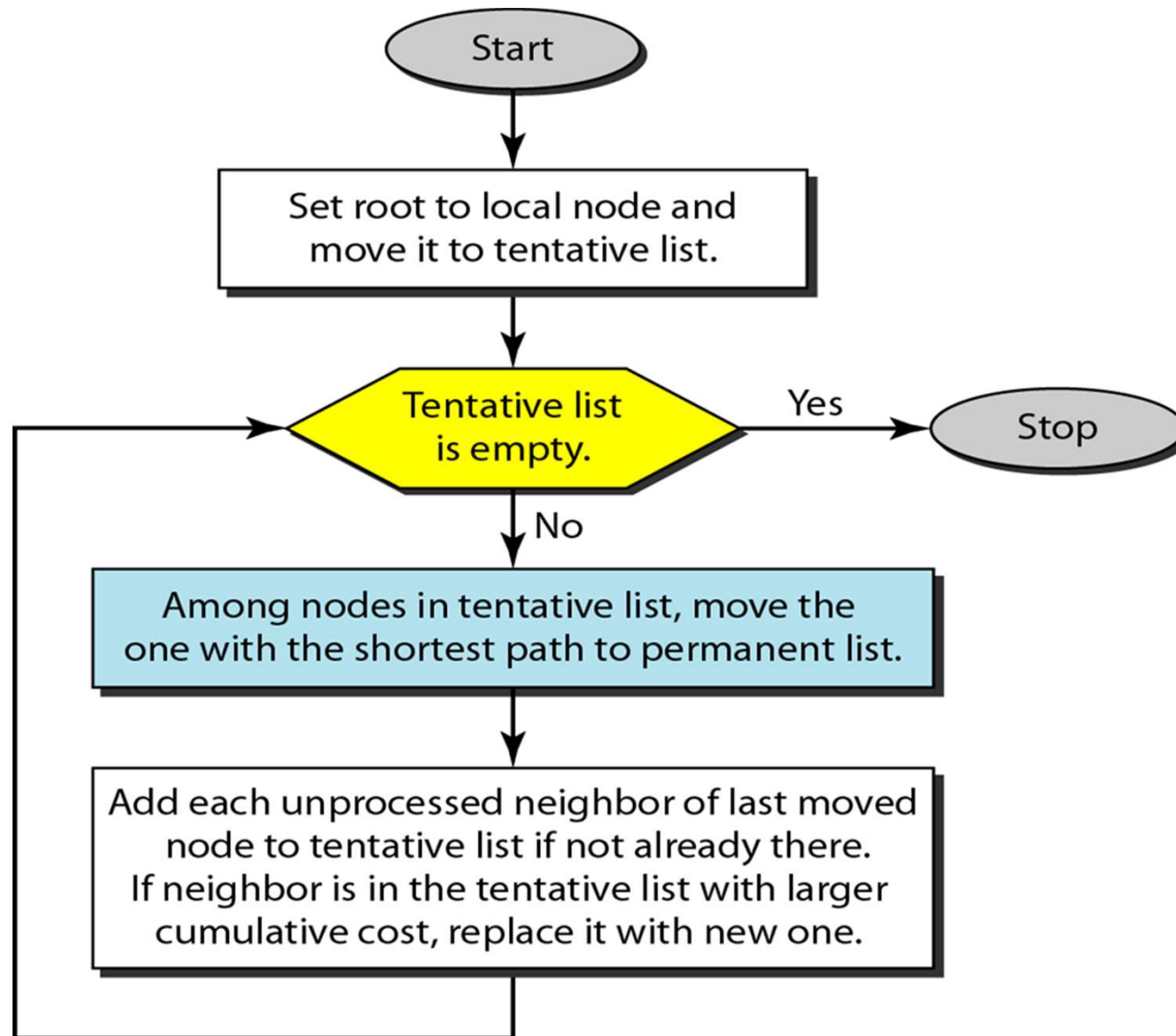
Dijkstra

- Find **shortest paths** from given source node to all other nodes

- **Routing Algorithm**

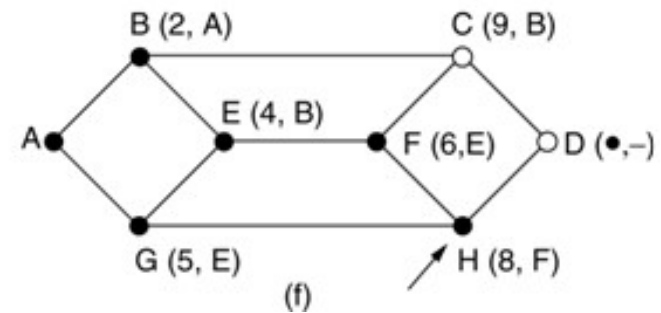
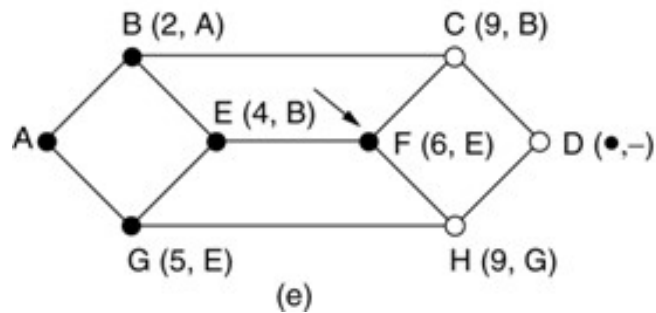
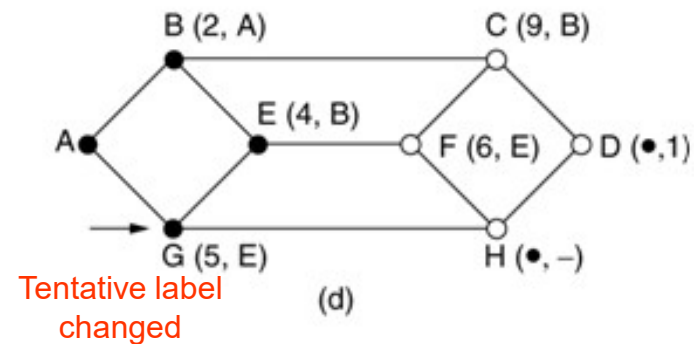
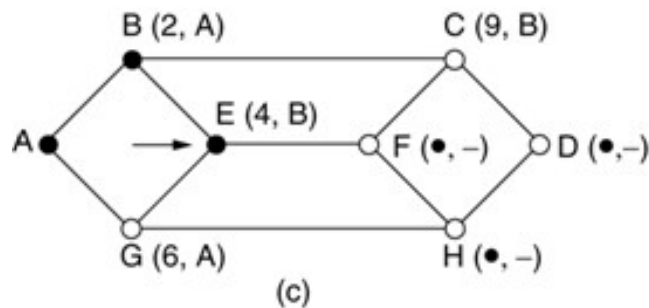
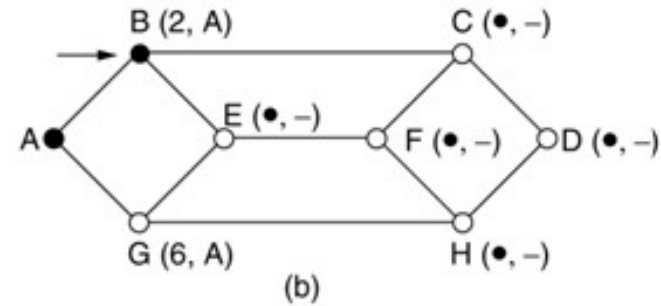
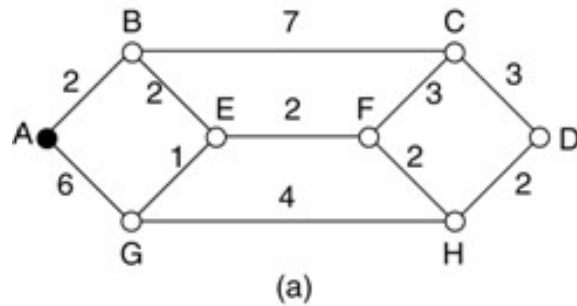
- ◆ 1. Starting with the **local node** (router): the **root** of the tree.
- ◆ 2. Assign a cost of 0 to this node and make it the first **permanent node**(永久节点).
- ◆ 3. Examine each neighbor node of the node that was the last permanent node.
- ◆ 4. Assign a **cumulative cost** (累计开销) to each node and make it **tentative** (临时节点) .
- ◆ 5. Among the list of tentative nodes
 - Find the node with the **smallest** cumulative cost and make it permanent.
 - If a node can be reached from more than one direction, Select the direction with the shortest cumulative cost.
- ◆ 6. Repeat steps 3 to 5 until **every** node becomes permanent.

Flow of Dijkstra

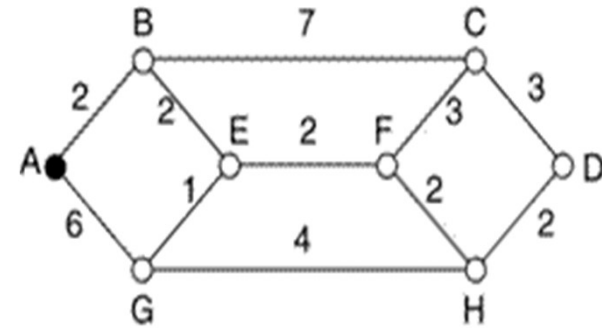


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Example of Dijkstra Algorithm



Example of Dijkstra

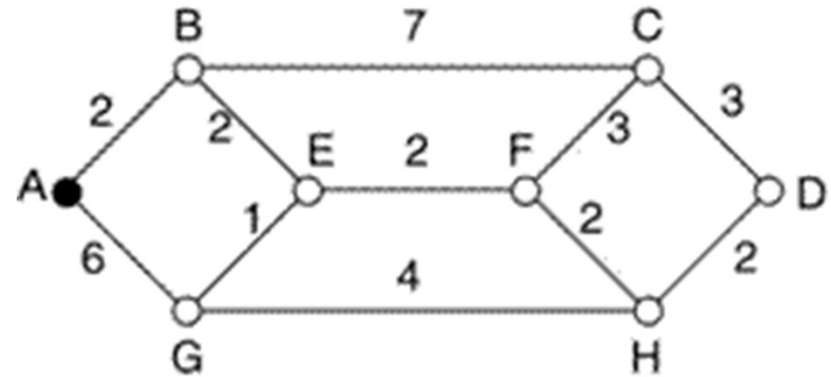


A	B	C	D	E	F	G	H	Permanent Node
∞	∞	∞	∞	∞	∞	∞	∞	
0	∞	∞	∞	∞	∞	∞	∞	A
0	2, A	∞	∞	∞	∞	6, A	∞	B
0	---	9, B	∞	4, B	∞	6, A	∞	E
0	---	9, B	∞	---	6, E	5, E	∞	G
0	---	9, B	∞	---	6, E	---	9, G	F
0	---	9, B	∞	---	---	---	8, F	H
0	---	9, B	10, H	---	---	---	---	C
0	---	---	10, H	---	---	---	---	D

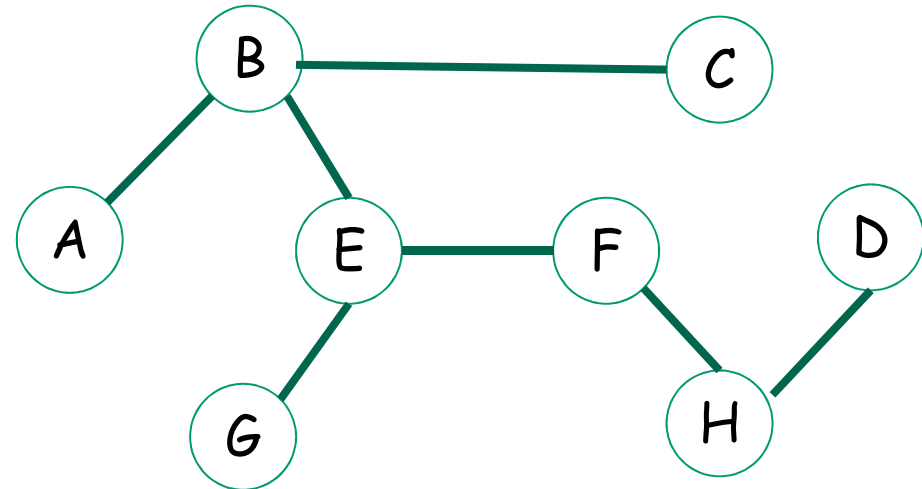
What we get by using Dijkstra?

A's routing table

Dest.	Cost	Next Hop
A	0	--
B	2	B
C	9	B
D	10	B
E	4	B
F	6	B
G	5	B
H	9	B



A's shortest path tree



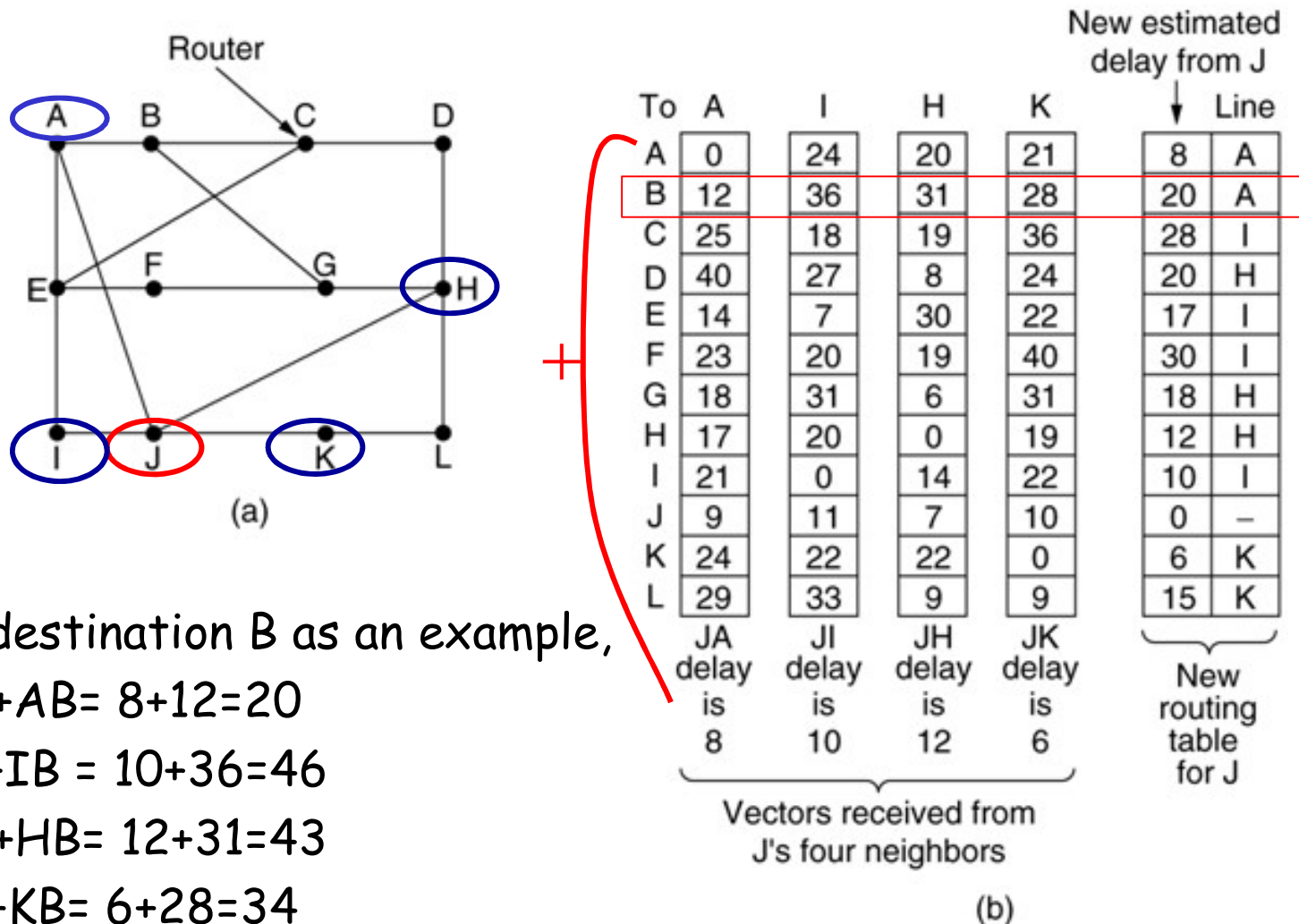
Routing Algorithms

- Concept of Routing and Routing Table
- Optimality principle
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- Dijkstra's "Shortest Path" Algorithm
- Distance Vector Routing (DVR)
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Distance Vector Routing(距离矢量选路)

- Each router maintains a routing table (Vector)
 - ◆ The best known distance to each destination(距离)
 - The distance might be number of hops, time delay in milliseconds, ...
 - ◆ Which line to use (下一跳/接口)
- The table is updated by exchanging information with neighbors
 - ◆ Routing information sent: (Destination, Distance)
 - Periodically (RIP:30s)
 - ◆ When routing information received
 - Router updates local table if receive a better route
 - Refreshes existing routes (old routing table is not used in the calculation.)
 - ◆ Delete routing table items if they time out (RIP: 180s)

DVR: Example in textbook



Taking destination B as an example,

$$JB = JA + AB = 8 + 12 = 20$$

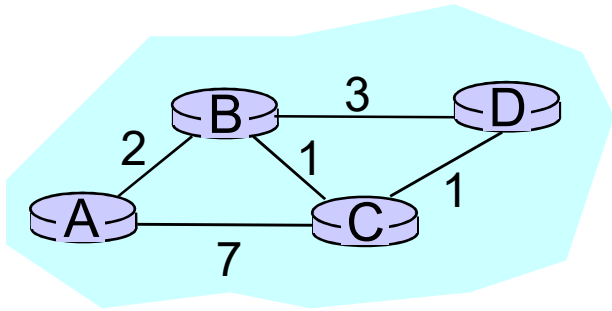
$$JB = JI + IB = 10 + 36 = 46$$

$$JB = JH + HB = 12 + 31 = 43$$

$$JB = JK + KB = 6 + 28 = 34$$

Thus, A is selected as next hop.

DVR Example 2: Initial Status



Node A

Dest.	Cost	NextHop
B	2	B
C	7	C
D	∞	-

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	3	D

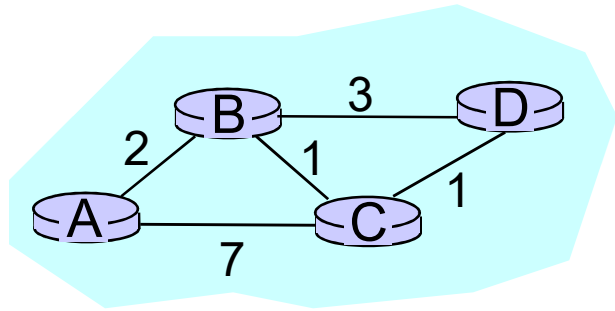
Node C

Dest.	Cost	NextHop
A	7	A
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	∞	-
B	3	B
C	1	C

Example 2: 1st Iteration ($C \rightarrow A$)



Node A

Dest.	Cost	NextHop
B	2	B
C	7	C
D	8	C

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	3	D

$$D(A, D) = D(A, C) + D(C, D) = 7 + 1 = 8$$

(D(C,A), D(C,B), D(C,D))

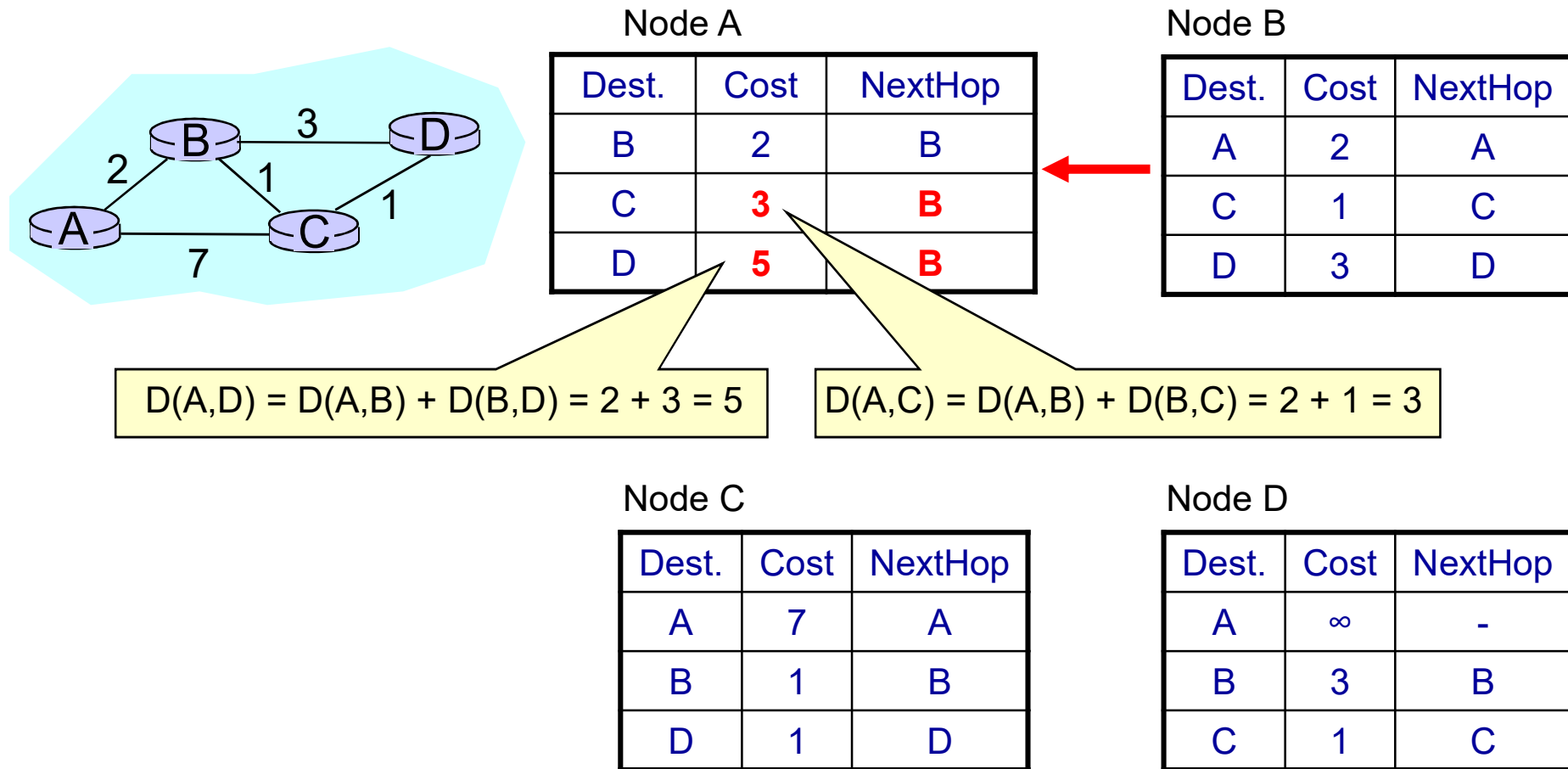
Node C

Dest.	Cost	NextHop
A	7	A
B	1	B
D	1	D

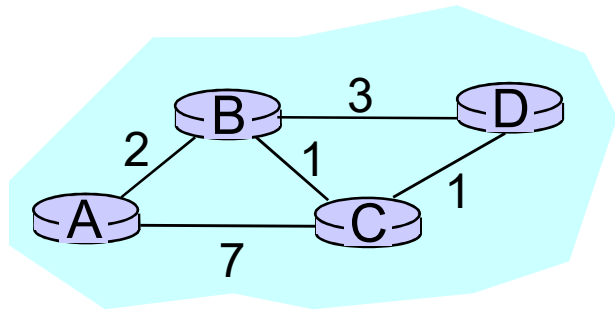
Node D

Dest.	Cost	NextHop
A	∞	-
B	3	B
C	1	C

Example 2: 1st Iteration ($B \rightarrow A$, $C \rightarrow A$)



Example 2: End of 1st Iteration



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	5	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	2	C

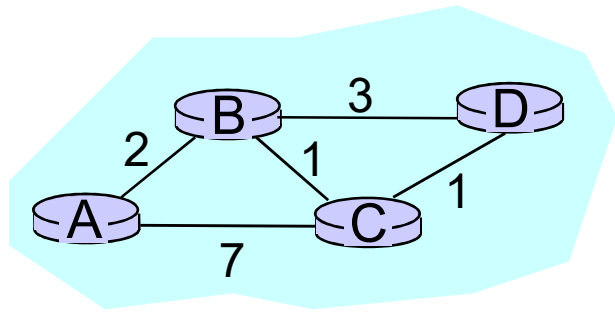
Node C

Dest.	Cost	NextHop
A	3	B
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	5	B
B	2	C
C	1	C

Example 2: End of 2nd Iteration



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	4	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	2	C

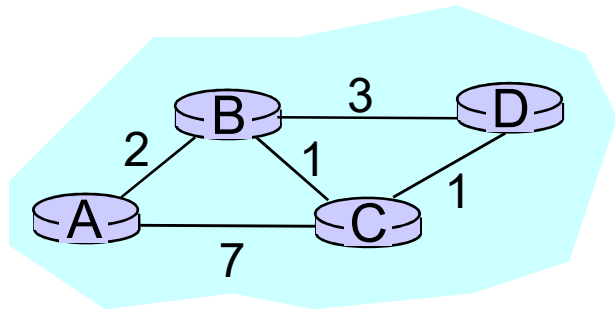
Node C

Dest.	Cost	NextHop
A	3	B
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	4	C
B	2	C
C	1	C

Example 2: End of 3rd Iteration



Node A

Dest.	Cost	NextHop
B	2	B
C	3	B
D	4	B

Node B

Dest.	Cost	NextHop
A	2	A
C	1	C
D	2	C

Node C

Dest.	Cost	NextHop
A	3	B
B	1	B
D	1	D

Node D

Dest.	Cost	NextHop
A	4	C
B	2	C
C	1	C

Nothing changes → algorithm terminates

Features and Problems of DVR

■ 3 key features

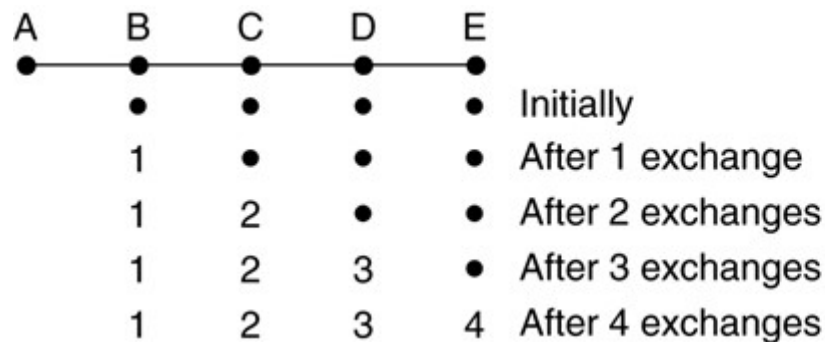
- ◆ Sharing knowledge about **the entire network** with neighbours.
- ◆ Sharing only with **immediate neighbours**. It sends whatever knowledge it has through all its interface.
- ◆ Sharing at **regular intervals**.

■ Drawbacks:

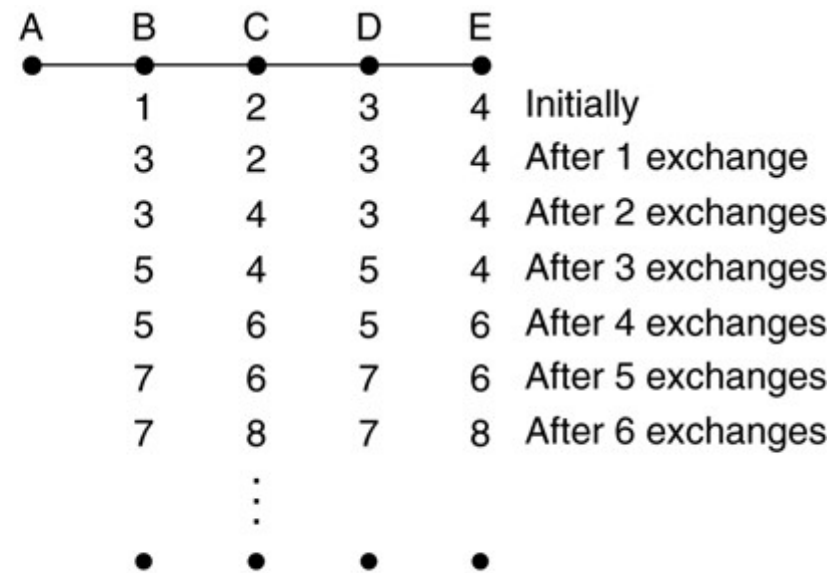
- ◆ Slow convergence(收敛慢)
- ◆ Reacts rapidly to good news, but leisurely to bad news

DVR: Count-to-Infinity(无穷计数) Problem

- Also regarded as "Infinite Loop Problem"



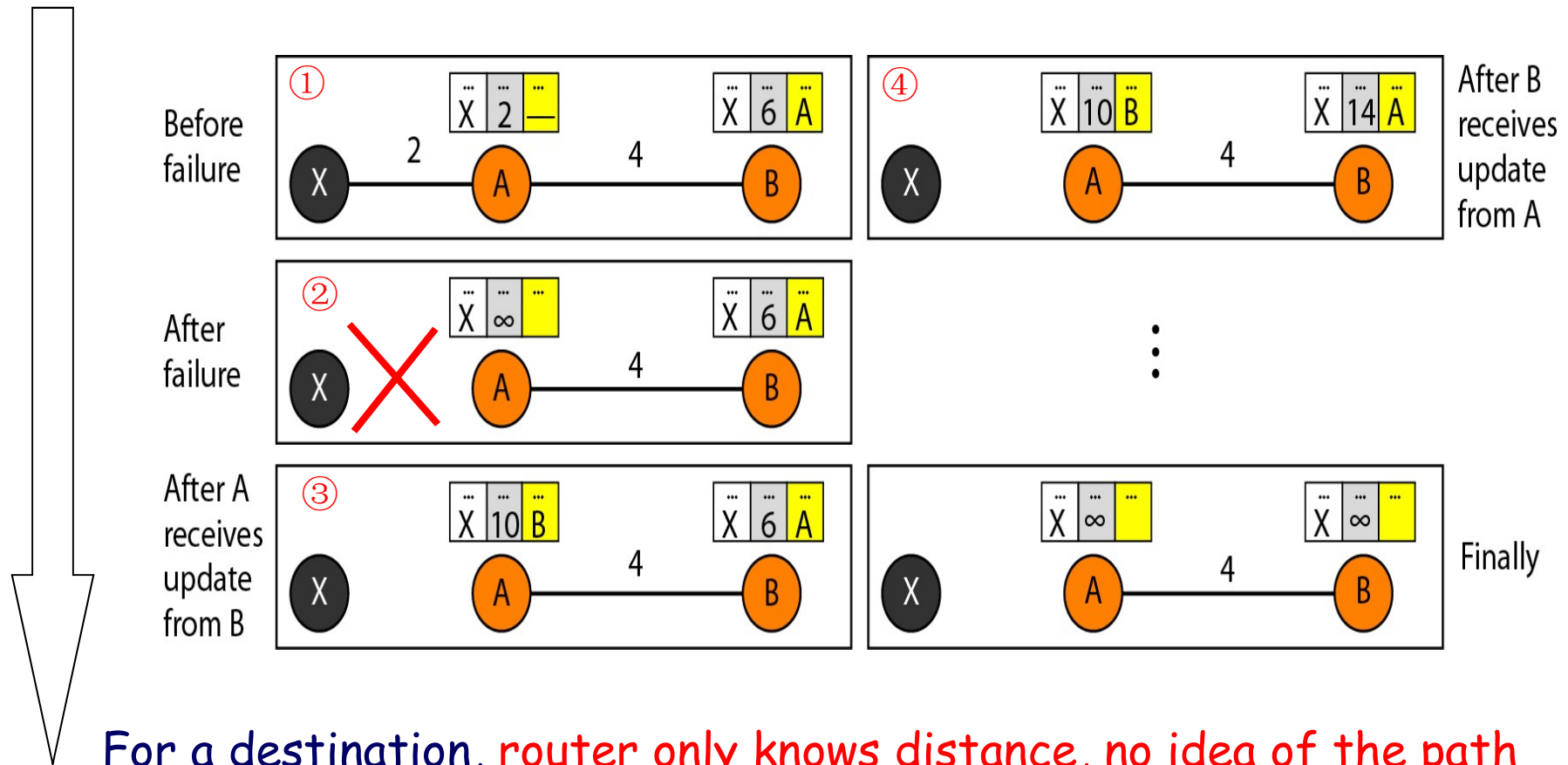
(a)



(b)

Bad news travels slowly

DVR: Infinite Loop Example



For a destination, router only knows distance, no idea of the path
(只知距离，不知路径)

DVR: Summary

■ Used in

- ◆ RIP (Routing Information Protocol)
- ◆ Cisco EIGRP (Enhanced Interior Gateway Routing Protocol)

■ Problem of DVR

- ◆ When X tells Y that it has a path somewhere, Y has no way of knowing whether itself is on the path(whether there is a loop)
- ◆ Rumor based routing

■ Improvement: Path Vector Routing

- ◆ BGP-4 (Border Gateway Protocol)
- ◆ IDRP (Inter-domain Routing Protocol)

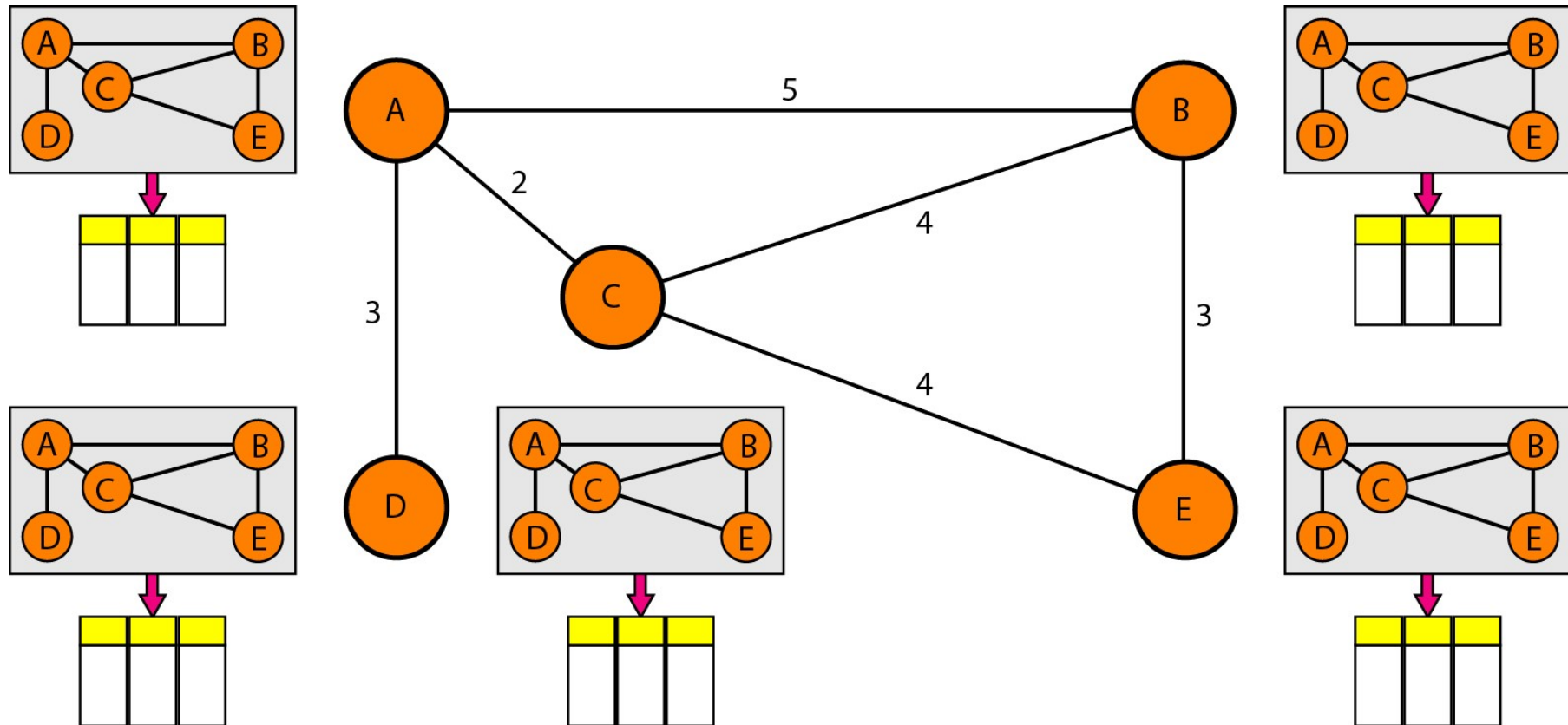
Routing Algorithms

- Concept of Routing and Routing Table
- Optimality principle
- Flooding
- Dijkstra's "Shortest Path" Algorithm
- Distance Vector Routing (DVR)
- Link State Routing (LSR)
- Hierarchical Routing

Link State Routing(链路状态选路)

- In LSR, each router shares its knowledge about the its neighbourhood with **every other** routers in the network.
- However, in LSR, the Link-State Packet (LSP) defines the **best known network topology** is sent to every routers after it is constructed locally.
- 3 keys to understand how LSR works:
 - ◆ Sharing knowledge about **network topology**.
 - ◆ Sharing **with every other routers**.
 - ◆ Sharing **when there is a change**.
- From the received LSPs and knowledge of **entire topology**, a router can then calculate the **shortest path** from itself to every other routers in the area.

LSR: What routers know?



The routers have knowledge of **entire** topology

Link State Routing: Operations

- Each router must do the following
 1. Discover its neighbors, learn their network address
 2. Measure the delay or cost to each of its neighbors
 3. Construct a packet (**LSP**: Link-state Packet, 链路状态包) telling all it has just learned
 4. Send LSP to **all other routers** (not only neighbors) reliably
 5. Compute the shortest path to every other router

Step1: Learning about the Neighbors

■ HELLO packet

- ◆ For a broadcast network, Broadcasting a special HELLO packet
- ◆ For a point-to-point network, Unicast HELLO to neighbors

■ Neighbors send back a reply telling who it is

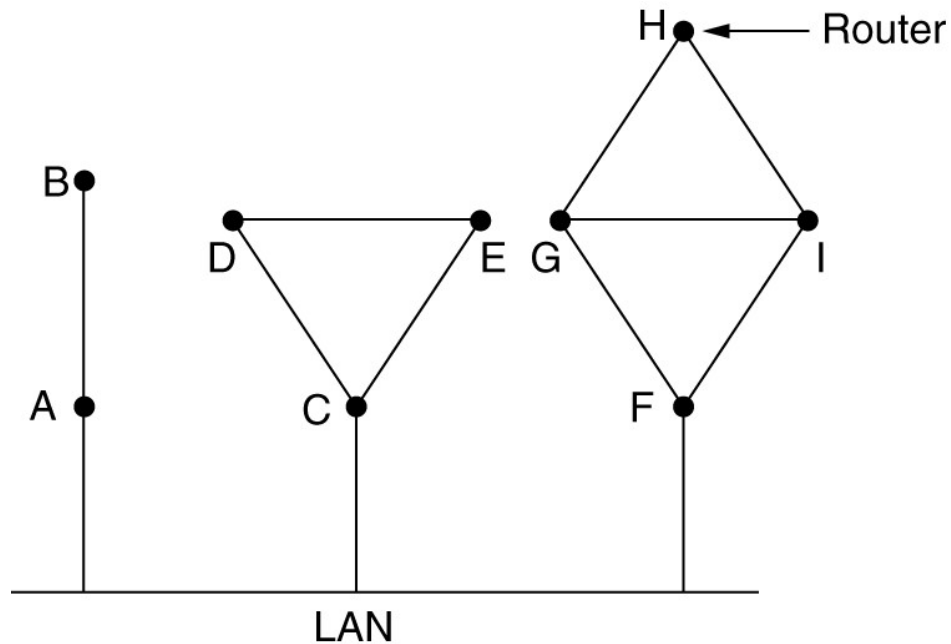
- ◆ Names (Route IDs) must be globally unique

■ Simplify the topology

- ◆ When two or more routers are connected by a LAN or other multi-access network

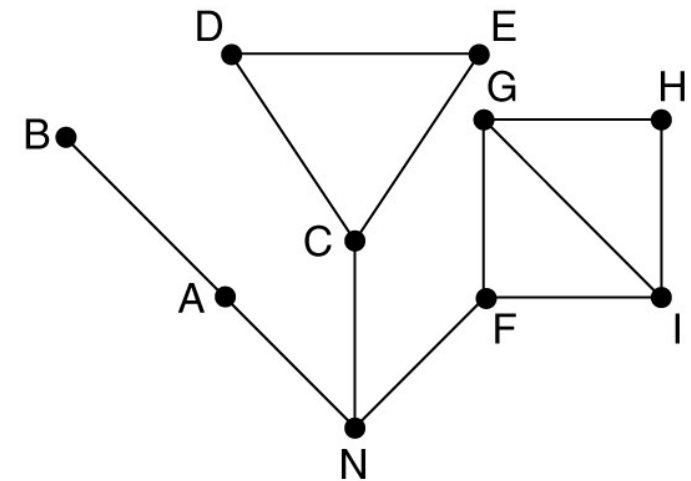
How to model the LAN

A LAN is considered as a Node(虚拟节点)



(a)

(a) Nine routers and a LAN.

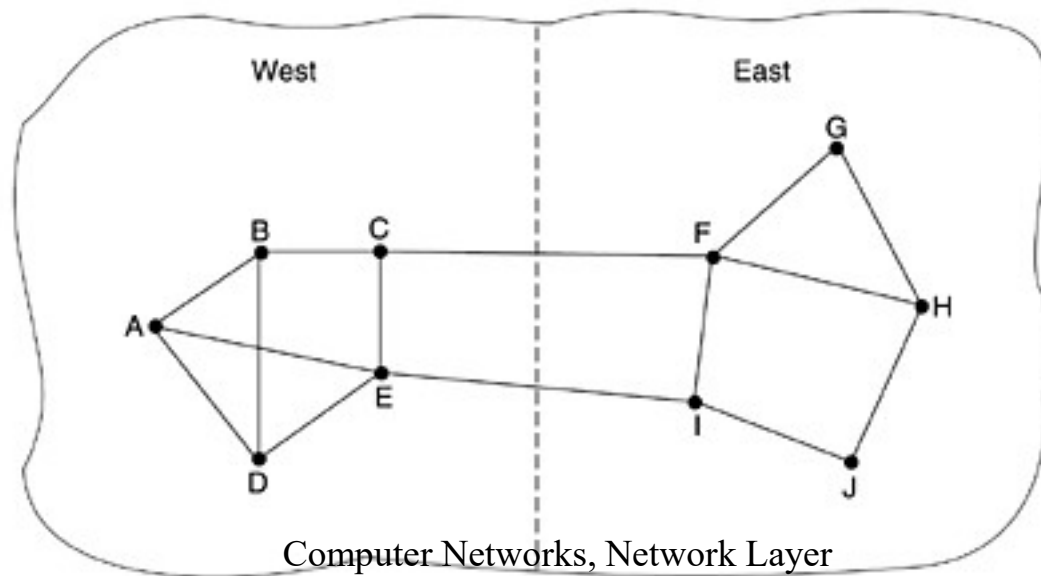


(b)

(b) A graph model of (a)

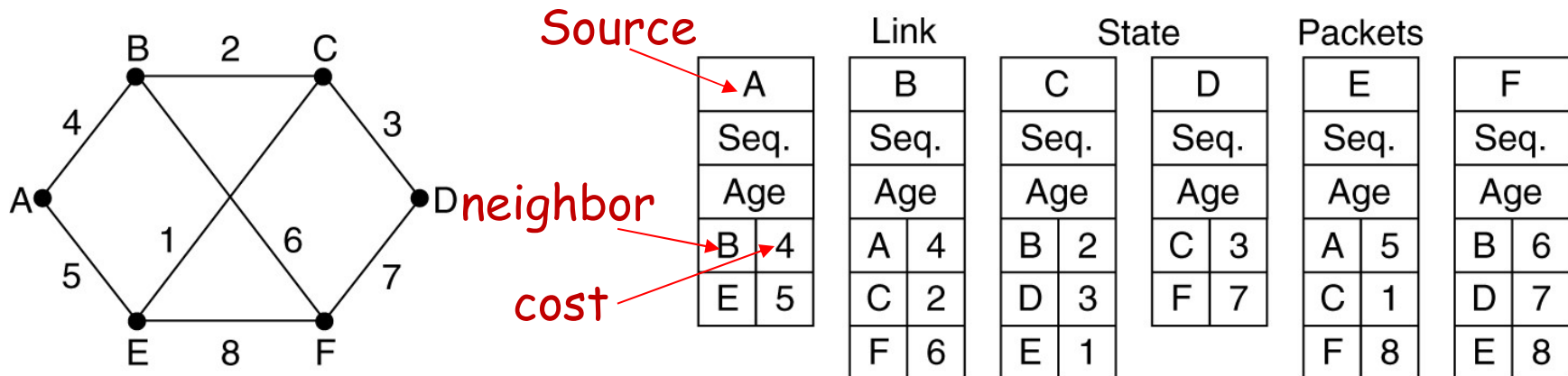
Step 2: Measuring Line Cost

- The delay to each of its neighbors
 - ◆ Measuring the round-trip time (ECHO Packet)
 - ◆ Bandwidth
 - ◆ An issue is whether to take the load into account when measuring the delay? (routing tables may oscillate, leading to erratic routing and many potential problems)



路由振荡，
导致不稳定

Step 3: Building Link State Packets



(a) A subnet. (b) The link state packets for this subnet

■ When to build LSP

- ◆ Build LSP **periodically**
- ◆ Build LSP when some **significant event** occurs
 - such as a line or neighbor going down or coming back up again or changing its properties appreciably

About Link State Packets

- Sequence number: used to check whether the received LSP is new
 - ◆ If LSP is new, forward with flooding
 - ◆ If LSP is a duplicate, it is discarded
 - ◆ If lower than the highest one seen so far ever arrives, it is rejected as being obsolete
 - ◆ If a router is down, and then restart, ...
- Age: used to delete looped LSP
 - ◆ If a router is down
 - ◆ If a router changed its routerID
 - ◆ If a router need to delete an LSP

Step 4: Distributing the Link State Packets

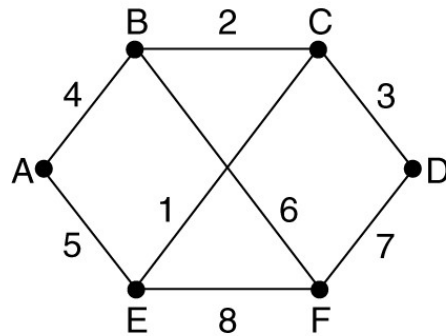
■ Responsibility

- ◆ Distributing the LSP **reliably**

■ The basic distribution algorithm

- ◆ Use **flooding** to distribute LSPs
- ◆ Guard against errors on the router-router lines, all LSP are acknowledged

LSP: Example



(a)

Link		State		Packets	
A		B		C	
Seq.		Seq.		Seq.	
Age		Age		Age	
B	4	A	4	B	2
E	5	C	2	D	3
		F	6	E	1

(b)

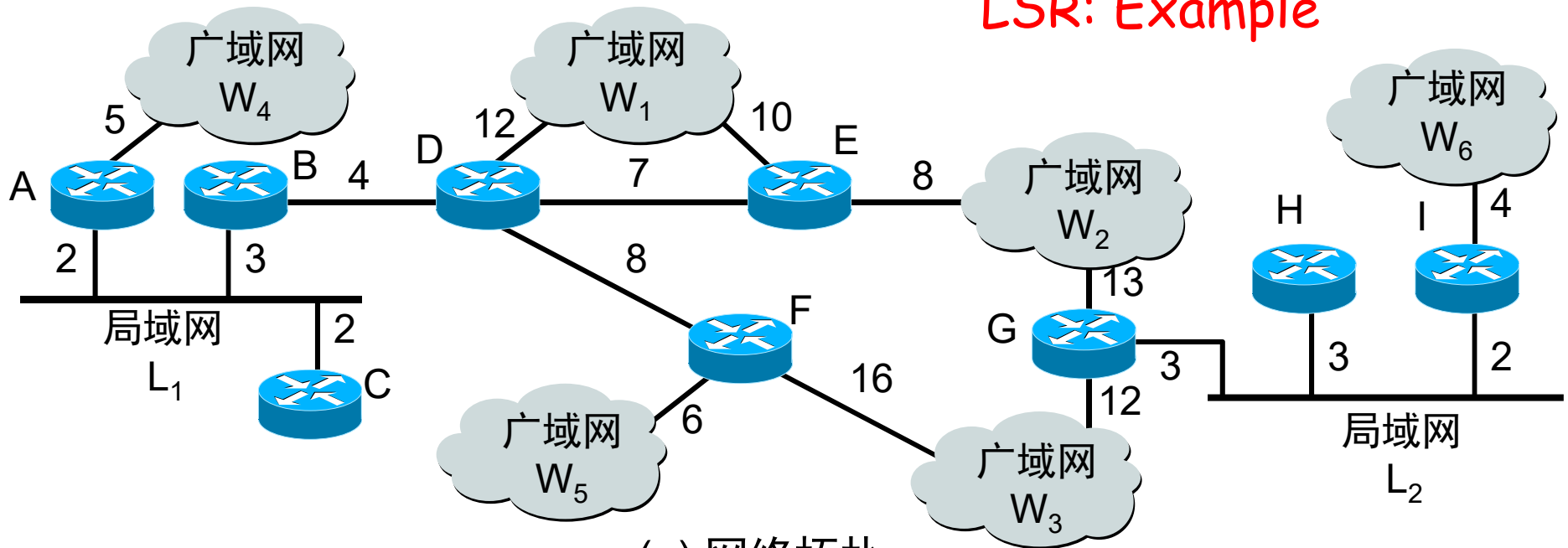
Source	Seq.	Age	Send flags			ACK flags			Data
			A	C	F	A	C	F	
A	21	60	0	1	1	1	0	0	
F	21	60	1	1	0	0	0	1	
E	21	59	0	1	0	1	0	1	
C	20	60	1	0	1	0	1	0	
D	21	59	1	0	0	0	1	1	

Record of LSPs received by B

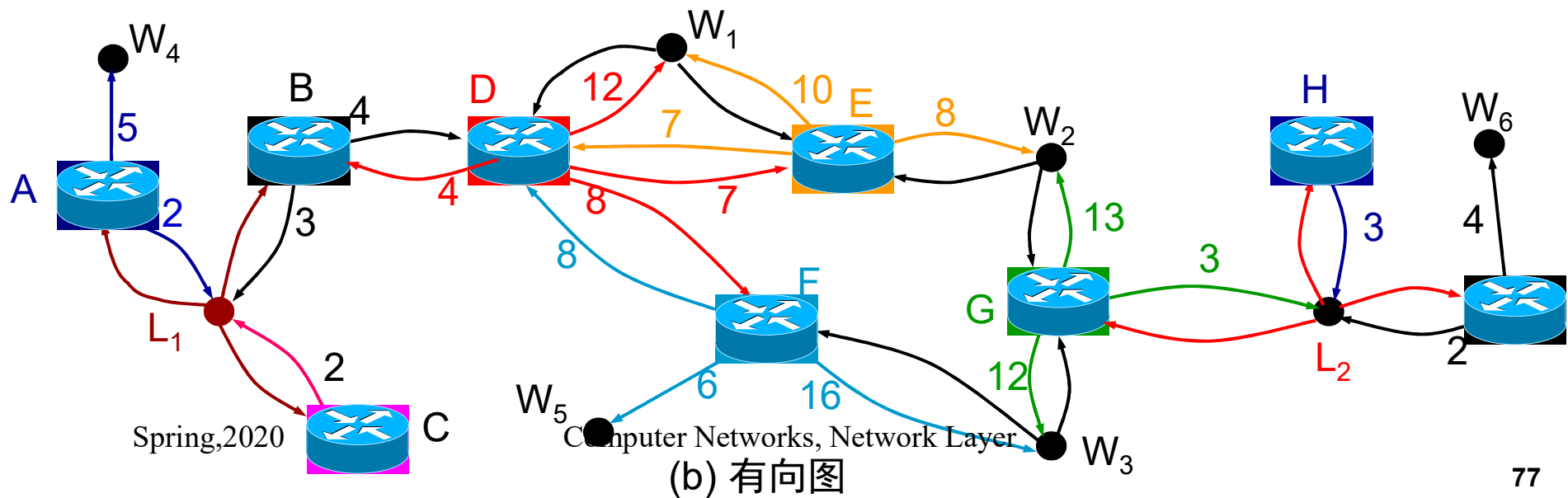
Step 5: Computing the New Route

- Once a router has accumulated a full set of LSPs, it can construct the entire graph
 - ◆ Every link is represented twice, once for each direction. The two values can be averaged or used separately.
- Run **Dijkstra's algorithm** locally to construct the shortest path to all possible destinations
- The results of this algorithm can be installed in the routing tables

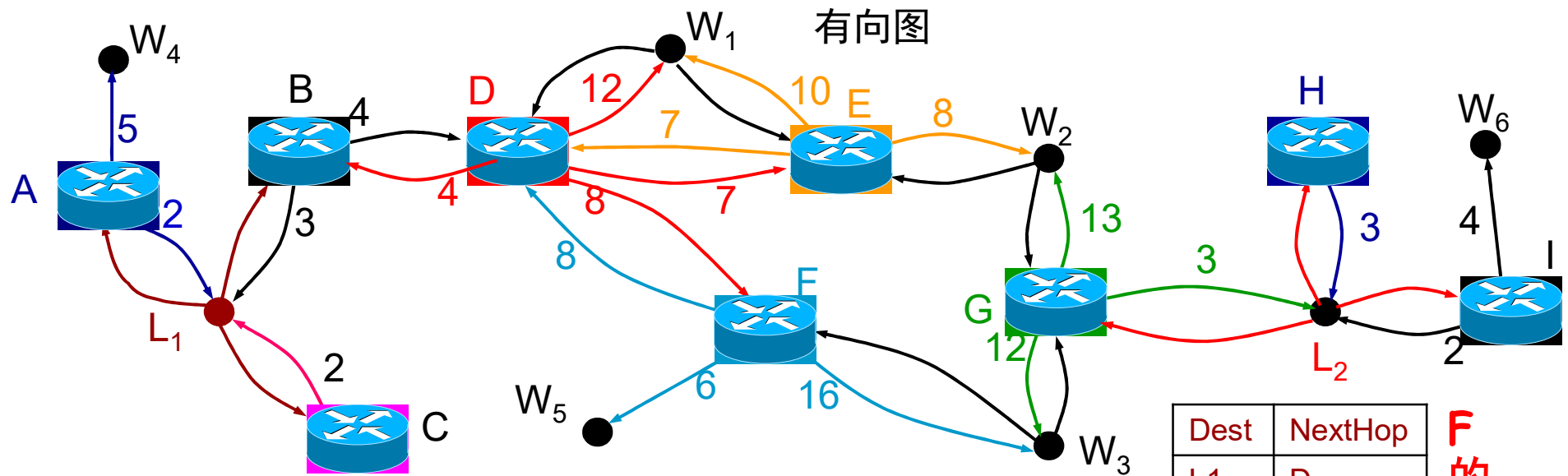
LSR: Example



(a) 网络拓扑



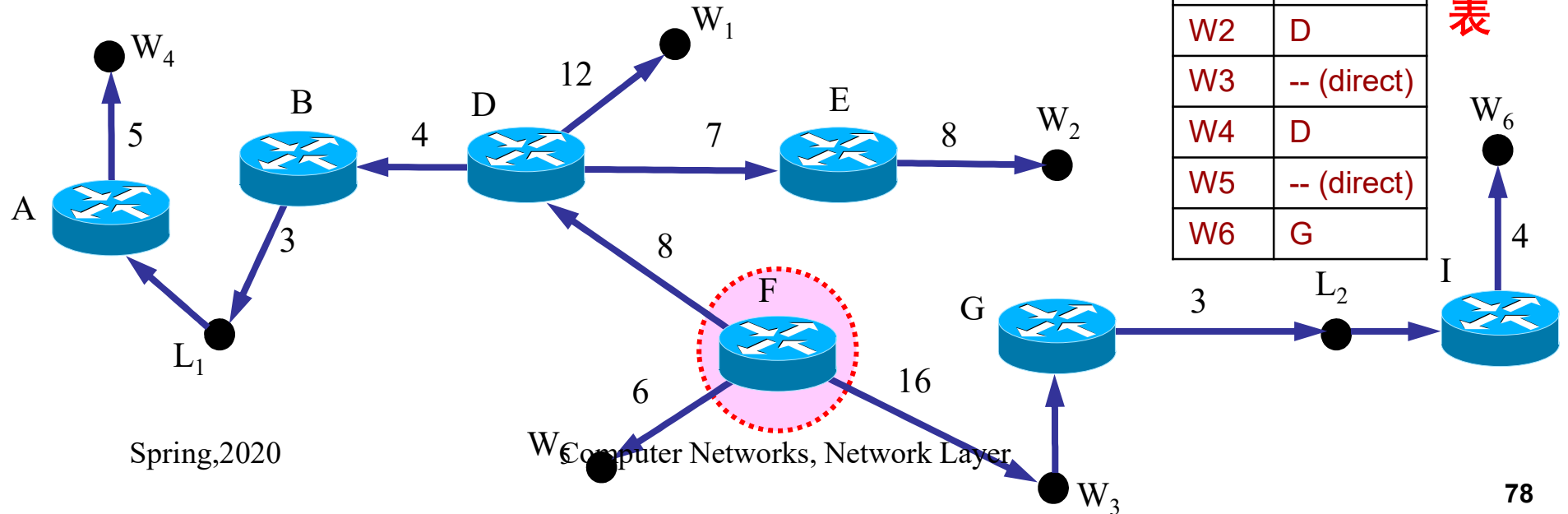
(b) 有向图



Dest	NextHop
L1	D
L2	G
W1	D
W2	D
W3	-- (direct)
W4	D
W5	-- (direct)
W6	G

F 的路由表

以路由器F为根的最短路径树



Spring, 2020

Computer Networks, Network Layer

Adoptions of Link State Routing

■ IS-IS (Intermediate System-Intermediate System):

- ◆ support multiple network layer protocols at the same time
- ◆ DECnet, ISO CLNP, IP, AppleTalk, Novell NLSP (IPX)

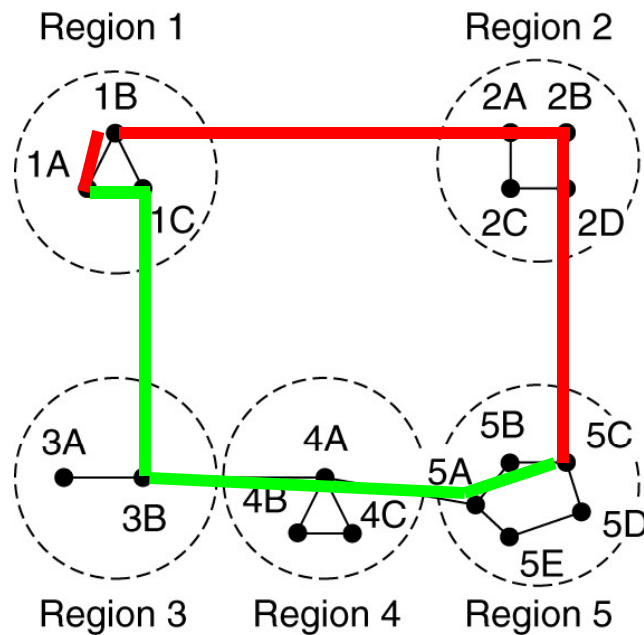
■ OSPF (Open Shortest Path First)

- ◆ designed several years after IS-IS
- ◆ Only for IP

Routing Algorithms

- Concept of Routing and Routing Table
- Optimality principle
- Flooding
- Dijkstra's "Shortest Path" Algorithm
- Distance Vector Routing (DVR)
- Link State Routing (LSR)
- Hierarchical Routing

Hierarchical Routing(层次选路/分级选路) (1)



(a)

Full table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	—	—
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

May take
sub-optimal
route
(次优路由)

(c)

Hierarchical Routing (2)

Consider a subnet with 720 routers.

- If there is no hierarchy
 - ◆ 720 routing table entries
- If the subnet is partitioned into 24 regions of 30 routers each (Total:53)
 - ◆ 30 local entries
 - ◆ 23 remote entries
- If a three-level hierarchy is chosen, with 8 clusters, each containing 9 regions of 10 routers (Total: 25)
 - ◆ 10 entries for local routers
 - ◆ 8 entries for routing to other regions within its own cluster
 - ◆ 7 entries for distant clusters

Summary of Routing Algorithms

■ Distance Vector Routing vs Link State Routing

	DVR	LSR
What information to share	routing table(destination and distance)	network topology (LSP)
Whom to share with	neighbors	All routers in the network
When to share	periodically	When there is change
What does router knows	distance, not path	Entire network topology
Count-to-infinity problem?	Yes	No

■ Hierarchical Routing(分级选路): 简化路由表

◆ 域内: 选路到路由器; 域间: 选路到域



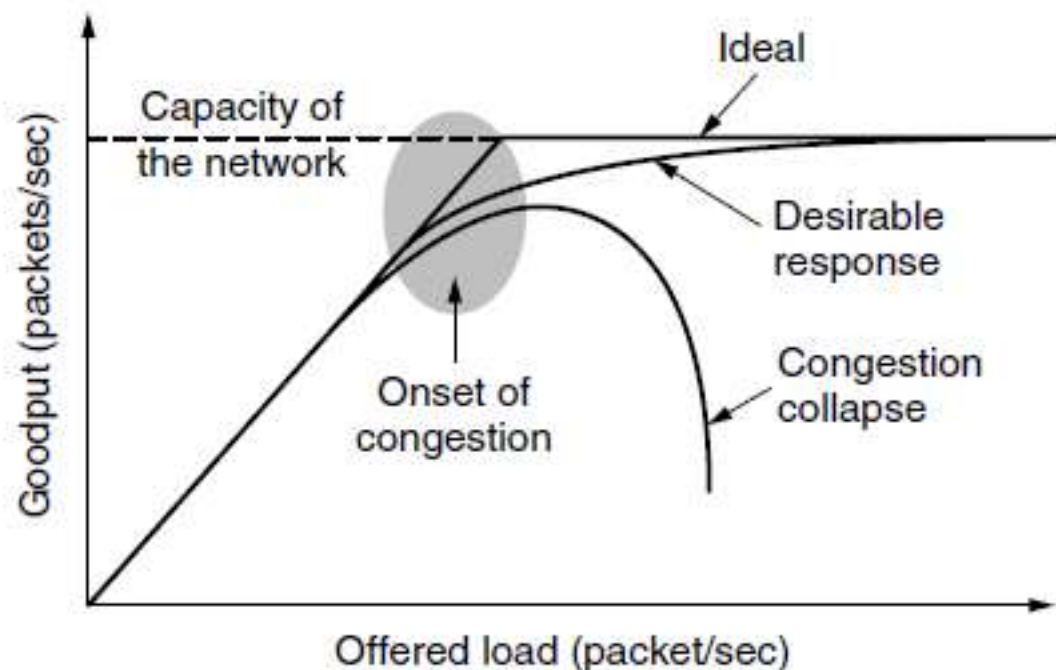
Outline

- Introduction to Network Layer
- Routing Algorithms
- Congestion Control and Internetworking
 - ◆ Congestion Control Algorithms
 - ◆ Quality of Service
 - ◆ Internetworking
- The network layer in the Internet

What is Congestion(拥塞)?

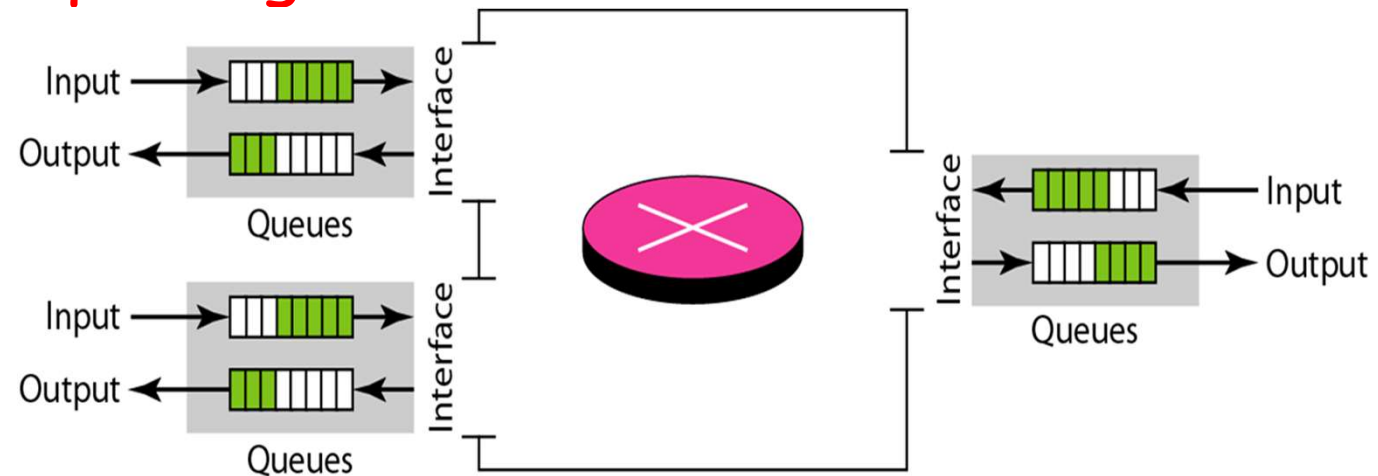
- Informally, “too many sources sending too much data too fast for **network** to handle” (资源不足)
- Congestion may occur if the **load** on the network (the number of packets sent to the network) is **greater than** the **capacity** of the network (the number of packets a network can handle)

When **too much traffic** is offered, congestion sets in and performance degrades sharply



Reasons of Congestion

- Congestion happens in any system that involves waiting or **queuing**



- If the rate of packet arrival is higher than the packet processing rate, the input queues become longer and longer
- If the packet departure rate is lesser than the packet processing, the output queues become longer and longer.

Symptoms of Congestion

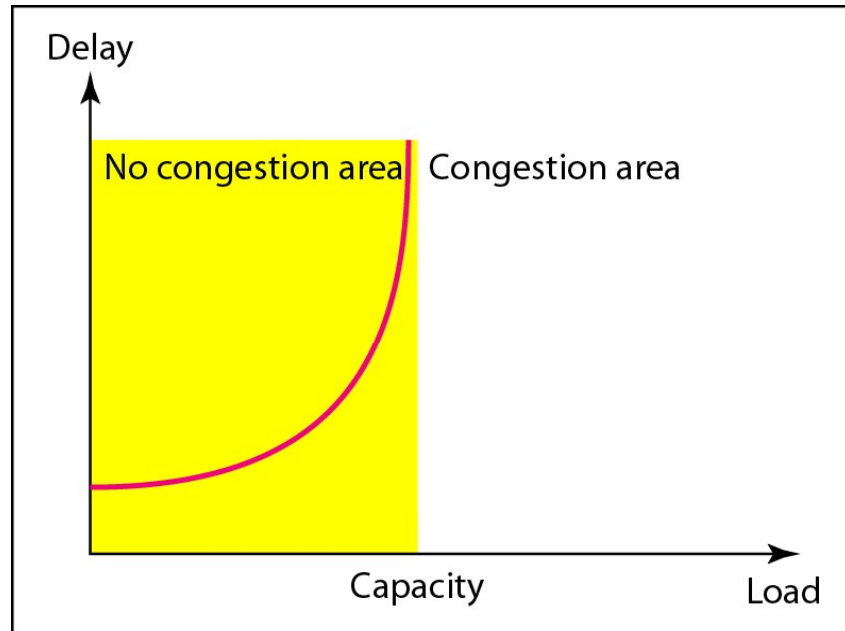
■ 导致拥塞的次要因素:

- ◆ CPU处理速度慢
- ◆ 链路带宽低
- ◆ 缓冲队列短

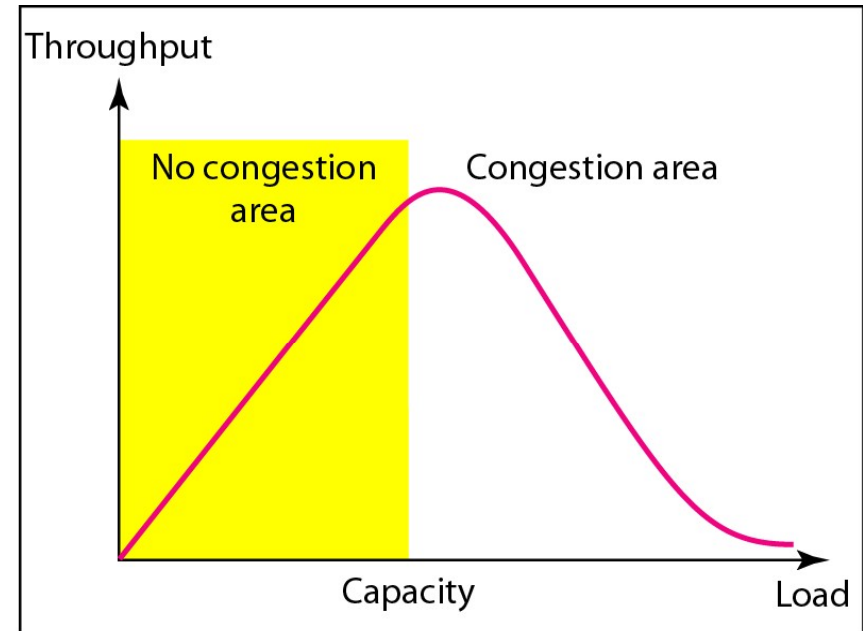
■ Symptoms:

- ◆ Long delays (Queuing in router buffers)
- ◆ Lost packets (Buffer overflow at routers)

Network performance: delay(时延) and throughput(吞吐量)



a. Delay as a function of load

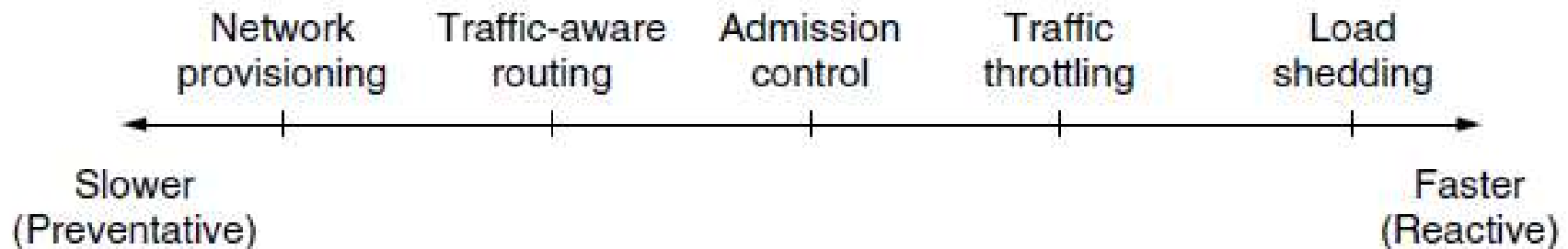


b. Throughput as a function of load

- Congestion Control involves two factors that measure the performance of a network:
 - ◆ **Delay**: queuing delay, processing delay and propagation delay
 - ◆ **Throughput**: number of packets passing through in a unit of time

Approaches to Congestion Control

- Two types of solutions
- Increasing the resources
 - ◆ Network provisioning(网络提供)
 - ◆ Traffic aware-routing(业务量感知的路由)
- Decreasing the load
 - ◆ Admission control(接纳控制)
 - ◆ Traffic throttling(业务量减速)
 - ◆ Load shedding(负载掉落)



Network provisioning (网络提供)

- 规划、调整网络，以适应业务量(负载)
- To build a network that is well matched to the traffic that it carries
- Resources are added dynamically when there is serious congestion
- Links and routers that are regularly heavily utilized are upgraded at the earliest opportunity

Traffic-aware routing(业务量感知的选路)

■ Normal routing schemes

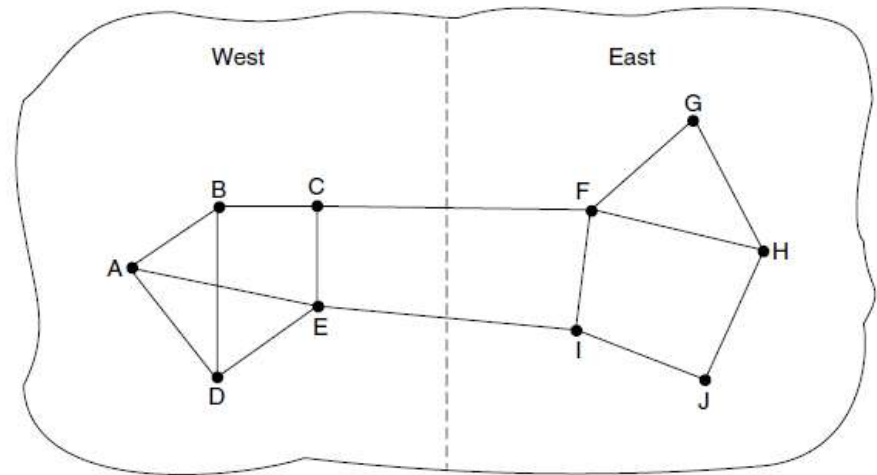
- ◆ Used **fixed link weights**, adapted to changes in topology, **not** to changes in load

■ Routes can be tailored to traffic patterns(基于业务量模式选路)

- ◆ **Link weight**: a function of link bandwidth, propagation delay, plus measured load or average queuing delay

- ◆ Taking **load** into account to shift traffic away from hotspots paths

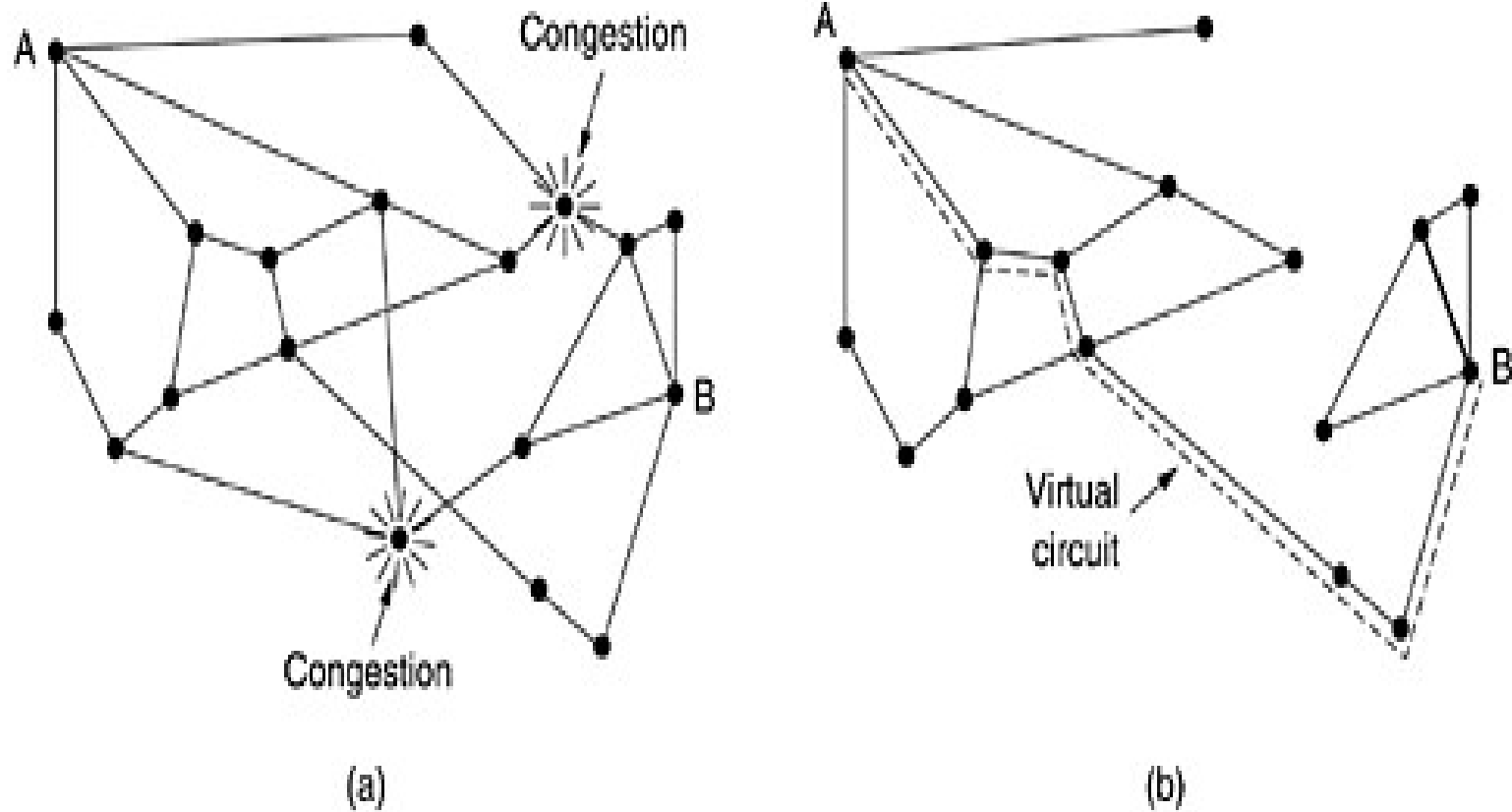
■ May lead to routing oscillations(路由震荡, 不稳定)



Admission control(接纳控制)

- Widely used in virtual circuit networks
- Idea: do not set up a new virtual circuit unless the network can carry the added traffic without becoming congested. 没有足够资源，就不允许建立VC
- Key issue: how to describe the traffic?
- Armed with traffic descriptions(by leaky bucket 漏桶 or token bucket 令牌桶), the network can decide whether to admit the new virtual circuit.
- Combined with traffic-aware routing: considering routes around traffic hotspots as part of the setup procedure

Route new VC around Congestion Areas



Traffic-aware routing

Traffic Throttling(业务量减速/节流)

■ Feedback based solution

■ Step 1: Router detects congestion by

- ◆ utilization of the output links (链路利用率)
- ◆ buffering of queued packets inside the router
- ◆ number of packets that are lost due to insufficient buffering

Estimate of the queuing delay:

EWMA (Exponentially Weighted Moving Average,
指数加权移动均值)

$$d_{\text{new}} = \alpha d_{\text{old}} + (1 - \alpha)s$$

- ◆ d : recent queue length
- ◆ s : sample of the instantaneous queue length
- ◆ $0 < \alpha < 1$ 平衡因子

Traffic Throttling

■ Step 2: Transfer Information about Congestion

- ◆ Sending a packet(**Choke Packet**, 抑制分组) to the source host, giving it the destination found in the packet
- ◆ The original **packet is tagged** so that it will not generate any more choke packets
- ◆ To avoid increasing load on the network, the router may only send choke packets at a low rate

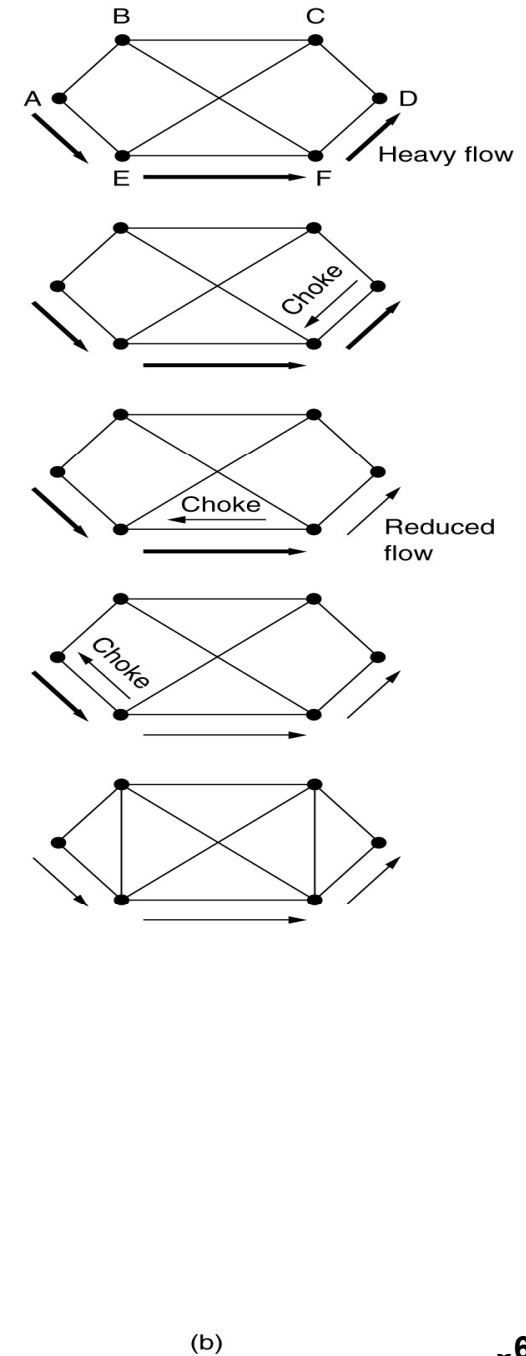
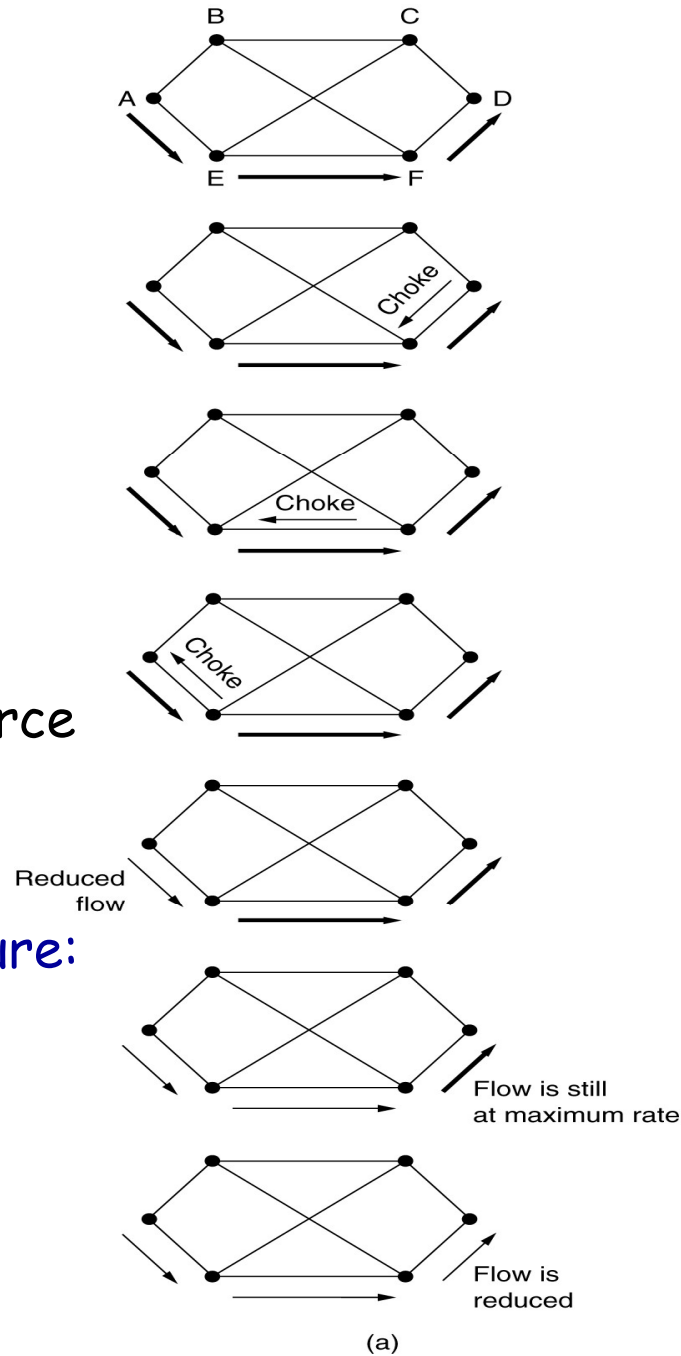
■ Step 3: Adjust System Operation

- ◆ Receiving choke packet, the source may **reduce the traffic** sent to the specified destination by x percent (e.g. 50%)

Other Feedback Mechanisms: Hop-by-Hop Backpressure 逐跳反压

a) Normal choke packet:
affecting only the source

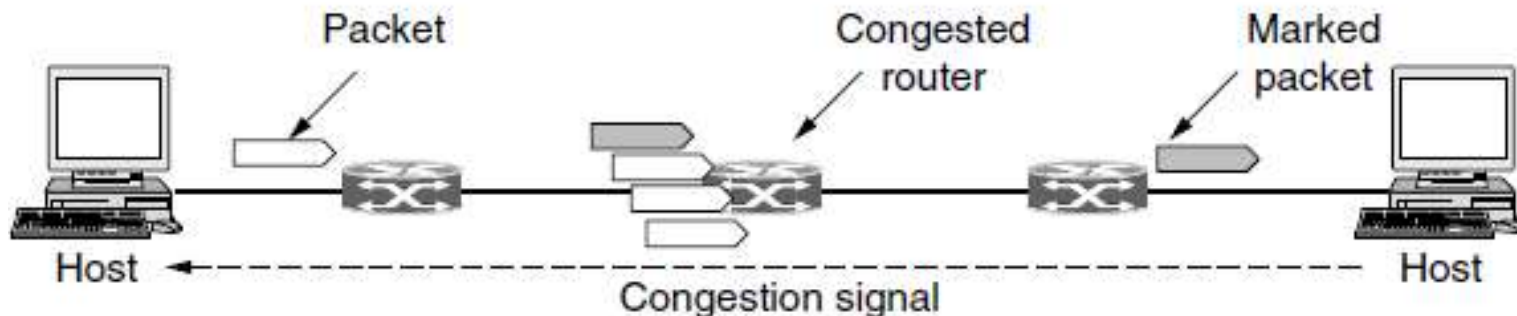
b) Hop-by-hop backpressure:
affecting each hop it
passes through



Other Feedback Mechanisms: ECN

■ Explicit Congestion Notification, 显式拥塞通知

- ◆ Used by IP and TCP
- ◆ Two bits in the IP packet header are used to record whether the packet has experienced congestion.
- ◆ Router detecting congestion marks the packet as it is forwarded.
- ◆ The destination will then echo any marks back to the sender as an explicit congestion signal in its next reply packet.



Loading Shedding(负载掉落)

■ 丢包

- ◆ When routers are being inundated by packets, just throw them away

■ Policies: which packets to drop?

- ◆ Pick packets at random

◆ Wine vs. Milk

- Old vs. New, which is better?
- e.g. file transfer vs. multimedia

◆ Intelligent discard

- e.g. Packets carrying routing information are more important
- e.g. algorithms for compressing video, like MPEG
 - ▣ Entire frame vs. subsequent frames as differences
- Requiring cooperation from the senders to mark packets

Load shedding in Internet: RED

- Random Early Detection (随机早期检测)
- Drop packets before the situation has become hopeless
 - ◆ When the **average queue length** on some line exceeds a threshold, the line is said to be congested, and a small fraction of the packets are dropped at random(随机丢包)
- How should the router tell the source about the problem?
 - ◆ Router just **discards** the selected packet
 - ◆ The affected sender will notice the loss when there is no acknowledgement, **implicit notification** (隐式通知)
- Sources respond to lost packets by slowing down their transmission rate
 - ◆ In wireless networks, where most losses are due to noise on the air link, this approach cannot be used.

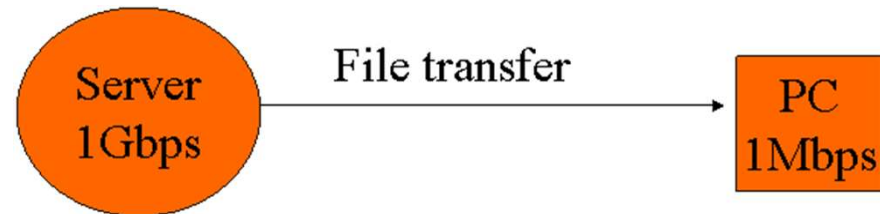
Review of Congestion Control

- Congestion may occur if the load on the is greater than the capacity of the network
- Symptoms:
 - ◆ Long delays (Queuing in router buffers)
 - ◆ Lost packets (Buffer overflow at routers)
- Congestion control approaches
 - ◆ Increasing the resources: Network provisioning, Traffic aware-routing
 - ◆ Decreasing the load
 - Admission control: only in virtual circuit networks
 - Traffic throttling: 1) Router detects congestion
2) Congestion notification: choke packets, ECN
3) Source reduces traffic
 - Load shedding: RED

Flow control vs. Congestion control

■ Flow Control(流量控制)

- ◆ To avoid overflowing the receiver
- ◆ Local issue, involves only 2 endpoints, the sender and the receiver



■ Congestion Control(拥塞控制)

- ◆ To avoid congesting the network
- ◆ Global issue, concerns all nodes/hosts along the path

Outline

- Introduction to Network Layer
- Routing Algorithms
- Congestion Control and Internetworking
 - ◆ Congestion Control Algorithms
 - ◆ Quality of Service
 - ◆ Internetworking
- The network layer in the Internet

Primary QoS Parameters(服务质量参数)

- Reliability

 - ◆ Packet error rate

 - ◆ Packet loss rate

- Delay: the time period from 1st bit of packet sent to last bit received

- Jitter(抖动): the variation in delay of packets belonging to the same flow

- Bandwidth: maximum data rate

QoS Requirements of Various Applications

Application	Reliability	Delay	Jitter	Bandwidth
E-mail	High	Low	Low	Low
File transfer	High	Low	Low	Medium
Web access	High	Medium	Low	Medium
Remote login	High	Medium	Medium	Low
Audio on demand	Low	Low	High	Medium
Video on demand	Low	Low	High	High
Telephony	Low	High	High	Low
Videoconferencing	Low	High	High	High

Techniques to improve QoS

- Traffic shaping
- Packet Scheduling
- Admission control
- Resource reservation

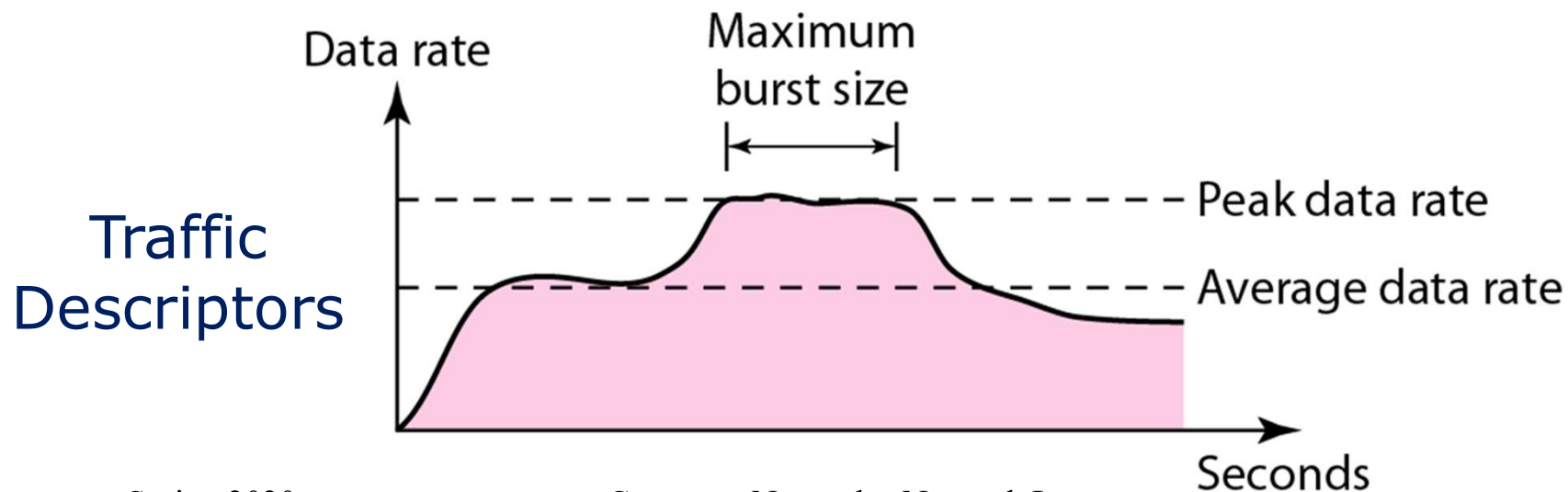
Technique to improve QoS: Traffic Shaping(业务量/流量整形)

■ Problem

- ◆ **Traffic burst**: Host emits the packets irregularly, which may cause congestion to occur in network

■ Traffic shaping

- ◆ Regulate the **average rate** (and burstiness) of data transmission, smooth out the traffic (去突发)



Leaky Buckets and Token Buckets

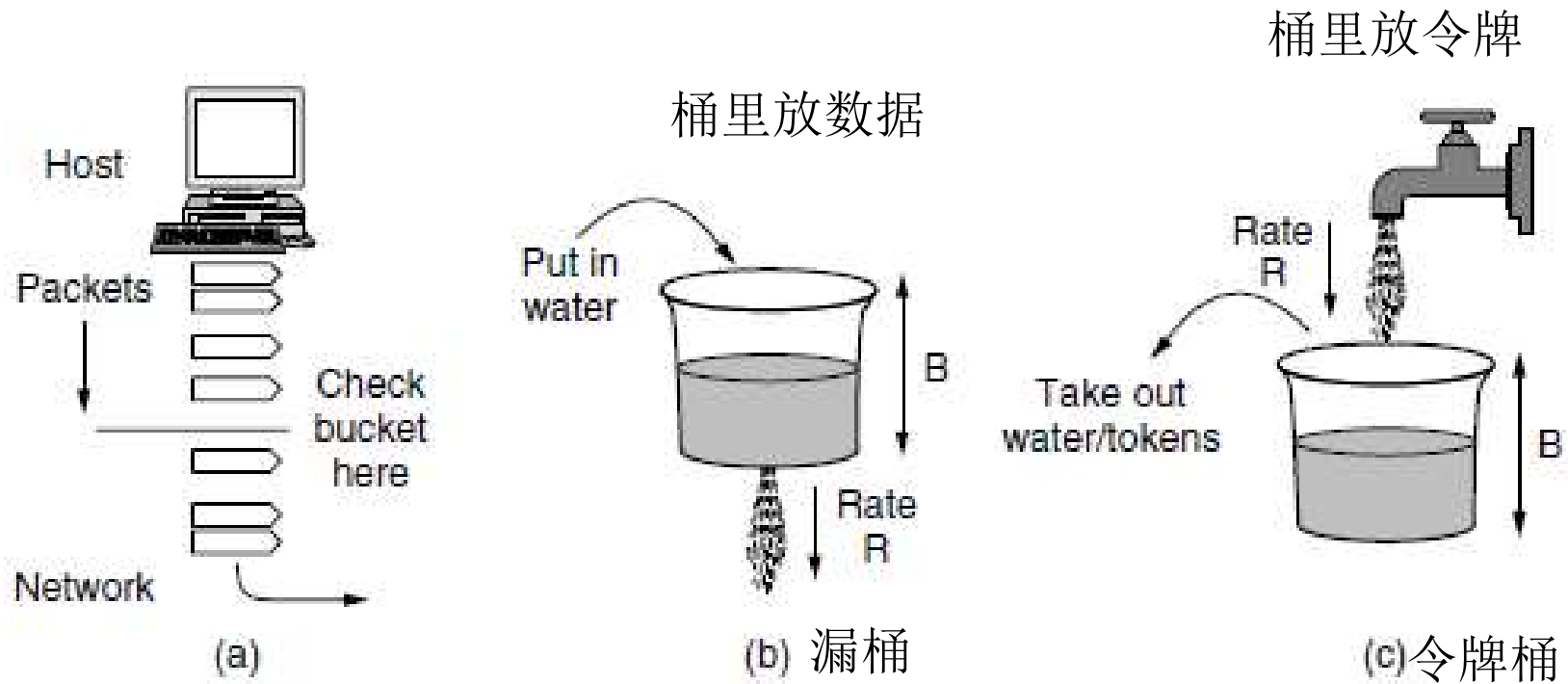
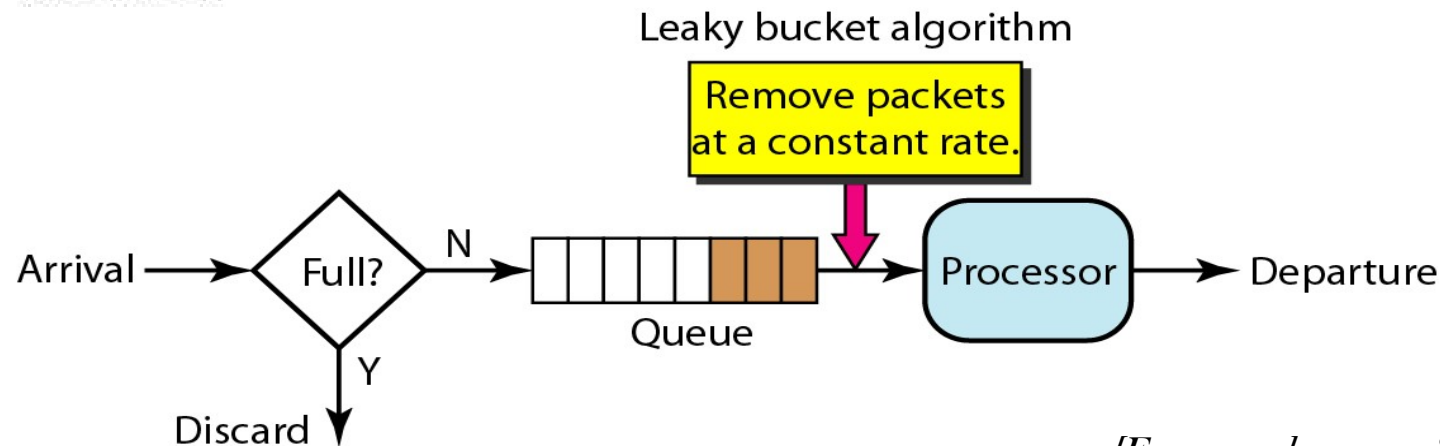
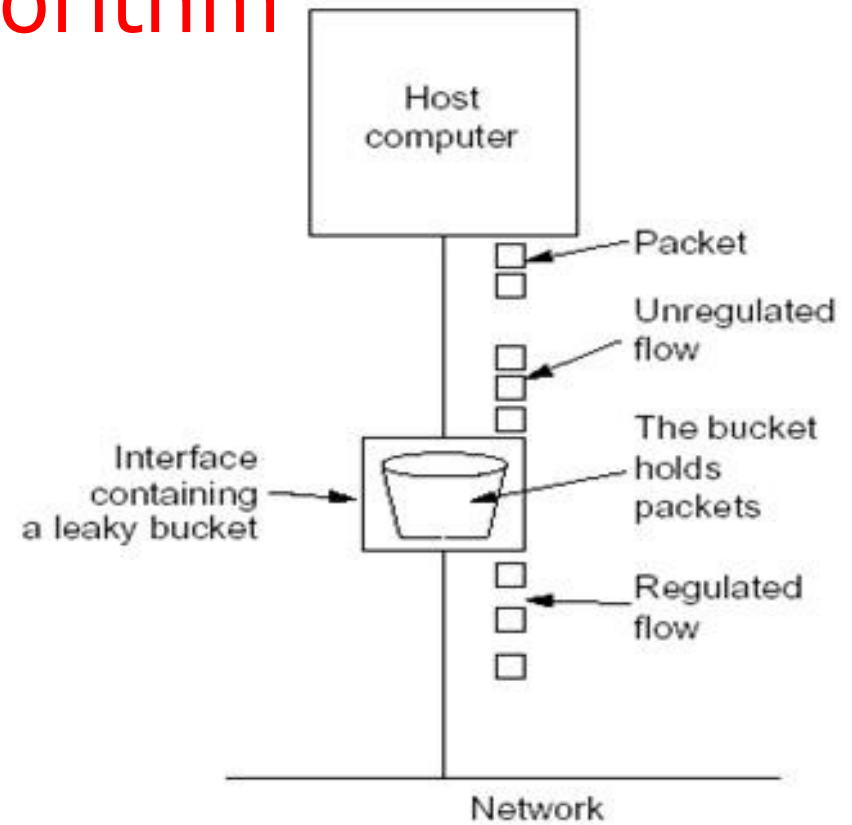
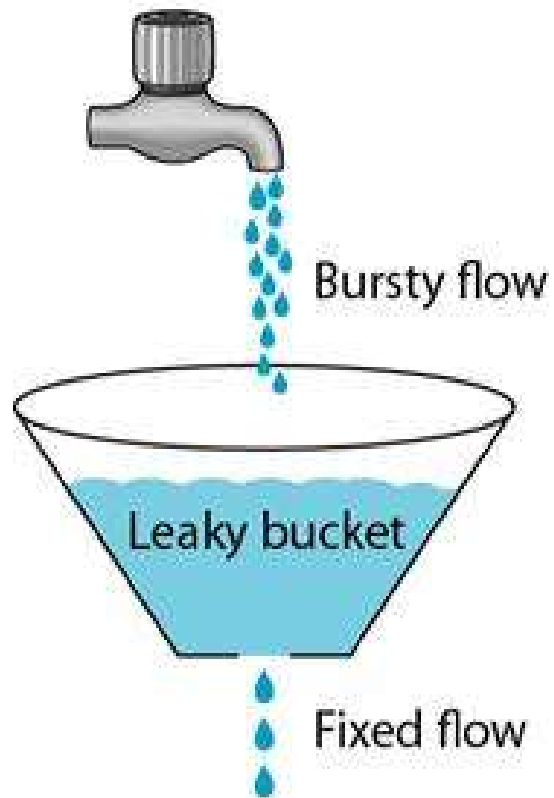


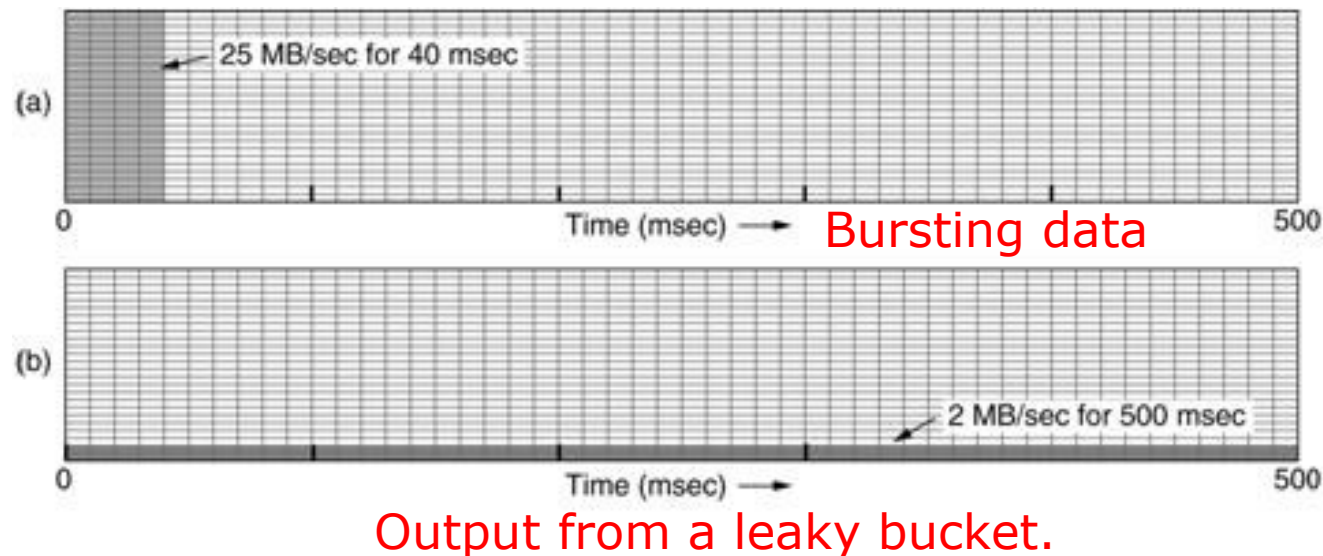
Figure 5-28. (a) Shaping packets. (b) A leaky bucket. (c) A token bucket.

Leaky Bucket (漏桶) Algorithm



Leaky Bucket: Example

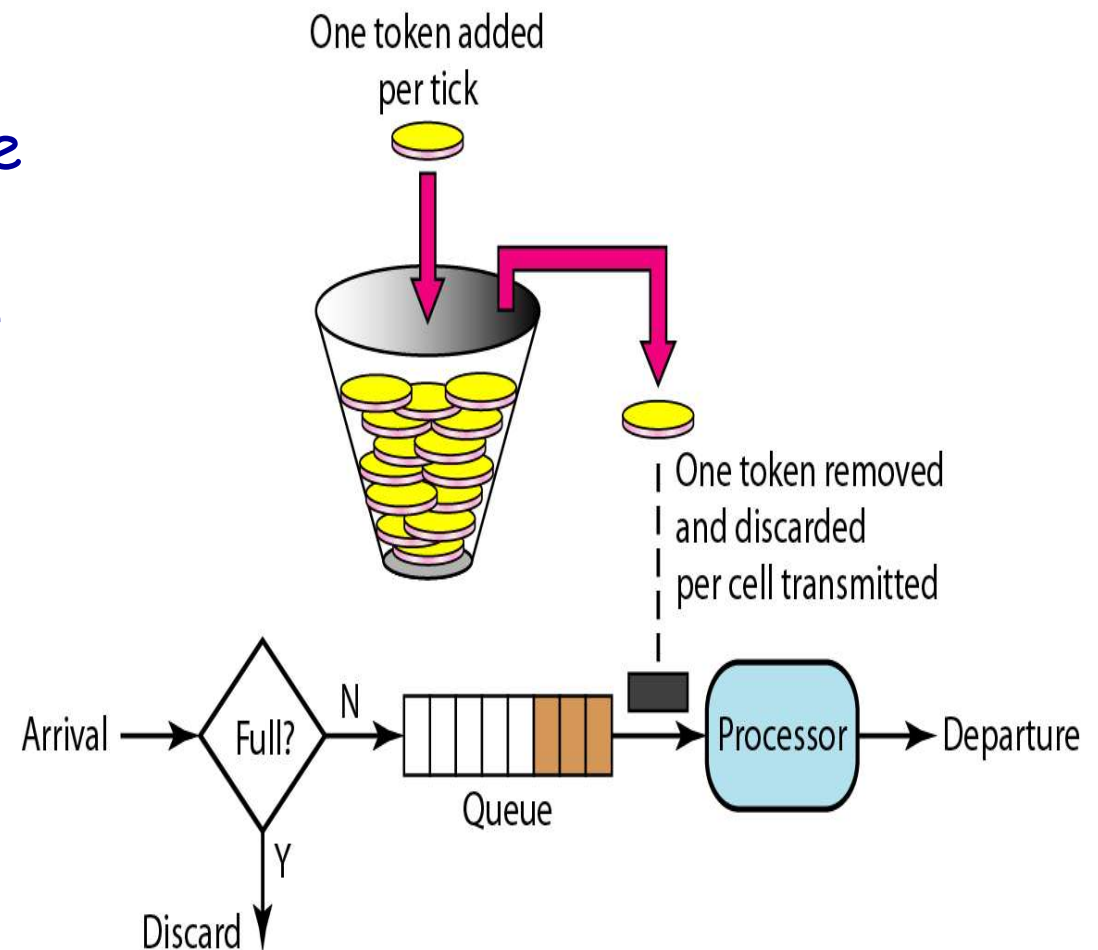
- Turns an uneven flow of packets into an even flow
 - ◆ Smoothing out bursts
 - ◆ Reducing the chances of congestion
- The byte-counting leaky bucket



- Limitation: (桶满则数据丢失)
When bucket is full, subsequent packets are dropped

Token Bucket (令牌桶) Algorithm

- Token bucket allows idle hosts to **accumulate credit** for the future in the forms of token.
- Token bucket allows **bursts**, but up to a regulated maximum length



Token to be collected based on idle time and can be used later to send bursty packets as long as it is allowed by the bucket.

Leaky Bucket vs. Token Bucket

■ Burstiness

- ◆ **Leaky bucket** algorithm enforces a rigid output pattern at the **average rate**, no matter how bursty the traffic is
- ◆ **Token Bucket** allowing **some burstiness** in the output stream

■ When the bucket fills up

- ◆ Token bucket algorithm throws away tokens but **never discards packets**
- ◆ Leaky bucket algorithm discards packets

Token Bucket: Example

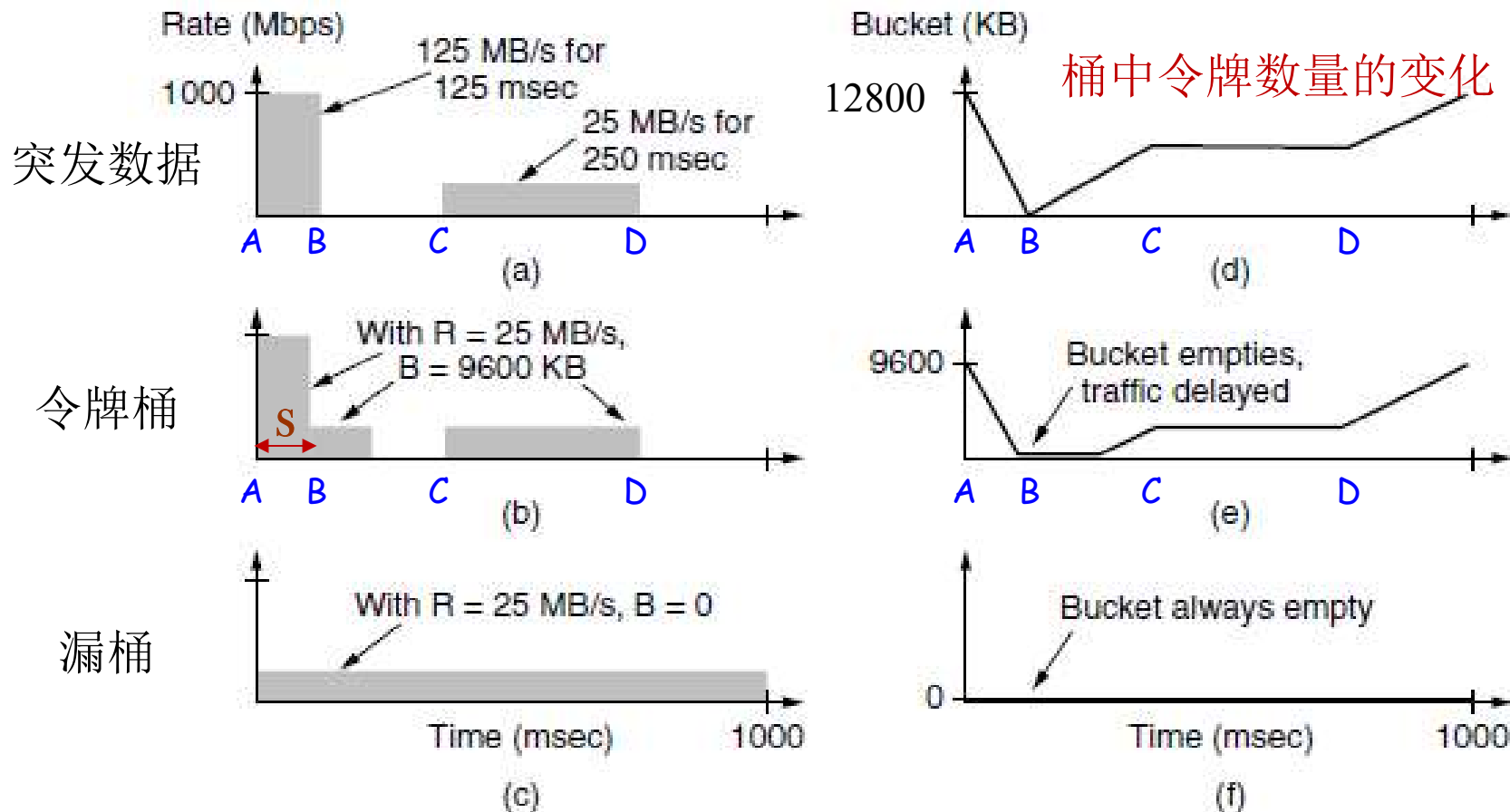


Figure 5-29. (a) Traffic from a host. Output shaped by a token bucket of rate 200 Mbps and capacity (b) 9600 KB and (c) 0 KB. Token bucket level for shaping with rate 200 Mbps and capacity (d) 12800KB (e) 9600 KB, and (f) 0 KB.

The Length of the Maximum Rate Burst

经过令牌桶之后的突发数据持续时间:

burst length **S** sec

token bucket capacity **B** bytes

token arrival rate **R** bytes/sec

maximum output rate **M** bytes/sec

An output burst contains a maximum of **B + RS** bytes

the number of bytes in a maximum-speed burst of length **S** seconds is **MS**

$$B + RS = MS$$

$$S = B / (M - R)$$

If $B = 9600\text{KB}$, $M = 125 \text{ MB/sec}$, and $R = 25 \text{ MB/sec}$
then $S = 96 \text{ ms}$

Packet Scheduling(分组/包调度)

■ Flow

- ◆ A sequence of packets that are sent from a particular source to a destination.
- ◆ Assuming packets of a flow follow the same route

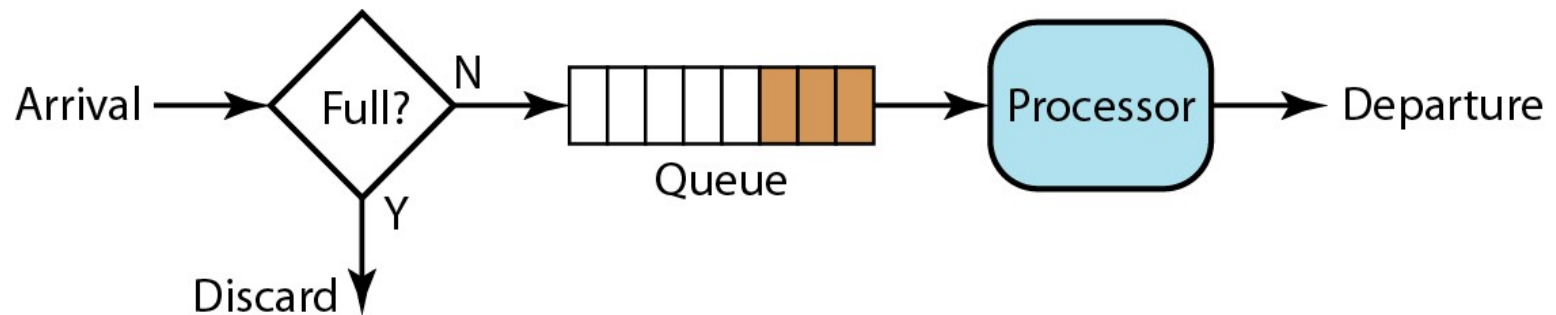
■ Packet scheduling: allocate router resources among the packets of a flow and between competing flows

■ Algorithms

- ◆ FIFO Queuing
- ◆ Priority Queuing
- ◆ Fair Queuing
- ◆ Weighted Fair Queuing

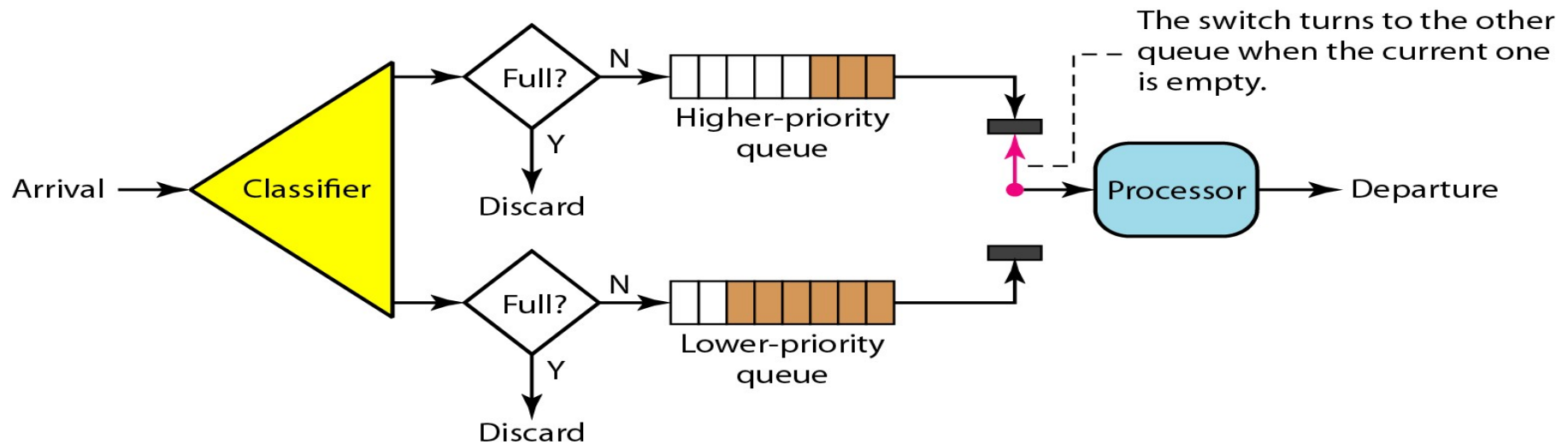
FIFO Queuing(先进先出队列)

- First-In First-Out, or FCFS (First-Come First-Serve)
- Tail drop: drop newly arriving packets when the queue is full



FIFO: First In, First Out

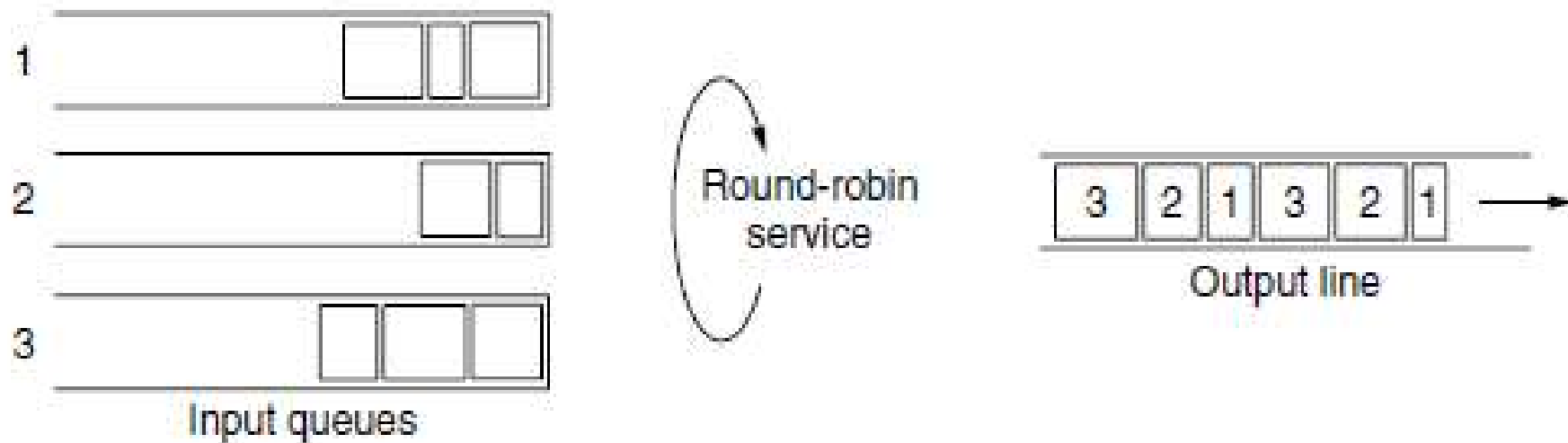
Priority Queuing (优先级队列)



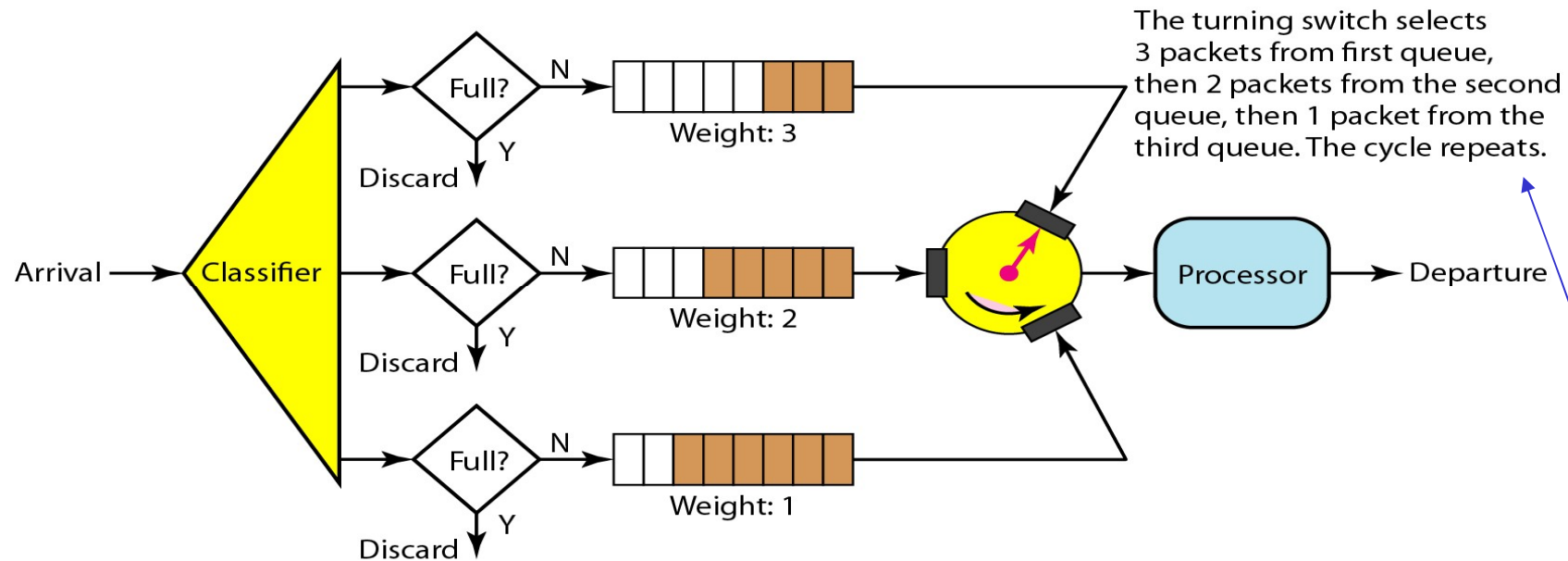
- Packets are first assigned a **priority** class. Each priority class has its **own queue**.
- When a line is idle, packets in the highest priority queue are processed first. 高优先级数据先发送
- **Better QoS** because higher priority traffic such as multimedia can reach the destination with less delay.
- Potential drawback: starvation of lower priority traffic, not fair .对低优先级数据不公平

Fair Queuing (公平队列)

- Packet are processed in a round robin(循环) way.



Weighted Fair Queuing(公平加权队列)



- Priority Queue+ Fair Queue (按权值轮流)
- Queues are weighted based on priority; higher priority means higher weights
- The system processes packets in each queue in a round-robin fashion with the number of packets selected from each queue based on corresponding weight.

WFQ Example

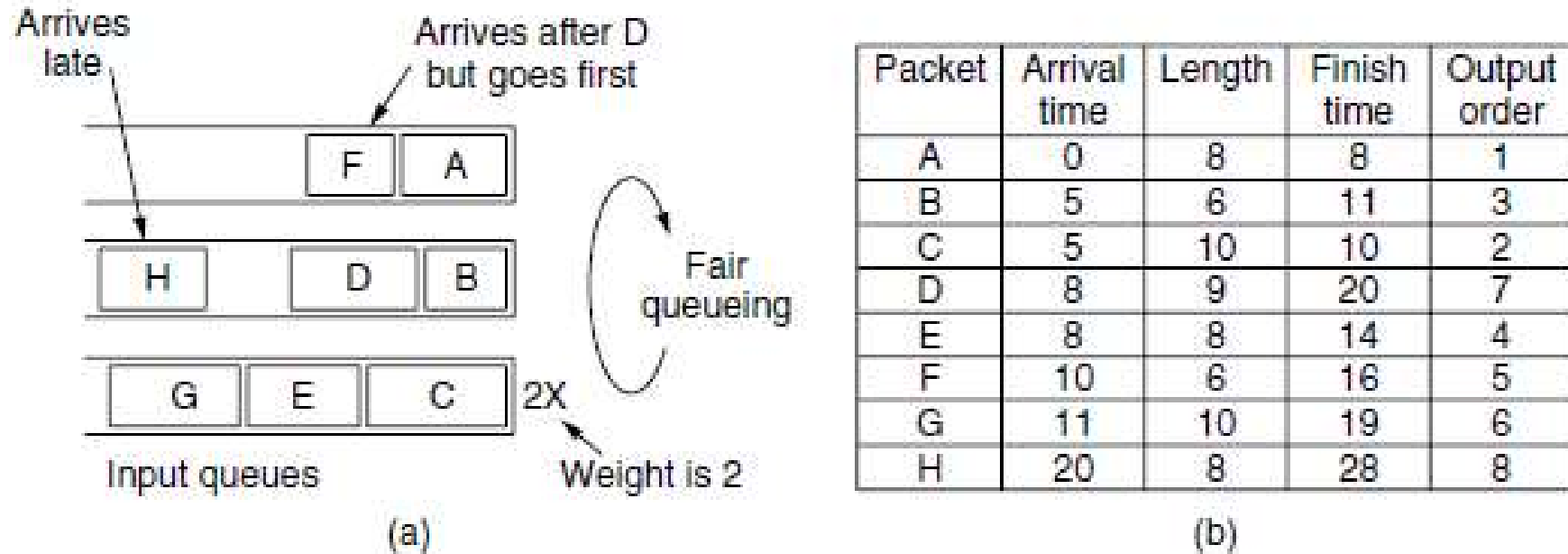


Figure 5-31. (a) Weighted Fair Queueing. (b) Finishing times for the packets.

$$F_i = \max(A_i, F_{i-1}) + \frac{L_i}{W}$$

F: finish time, A: Arrival time, L: length of packet, W: weight

Resource Reservation(资源预留) & Admission Control(接纳控制)

■ Resource Reservation

- ◆ Bandwidth.
- ◆ Buffer space.
- ◆ CPU cycles.

Flow: 同一对源/目的主机之间的多个数据包

■ Flow negotiation

- ◆ Sender, receiver, and all routers along the path are involved in the flow negotiation

■ Flow specification (流描述)

- ◆ 描述一个流对网络的要求
- ◆ 源主机产生
- ◆ 沿途路由器和目的主机可修改

Parameter	Unit
Token bucket rate	Bytes/sec
Token bucket size	Bytes
Peak data rate	Bytes/sec
Minimum packet size	Bytes
Maximum packet size	Bytes

Relating flow specifications to router resources

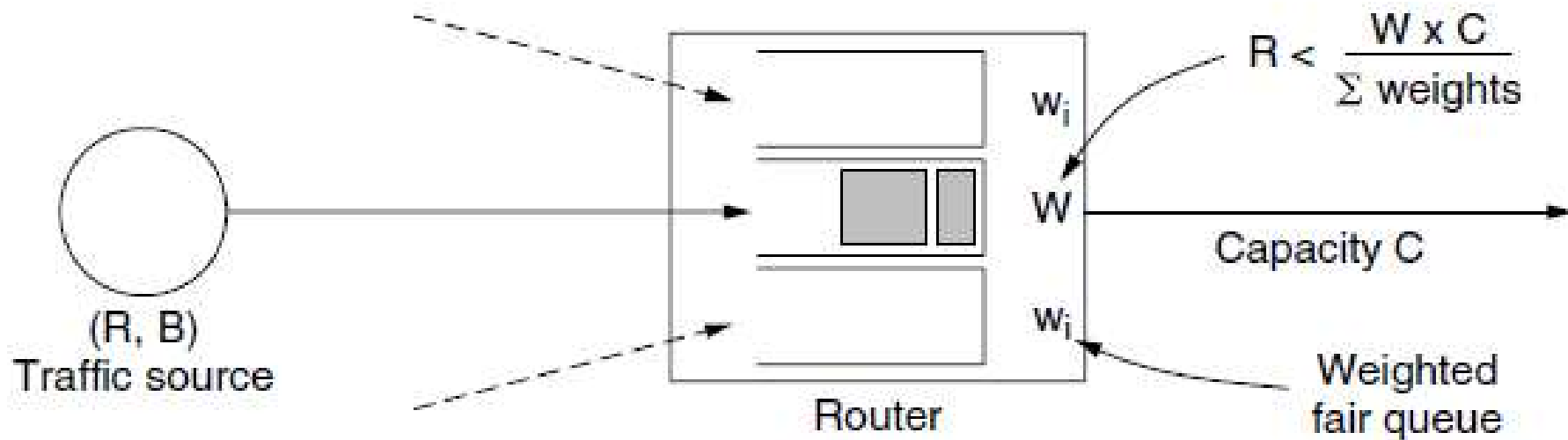


Figure 5-33. Bandwidth and delay guarantees with token buckets and WFQ.

给每个流的权值 W 足够大，保证流以 R 速率输出时不会排队

Review of QoS

- metrics: bandwidth, delay, jitter, reliability(packet error rate, loss rate)
- Ways to improve QoS
 - ◆ Traffic shaping(流量整形): 去突发
Leaky bucket(漏桶) vs. **Token bucket(令牌桶)**
漏桶也可用于traffic policing(流量监管)
 - ◆ Packet scheduling(分组/包调度):
 - FIFO(先进先出)
 - 优先级队列(PQ)
 - 公平队列(FQ)
 - 公平加权队列 (**WFQ**)

Outline

- Introduction to Network Layer
- Routing Algorithms
- Congestion Control and Internetworking
 - ◆ Congestion Control Algorithms
 - ◆ Quality of Service
 - ◆ Internetworking (网络互通)
- The network layer in the Internet

Connecting Networks

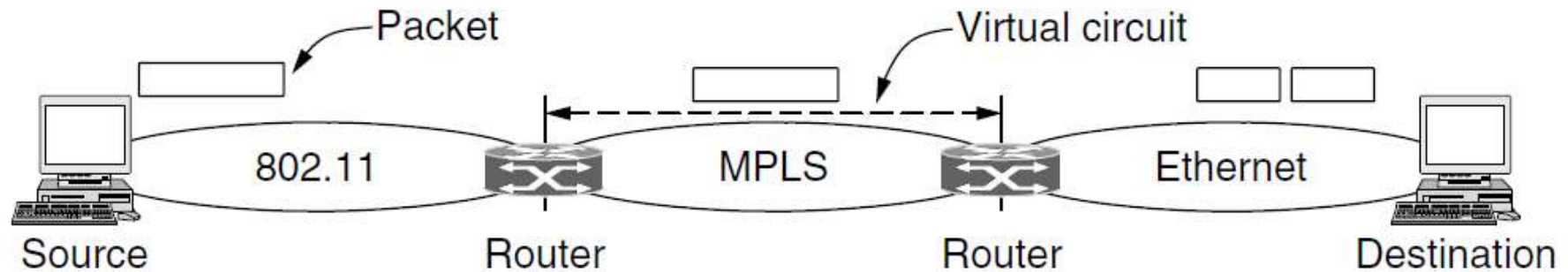
- Different networks & different protocols
 - ◆ Ethernet, Internet over cable, Fixed and mobile telephone networks, 802.11, 802.16, ATM, ...
- Purpose of interconnecting
 - ◆ Allow users on any network to communicate with users on all the other ones
- Internetwork(internet, 互联网): two or more networks are connected

How Networks Differ

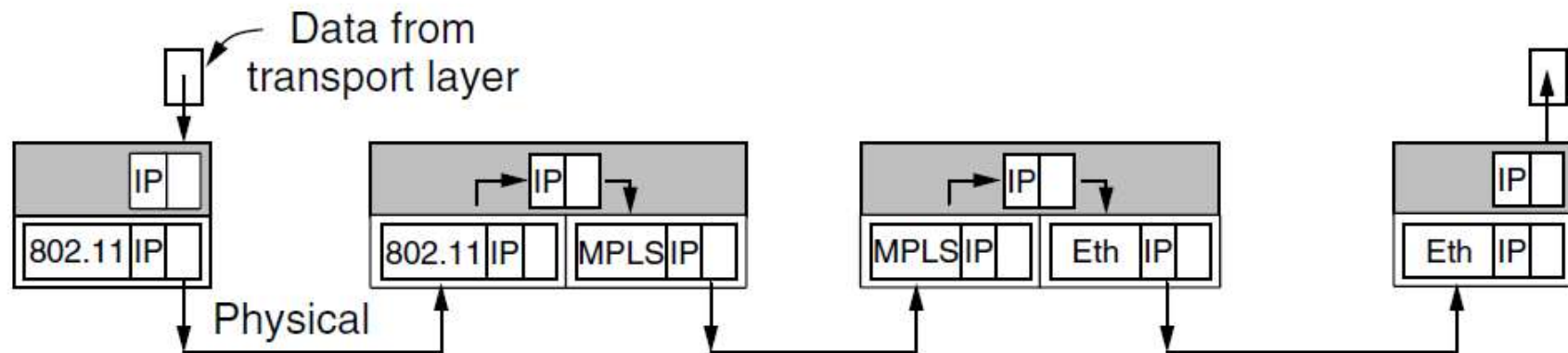
Item	Some Possibilities
Service offered	Connectionless versus connection oriented
Addressing	Different sizes, flat or hierarchical
Broadcasting	Present or absent (also multicast)
Packet size	Every network has its own maximum
Ordering	Ordered and unordered delivery
Quality of service	Present or absent; many different kinds
Reliability	Different levels of loss
Security	Privacy rules, encryption, etc.
Parameters	Different timeouts, flow specifications, etc.
Accounting	By connect time, packet, byte, or not at all

How Networks Can Be Connected

- common network layer interconnecting dissimilar networks



A packet crossing different networks



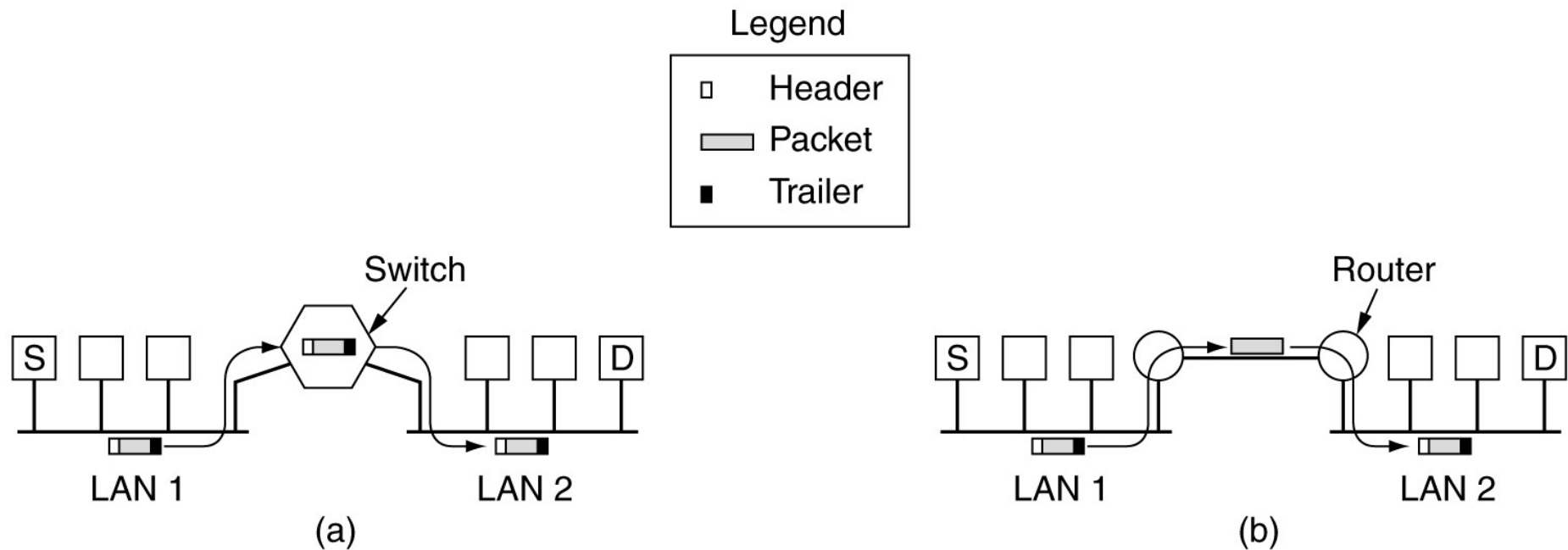
Network and link layer protocol processing

MPLS: Multi-Protocol Label Switching

Interconnection at different layers

- Layer 1: Repeater, HUB
- Layer 2: Bridges, Switches
 - ◆ Protocol translation: Ethernets/802.11
- Layer 3: Router (Packet translation)
- Layer 4: Transport gateway
 - ◆ TCP/Wireless TCP
- Above layer 4: Application gateway
 - ◆ RFC 5822/X.400
 - ◆ Instant message gateway

Switch vs. Router

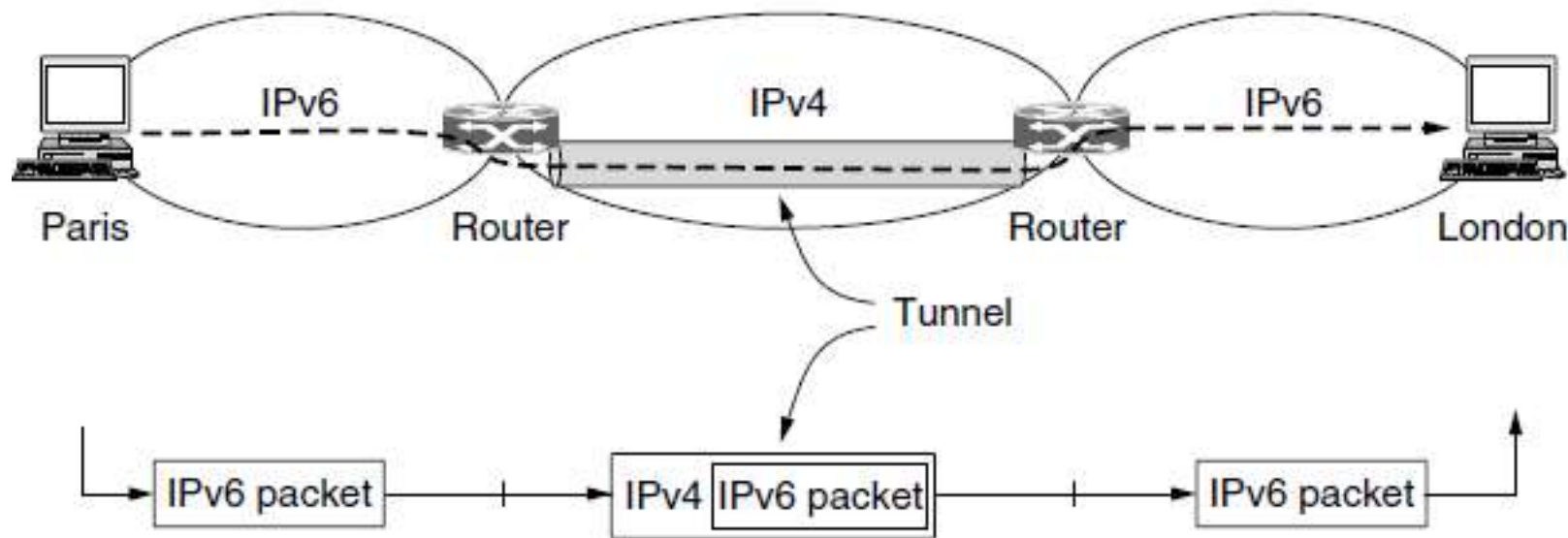


Switch: the entire frame is transported on the basis of its **MAC address**

Router: packet is extracted from the frame and the **Layer-3 address** is used for deciding where to send it

Tunneling(隧道)

- ◆ Source and destination hosts are on the same type of network, but there is a different network in between
- ◆ Packet from source is **encapsulated** inside packet of the intermediate network



源主机与目的主机网络协议相同，中间网络不同
用中间网络的协议封装源主机发送的数据包

Internetwork Routing

- A two-level routing algorithm is used
 - ◆ Within each network, an intradomain(域内) or interior gateway protocol is used for routing.
 - ◆ Across the networks that make up the internet, an interdomain(域间) or exterior gateway protocol is used.
- See how it is done in Internet

Fragmentation(分段/分片), Why?

Different MTUs

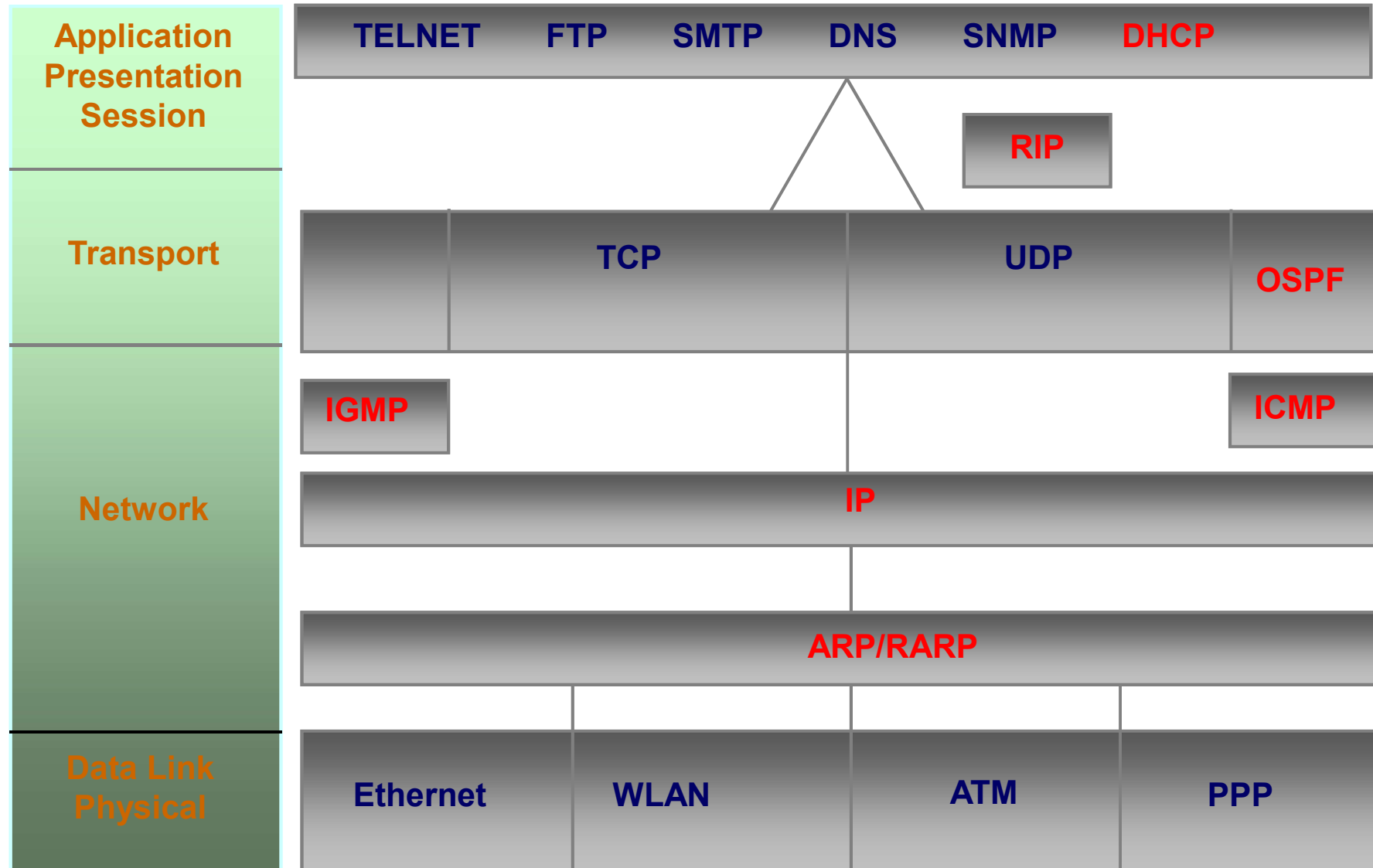
- **MTU**: Maximum Transmission Unit (最大传输单元)
- MTU is determined by
 - ◆ Hardware (e.g., the size of an Ethernet frame)
 - ◆ Operating system (e.g., all buffers are 512 bytes).
 - ◆ Protocols (e.g., the number of bits in the packet length field)
 - ◆ Compliance with some (inter)national standard.
 - ◆ Desire to reduce error-induced retransmissions to some level
 - ◆ Desire to prevent one packet from occupying the channel too long
- How to do fragmentation?
 - ◆ See **how IP does**

Outline

- Introduction to Network Layer
- Routing Algorithms
- Congestion Control and Internetworking
- *The network layer in the Internet*
 - ◆ IP Addressing
 - ◆ IP Packet format
 - ◆ IPv6
 - ◆ ICMP、 ARP and DHCP
 - ◆ Routing Protocols: RIP, OSPF and BGP
 - ◆ IP Multicast
 - ◆ Mobile IP

OSI

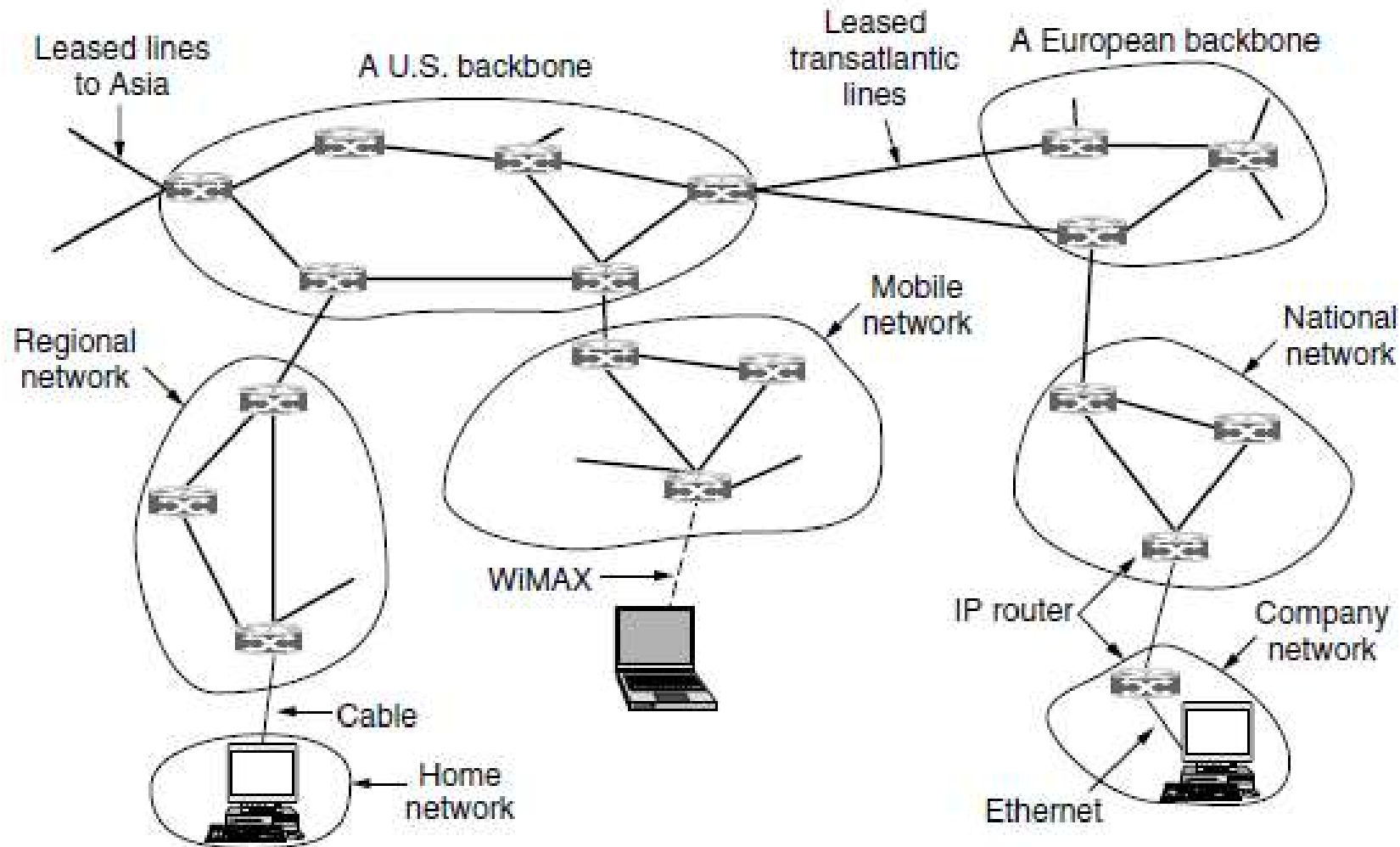
TCP/IP



Design Principles of Internet's network layer

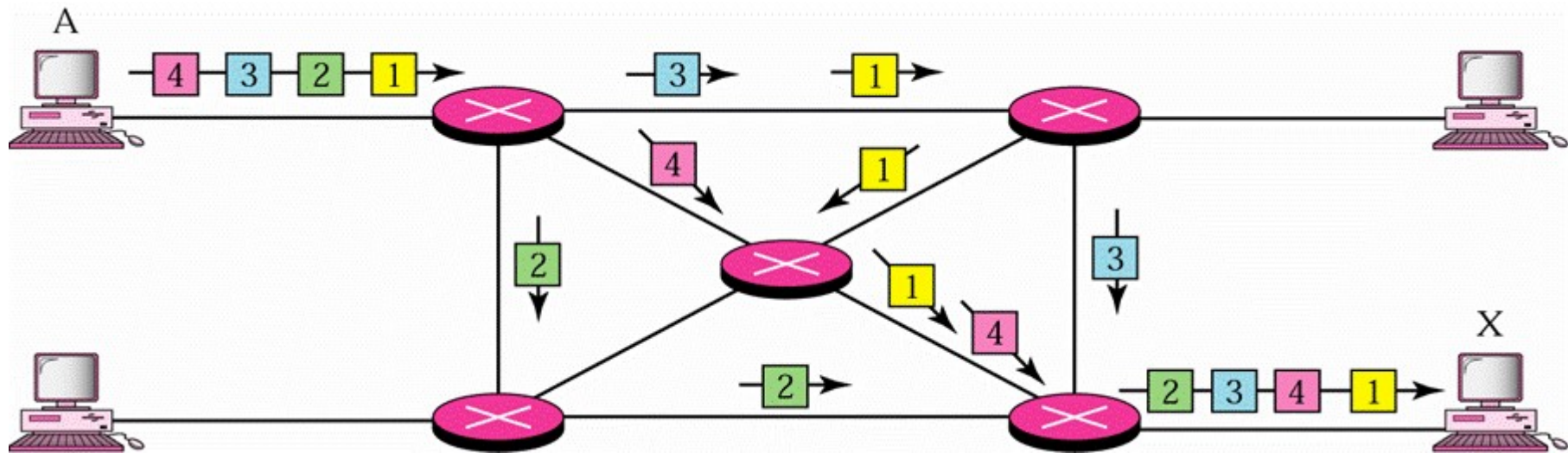
- Make sure it works
- Keep it simple
- Make clear choices
- Exploit modularity (模块化)
- Expect heterogeneity (异构性)
- Avoid static options and parameters
- Look for a good design; it need not be perfect
- Be strict when sending and tolerant when receiving
- Think about scalability (可扩展性)
- Consider performance and cost

Quasihierarchical Organization of Internet (准体系结构)



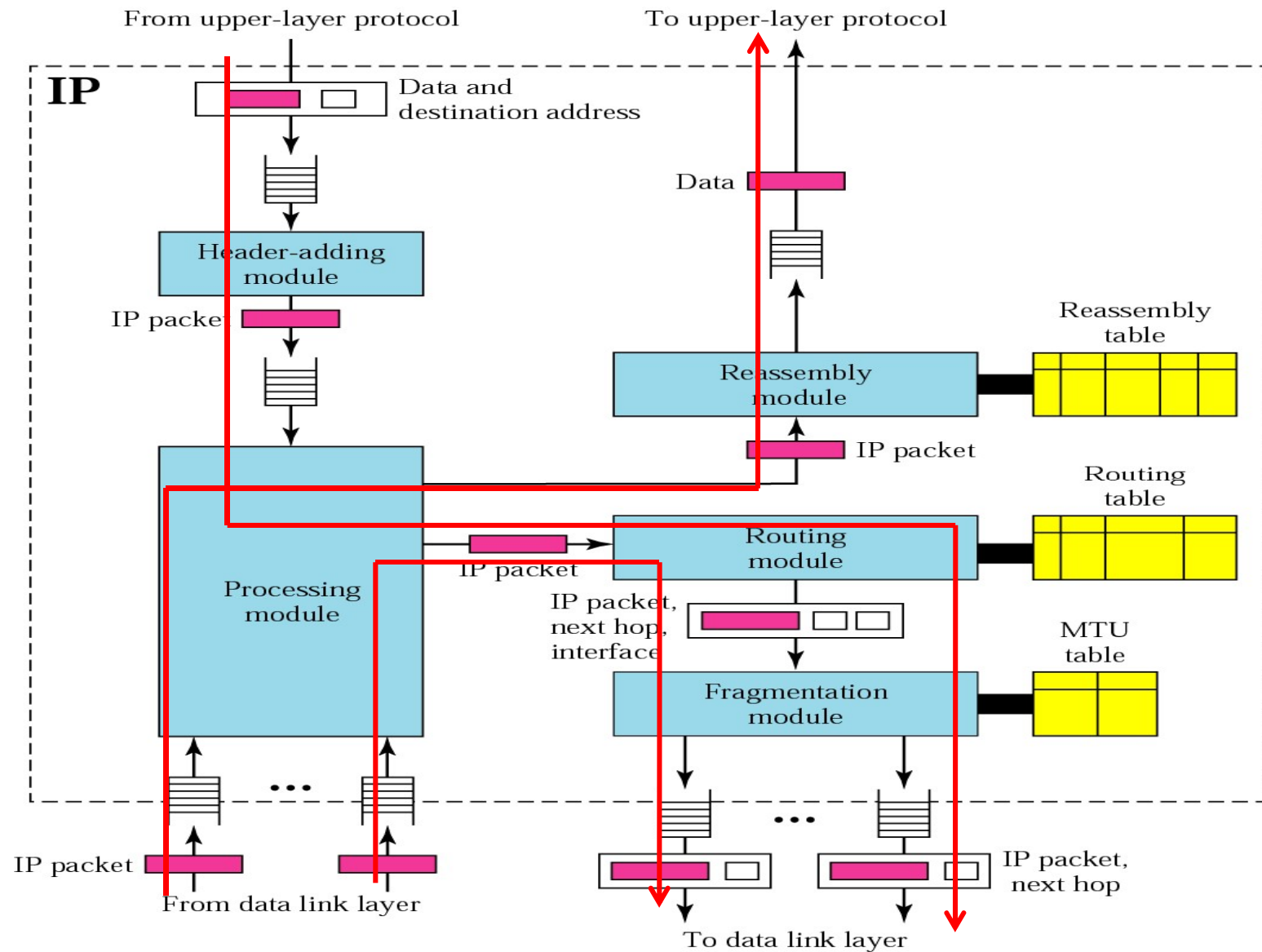
Internet Protocol (IP)

- The most-widely used Internetworking protocol
- A connectionless network-layer protocol.
- Based on datagram routing.
- IP provides **best-effort** service, unreliable !!
- Routing in IP is a mixture of hierarchical, distance vector and link state routing algorithms.



[Forouzan]

IP components



Design Issues of IP

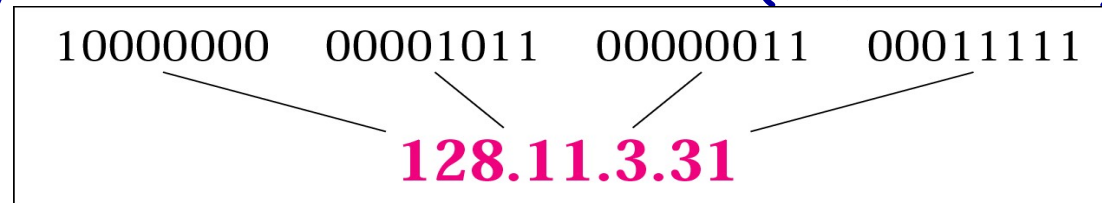
- Design requirements: accommodate differences in constituting networks:
 - ◆ Different addressing schemes:
 - using a **global** address
 - ◆ Different maximum packet size
 - ◆ Different network access mechanism
 - ◆ Different timeouts
 - ◆ Different transmission modes (connection-oriented, connectionless)
 - ◆ Error control
 - ◆ Flow control

IPv4 Addressing

- Classful Addressing(分类地址): 1981
- Subnetting(子网): 1985
- Supernetting(超网): 1993
- Classless Addressing(无类别地址): 1995

IPv4 Address

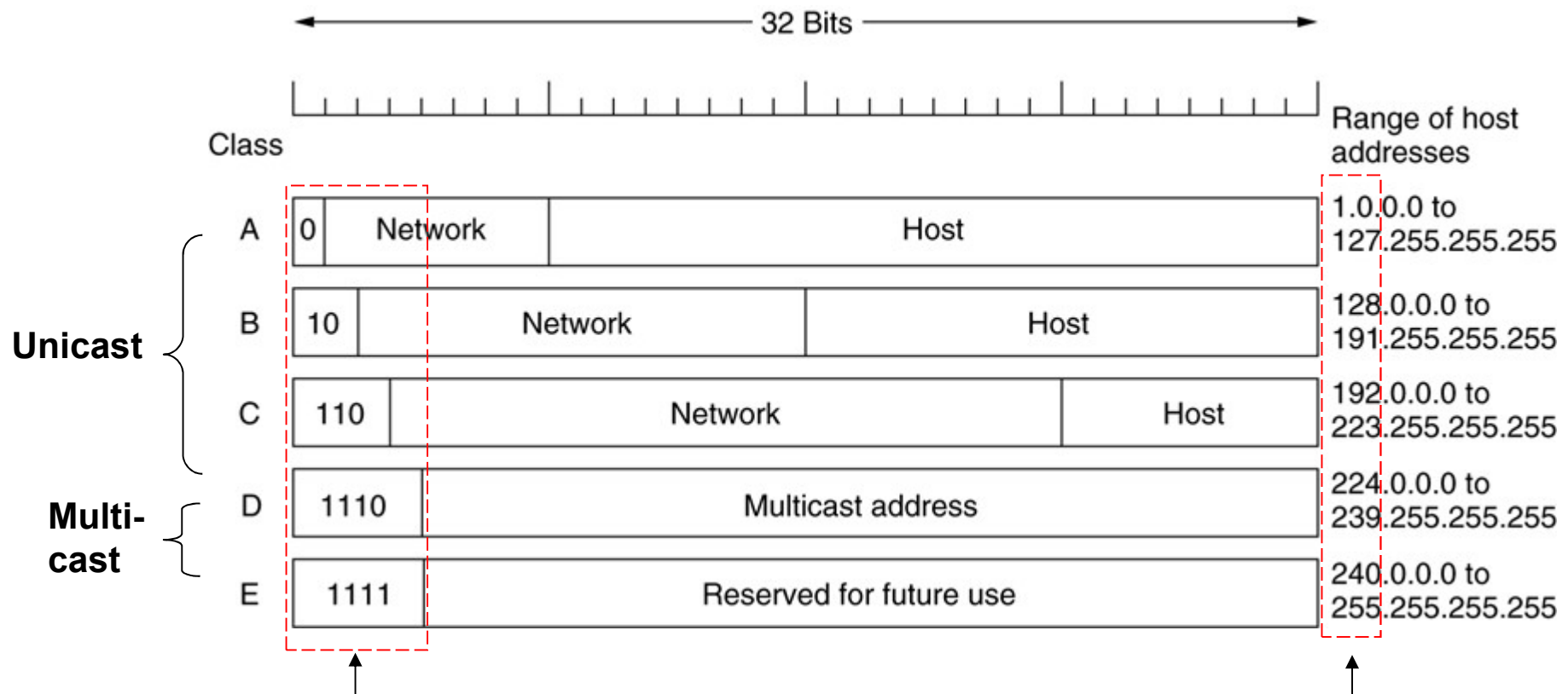
- Identifier used in network layer to identify each device connected to the Internet is called the **Internet address** or **IP address**.
- 32-bit, Dotted-decimal notation(点分十进制)



- Unified addressing
- Unique for each **network interface**, irrelative with its physical address, which universally defines **1** connection to Internet.
- If a host/router has 2 interfaces connecting to Internet, then it has 2 IP addresses
- Two types of IP addressing: **Classful** vs. **Classless**

Classful IP Addressing(分类地址)

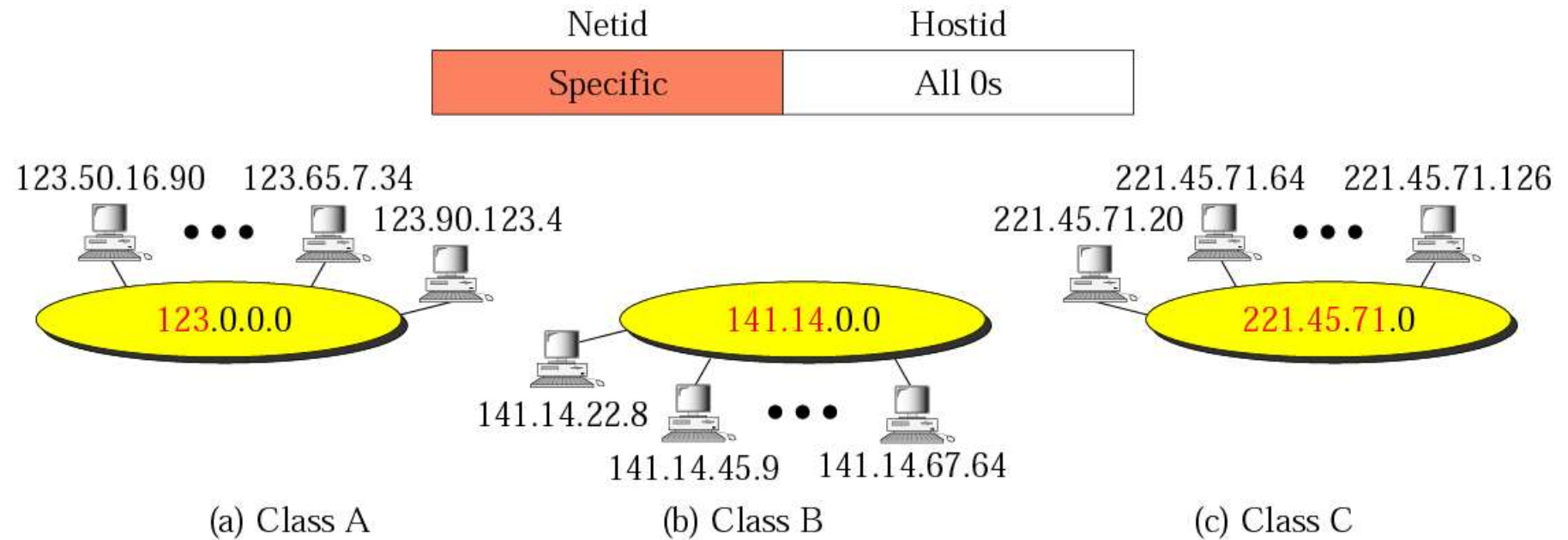
- IP has a two-level (at least, traditionally) hierarchical addressing mechanism in class A, B and C: **Network ID(net-id) + Host ID (host-id)**



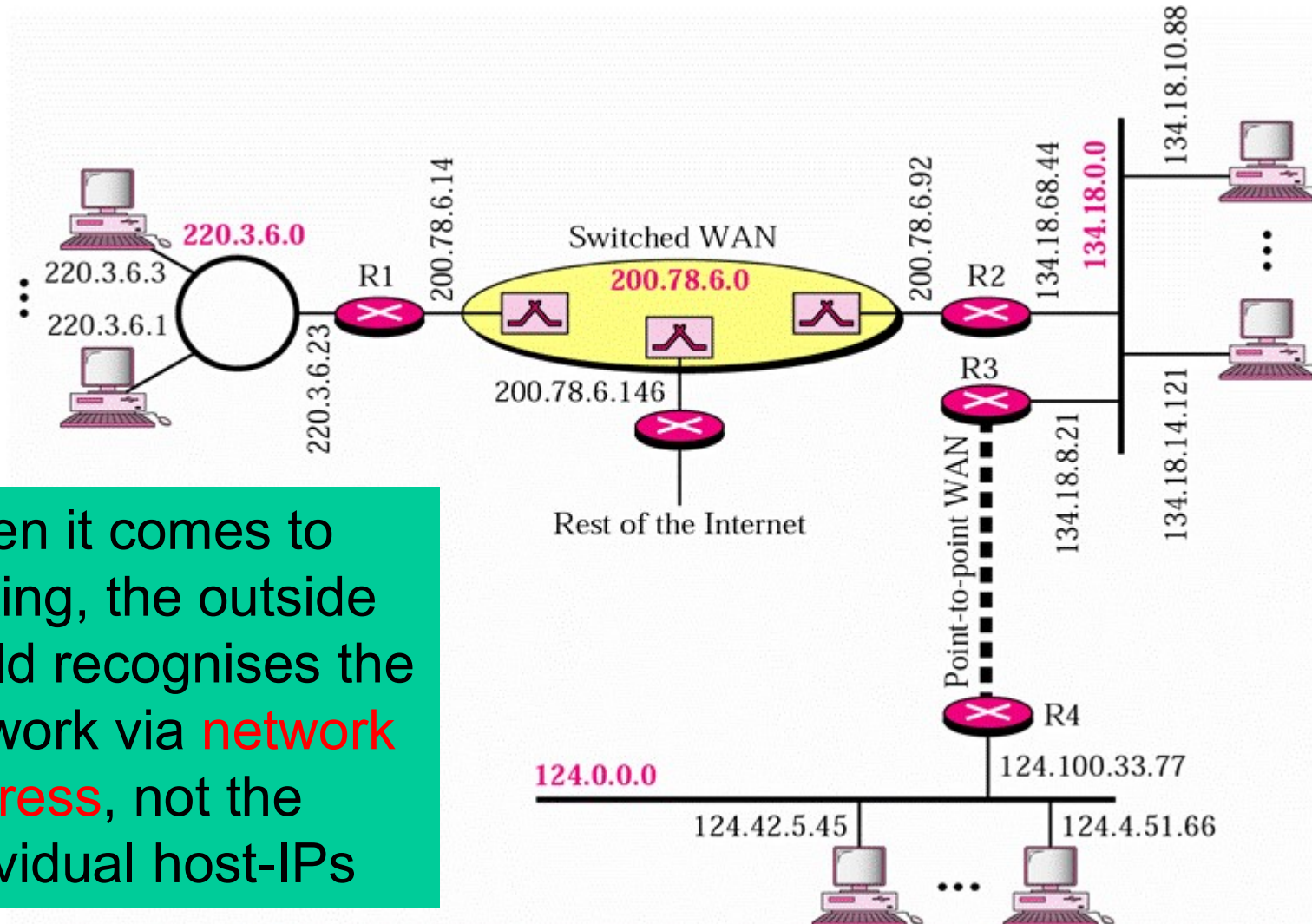
Network address

- Network address is the **first address** in the block, is the one that is assigned to the organization.
- 网络号部分保留，主机号部分为0 
- It can be found by applying the **default mask**(默认/缺省掩码) to any of the IP addresses in the network
 - ◆ IP address bit AND with default mask
- Default masks
 - ◆ Class A: 255.0.0.0
 - ◆ Class B: 255.255.0.0
 - ◆ Class C: 255.255.255.0 

Network address: example



Network Addresses: Usage



When it comes to routing, the outside world recognises the network via **network address**, not the individual host-IPs

路由选择使用网络地址



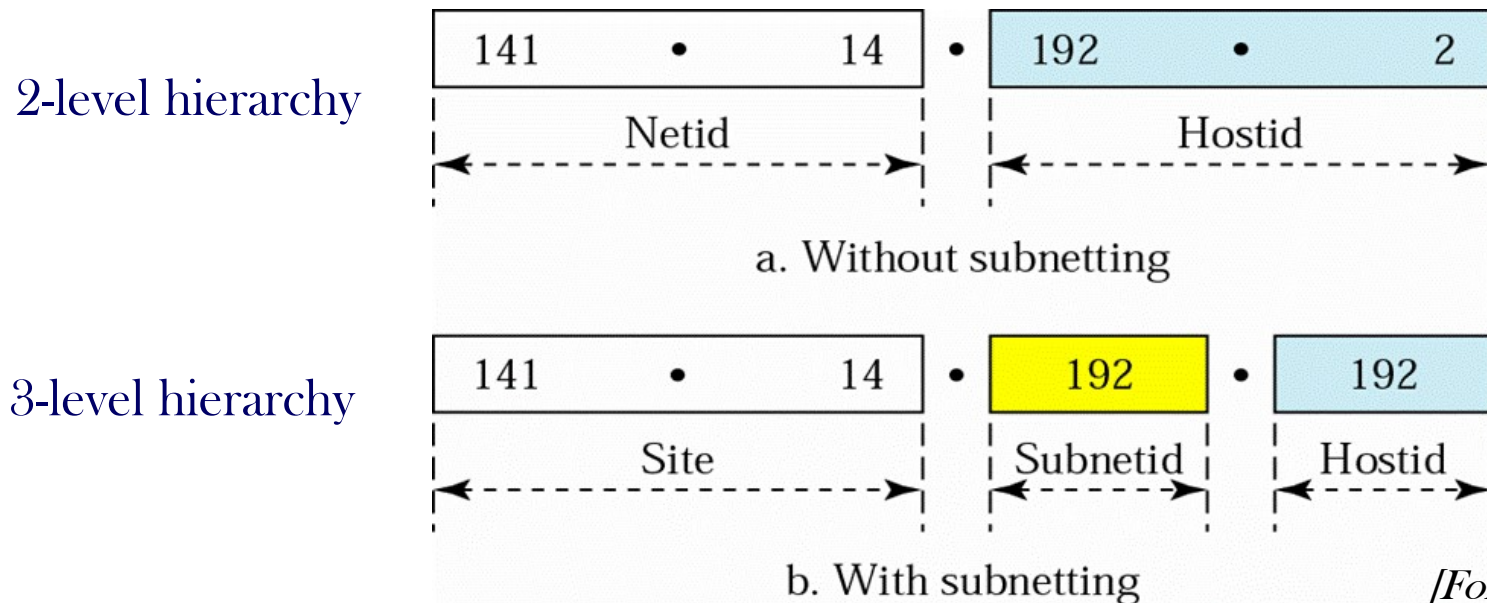
[Forouzan]

Special IP addresses

0 0	This host
0 0 ... 0 0 Host	A host on this network
1 1	Broadcast on the local network
Network 1 1 1 1 ... 1 1 1 1	Broadcast on a distant network
127 (Anything)	Loopback 环回地址

Subnetting(子网划分)

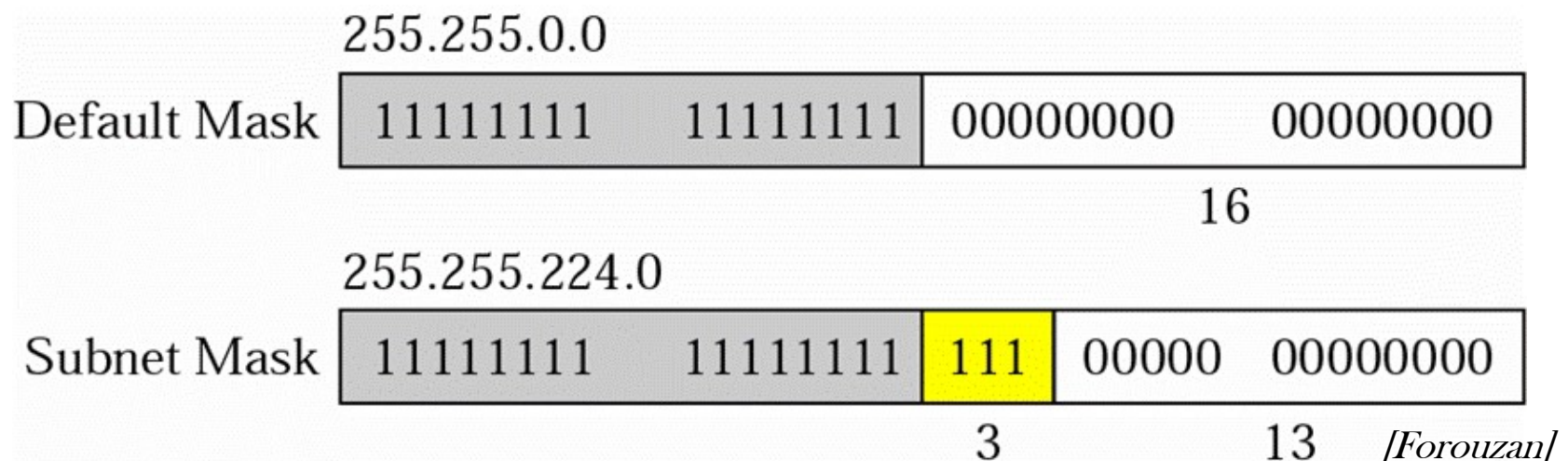
- To manage IP addresses and routing packets more efficiently
- A network is split into several parts for internal use, but still act like a single network to outside world identified by **network address**
 - ◆ Subnetting is invisible to outside world 子网划分对于外部不可见
- Subnet is identified by using part of **host portion** to identify physical network



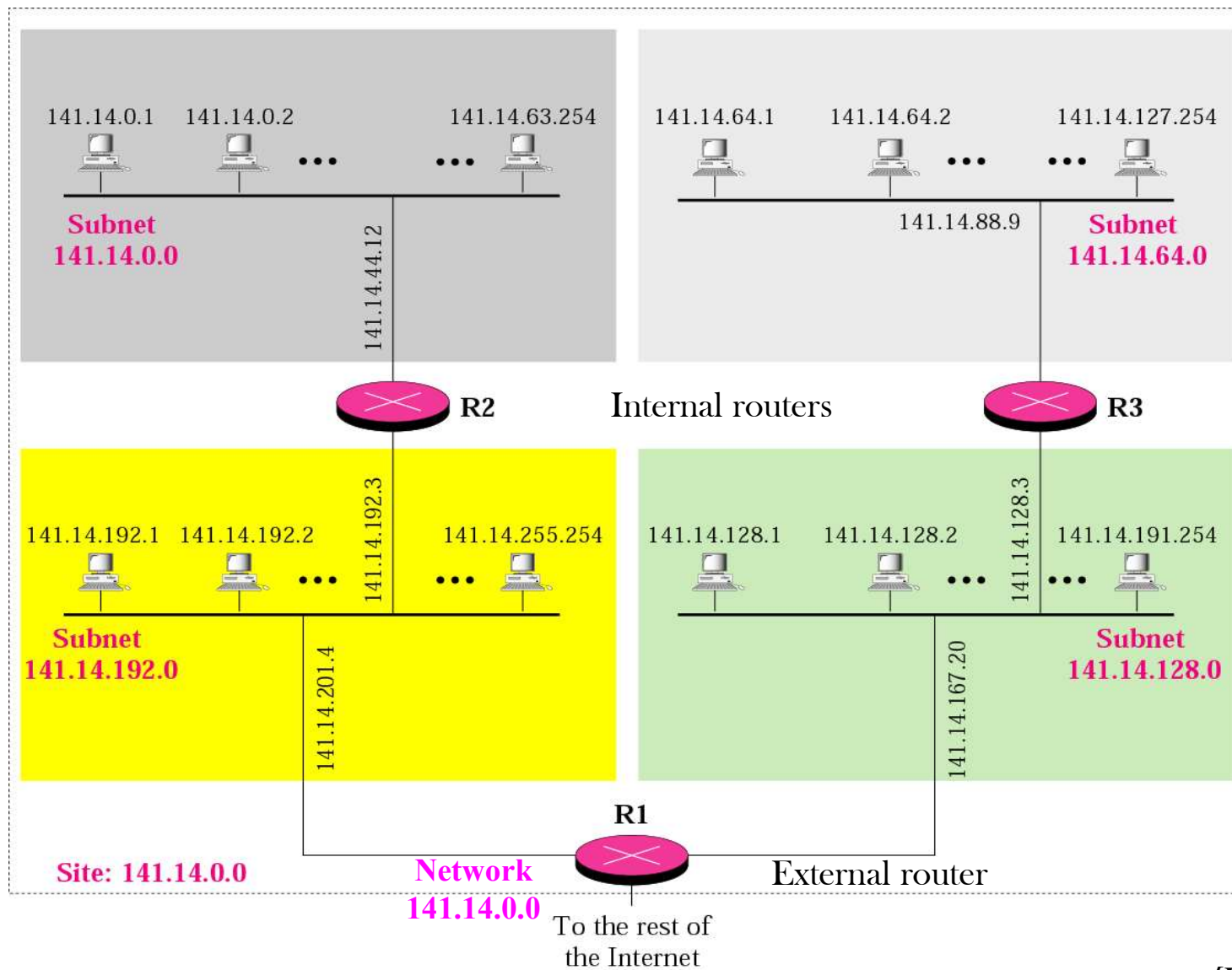
[Forouzan]

Subnet Mask(子网掩码)

- How a router knows the network address?
 - ◆ Using the **subnet mask**.
- **Boolean ANDed** (按位“与”) with destination address when routing
- If IP addresses of source and destination, and subnet mask are known, then we can find that
 - ◆ IP packet is sent to a host in same subnet, or
 - ◆ A host in another network



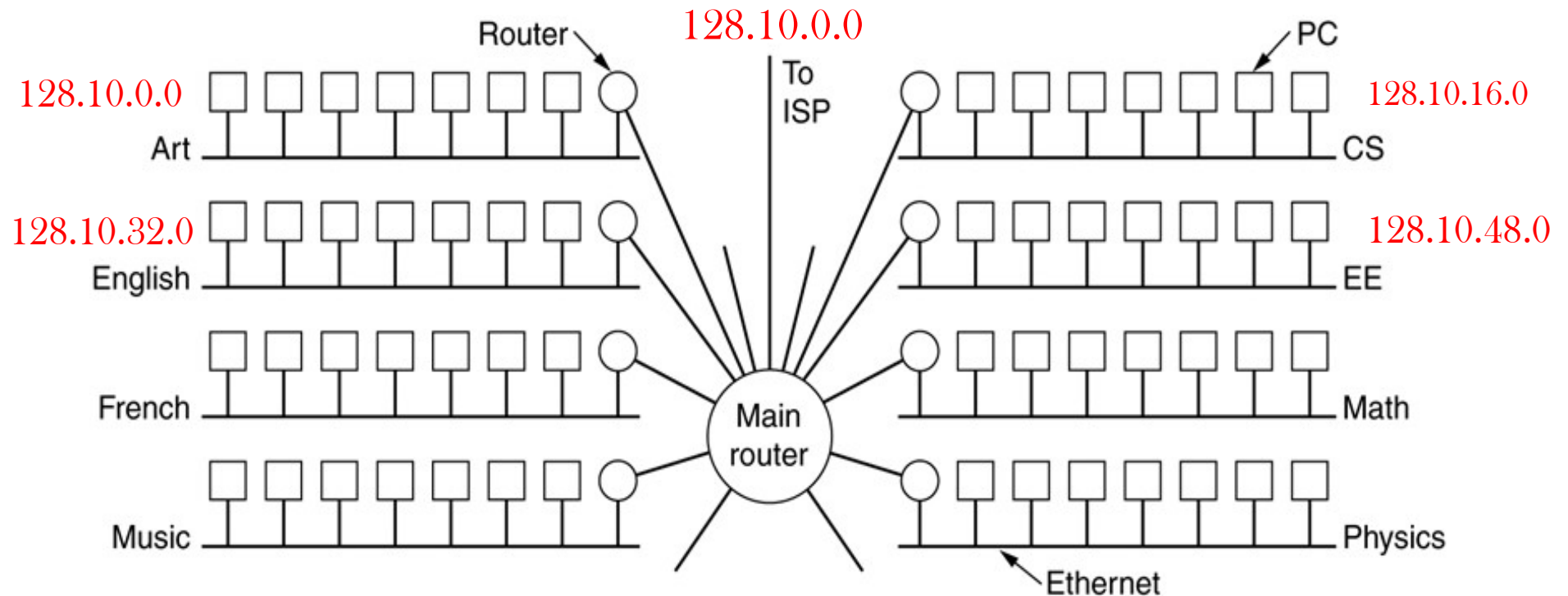
Example of Subnetting (1)



[Forouzan]

Example of Subnetting (2)

A campus network consisting of LANs for various departments



Subnet mask: 255.255.240.0

What can we get from subnet mask?

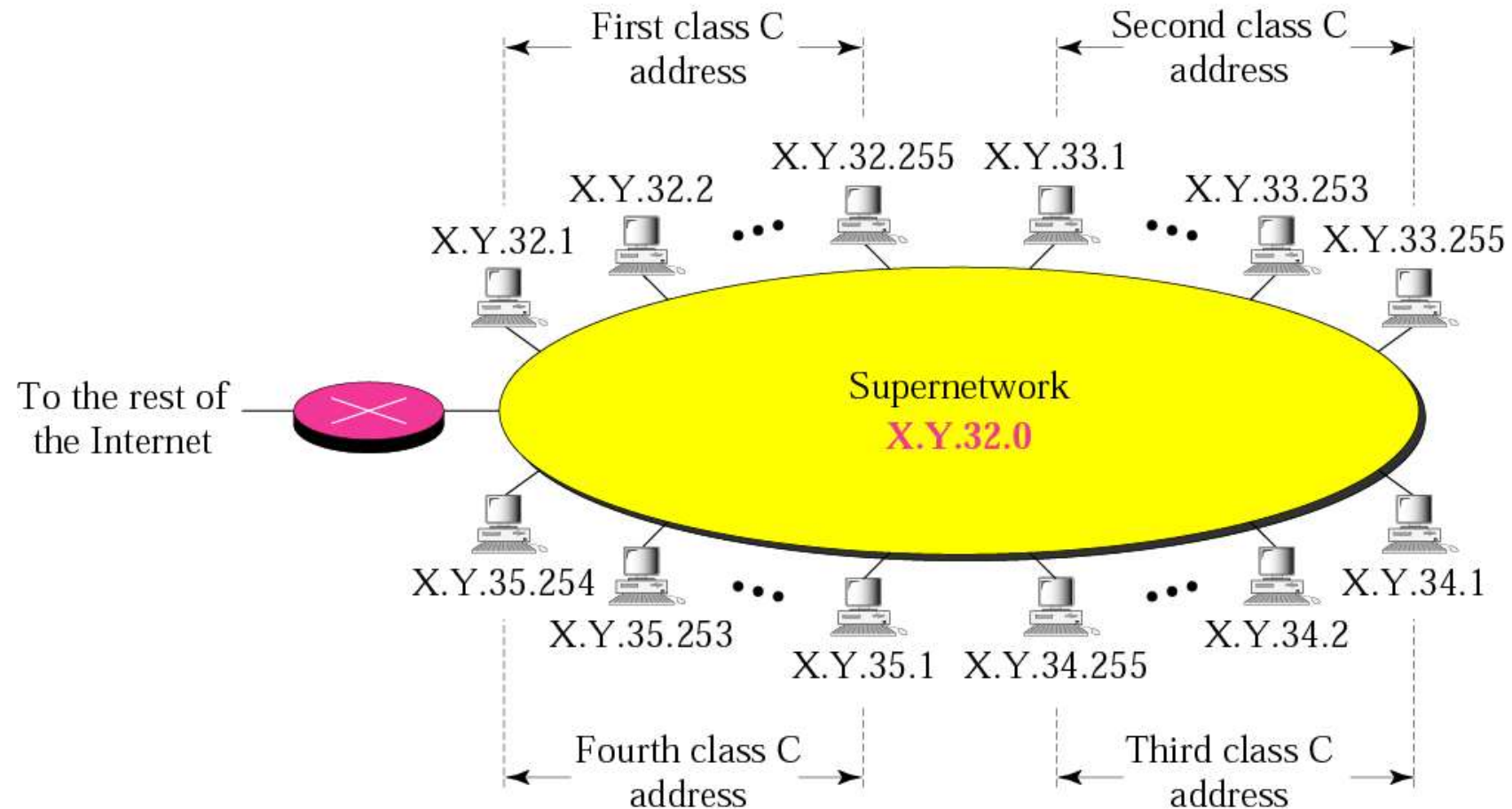
- Number of IP addresses in the subnet.



Supernetting(超网划分)

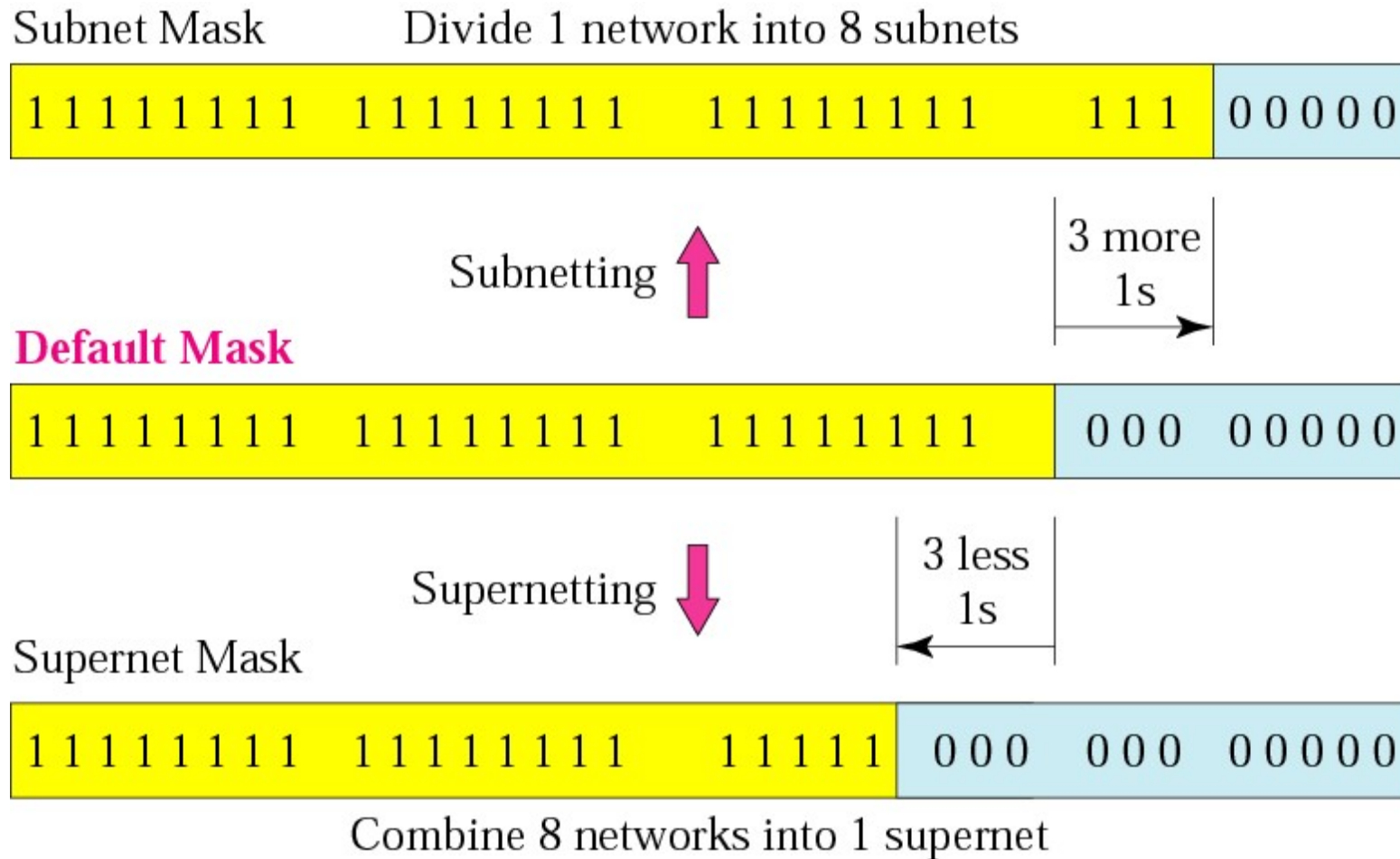
- Problem of **Class C** addresses: the size of a block is too small to satisfy the needs of an organisation
- Solution: supernetting
 - ◆ To combine several continuous class C blocks to create a large range of addresses
 - ◆ i.e. several networks are combined to create a **supernetwork** (超网)
- The number of class C blocks must be a power of 2
- The blocks must be **contiguous** in the address space (no gaps between the blocks).

Example of a Supernetwork



Supernet mask: 255.255.252.0

Comparison of subnet, default, and supernet masks

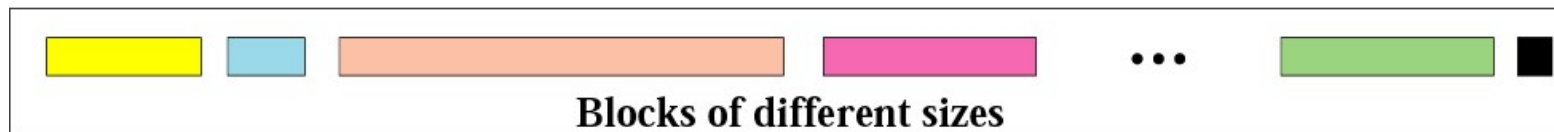


[Forouzan]

Classless Addressing(无类别地址)

- Until mid-1990s, a range of addresses meant a block of addresses in class A, B or C.
- Problems of Classful addressing
 - ◆ Continued exponential Internet growth causes address space exhaustion
 - ◆ Inefficient use of address space
 - ◆ Explosion in the size of the Internet routing tables
- Solution: Classless addressing (from 1996)
- The idea is to have **variable-length blocks** that belongs to **no class**

Address Space



Classless InterDomain Routing (CIDR, 无类别域间选路)

■ CIDR Address Format: slash notation

a.b.c.d/x, where x is # bits in network portion of address



■ Rules of addressing

- ◆ The number of addresses in a network(subnet) **must be a power of 2**
- ◆ The beginning address must be **evenly divisible** by the number of addresses.
 - For example, if a subnet contains 4 addresses, the beginning address must be divisible by 4.

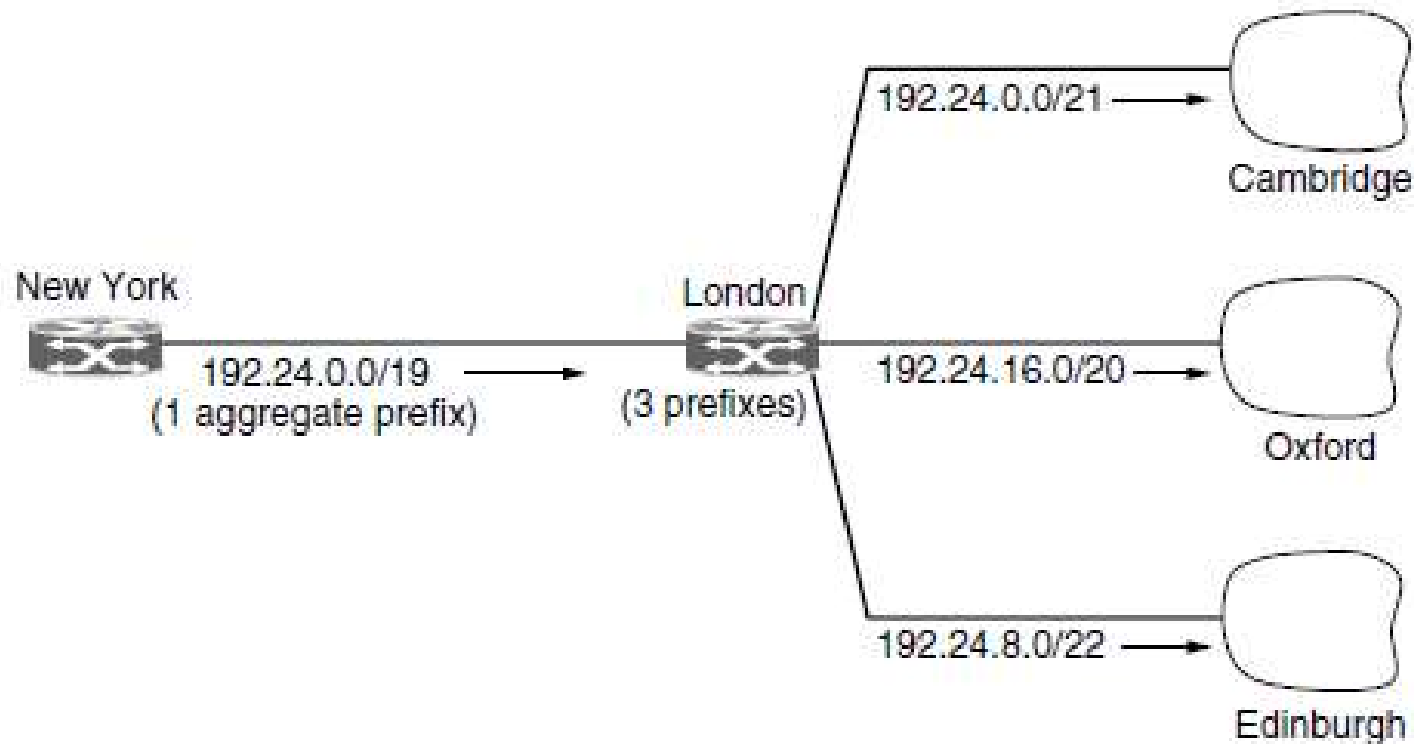
CIDR Prefixes: Dotted Decimal Equivalents

CIDR Notation	Dotted Decimal	CIDR Notation	Dotted Decimal
/1	128.0.0.0	/17	255.255.128.0
/2	192.0.0.0	/18	255.255.192.0
/3	224.0.0.0	/19	255.255.224.0
/4	240.0.0.0	/20	255.255.240.0
/5	248.0.0.0	/21	255.255.248.0
/6	252.0.0.0	/22	255.255.252.0
/7	254.0.0.0	/23	255.255.254.0
/8	255.0.0.0	/24	255.255.255.0
/9	255.128.0.0	/25	255.255.255.128
/10	255.192.0.0	/26	255.255.255.192
/11	255.224.0.0	/27	255.255.255.224
/12	255.240.0.0	/28	255.255.255.240
/13	255.248.0.0	/29	255.255.255.248
/14	255.252.0.0	/30	255.255.255.252
/15	255.254.0.0	/31	255.255.255.254
/16	255.255.0.0	/32	255.255.255.255

Figure 10.11 Dotted decimal mask values for all possible CIDR prefixes.

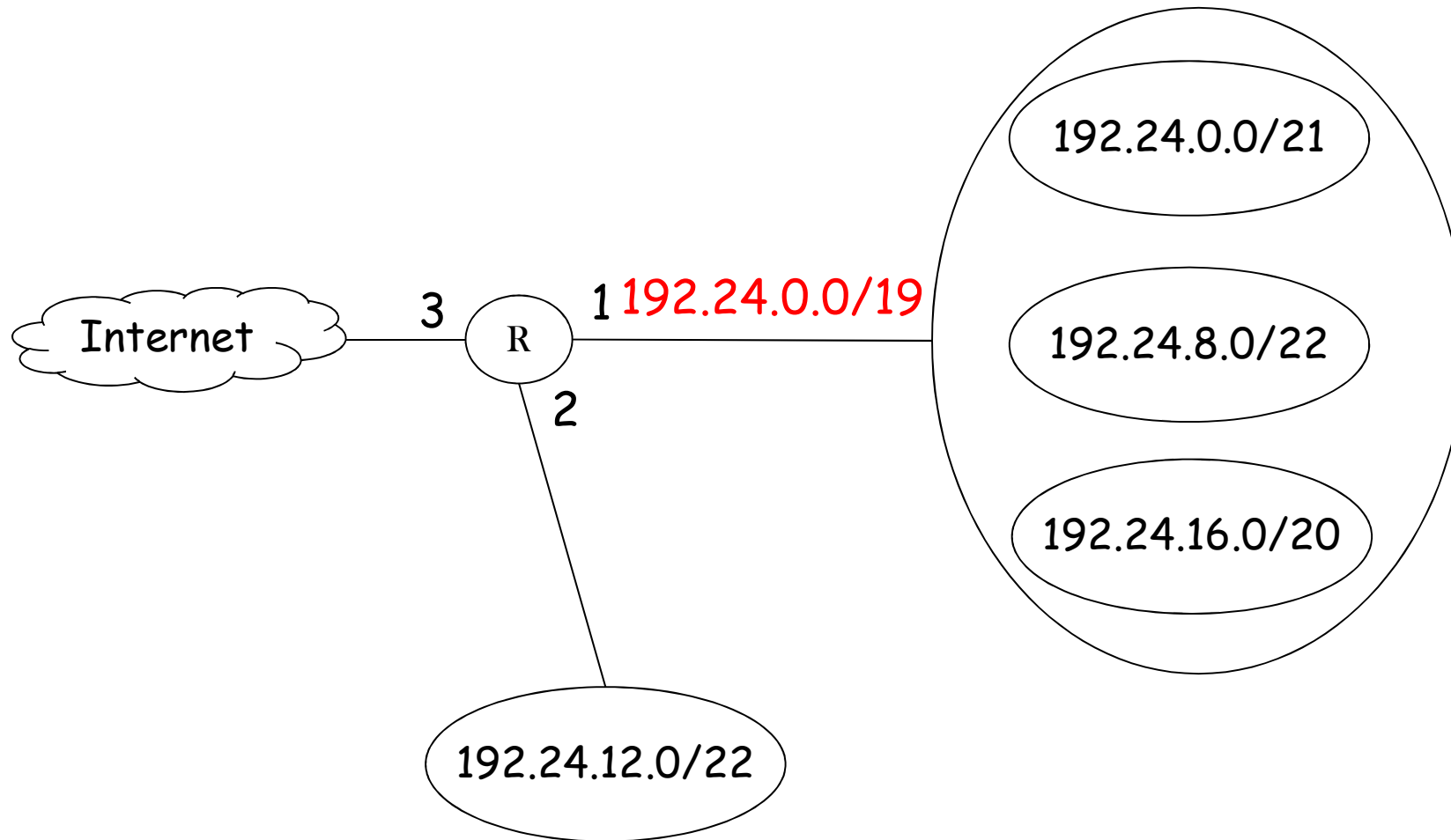
CIDR: Route Aggregation(路由聚合/汇聚)

- CIDR permits route aggregation
 - ◆ reduce the Internet routing table



问题：地址空间被放大

Example of routing aggregation



网内的地址数与可分配(可用)地址

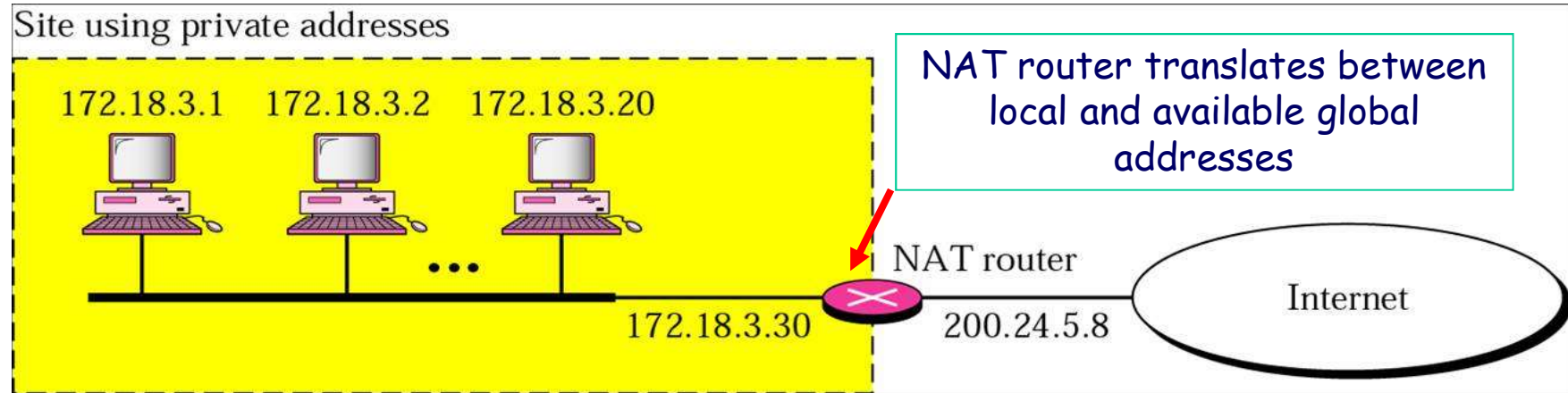
- 网络地址: $x.x.x.x/m$
- 网络号 m 位, 主机号 $=32-m=n$ 位
- 该网络内共有 2^n 个地址
- 其中第一个地址 (主机号为全0) 为网络地址
最后一个地址 (主机号为全1) 为广播地址
- 可分配 (可用) 地址数: 2^n-2

NAT: Network Address Translation

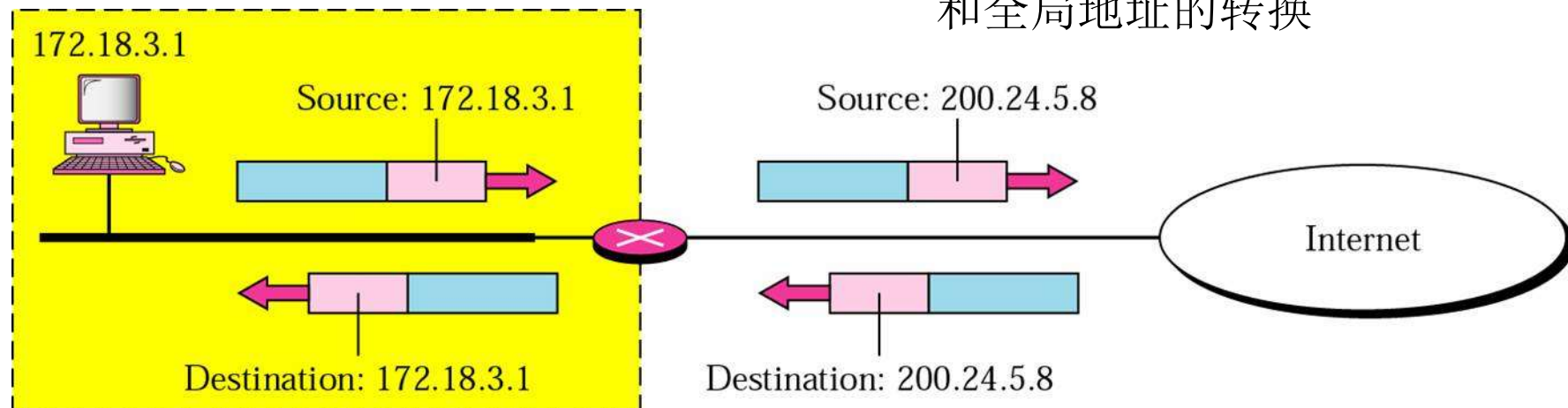
(网络地址翻译)

- Problem: IP addresses are scarce
- Solution: using reserved Private IP addresses(私有/专用地址)
 - ◆ 10.0.0.0 - 10.255.255.255/8
 - ◆ 172.16.0.0 - 172.31.255.255/12
 - ◆ 192.168.0.0 - 192.168.255.255/16
- NAT router converts internal IP source address to another global IP address
- NAT box is often combined with a firewall in a single device
- Source TCP/UDP port is also converted by NAT box

NAT Sample

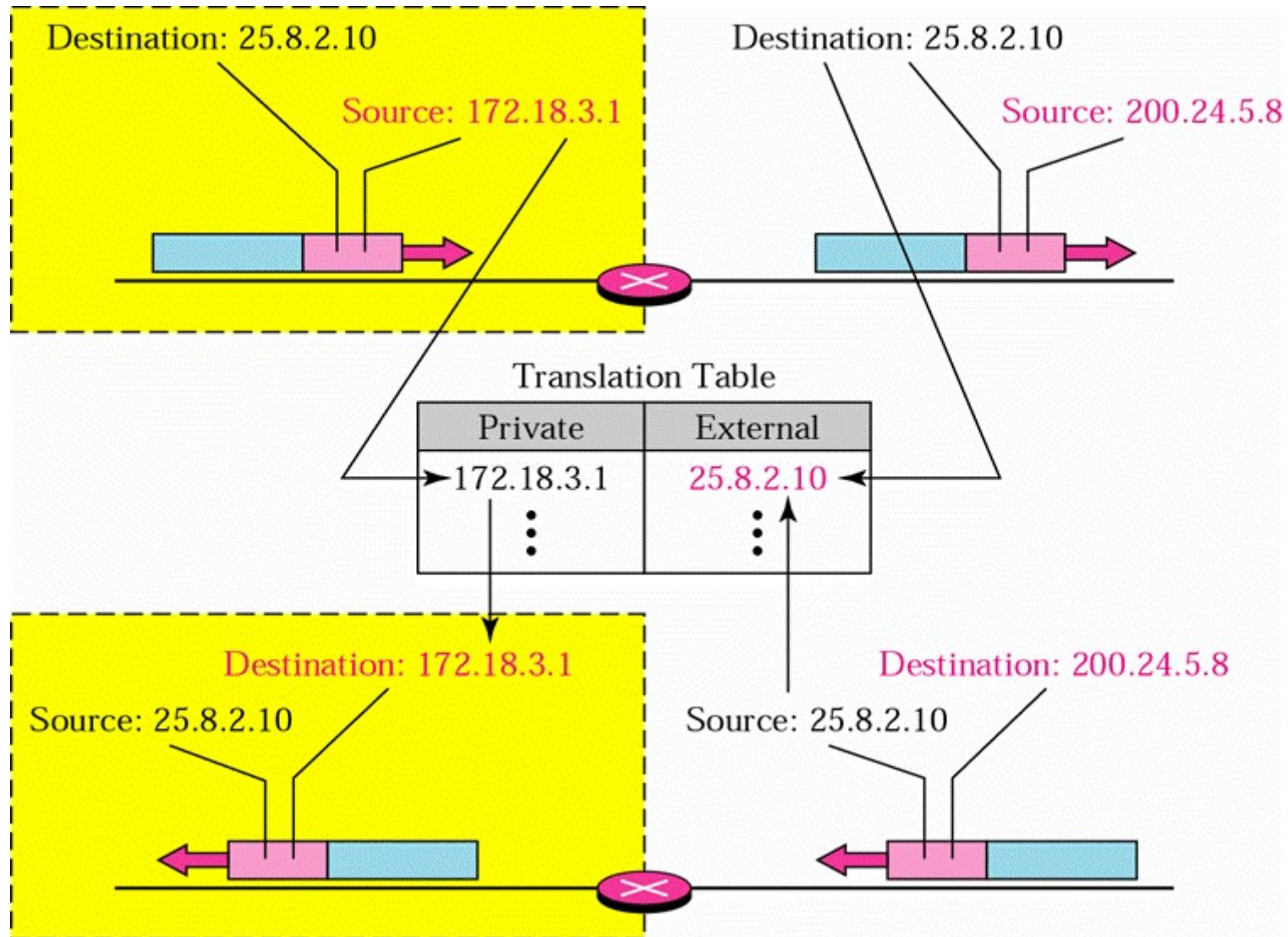


NAT路由器负责私有地址
和全局地址的转换



[Forouzan]

Address Translation



[Forouzan]

What if 2 local hosts talking to same destination?

- Using port number
 - ◆ Layering discipline is violated

Translation Table

<i>Private Address</i>	<i>Private Port</i>	<i>External Address</i>	<i>External Port</i>	<i>Transport Protocol</i>
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
...	...	...	...	...

[Forouzan]

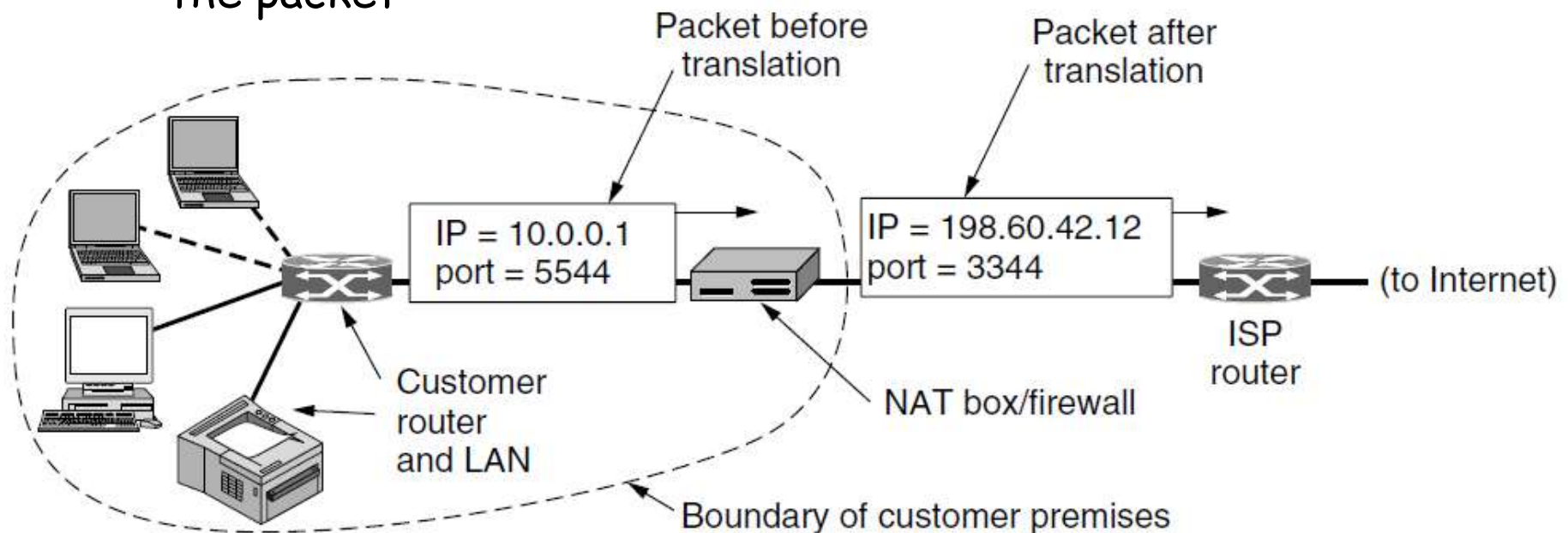
Placement and Operation of a NAT box

■ For outgoing packet

- ◆ 10.x.y.z source address → customer's **true IP address**
- ◆ **Source port** → **an index** into the NAT box's 65,536-entry translation table.

■ For incoming packet

- ◆ **Source port** is used **as an index** into the NAT box's mapping table
- ◆ **Internal IP address** and **original Source port** are inserted into the packet



Review of IP Addressing

- 32bits, dotted-decimal notation, 2 levels(net-id + host-id)
- **Network address**: used in routing
- Classful addressing: A, B and C classes
 - ◆ Fixed size of each class, inflexible and waste of addresses
 - ◆ Default mask
- Subnetting: using portion of host-id
 - ◆ **Subnet mask**
- Supernetting: using portion of net-id
 - ◆ Supernet mask
- How to prevent address exhaustion?
 - ◆ Classless addressing: **CIDR, Slash notation**
 - ◆ Re-use private addresses: **NAT**



关于**IP**地址我们要掌握什么？

- “网络地址/前缀”的含义
- 已知网络需要的地址数（主机数），能够分配**IP**地址空间
- 把多个网络聚合成一个网络
- 子网掩码的作用：计算出网络地址
- 已知子网掩码(“/前缀”), 可以知道网络中有多少个**IP**地址，多少个可用地址
- 已知一个数据报的目的**IP**地址和子网掩码，可以获得目的网络地址，并根据该地址查路由表确定如何转发该**IP**包
- 注意**最长前缀匹配法**



Design Issues of IP

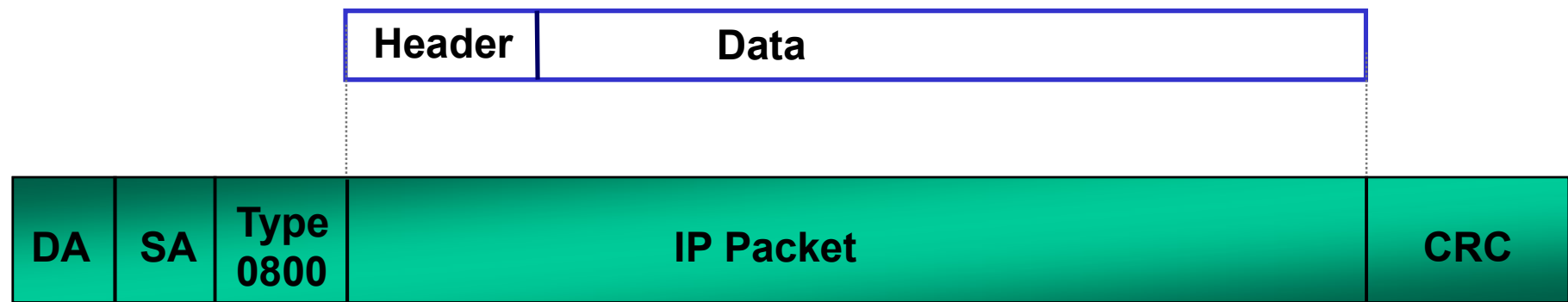
- Design requirements: accommodate differences in constituting networks:
 - ◆ Different addressing schemes
 - ◆ Different maximum packet size:
Fragmentation/Reassembly
 - ◆ Different network access mechanism
 - ◆ Different timeouts
 - ◆ Different transmission modes (connection-oriented, connectionless)
 - ◆ Error control
 - ◆ Flow control

The Network Layer in Internet

- IP Addressing
- IP Packet format
- IPv6
- ICMP, ARP and DHCP
- Routing Protocols: RIP, OSPF and BGP
- IP Multicast
- Mobile IP

IP Datagram

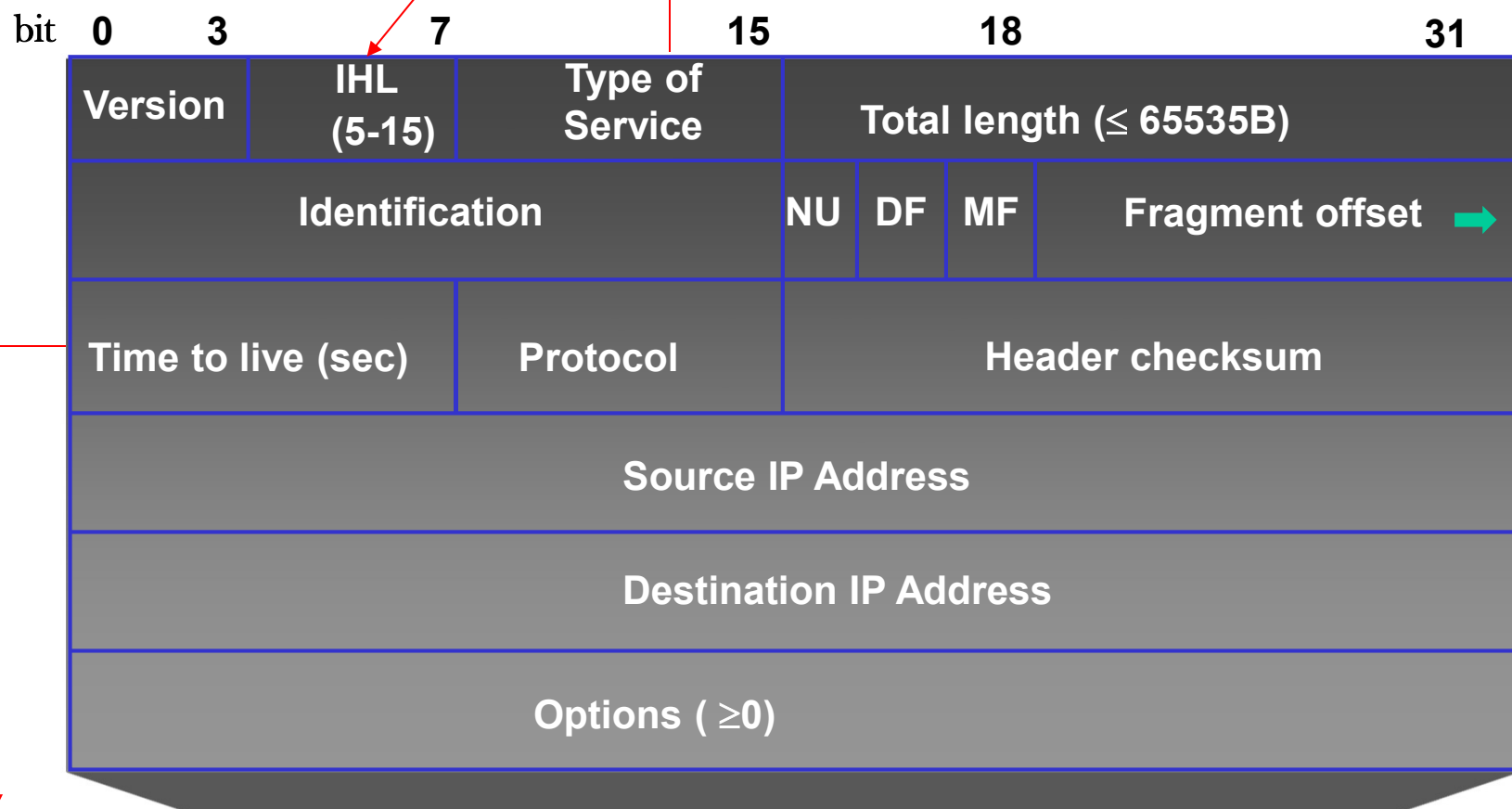
■ Independent of Hardware



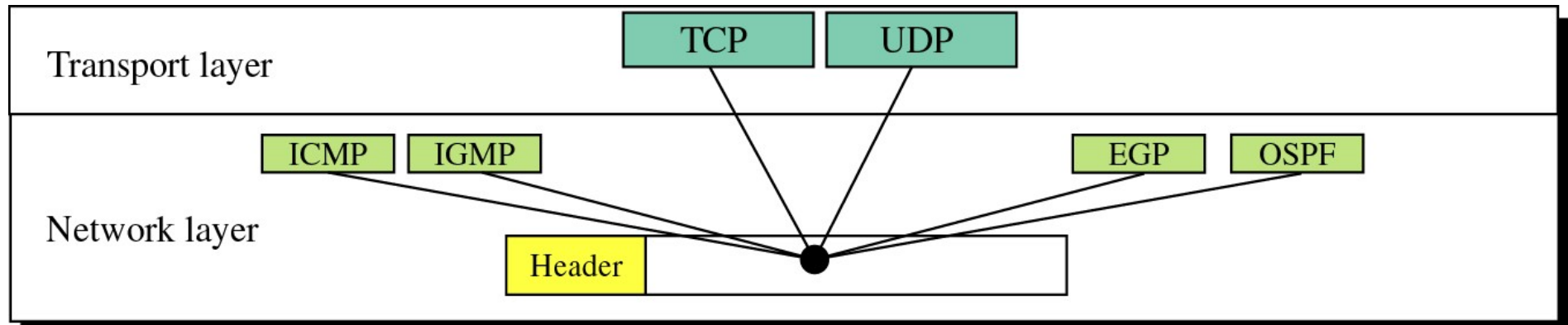
↑
Indicating data in Ethernet
frame is IP packet

Encapsulation of IP Datagram in Ethernet Frame

IP Header



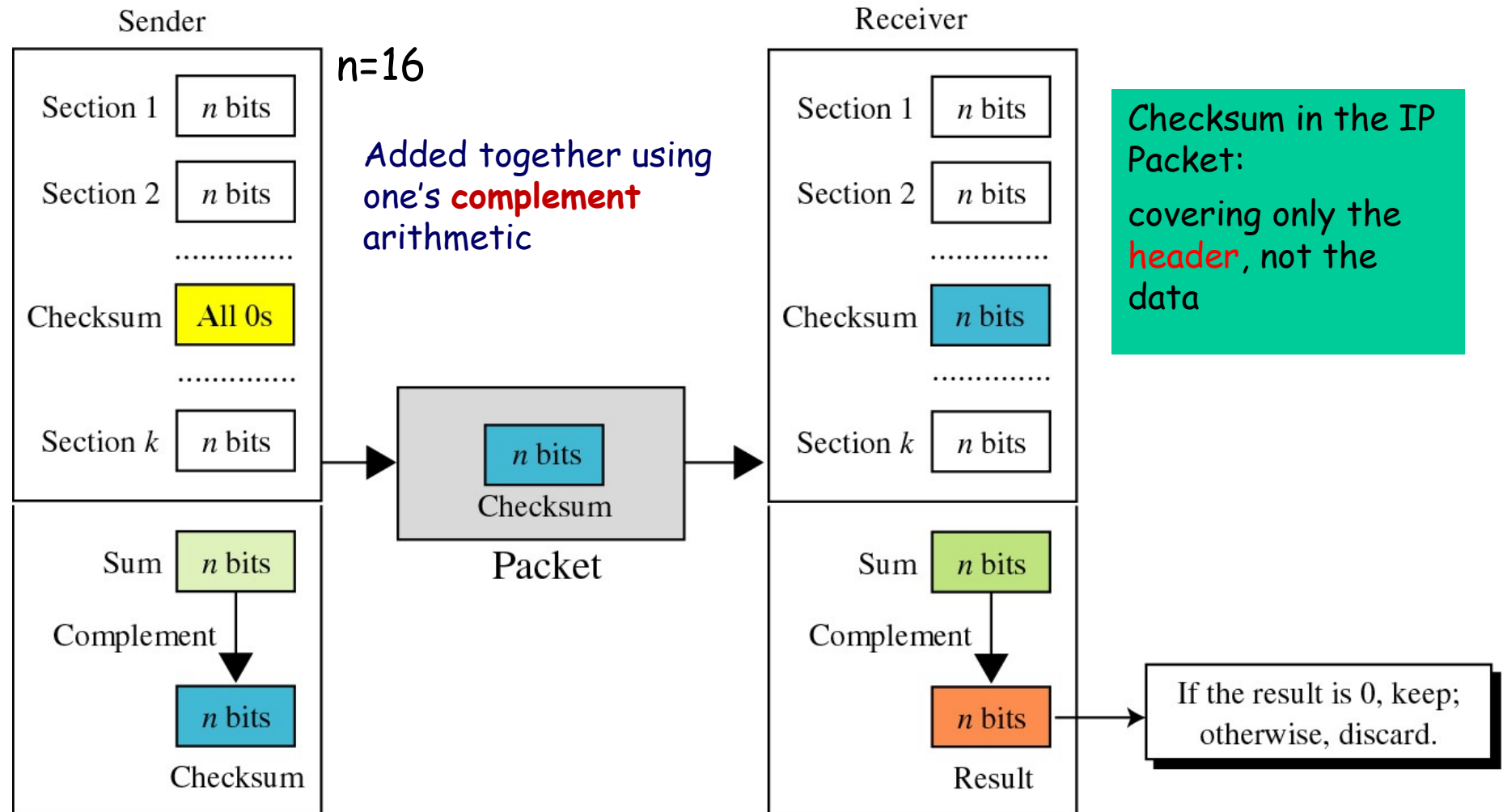
Protocol Field



- Defines the higher level protocol (transport-layer) that uses the services of IP-layer.
- An IP datagram can encapsulate data from several higher-level protocols such as: TCP=6, UDP=17, ICMP=1, IGMP=2, OSPF=89
- Specifies final destination protocol to which IP datagram should be delivered

[Forouzan]

Checksum(校验和/检验和) Concept(RFC 1071)



Words from RFC1071

- In outline, the Internet checksum algorithm is very simple:
 - ◆ (1) Adjacent octets to be checksummed are paired to form 16-bit integers, and the 1's complement sum of these 16-bit integers is formed.
 - ◆ (2) To generate a checksum, the checksum field itself is cleared, the 16-bit 1's complement sum is computed over the octets concerned, and the 1's complement of this sum is placed in the checksum field.
 - ◆ (3) To check a checksum, the 1's complement sum is computed over the same set of octets, including the checksum field. If the result is all 1 bits (-0 in 1's complement arithmetic), the check succeeds.

Example of Checksum Calculation

4	5	0	28	
1			0	0
4	17		0	
10.12.14.5				
12.6.7.9				

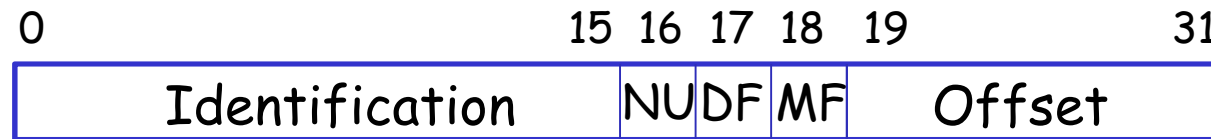
4, 5, and 0	→	01000101	00000000
28	→	00000000	00011100
1	→	00000000	00000001
0 and 0	→	00000000	00000000
4 and 17	→	00000100	00010001
0	→	00000000	00000000
10.12	→	00001010	00001100
14.5	→	00001110	00000101
12.6	→	00001100	00000110
7.9	→	00000111	00001001
Sum	→	01110100	01001110
Checksum	→	10001011	10110001
		11111111	11111111



Fragmentation and Re-assembly (分段/分片和重装)

- Problem: A datagram can travel through different networks. different networks have different maximum frame sizes.
 - ◆ **MTU**: maximum transmission unit(最大传输单元)
 - Ethernet: MTU = 1500 bytes.
 - WiFi(IEEE802.11): MTU = 2312 bytes.
 - ◆ MTU may be as small as 128 bytes or less.
- Consequence: IP should adapt to different MTUs.
 - ◆ Choose convenient initial datagram size.
 - ◆ Divide datagrams when they traverse network with smaller MTU.
- IP datagram is fragmented by the **source host** or any **router** in the path.
- IP datagram is re-assembles **at destination** only
- Size of fragments determined by network with minimum MTU

Fragmentation and Re-assembly



■ Uses fields in header

◆ Data Unit Identifier (Identification)

- Unique integer that identifies datagram
- Copied into each fragment when datagram is fragmented

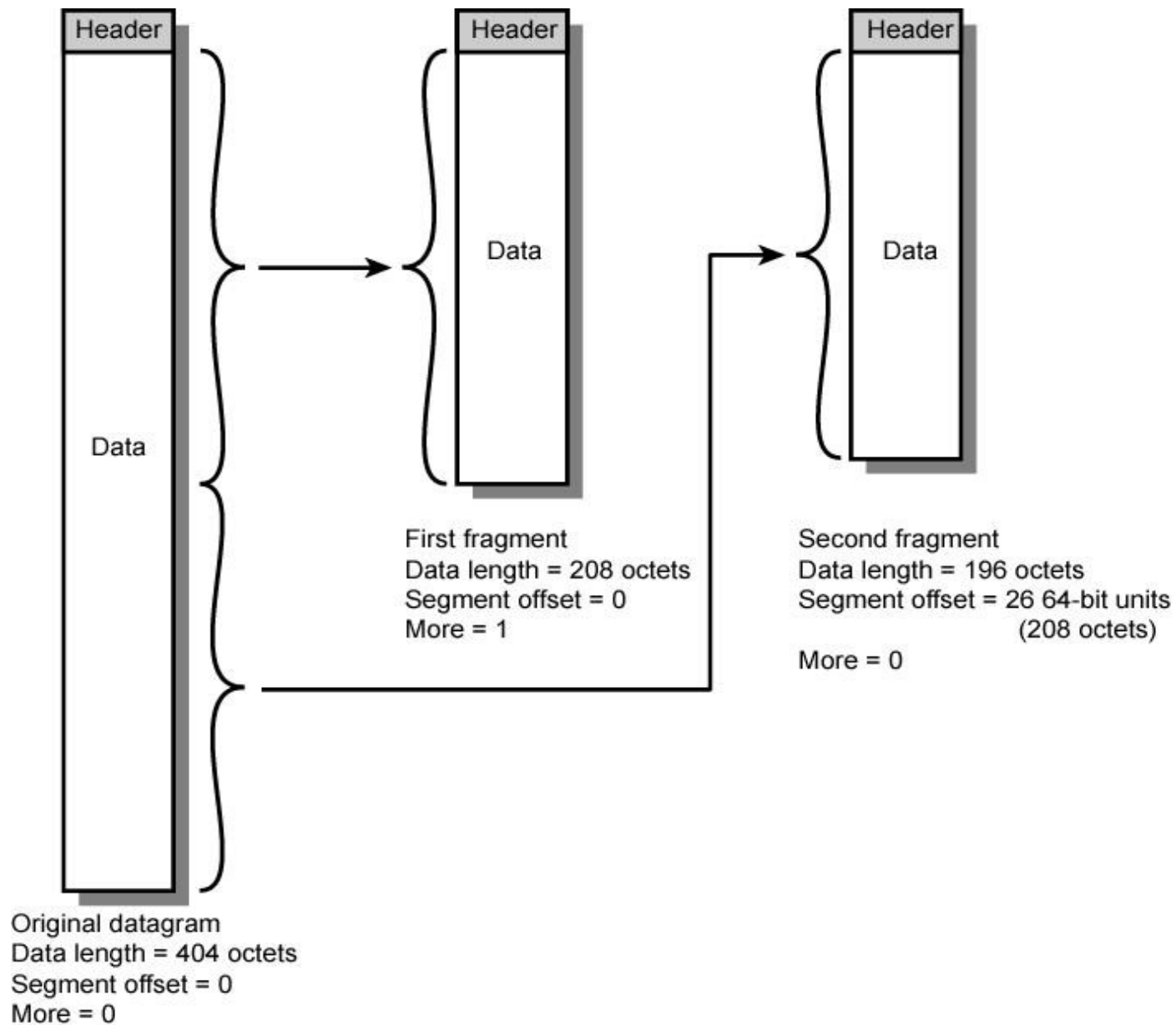
◆ DF: Do not Fragment datagram when DF=1

◆ MF(More Fragments): it is the last fragment when MF=0

◆ Offset

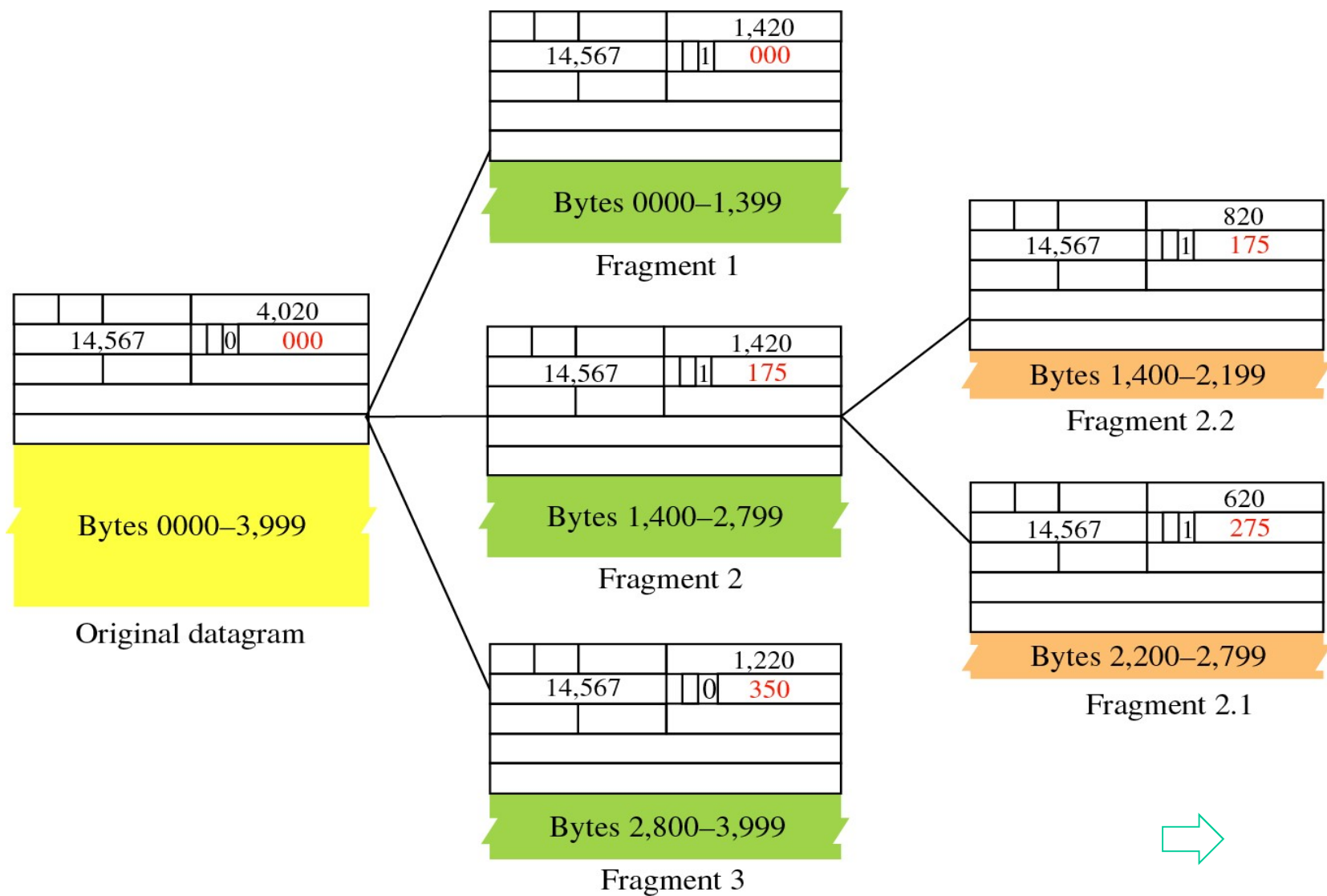
- Position of fragment of data in original datagram
- In multiples of 64 bits (8 octets)

Fragmentation Example (1)



[Stallings]

Fragmentation Example (2)



Some of the IP options

- Security option
- Strict source routing option 严格源选路
 - ◆ Gives the complete path from source to destination as a sequence of IP addresses
- Loose source routing option 宽松源选路
 - ◆ Requires the packet to traverse the list of routers specified, and in the order specified, but it is allowed to pass through other routers on the way
- Record route option 记录路由
 - ◆ Tells the routers along the path to append their IP address to the option field
- The Timestamp option 时间戳
 - ◆ Like the Record route option, except that in addition to recording its 32-bit IP address, each router also records a 32-bit timestamp

Option	Description
Security	Specifies how secret the datagram is
Strict source routing	Gives the complete path to be followed
Loose source routing	Gives a list of routers not to be missed
Record route	Makes each router append its IP address
Timestamp	Makes each router append its address and timestamp

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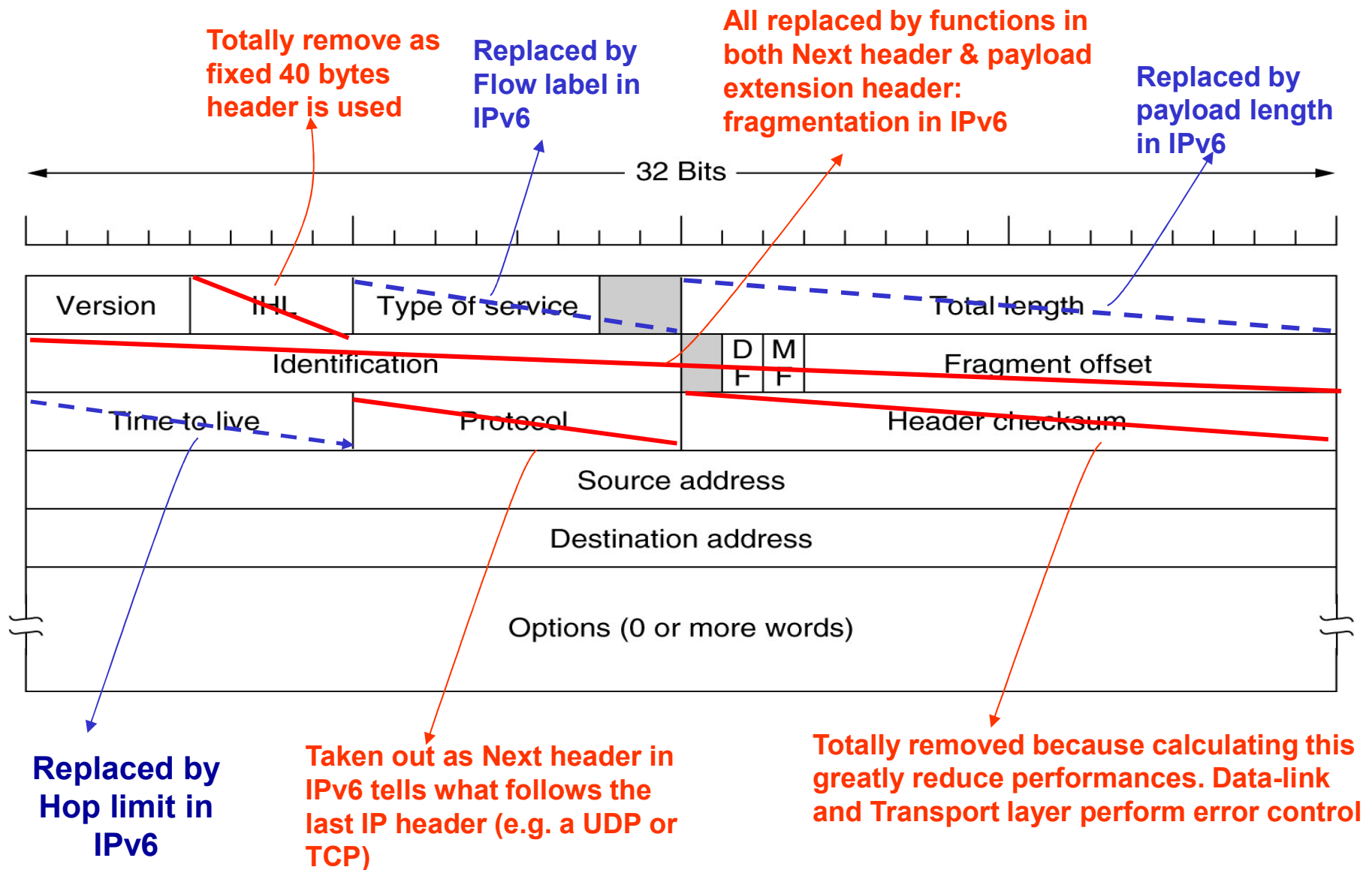
IPv6: Motivation

- IPv4 is becoming insufficient for the fast growing Internet and the needs:
 - ◆ The use of addressing space is becoming insufficient for conventional netid & hostid two-level addressing structure in Classful addressing
 - ◆ Real-time application i.e. video transmission requires minimum delay strategies and reservation of resource is not provided
 - ◆ Lack of security mechanism
 - ◆ Limitation of flexibility and mobility supported

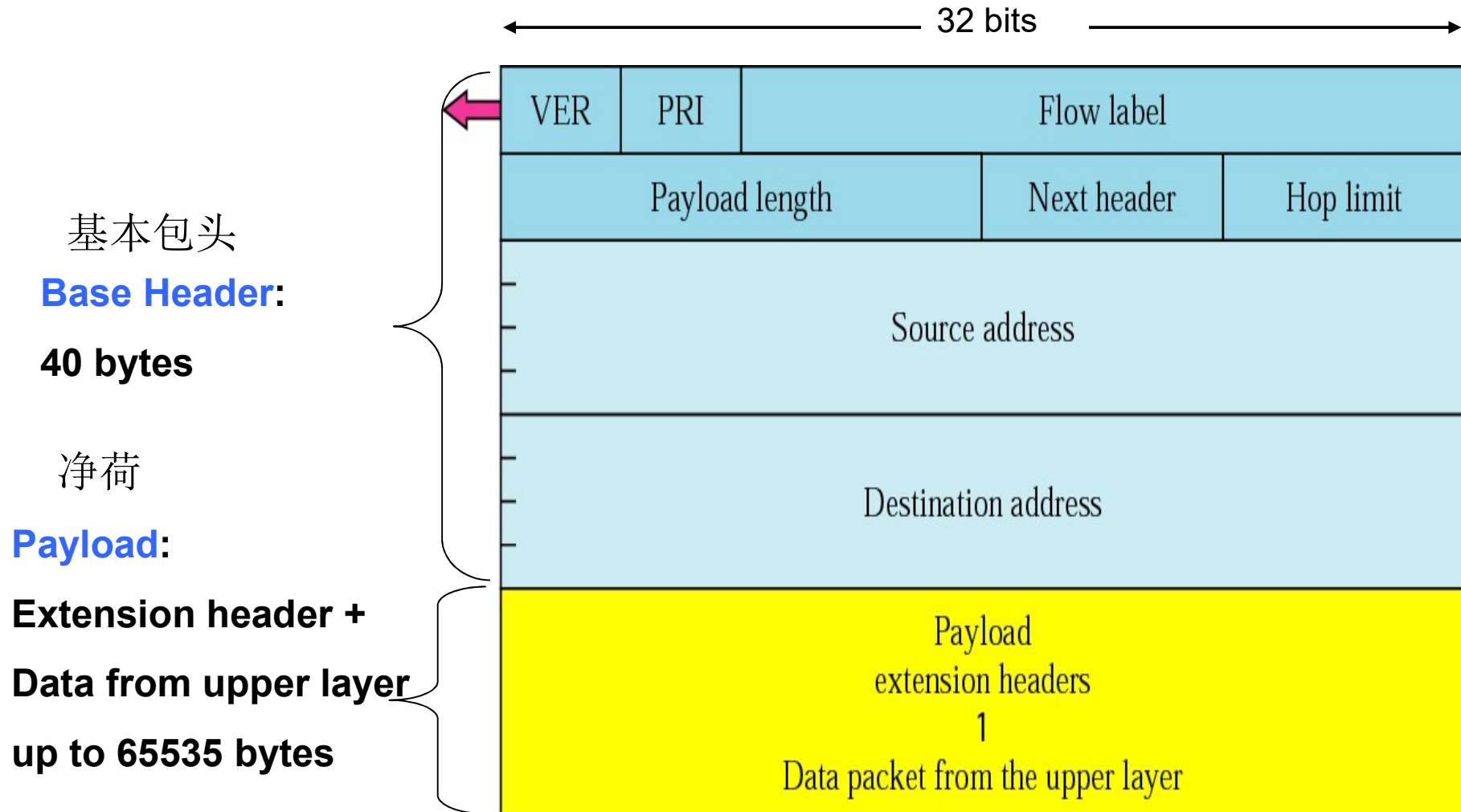
IPv6: Improvement

- **Longer addresses** : 128 bits long > 32 bits currently in IPv4, a huge increase of addressing space
- **Simplification in Header**: contains only 8 vital fields < 13 in IPv4. (speeds up routing process). - 5 fields becomes options.
- **Better support for options**: simple for routers, speeding up packet processing time
- **Security**: Authentication & privacy are key features of new IP
- **Improved QoS**: to handle special/real-time multimedia traffic.

IPv6: Revised IPv4 Header



IPv6 Datagram



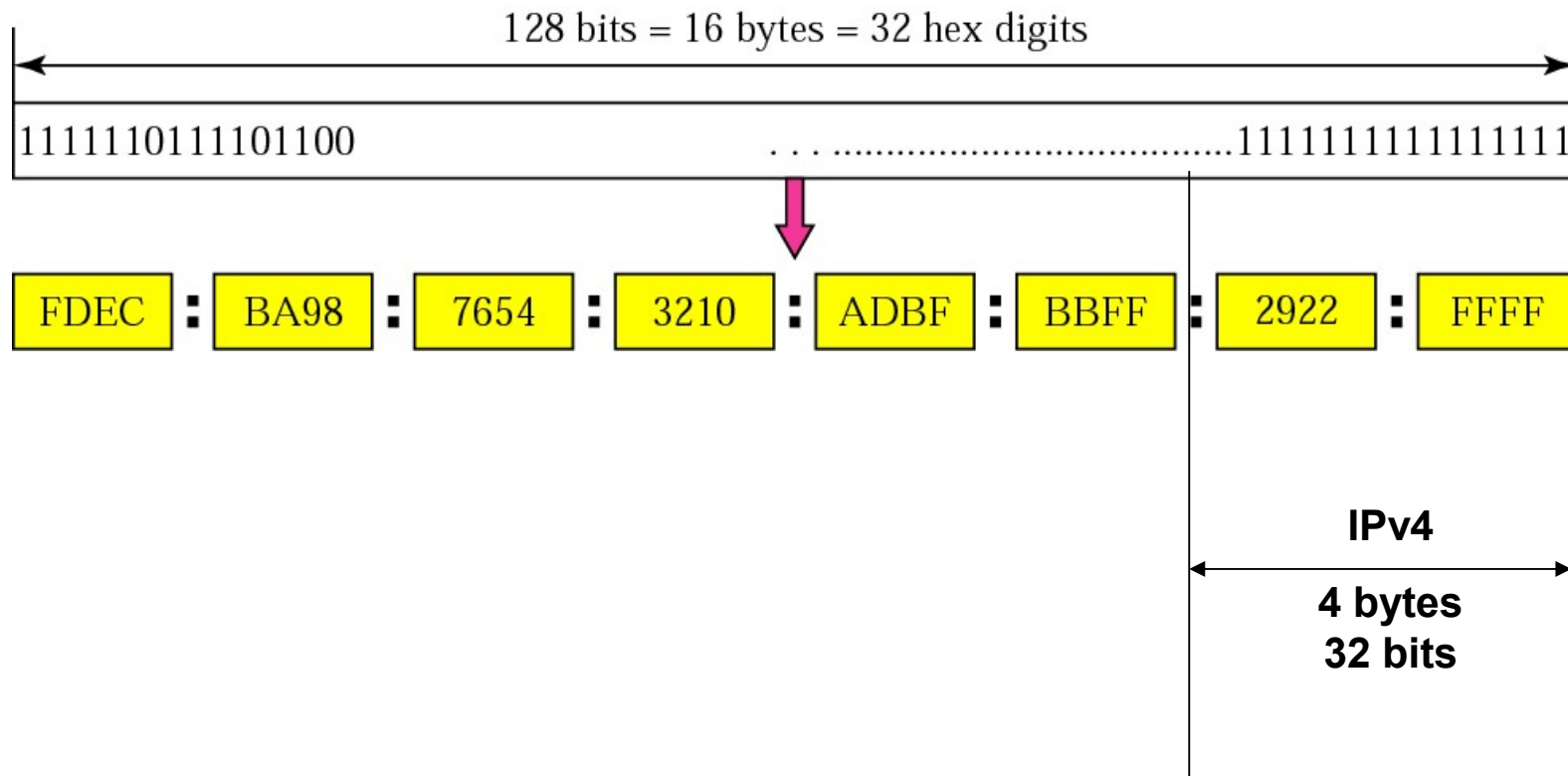
IPv6 Header: Fields

- **VER:** IP version; 4 for IPv4 and 6 for IPv6
- **PRI:** priority of packets; 0-7: for packets that can slow down in event of congestion; 8-15 for real-time traffic with constant sending rate - Audio & Video.
- **Flow label:** pseudo-connection between source and destination. for reserved bandwidth and stringent delay requirement. (Still experimental).
- **Payload length:** tells how many payload bytes follow the 40-bytes header
- **Next header:** the subsequent headers that follow the base header; either **optional extension headers** used by IP or **header for an upper-layer protocol**, e.g. UDP & TCP.
- **Hop limit:** serves the same purpose as the TTL in IPv4.
- **Source and Destination address field:** both 128 bits
- **Extension Headers:** 6 extension headers which is optional in IPv4.

The IPv6 Extension Headers

Extension header	Description
Hop-by-hop options	Miscellaneous information for routers
Destination options	Additional information for the destination
Routing	Loose list of routers to visit
Fragmentation	Management of datagram fragments
Authentication	Verification of the sender's identity
Encrypted security payload	Information about the encrypted contents

IPv6 Address(冒分十六进制)



[Forouzan]

Abbreviated Address

Unabbreviated

FDEC : BA98 : 0074 : 3210 : 000F : BBFF : 0000 : FFFF



FDEC : BA98 : 74 : 3210 : F : BBFF : 0 : FFFF

Abbreviated

Abbreviated

FDEC : 0 : 0 : 0 : 0 : BBFF : 0 : FFFF



FDEC :: BBFF : 0 : FFFF

More Abbreviated

关于IP包头

- 长度：20-60字节
- 分段：源主机/路由器分段，目的主机重装
使用“段标识、偏移量（单位为8字节）、MF”实现
- TTL：包可以继续传递的跳数，防止IP包兜圈子
- 校验和：分组累加取反，源主机/路由器/目的主机均校验，只校验IP包头

IPv6:

- 简化包头，加快路由器处理；增加对实时业务和安全支持
- 地址长度：16字节（128位），冒分十六进制

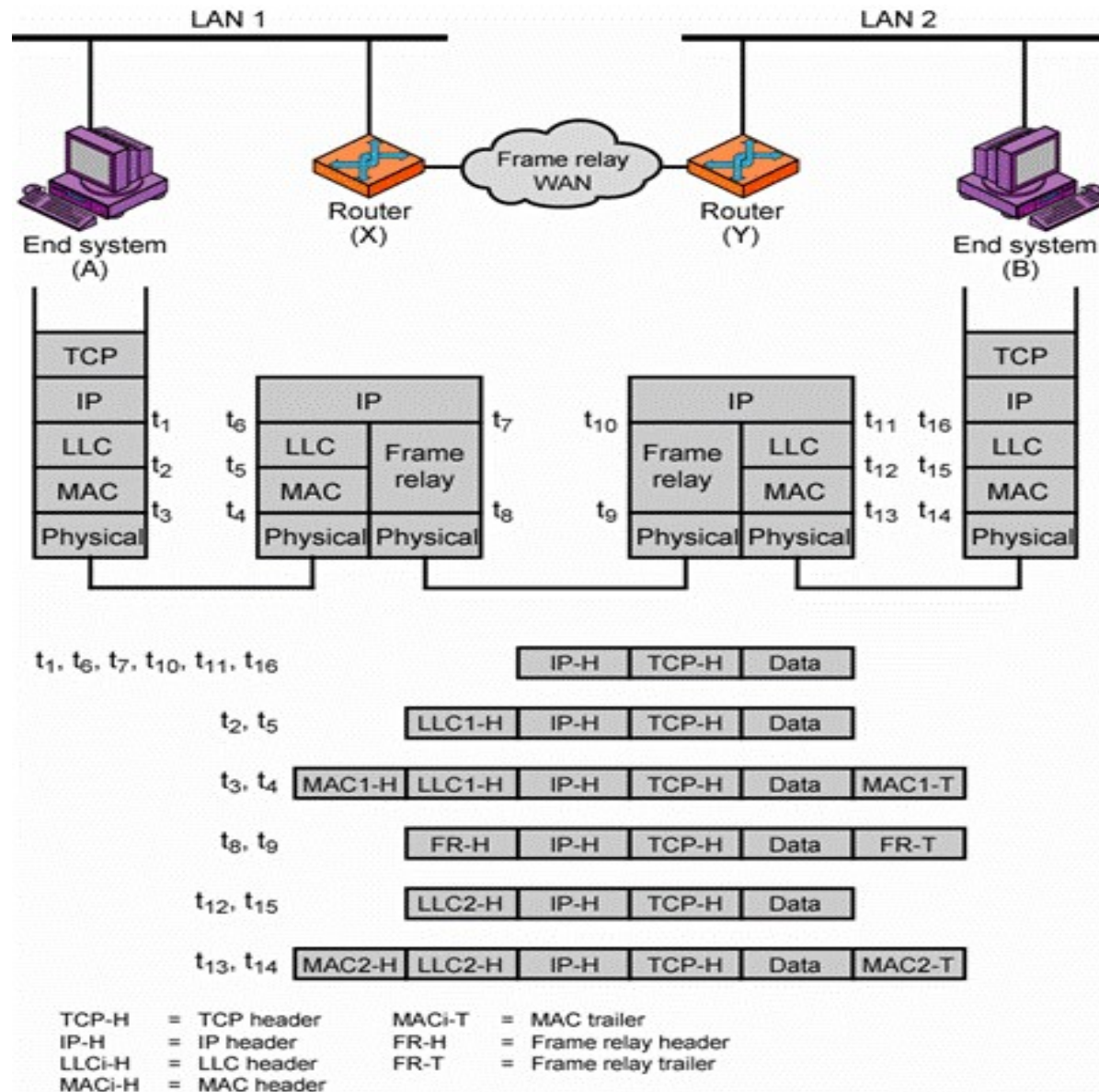
Design Issues of IP

- Design requirements: accommodate differences in constituting networks:
 - ◆ Different addressing schemes
 - ◆ Different maximum packet size
 - ◆ Different network access mechanism
 - ◆ Different timeouts
 - ◆ Different transmission modes (connection-oriented, connectionless)
 - ◆ Error control
 - ◆ Flow control

Different Network Access Mechanism

- Solved by abstracting networking functions in the Networking layer, and **relying on Data Link Layer** for networking access
- **Routers** handle the difference in network access mechanism

Operations of IP



[Stallings]

Different Timeout

- Problem: different networks use different timeout mechanisms.
- Solution: Introduce the concept of **Datagram Lifetime**
- Datagrams could loop indefinitely
 - ◆ Consumes resources
 - ◆ Transport protocol may need upper bound on datagram life
- Datagram marked with lifetime
 - ◆ **Time To Live (TTL, 生命期)** field in IP datagram
 - ◆ Once lifetime expires, datagram is discarded (not forwarded)
 - ◆ Hop count

Different Transmission Modes

- Connection-oriented and connectionless modes
- A **connectionless** architecture is adopted by IP

Error Control

- Minimal error control, error check only for the header
- Router should attempt to inform source if packet discarded (using **ICMP**, Internet Control Message Protocol, 因特网控制报文协议)
 - ◆ ICMP messages are encapsulated in IP
 - ◆ Error messages from a router or destination host
 - Destination host is unreachable,
 - TTL reached 0
 - IP header checksum failed
 - ICMP redirect, which tells the source of a better way to a destination, ...
- No guaranteed delivery
- May inform higher layer protocol

Flow Control

- Allows routers and/or destination to limit rate of incoming data
- Limited in connectionless systems
- Send flow control packets
 - ◆ Requesting reduced flow
- Some **ICMP** messages are used for flow control
 - ◆ Source quench messages(源抑制报文, choke packet)

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ICMP(因特网控制报文协议)

■ Reporting unexpected event to the source

◆ Destination Unreachable

- a router cannot locate the destination
- a packet with the **DF** bit cannot be delivered because a "small-packet" network stands in the way

◆ Time Exceeded

- a packet is dropped because its counter has reached zero

◆ Source Quench(源抑制)

- When a host received this message, it was expected to slow down

◆ Redirect

- a router notices that a packet seems to be routed wrong
- It is used by the router to tell the sending host about the probable error

■ Also used to test the Internet

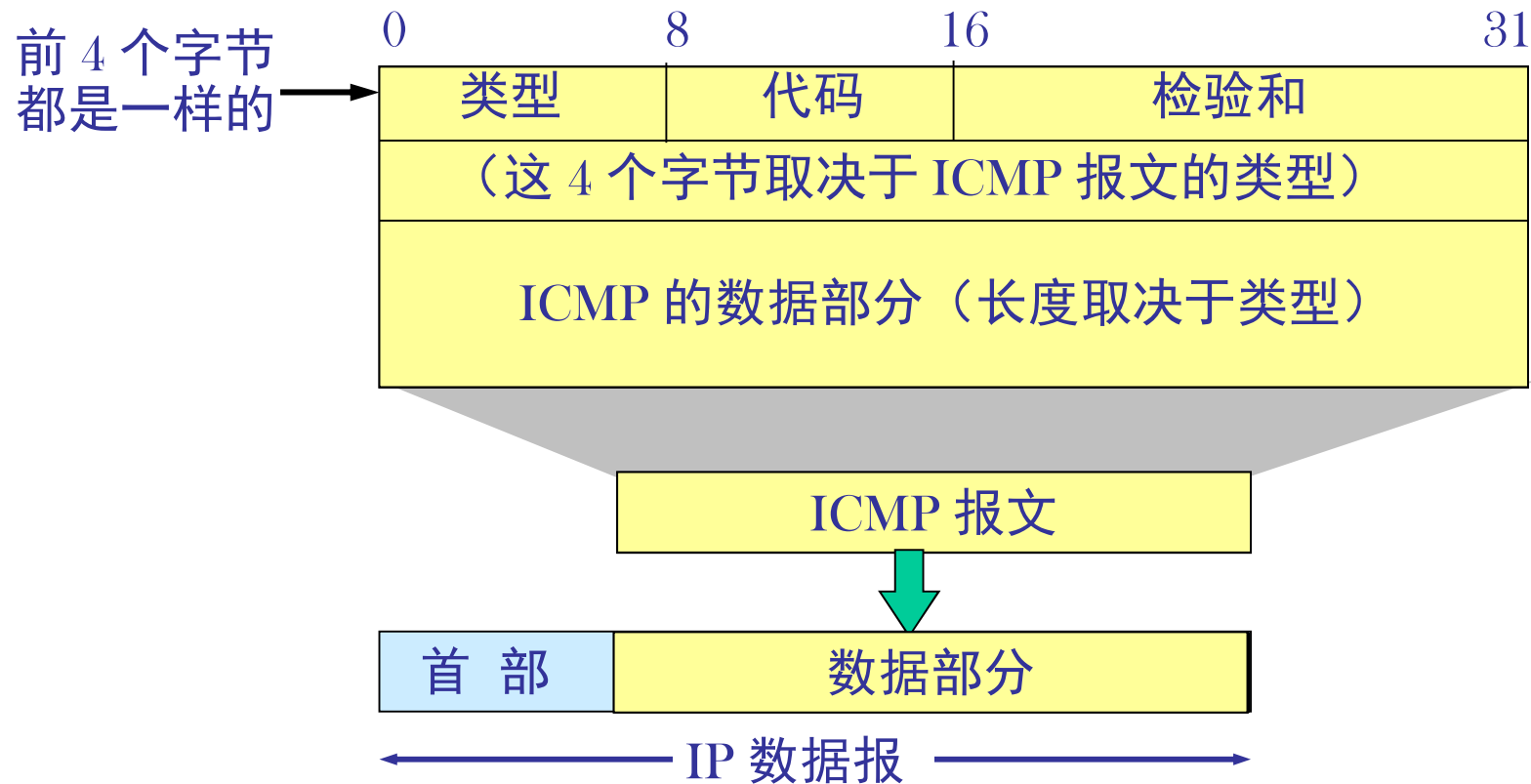
◆ ECHO and ECHO Reply: **ping**

◆ Time Stamp Request and Reply

◆ Router advertisement/solicitation: letting hosts find nearby routers

■ An ICMP message is encapsulated in an IP packet.

ICMP报文的格式



- ◆ ICMP报文分两类，即ICMP差错报告报文和ICMP询问报文
- ◆ ICMP 报文的前4个字节是统一的格式，共有三个字段：即类型、代码和检验和。接着的4个字节的内容与ICMP的类型有关

/谢/

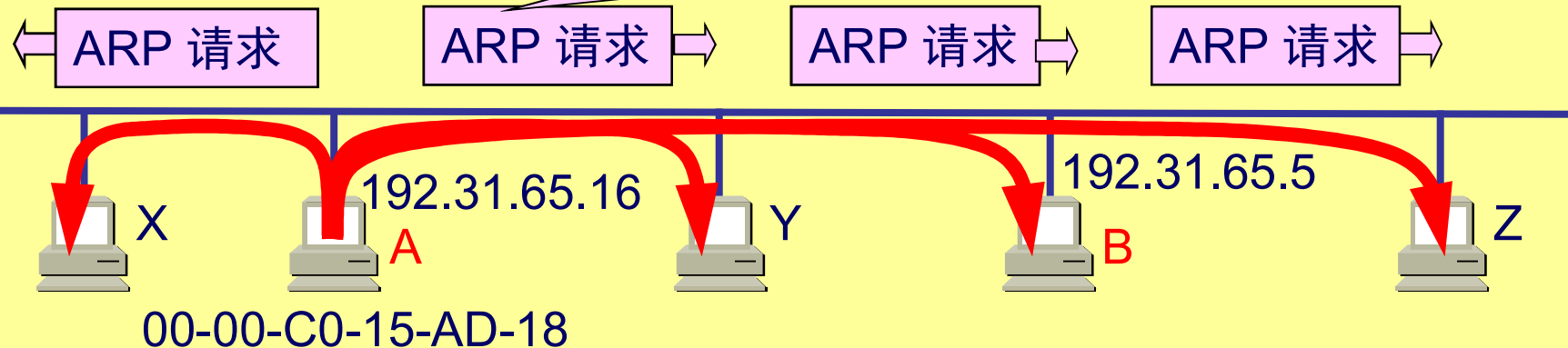
ARP: Address Resolution Protocol

(地址解析协议)

- How do IP addresses get mapped onto data link layer addresses, such as Ethernet?
 - ◆ One solution is to have a configuration file somewhere in the system that maps IP addresses onto Ethernet addresses
 - ◆ A better solution is to output a **broadcast** packet onto the Ethernet asking: Who owns IP address 192.31.65.5?

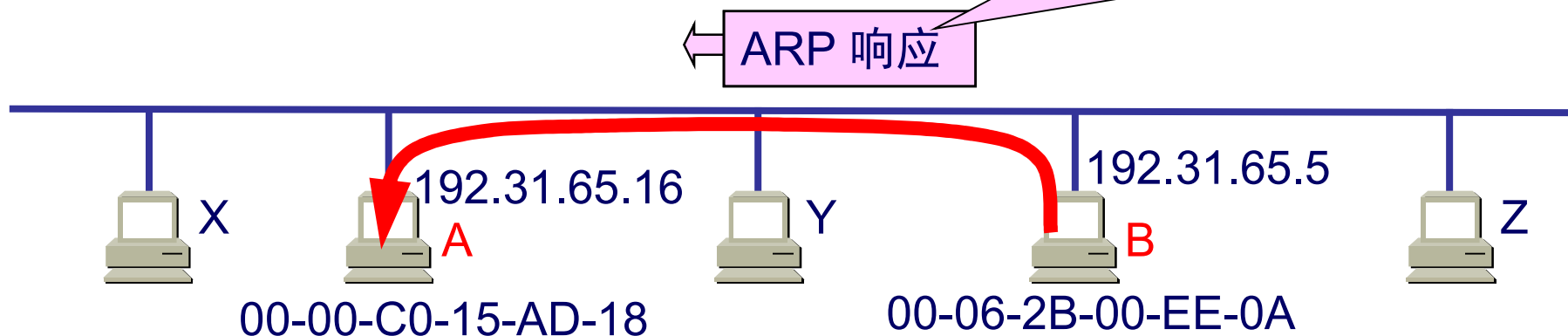
A Broadcasts ARP Request

我是 192.31.65.16，硬件地址是 00-00-C0-15-AD-18
我想知道主机 192.31.65.5 的硬件地址

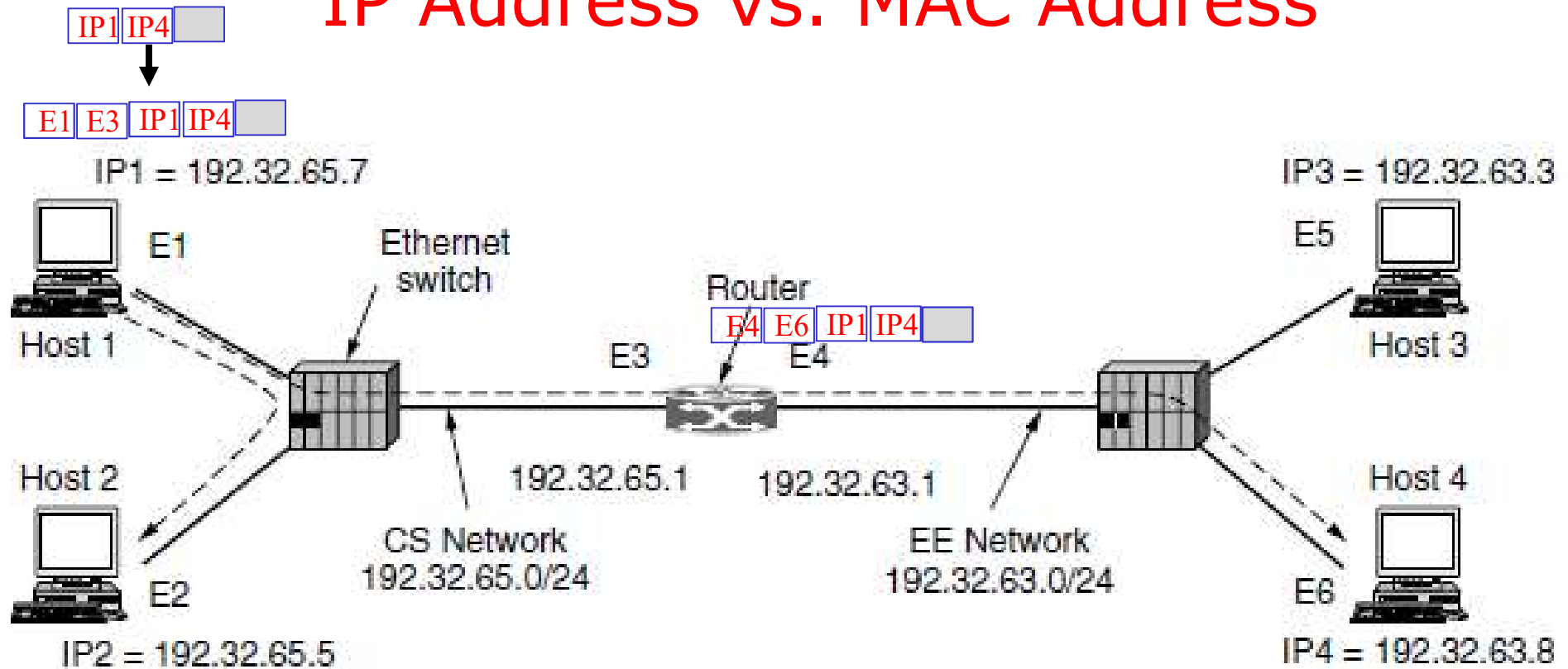


B unicasts to A ARP Response

我是 192.31.65.5
硬件地址是 00-06-2B-00-EE-0A



IP Address vs. MAC Address



Frame	Source IP	Source Eth.	Destination IP	Destination Eth.
Host 1 to 2, on CS net	IP1	E1	IP2	E2
Host 1 to 4, on CS net	IP1	E1	IP4	E3
Host 1 to 4, on EE net	IP1	E4	IP4	E6

IP地址全程有效，**MAC**只在一个子网内有效

RARP(逆向ARP)

■ Requirement

- ◆ Given an Ethernet address, what is the corresponding IP address?
- ◆ Example: when a diskless workstation is booted

■ RARP (Reverse Address Resolution Protocol)

- ◆ Allows a workstation to broadcast on Ethernet:
My Ethernet address is 14-04-05-18-01-25. Does anyone out there know my IP address?

BOOTP(启动协议)

■ BOOTP

- ◆ Used to configure diskless workstations
- ◆ Unlike RARP, BOOTP uses **UDP messages**, which are forwarded over routers
- ◆ It also provides the IP address of the file server, the IP address of the default router, and the subnet mask to use

■ Problem

- ◆ It requires **manual** configuration of tables mapping IP address to Ethernet address

DHCP

■ Dynamic Host Configuration Protocol (动态主机配置协议)

- ◆ Provides **automatic configuration** of remote hosts
- ◆ Including IP address, Router's address, subnet mask, DNS server's address
- ◆ An **application layer** protocol

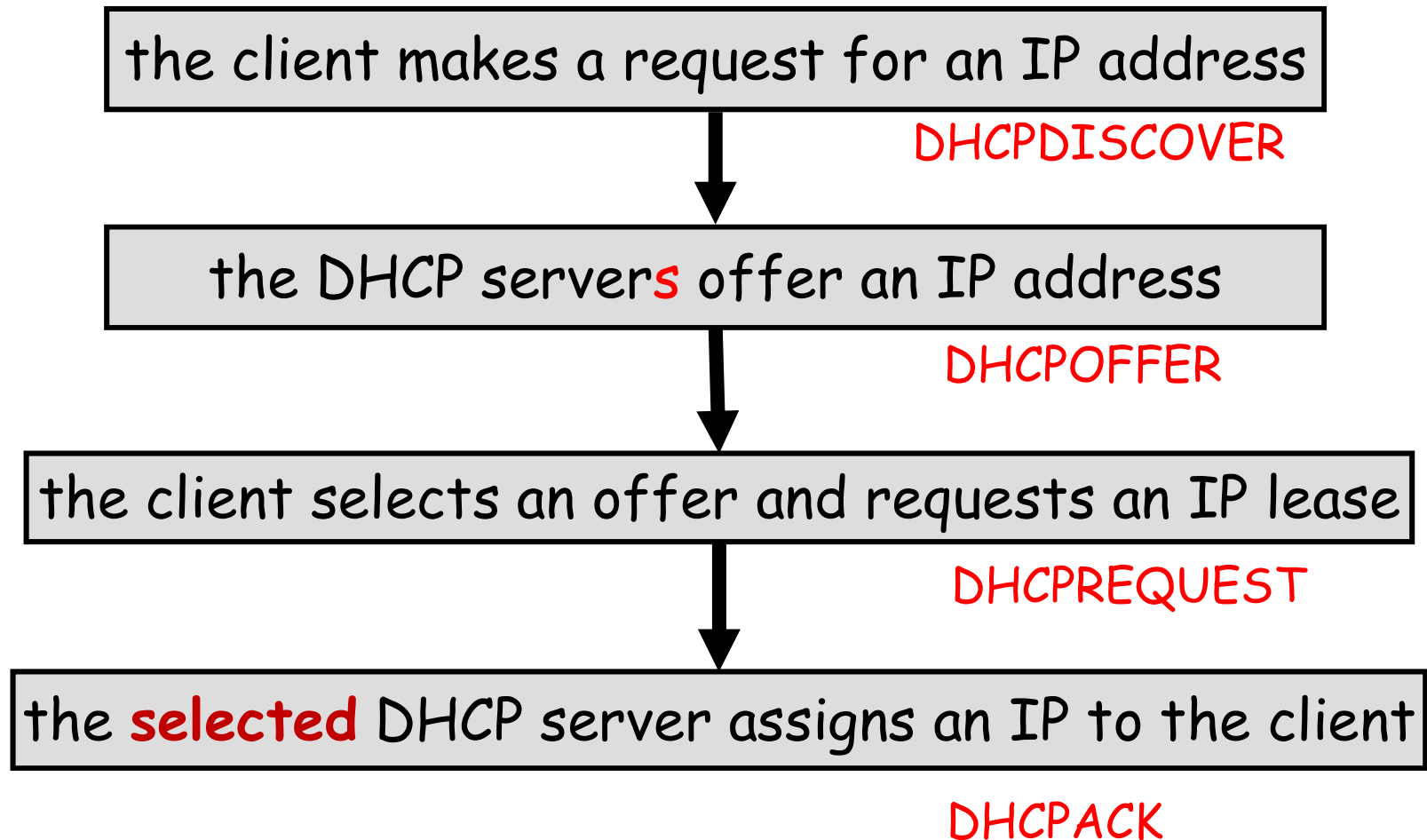
■ An extension to the BOOTP protocol

- ◆ In most systems, it has largely replaced RARP and BOOTP

■ Following client-server paradigm

■ Leasing and Renewal

DHCP: Phases of IP Assignment

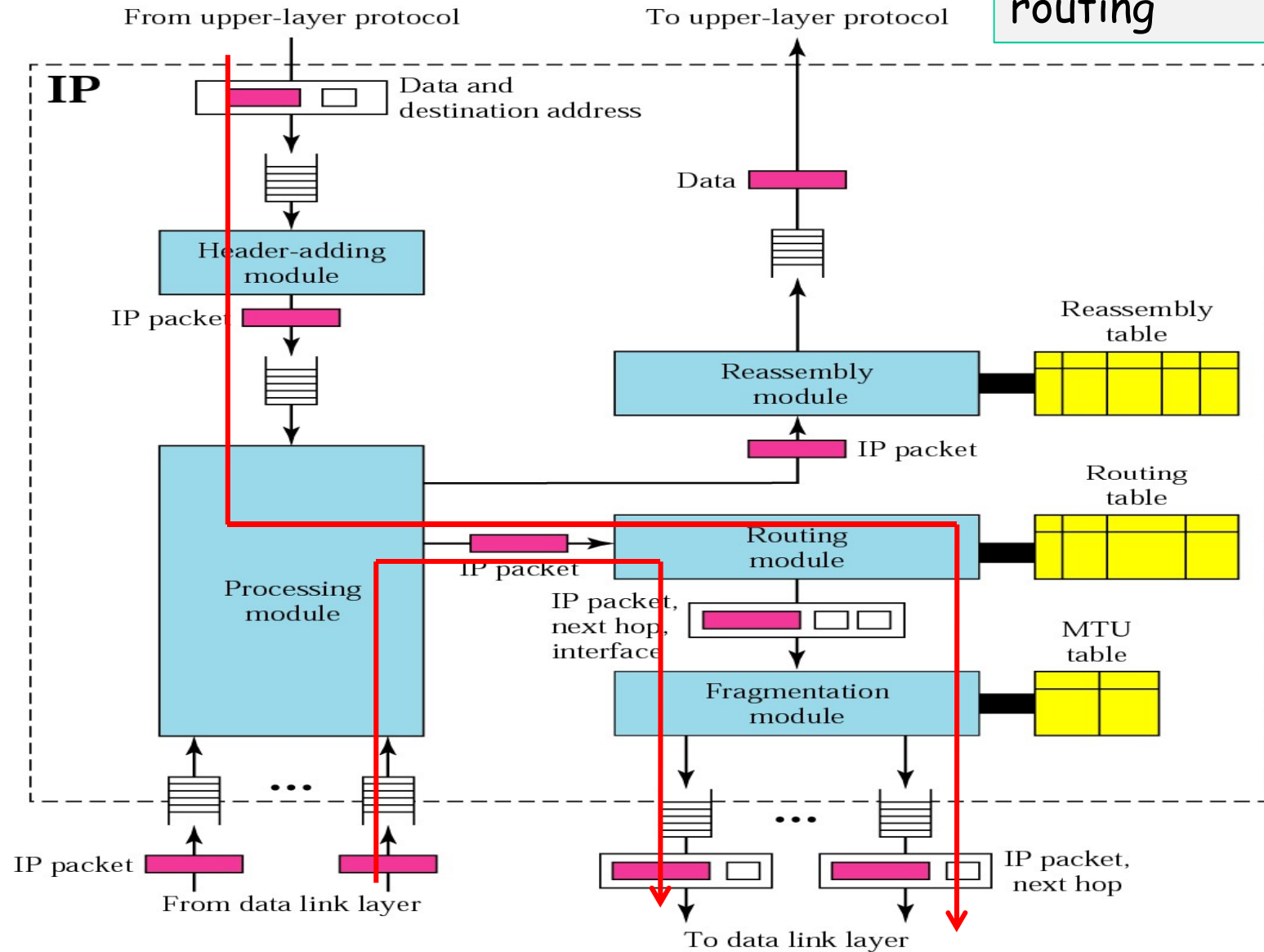


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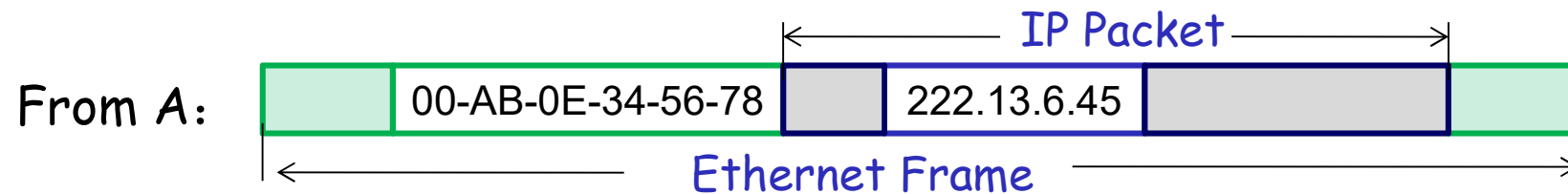
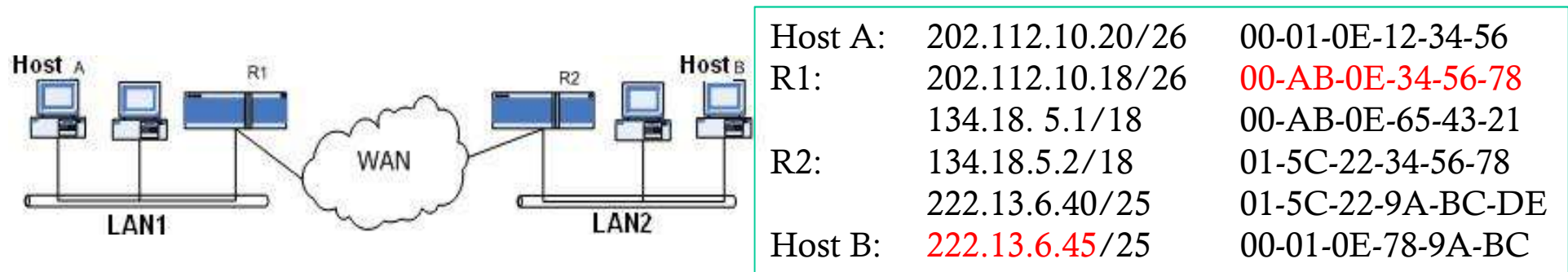
IP components

Both **source** and **routers** consider routing



How does Source make routing?

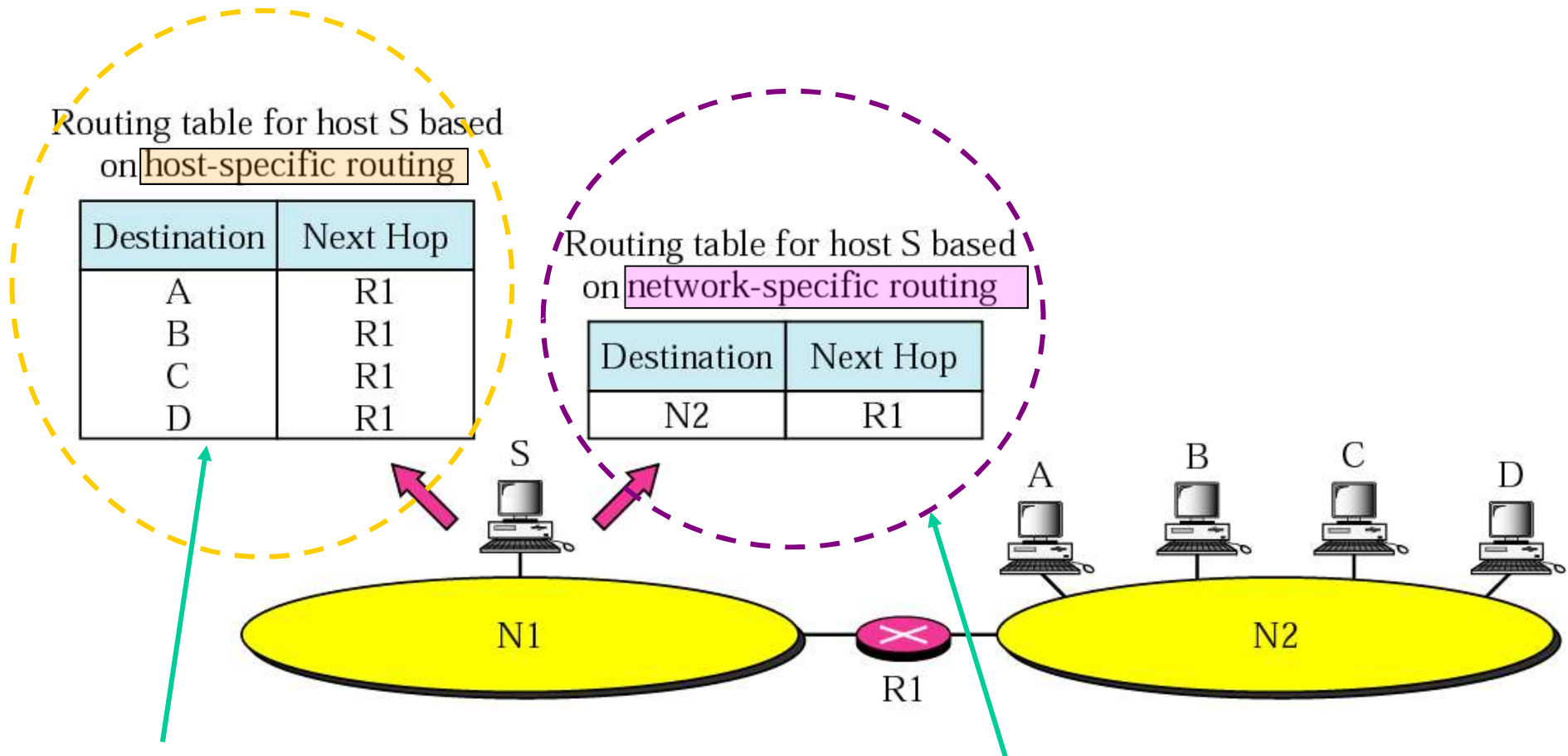
- Source host applies the **mask** to the destination address of the packet in order to find the corresponding **destination network address**.
 - ◆ If it is local network, send directly to destination;
 - ◆ Otherwise, send to the local router



How do routers make routing?

- Generally, a routing table needs a minimum of 4 columns: **mask**, **destination network address**, **next hop address** and **outgoing interface**.
- When a packet arrives, the router applies the **mask** to the destination address of the packet in order to find the corresponding **destination network address**.
- If found **matching entry** in routing table, the packet is sent out from the corresponding **interface** in the table.
- If not found, the packet is delivered to the default interface which carries the packet to **default** router.

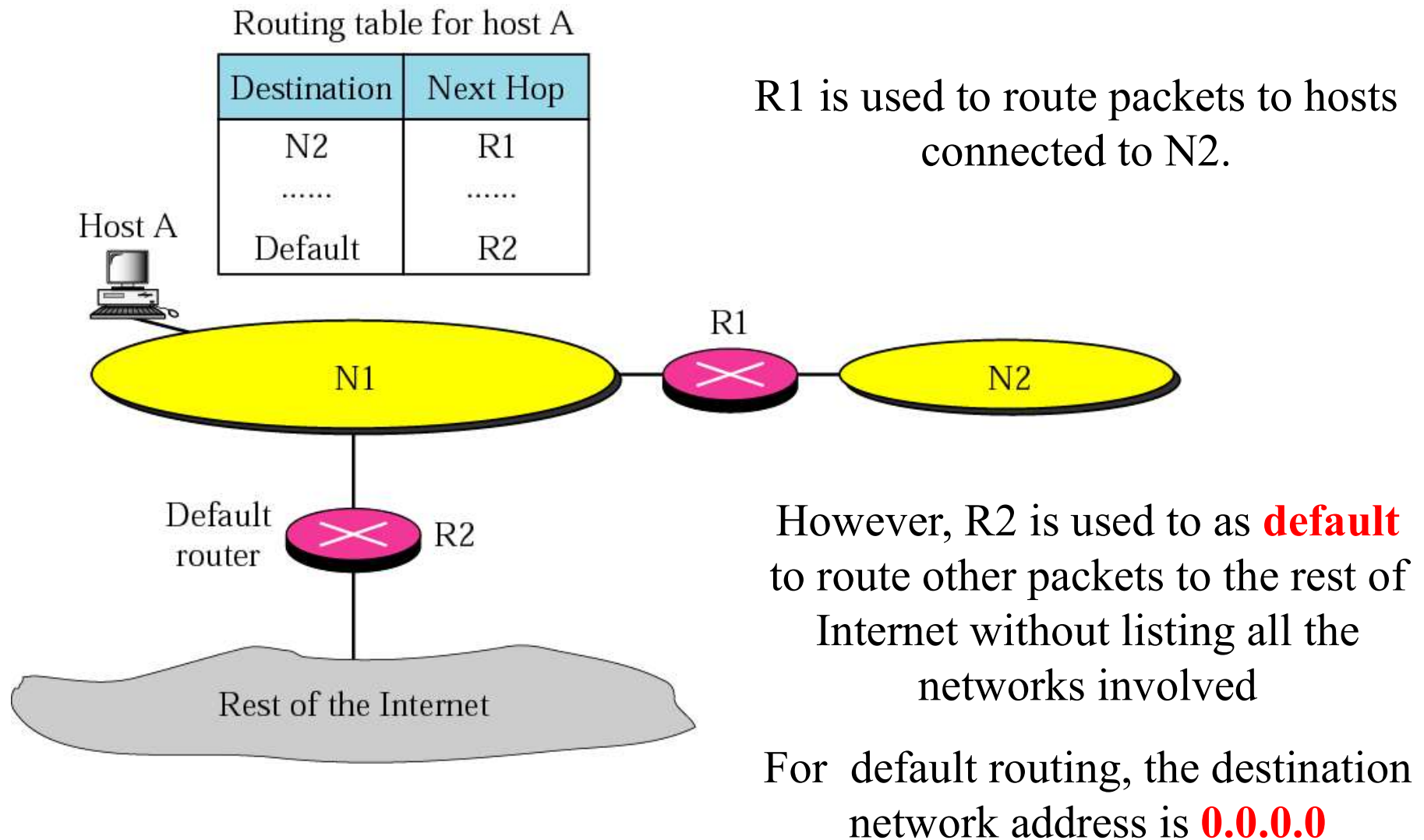
Network-specific & host-specific routing



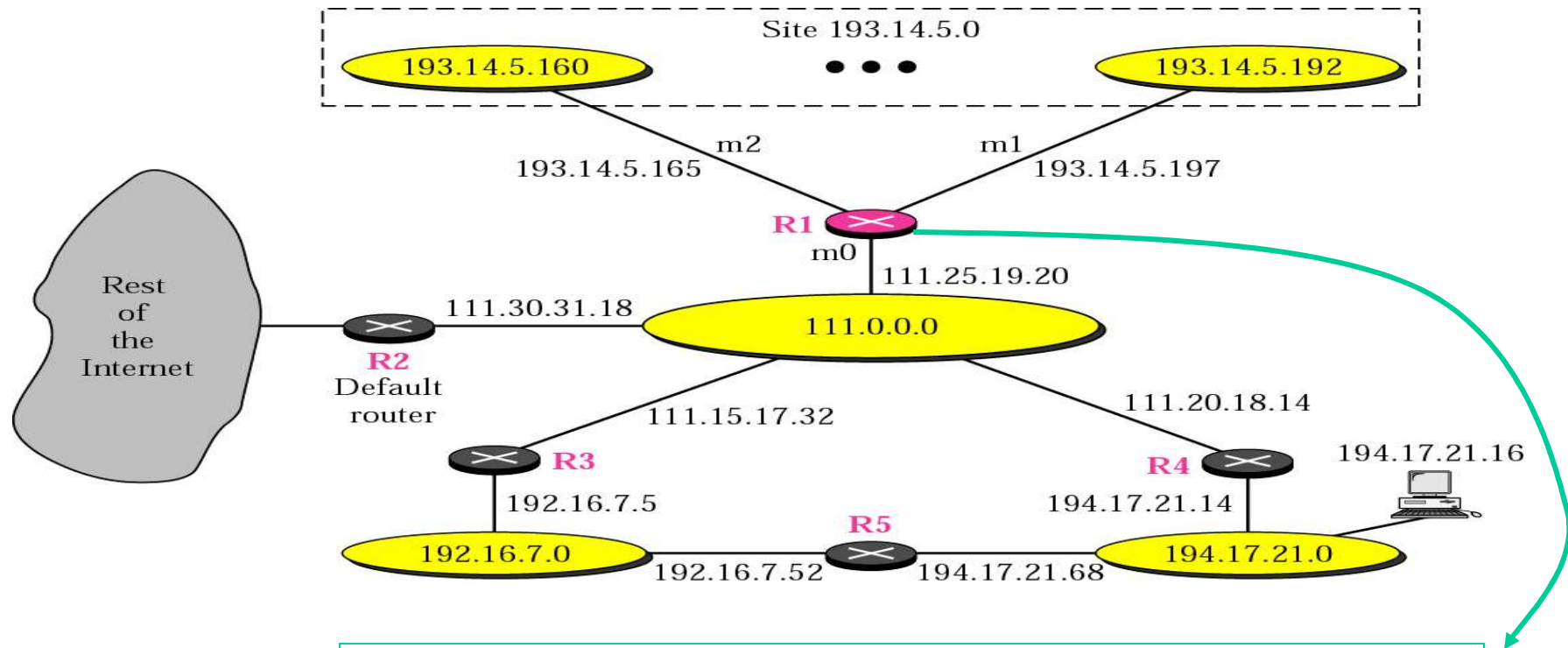
The **destination host address** is given in the routing table; to have greater control over routing.

Only **one entry** is needed to defined the address of the **network** itself. All host connected to the same network as one single entity.

Default routing (默认/缺省路由)



Configuration for routing example



	<u>Mask</u>	<u>Dest</u>	<u>Next Hop</u>	<u>Int.</u>
Standard delivery	255.0.0.0	111.0.0.0	--	m0
	255.255.255.224	193.14.5.160	-	m2
	255.255.255.224	193.14.5.192	-	m1
Host-specific	255.255.255.255	194.17.21.16	111.20.18.14	m0
Network-specific	255.255.255.0	192.16.7.0	111.15.17.32	m0
	255.255.255.0	194.17.21.0	111.20.18.14	m0
	0.0.0.0	0.0.0.0	111.30.31.18	m0

Example 1

Router R1 receives 500 packets for destination 192.16.7.14; the algorithm applies the masks row by row to the destination address until a match is found:

Solution

Rule of thumb: Apply the individual mask (from Routing table) to the received destination address (row-by-row) and see if it matches any of the DEST address stated in its routing table. If match is found, then stop

Direct delivery

192.16.7.14 & 255.0.0.0	→ 192.0.0.0	no match	to 111.0.0.0
192.16.7.14 & 255.255.255.224	→ 192.16.7.0	no match	to 193.14.5.160
192.16.7.14 & 255.255.255.224	→ 192.16.7.	no match	to 193.14.5.192

Host-specific

192.16.7.14 & 255.255.255.255	→ 192.16.7.14	no match	to 194.17.21.16
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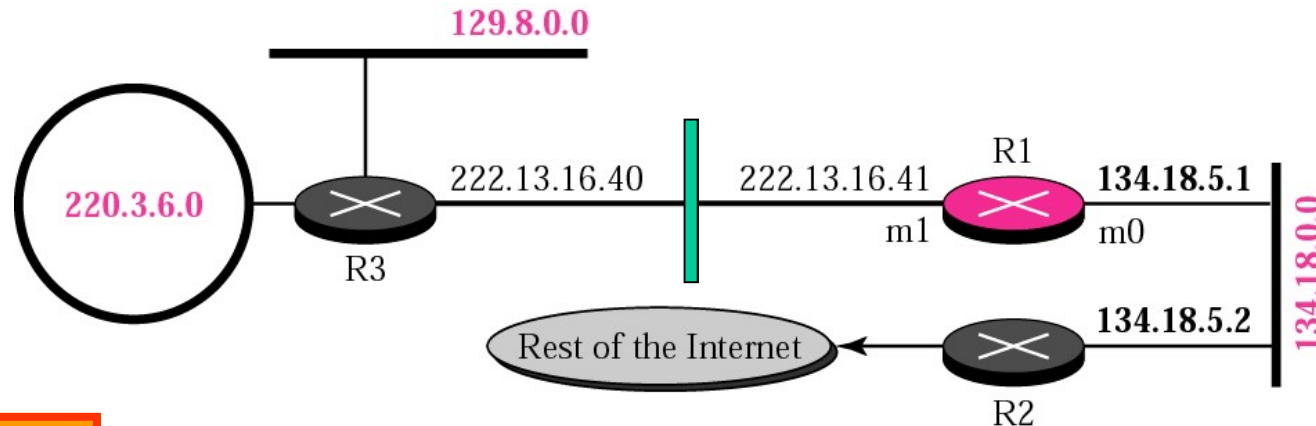
Network-specific

192.16.7.14 & 255.255.255.0	→ 192.16.7.0	match	to 192.16.7.0
-----------------------------	--------------	--------------	----------------------



Example 2

Make the routing table for router R1 in figure below



Solution

Mask	Destination	Next Hop	Int.
255.255.0.0	134.18.0.0	--	m0
255.255.0.0	129.8.0.0	222.13.16.40	m1
255.255.255.0	220.3.6.0	222.13.16.40	m1
0.0.0.0	0.0.0.0	134.18.5.2	m0

Routing Tables in IP with CIDR

- Longest-Prefix Matching(最长前缀匹配): Longest mask is used for multiple matches
- For each entry in the routing table:
 - ◆ MaskedAddress := EntryMask (bitAND) IPDatagramDestinationAddress;
 - ◆ if (MaskedAddress == EntryDestination), **Mark** the entry

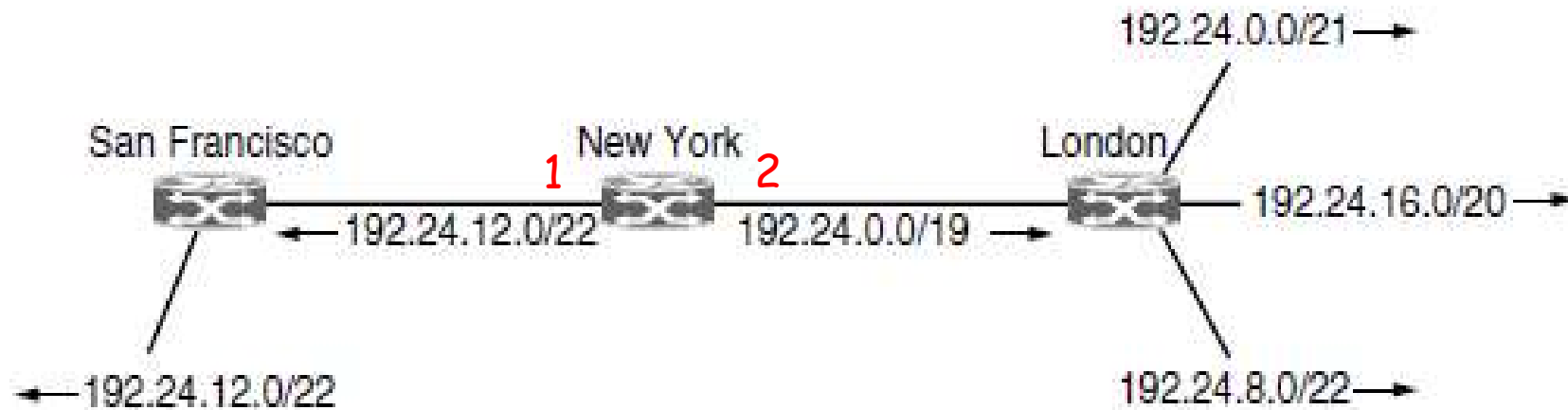
Mask	Destination	Next Hop
/12	128.96.0.0	145.12.56.29
<u>/17</u>	<u>128.125.0.0</u>	<u>153.202.12.128</u>
<u>/12</u>	<u>128.112.0.0</u>	<u>153.202.14.1</u>
/26	128.105.14.64	153.2.45.101
/32	128.105.14.66	153.2.45.101

- ◆ e.g. Destination: 128.125.1.3, what is next hop?

CIDR Example

Routing table of router in New York

Mask	Destination	Interface
/19	192.24.0.0	2
/22	192.24.12.0	1



小结：路由表查找转发过程

■ 路由表结构

目的网络IP地址	子网掩码	接口	下一跳IP地址
----------	------	----	---------

■ 路由表存在多个前缀匹配的表项(如目的IP地址192.24.12.3)

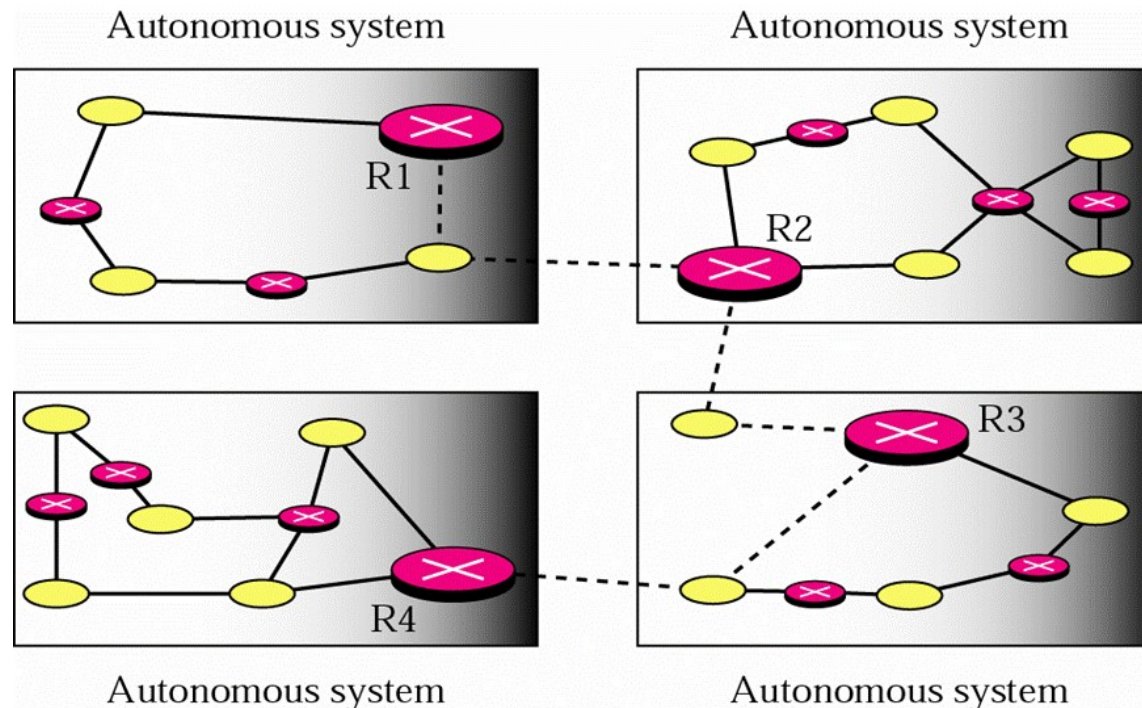
■ 扫描路由表，查找匹配的网络地址(最长匹配原则)

- ◆0项：采用默认路由，否则丢弃IP数据包并报错(ICMP)
- ◆1项：根据下一跳和接口转发
- ◆多项：选择掩码最长的一项的下一跳和接口转发

IP address/mask	Outgoing interface	Next hop
192.24.0.0/19	Port 1	XXX
192.24.8.0/21	Port 3	XXX
192.24.12.0/22	Port 2	XXX

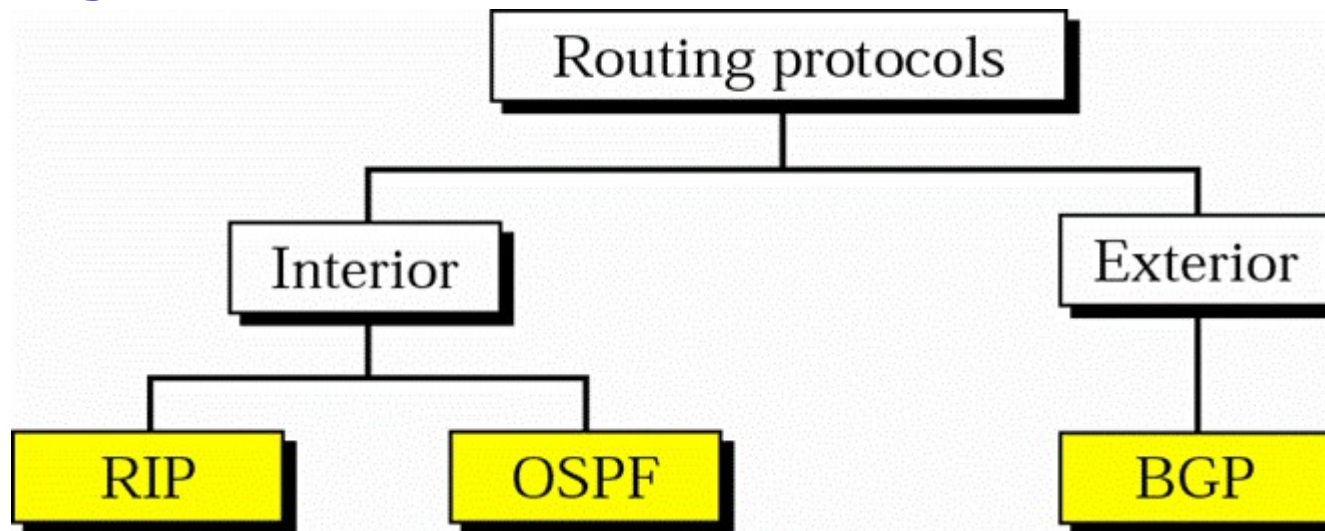
Routing Architecture in the Internet

- Internet is divided in *Autonomous Systems* (AS, 自治系统) that are independently administered by individual organizations.
- Each administrative authority use its own routing protocol within the AS, i.e. Interior Routing Protocol
- One routing protocol is used to exchange routing information among AS, i.e. Exterior Routing Protocol



Popular Routing Protocols in Internet

- RIP: Distance Vector Routing
- OSPF: Link State Routing
- BGP: Revised Distance Vector, Path Vector Routing

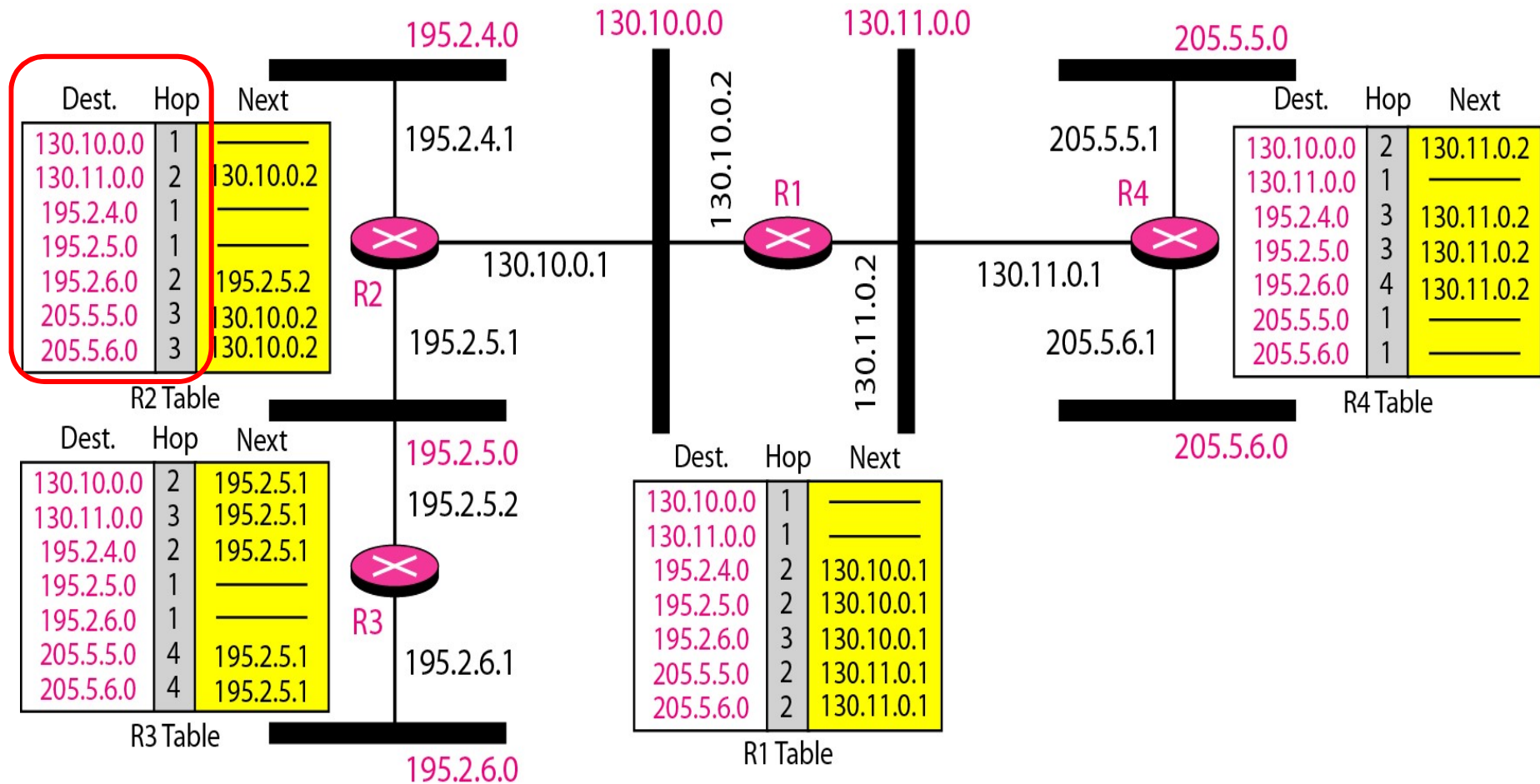


[Forouzan]

RIP: Features

- Routing Information Protocol (路由信息协议)
- Direct implementation of Distance Vector routing
- Path costs are based on number of hops
 - ◆ RIP treats all network equals; the cost of passing thru a network is the same: one hop count per network
- Number of hops is limited to 16, lack of Scalability
- RIP messages are transmitted in UDP packets
- RIPv1 cannot contain subnet masks
- RIPv2 is an improvement to contain subnet masks and next-hop addresses

Example of a domain using RIP

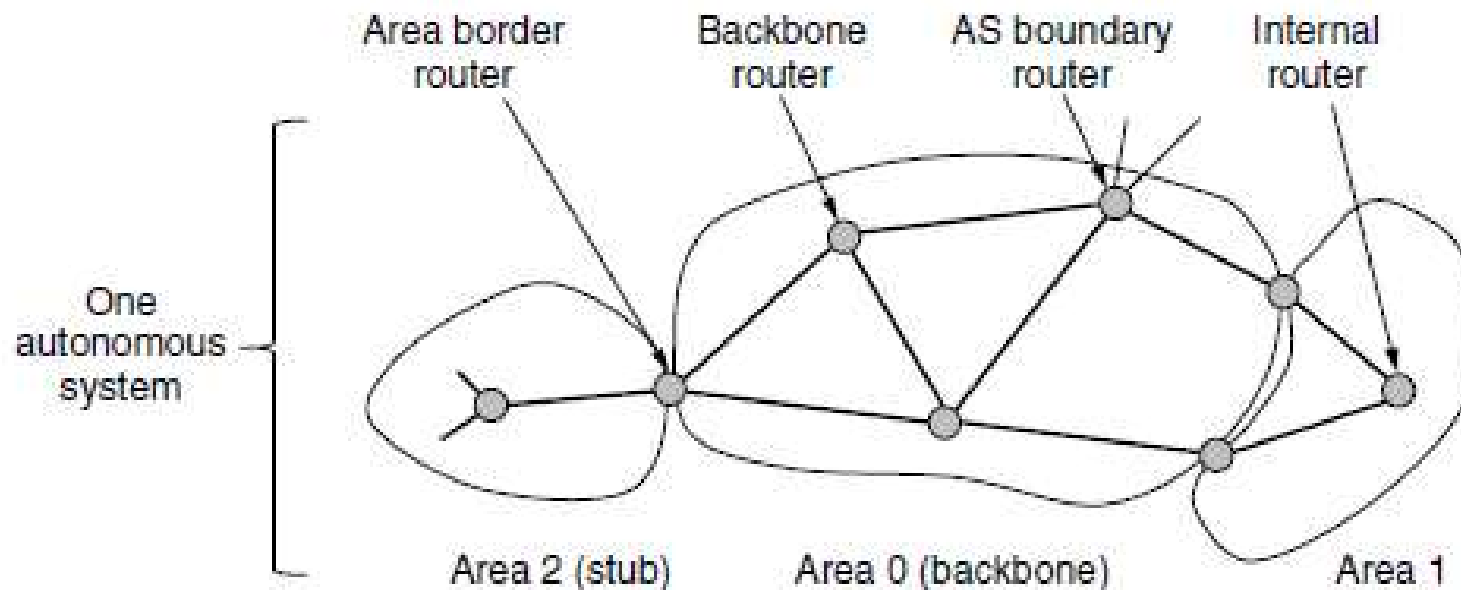


OSPF: Features

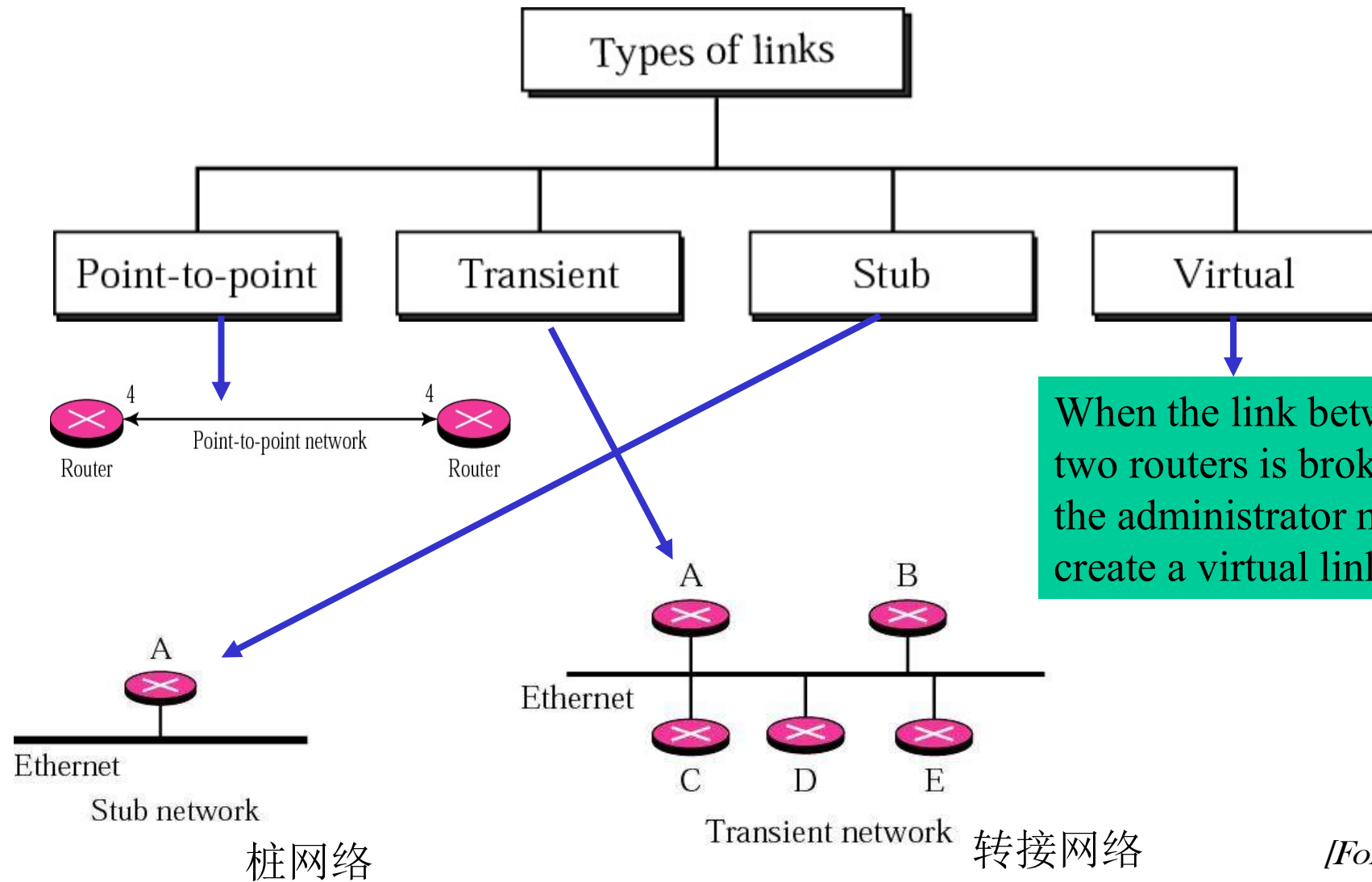
- Open Shortest Path First (开放的最短路径优先)
- Based on **Link State Routing**
- OSPF allows the administrator to (only) assign a **cost** for passing through a network based on the **type of service** required.
- Each router should have the **exact topology of the internet** at every moment.
- Each router needs to *advertise* to the neighbourhood of every other routers involved in an **Area**.
- OSPF standard supports novel concepts such as type of service routing, load balancing and authentication.
- OSPF messages are transported directly in **IP** packets

OSPF: Areas

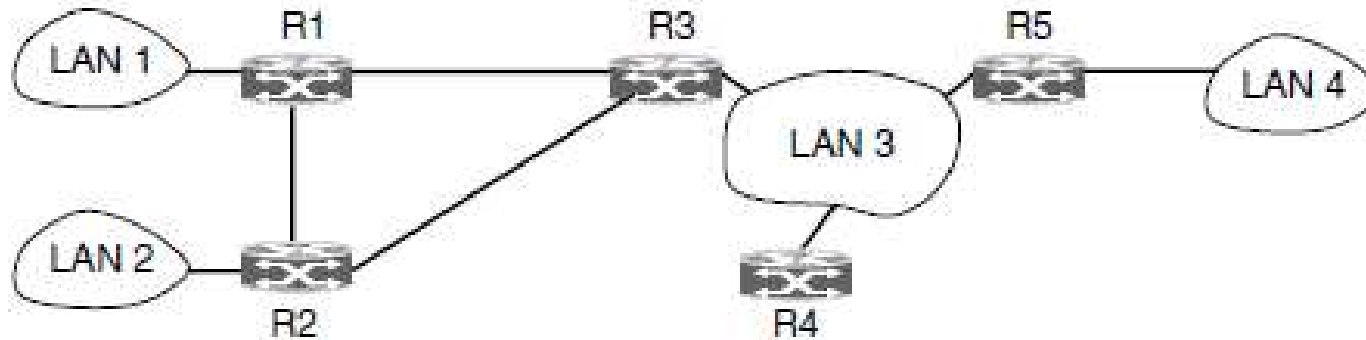
- OSPF divides an AS(自治系统) into **areas**, a collection of network, hosts and routers all contained within an AS.
- Backbone Router, Boundary Router, Border Router and Internal Router



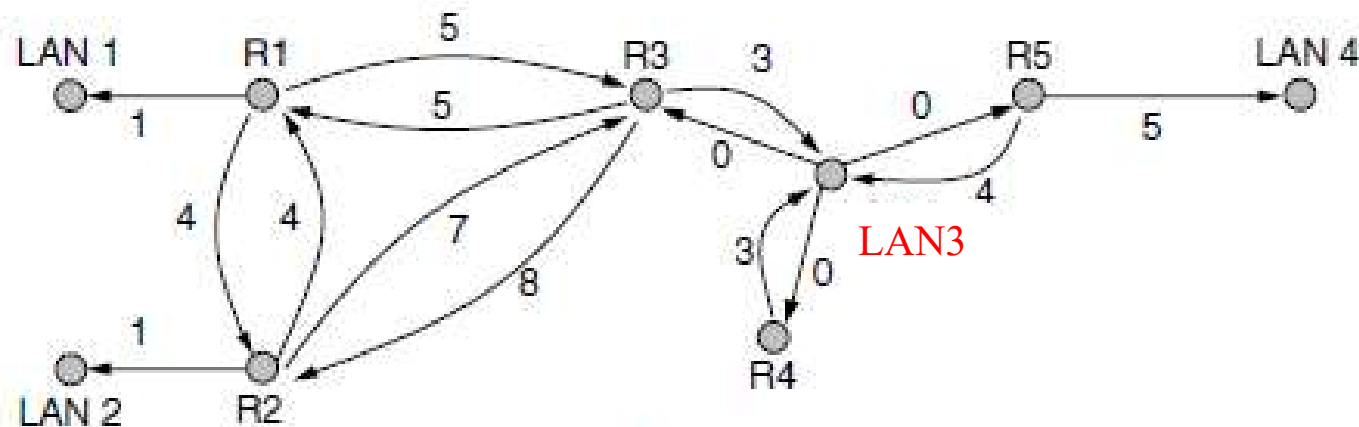
OSPF: Types of Links



OSPF: Graph Representation

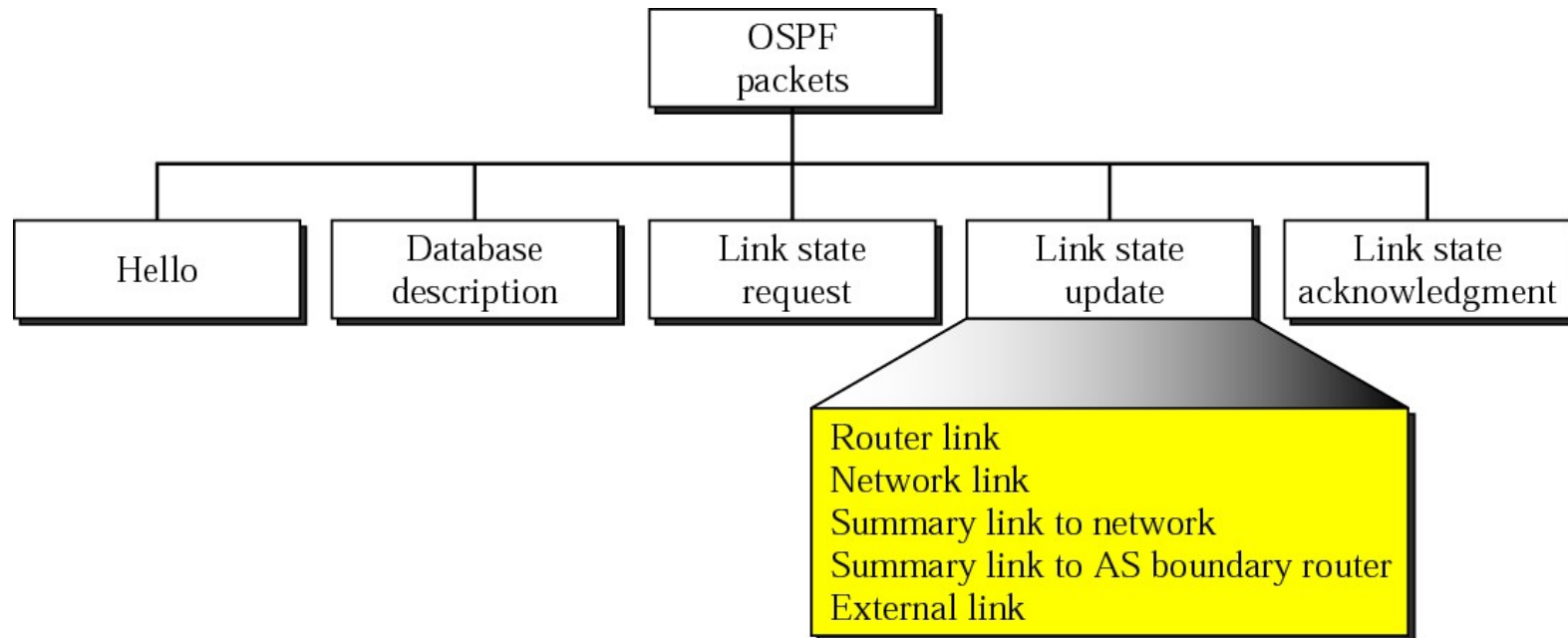


(a)



(b)

OSPF Packets



Message type	Description
Hello	Used to discover who the neighbors are
Link state update	Provides the sender's costs to its neighbors
Link state ack	Acknowledges link state update
Database description	Announces which updates the sender has
Link state request	Requests information from the partner

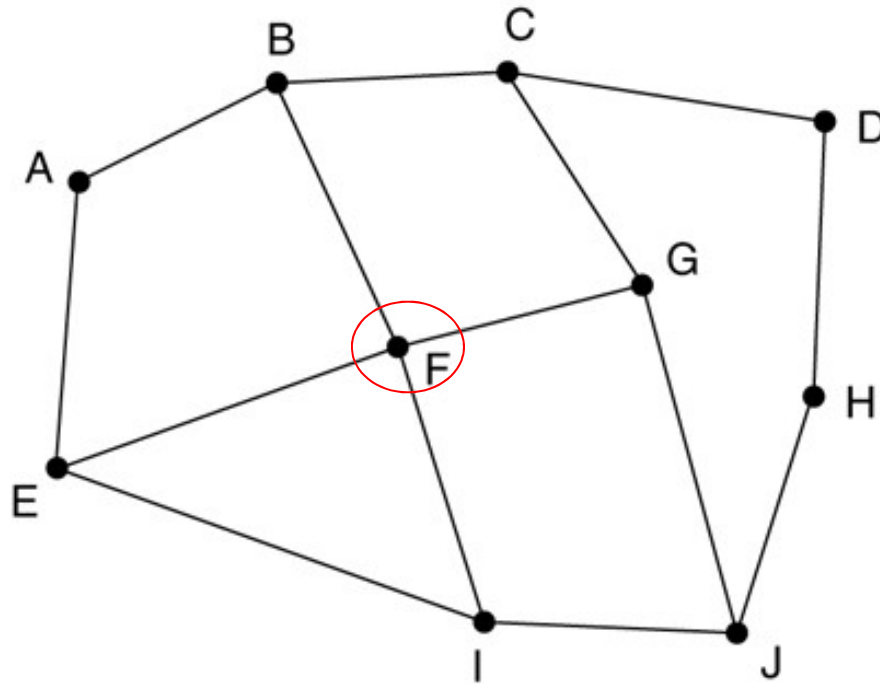
RIP vs. OSPF

Features	RIPv1	RIPv2	OSPF
RFC	RFC1058	RFC1723	RFC2178
Scheme	Distance-Vector		Link-state
Range of Metrics	16		65,535
Update Frequency	30s		Changes or 30mins
Dead interval	180s		Variable setting
Authentication	No	Yes	Yes
Convergence	Slow	slow	Fast
Support CIDR	No	Yes	Yes
Type of Service	No	No	Yes
Multiple paths	No	No	Yes
Ease of Use	Yes		Complex Setup

BGP

- Border Gateway Protocol (边界网关协议)
- Path Vector Routing 路径矢量选路
- Distance vector algorithm with extra information
 - ◆ For each route, store the complete path (ASs)
 - ◆ No extra computation, just extra storage
- Advantages
 - ◆ Enables router to perform policy routing
 - E.g. avoid path to avoid transiting particular AS
 - E.g. link speed, capacity, tendency to become congested, and overall quality of operation, security
 - E.g. minimizing number of transit ASs
 - ◆ Easily avoid loops
- BGP using TCP to carry its messages, reliable delivery!

BGP Example



(a)

Information F receives
from its neighbors about D

From B: "I use BCD"
From G: "I use GCD"
From I: "I use IFGCD"
From E: "I use EFGCD"

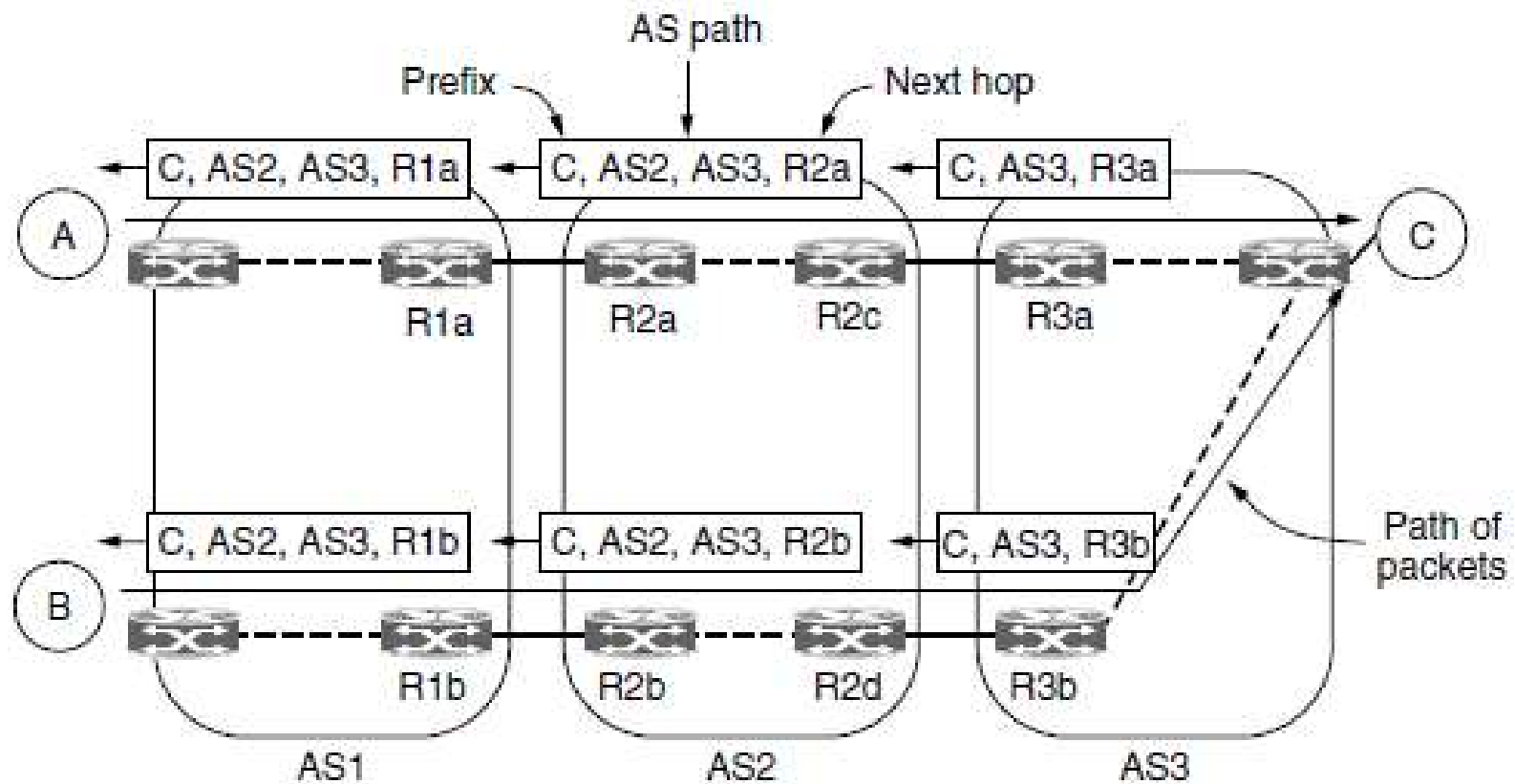
(b)

BGP: Routing Table Example

Network	Next Router	Path
N01	R01	AS14, AS23, AS67
N02	R05	AS22, AS67, AS05, AS89
N03	R06	AS67, AS89, AS09, AS34
N04	R12	AS62, AS02, AS09

[Forouzan]

BGP: Propagation of BGP route advertisements



因特网的选路协议

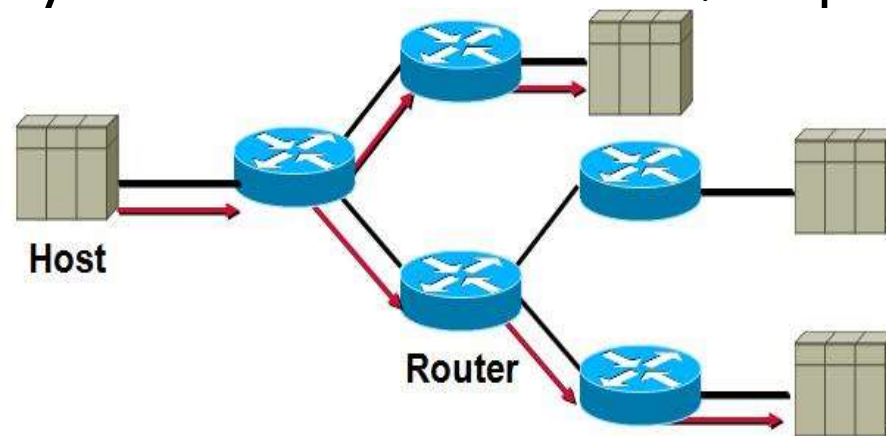
	RIP	OSPF	BGP
应用范围	AS内部(IGP)	AS内部	AS之间(EGP)
选路算法	距离矢量选路	链路状态选路	路径矢量选路
路由器之间交换的信息	到各网络的跳数	网络的拓扑结构	到各网络的路径
无穷计数 (路径环路) 问题	有	无	无
路由报文 承载协议	UDP	IP	TCP

The Network Layer in Internet

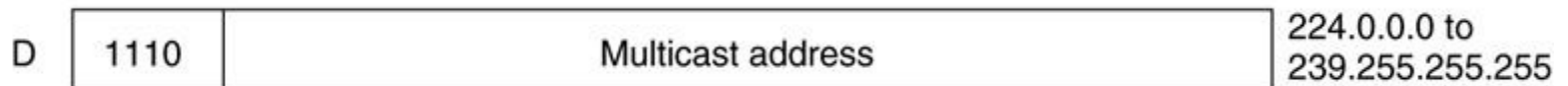
- IP Addressing
- IP Packet format
- IPv6
- ICMP, ARP and DHCP
- Routing Protocols: RIP, OSPF and BGP
- IP Multicast
- Mobile IP

IP Multicast(Textbook 5.6.6)

- Multicast(组播/多播): a network technology for the delivery of information to **a group of destinations** simultaneously in the most efficient way
 - ◆ deliver the messages over each link of the network only once, creating copies only when the links to the multiple destinations split.



- Group Address: **class D addresses**, with 28bits, over 250 million groups



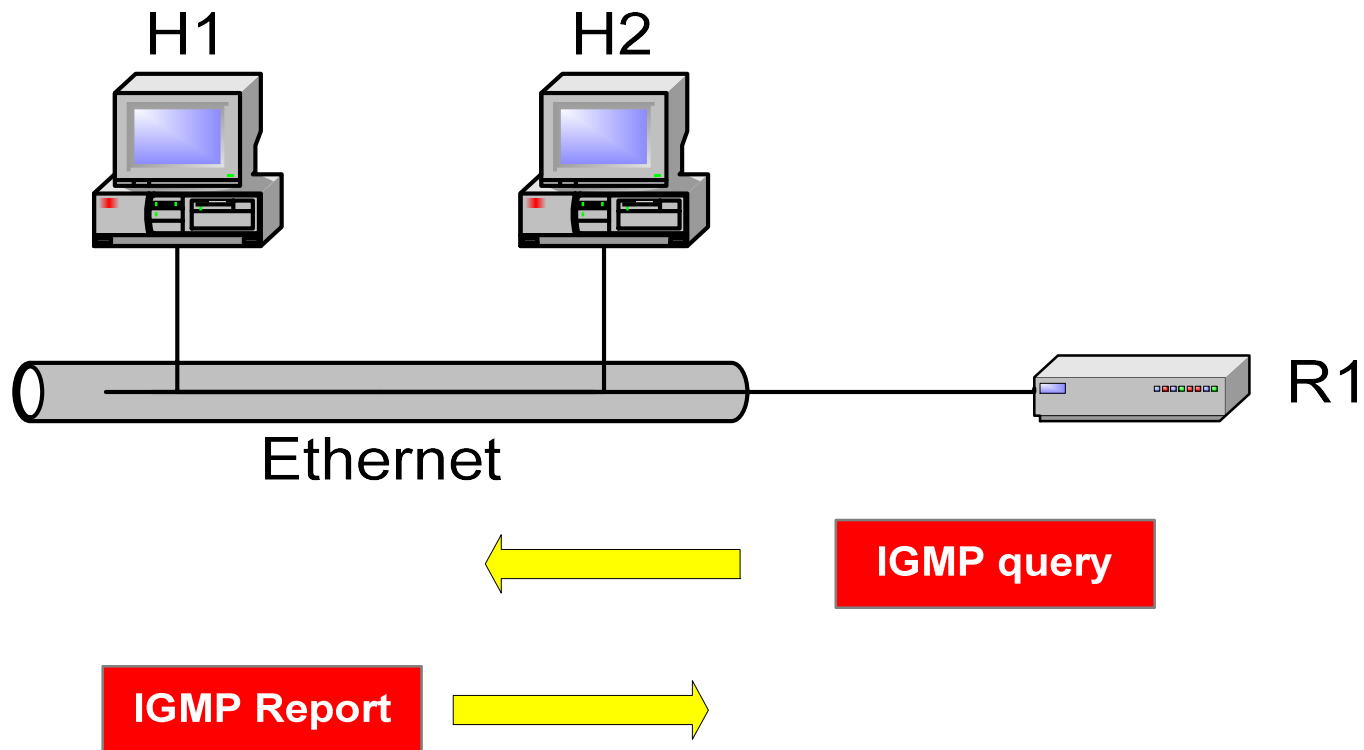
- Best Effort service

[Cisco]

IGMP

- Internet Group Management Protocol（因特网组管理协议）
- Used to establish host memberships in particular multicast groups on a single network.
 - ◆ A host sends an **IGMP report** to local router when it joins a multicast group
 - ◆ A multicast router **periodically** multicasts **IGMP query** (to 224.0.0.1), asking all hosts on its local network to report groups they belong to
 - ◆ Each host sends back an **IGMP report**, informing all the class D addresses it is interested in
- Defined in RFC 3376
- IGMP message is encapsulated in **IP** packet.

IGMP Messages



[J. Liebeherr]

The Network Layer in Internet

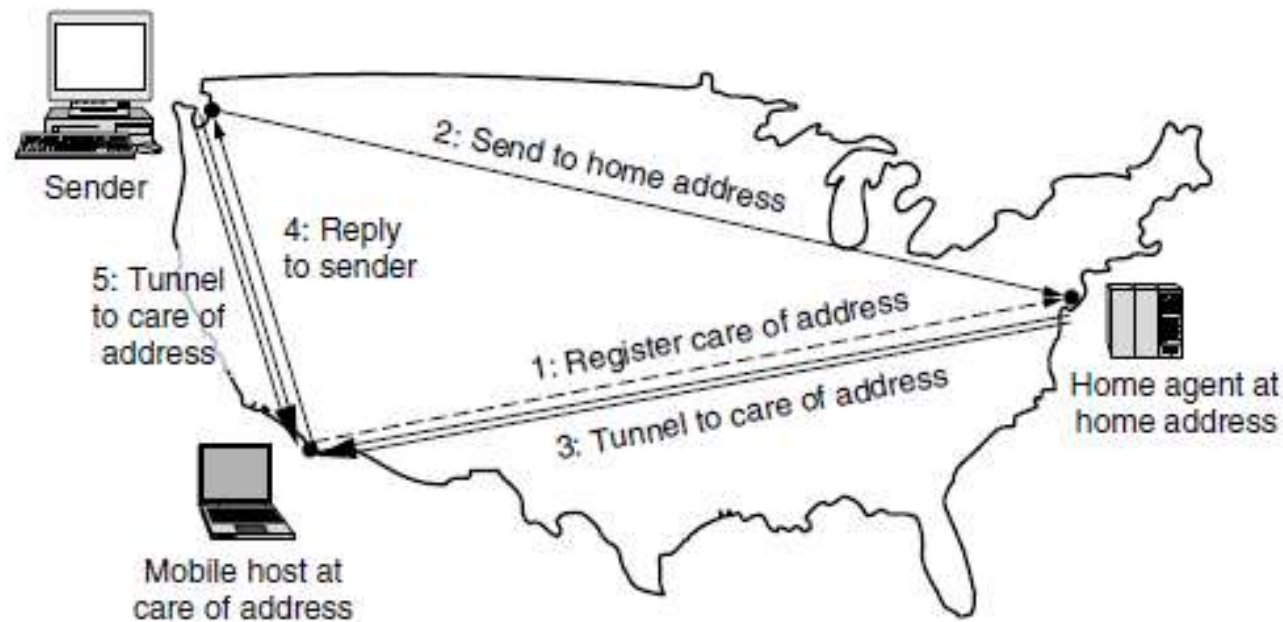
- IP Addressing
- IP Packet format
- IPv6
- ICMP, ARP and DHCP
- Routing Protocols: RIP, OSPF and BGP
- IP Multicast
- Mobile IP

Key Terms

- Mobile Node(MN, 移动节点): the host moving to another network(foreign network)
- Home Agent(HA, 本地/归属/家乡代理): the router on home network of MN
- Foreign Agent(FA, 外地代理): the router on foreign network
- Care of Address(CoA, 关照地址): IP address of FA, sent by FA to HA after MN registered
- Corresponding Node(相关节点): the host talking with MN

目标：移动到外地网络中，仍然能够接收到发给原归属地**IP**地址的数据包

Packet routing for mobile hosts(Fig 5-19)



- Traffic is sent as usual to the **home subnet**
- The **home agent** intercepts the traffic while the **Mobile Node** is registered as away
- Traffic is **tunneled** to the **Care-of-Address** of the Mobile Node and forwarded to Mobile Node
- Traffic from the Mobile Node can go **directly** to the correspondent host = "**Triangle Routing**" (三角路由)

[J. Liebeherr]

Key Points(1)

- Introduction to Network Layer
 - ◆ Position and functions of network layer
 - ◆ Virtual Circuit Switching vs. Datagram Routing
- Routing Algorithms
 - ◆ Static routing vs. Dynamic routing
 - ◆ Sink tree as benchmark
 - ◆ Flooding
 - ◆ Dijkstra algorithm
 - ◆ Distance Vector Routing (DVR): features, how to calculate routing table, drawbacks
 - ◆ Link State Routing (LSR): features, LSP vs. topology
 - ◆ Hierarchical Routing: idea

Key points(2)

■ Congestion Control:

- ◆ Virtual circuit network: Admission control
- ◆ Datagram network: find, notification and decrease sending rate
 - Choke packets, ECN, RED

■ QoS Issues: Token Bucket, Packet Scheduling

■ IP Addressing

- ◆ Classful address: class A, B and C
- ◆ Hierarchical address: Network-id + host-id
- ◆ Network address used in routing
- ◆ Subnetting
- ◆ Supernetting
- ◆ CIDR: slash notation
- ◆ How to calculate subnet mask

Key Points(3)

- IP Packet format
 - ◆ Function of each field
 - ◆ Checksum
 - ◆ Fragmentation
- Routing Protocols: RIP, OSPF and BGP
 - ◆ Features
- Other relative protocols
 - ◆ IPv6: features, addressing, header conversion
 - ◆ ICMP: error report and test
 - ◆ ARP: IP address → MAC address
 - ◆ DHCP: allocate IP address dynamically
 - ◆ IGMP: managing group members
 - ◆ Mobile Routing: idea

Terms

缩写	英文全称	中文
	adaptive algorithms	自适应算法
	addressing	寻址
	address resolution	地址解析
ARP	Address Resolution Protocol	地址解析协议
AS	Autonomous Systems	自治系统
	Admission control	接纳控制
BGP	Border Gateway Protocol	边界网关协议
	backward learning	逆向学习
	broadcast storm	广播风暴
	bridge	网桥
	broadcasting	广播
	checksum	校验和/检查和
	classful addressing	分类地址
	classless addressing	无类别地址
CIDR	Classless InterDomain Routing	无类别域间选路

缩写	英文全称	中文
	Choke Packet	抑制分组
	circuit switching	电路交换
	count-to-infinity	无穷计数
	congestion control	拥塞控制
	connectionless service	无连接服务
	connection-oriented service	面向连接的服务
	cut-through switching	直通交换
	datagram	数据报
	datagram network	数据报网络
DHCP	Dynamic Host Configuration Protocol	动态主机配置协议
	default mask	默认掩码
DVR	Distance Vector Routing	距离矢量选路
ECN	Explicit Congestion Notification	显式拥塞通知

缩写	英文全称	中文
	Ethernet	以太网
	Fair Queue	公平队列
	FIFO Queuing	先进先出队列
	Fixed Routing	固定查表选路
	flooding	洪泛
	flow control	流量控制
	forwarding	转发
	fragmentation	分段/分片
	fragmenting	分段/分片
	gateway	网关
	Hop-by-Hop	逐跳
	Hierarchical Routing	分级选路
ICMP	Internet Control Message Protocol	因特网控制报文协议
IGMP	Internet Group Management Protocol	因特网组管理协议

缩写	英文全称	中文
	implicit notification	隐式通知
	incoming interface	输入接口
	infinite loop	无限循环
	internet	互联网
	Internet	因特网
	internetworking	互通
IP	Internet Protocol	网际协议
	jitter	时延抖动
LAN	Local Area Network	局域网
	Leaky Buckets	漏桶
LSP	Link State Packet	链路状态包
LSR	Link State Routing	连路状态选路
	Longest-Prefix Matching	最长前缀匹配
MPLS	Multi-Protocol Label Switching	多协议标签交换
MTU	Maximum Transmission Unit	最大传输单元
	multicasting	多播/组播

缩写	英文全称	中文
	nonadaptive algorithms	非自适应算法
	network address	网络地址
NAT	Network Address Translation	网络地址翻译
	next hop	下一跳
	outgoing interface	输出接口
OSPF	Open Shortest Path First	开放的最短路径优先
	packet	分组/包
	packetizing	打包
	Packet Scheduling	分组调度
	packet switching	分组交换
	prefix	前缀
	Priority Queuing	优先级队列
PVR	Path Vector Routing	路径矢量选路
QoS	Quality of Service	服务质量
	re-assembly	重装

缩写	英文全称	中文
RARP	Reverse Address Resolution Protocol	逆向地址解析协议
RED	Random Early Detection	随机早期检测
	Random Routing	随机选路
	Reverse Path Forwarding	逆向路径转发
	round robin	循环
	round-trip time	环回时间
	Route Aggregation	路由聚合/会聚
RIP	Routing Information Protocol	选路信息协议
	router	路由器
	routing	选路
	routing algorithm	选路算法
	routing table	路由表
	Resource Reservation	资源预留
	Sink Tree	宿树/汇集树
	spanning tree	生成树)
	static routing	静态选路

缩写	英文全称	中文
	store-and-forward	存储转发
	subnet	子网
	subnet mask	子网掩码
	subnetting	子网划分
	supernetting	超网划分
	supernetwork	超网
	switch	交换机
	Token Buckets	令牌桶
	Traffic Shaping	业务量(流量)整形
TTL	Time-to-Live	生命期
	Tunneling	隧道
VC	Virtual Circuit	虚电路
VCI	Virtual Circuit Identifier	虚电路标识符
	virtual circuit network	虚电路网络
WAN	Wide Area Network	广域网
WFQ	Weighted Fair Queuing	公平加权队列

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 - ◆ "Data and Computer Communications", William Stallings, which are marked with [Stallings]
 - ◆ "计算机网络", 谢希仁, 第五版, 电子工业出版社, 2008年, which are marked with [谢]

Self-Study

Broadcast Routing

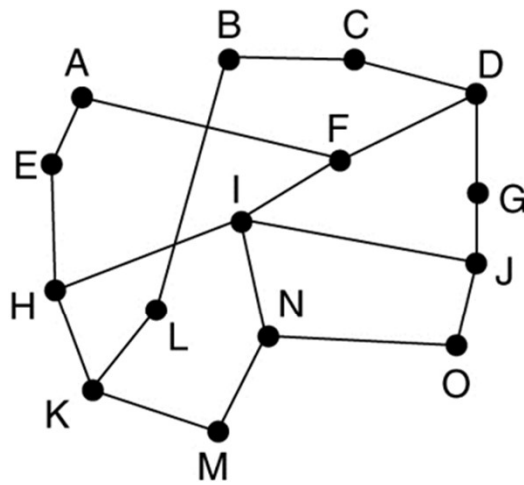
■ Using a spanning tree(生成树)

- ◆ Routers copy an incoming broadcast packet onto all the spanning tree lines except the one it arrived on
- ◆ Every router needs to know the spanning tree

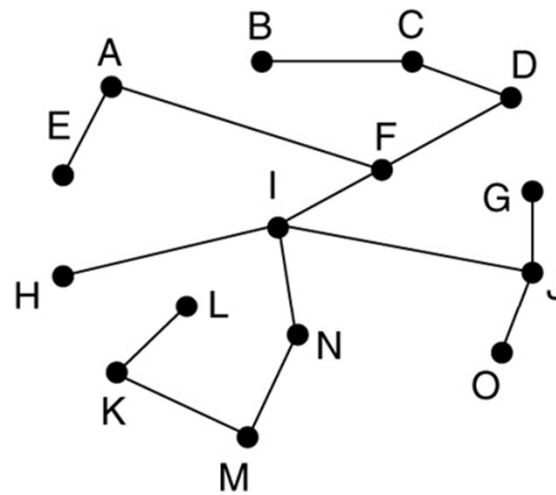
■ Reverse Path Forwarding(逆向路径转发)

- ◆ On receiving a broadcast packet, a router checks if the packet arrived on the line normally used for **sending packets to the source** of the broadcast.
- ◆ If so, the router forwards copies of it onto all lines except the one it arrived on.
- ◆ Otherwise, the packet is discarded
- ◆ Reasonably efficient and easy to implement, routers do not need to know about spanning tree

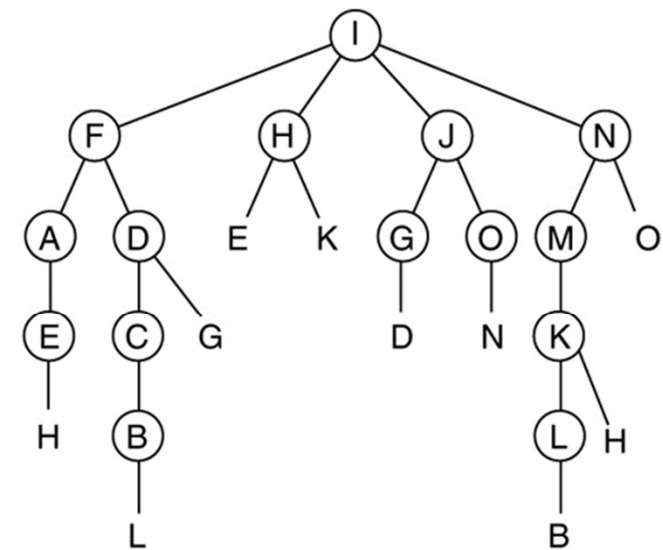
Example of Reverse Path Forwarding



A subnet



A Sink tree
(Destination is I)



The tree built by
reverse path
forwarding.

About IP Address

- 主机/路由器的每个网络接口有一个**IP**地址
- **IP**地址分成网络号和主机号两部分
- 一个网络内的所有**IP**地址的网络号部分一样，主机号部分不同
- 一个网络包含一组**IP**地址
 - 第一个地址（主机号部分为全**0**）是**网络地址**，代表整个网络；
路由选择使用网络地址
 - 最后一个地址（主机号部分为全**1**）是此网络的广播地址
- 分类网络：**A、B、C**三类，每类网络的大小固定，默认掩码固定
- 子网掩码：特殊的**IP**地址，网络号部分为全**1**，主机号部分为全**0**，其作用是：确定一个**IP**地址所在网络的网络地址
- 如何确定网络地址？子网掩码与已知**IP**地址“**按位与**”操作
- 划分子网：使用原网络地址中主机号的**若干高位**作为子网号

IPv6: Major Goals

- Address space
- Reduce the size of the routing tables
- Simplify the protocol, to allow routers to process packets faster
- Security
- Pay more attention to type of service, particularly for real-time data
- Multicasting
- Make it possible for a host to roam without changing its address
- Allow the protocol to evolve in the future
- Permit the old and new protocols to coexist for years

192.32.176.0/24和192.32.177.0/24聚合为一条路由表项

Routing Table

Host C

192.32.179.0/24,1,(direct)
192.32.176.0/23,1,192.32.179.1
192.32.178.0/24,1,192.32.179.1

Router R3

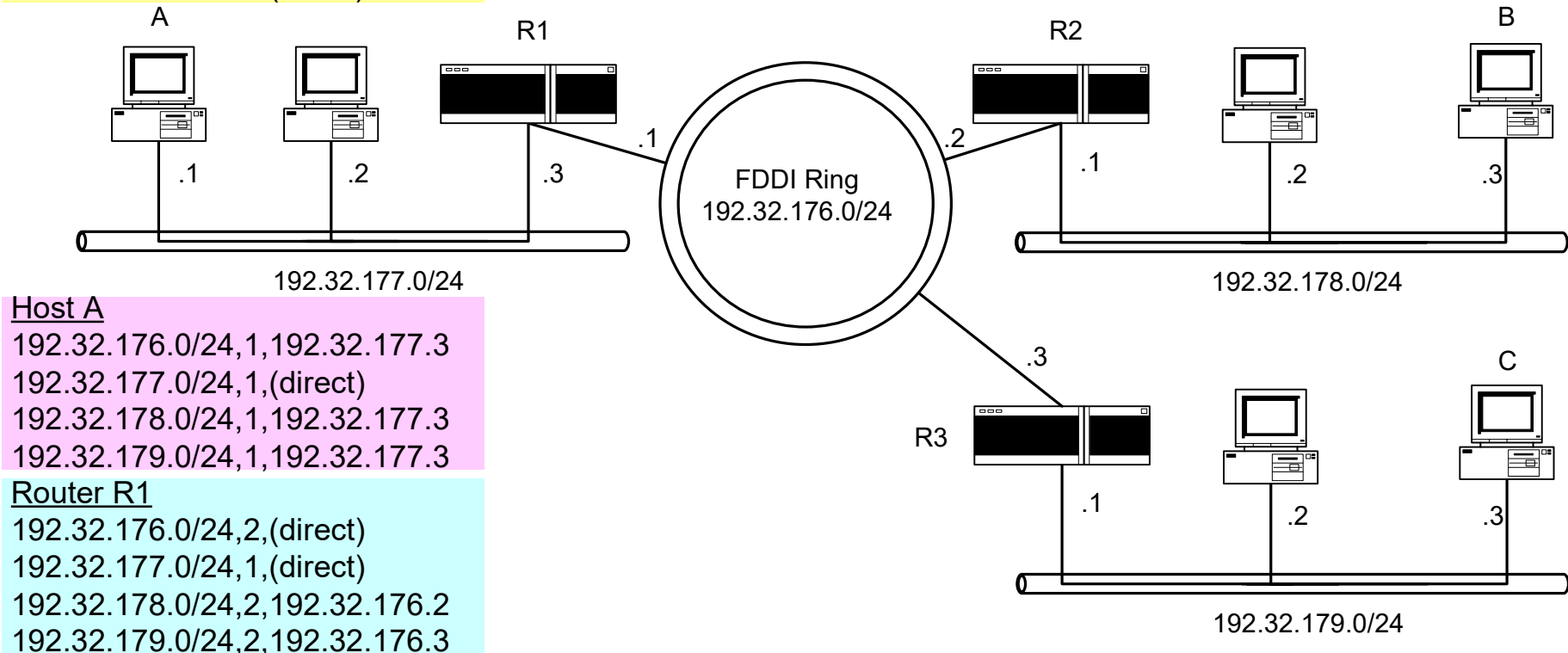
192.32.176.0/24,1,(direct)
192.32.177.0/24,1,192.32.176.1
192.32.178.0/24,1,192.32.176.2
192.32.179.0/24,2,(direct)

Host B

192.32.178.0/24,1,(direct)
0.0.0.0/0,1,192.32.178.1

Router R2

192.32.176.0/24,1,(direct)
192.32.177.0/24,1,192.32.176.1
192.32.178.0/24,2,(direct)
192.32.179.0/24,1,192.32.176.3



Host A

192.32.176.0/24,1,192.32.177.3
192.32.177.0/24,1,(direct)
192.32.178.0/24,1,192.32.177.3
192.32.179.0/24,1,192.32.177.3

Router R1

192.32.176.0/24,2,(direct)
192.32.177.0/24,1,(direct)
192.32.178.0/24,2,192.32.176.2
192.32.179.0/24,2,192.32.176.3

Routing Table

Host C

192.32.179.0/24,1,(direct)
 192.32.176.0/23,1,192.32.179.1 11000000.00100000.1011000x.xxxxxxxx
 192.32.178.0/24,1,192.32.179.1

Router R3

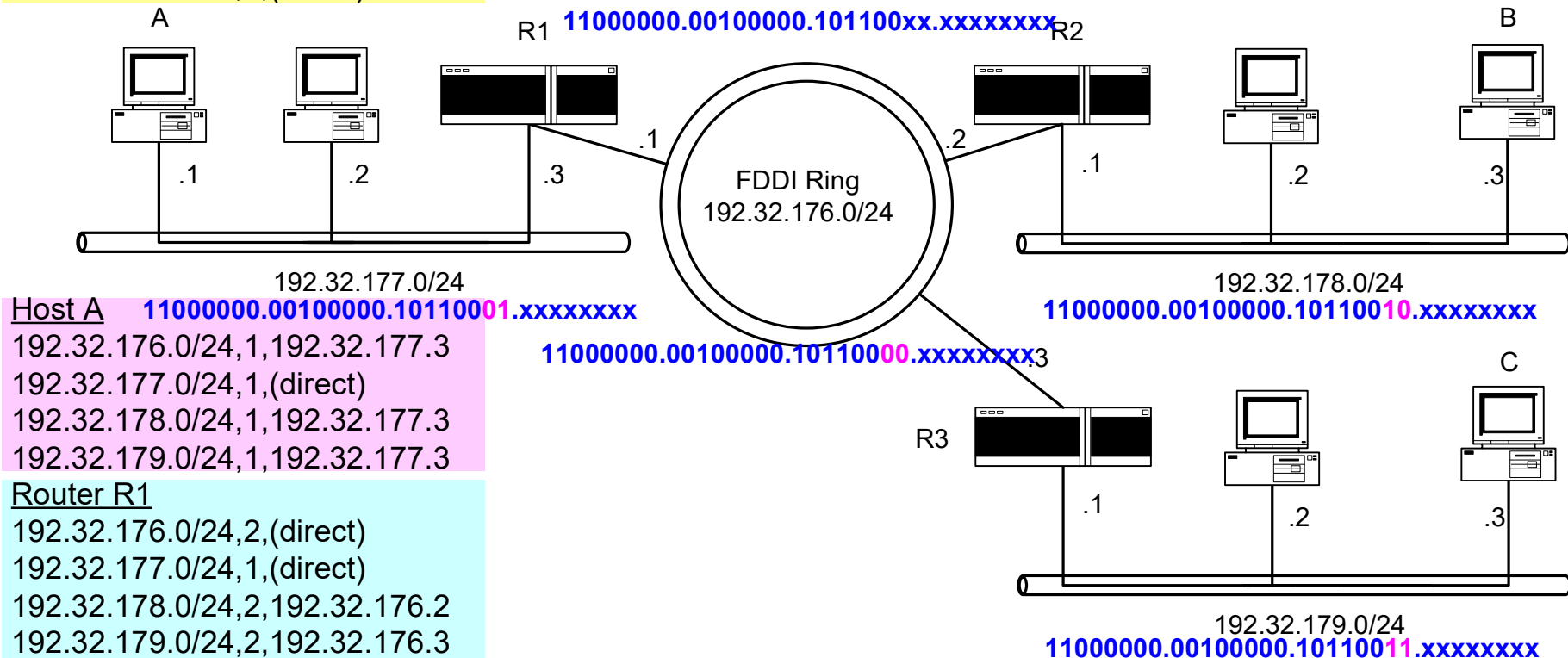
192.32.176.0/24,1,(direct)
 192.32.177.0/24,1,192.32.176.1
 192.32.178.0/24,1,192.32.176.2
 192.32.179.0/24,2,(direct)

Host B

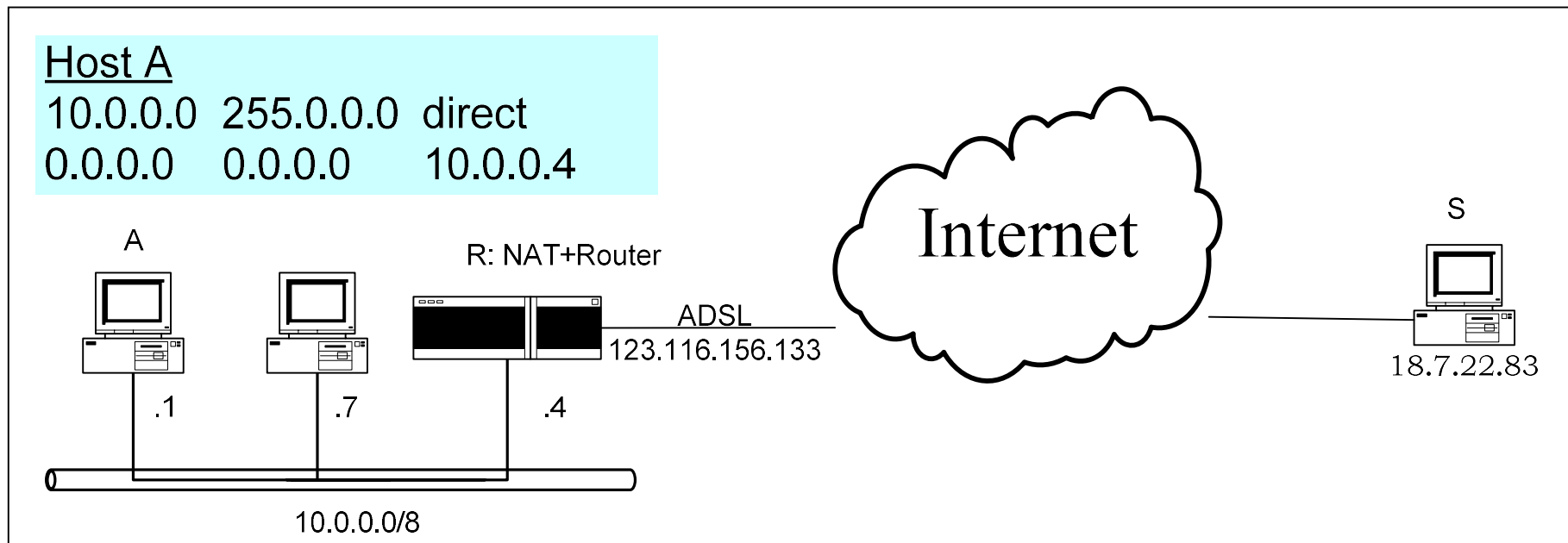
192.32.178.0/24,1,(direct)
 0.0.0.0/0,1,192.32.178.1
 11000000.00100000.10110010.xxxxxxxx
 xxxxxxxx.xxxxxxxx.xxxxxxxx.xxxxxxxx

Router R2

192.32.176.0/24,1,(direct)
 192.32.177.0/24,1,192.32.176.1
 192.32.178.0/24,2,(direct)
 192.32.179.0/24,1,192.32.176.3



举例：动态NAT表项



■ A: 发送TCP报 10.0.0.1:3723->18.7.22.83:80

■ R: 路由器动态创建**NAT**表项:

(Local)10.0.0.1:3723—(Global)123.116.156.133:33120—(Peer)18.7.22.83:80

改TCP报为123.116.156.133:33120->18.7.22.83:80

■ S: 服务器S回数据18.7.22.83:80->123.116.156.133:33120

■ R: 查NAT表: 改TCP报为18.7.22.83:80->10.0.0.1:3723

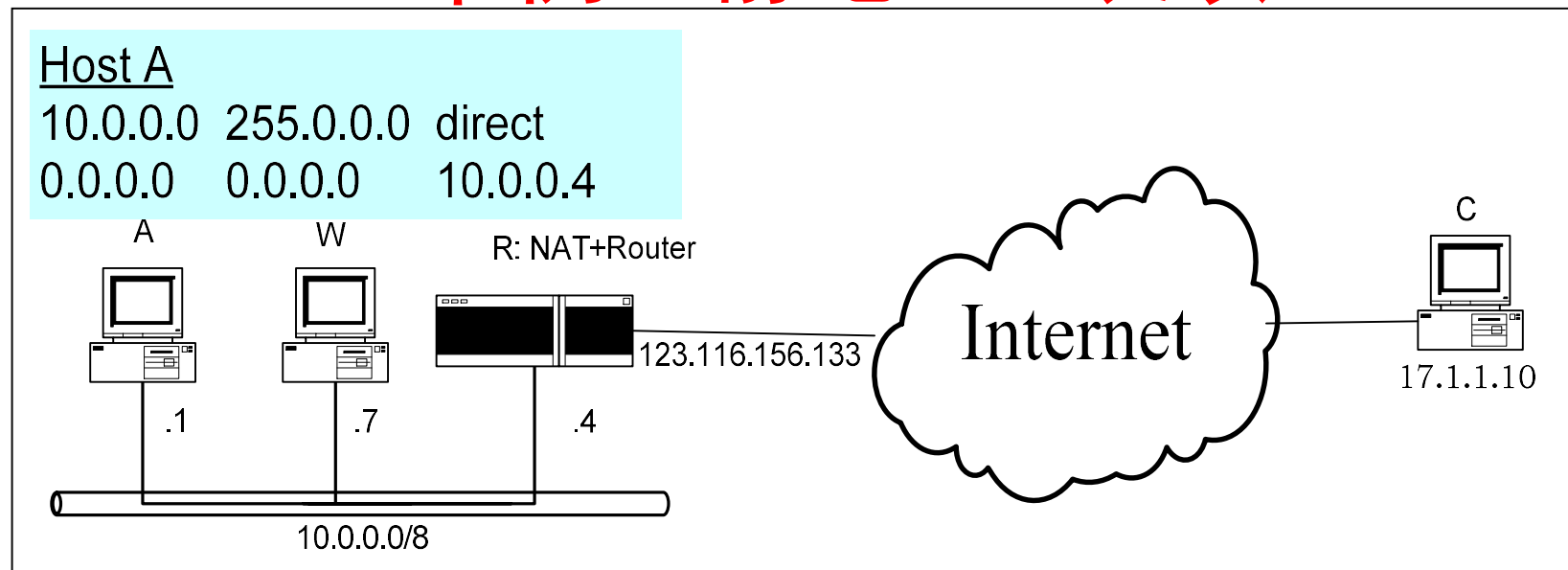
■ A: 收到数据包18.7.22.83:80->10.0.0.1:3723

(连接过程中使用该**NAT**表项, 当连接关闭时或长久未用的**NAT**表项, 删除)

OUT: (SRC)10.0.0.1:3723—(DST)18.7.22.83:80=>(SRC)123.116.156.133:33120

IN: (SRC)18.7.22.83:80—(DST)123.116.156.133:33120=>(DST)10.0.0.1:3723

举例：静态NAT表项

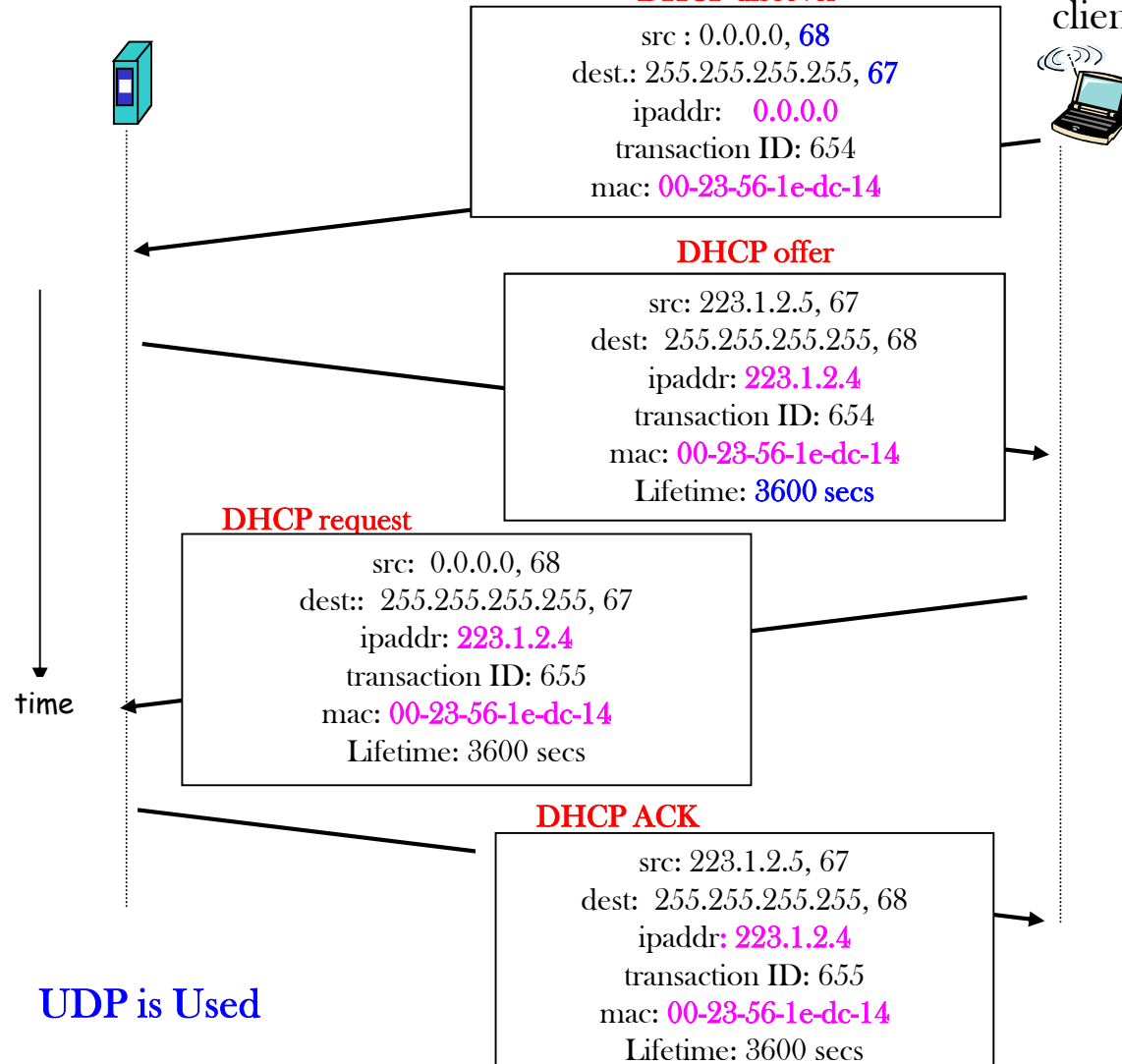


- R: 设10.0.0.7为Web服务器：建立静态**NAT**表
(Local)10.0.0.7:8080—(Global)123.116.156.133:80 —(Peer)*:*
- C: 因特网上的客户端C发送TCP数据包：
17.1.1.10:3598->123.116.156.133:80
- R: 查 NAT表，改为17.1.1.10:3598->10.0.0.7:8080
- W: Web服务器回数据10.0.0.7:8080->17.1.1.10:3598
- R: 查 NAT表，改为123.116.156.133:80->17.1.1.10:3598
- C: 收到返回TCP数据报123.116.156.133:80->17.1.1.10:3598

DHCP Client-Server Scenario

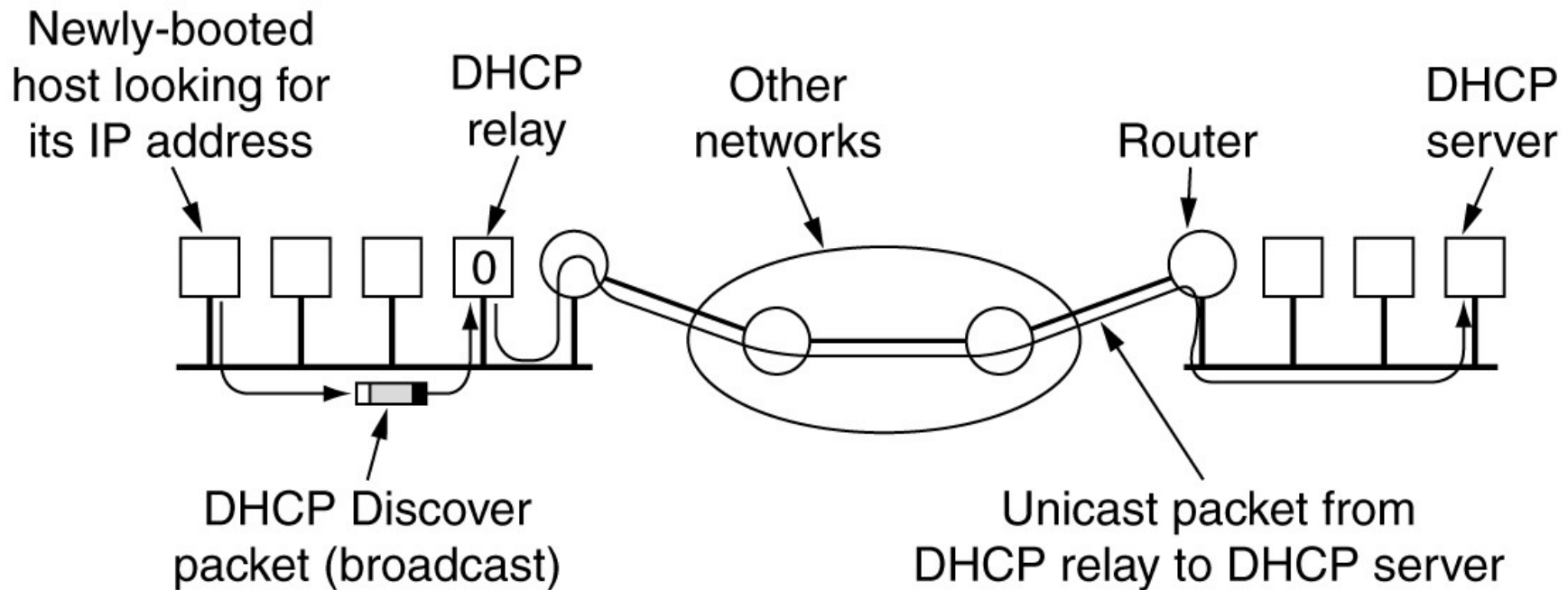
DHCP server: 223.1.2.5

Arriving
client

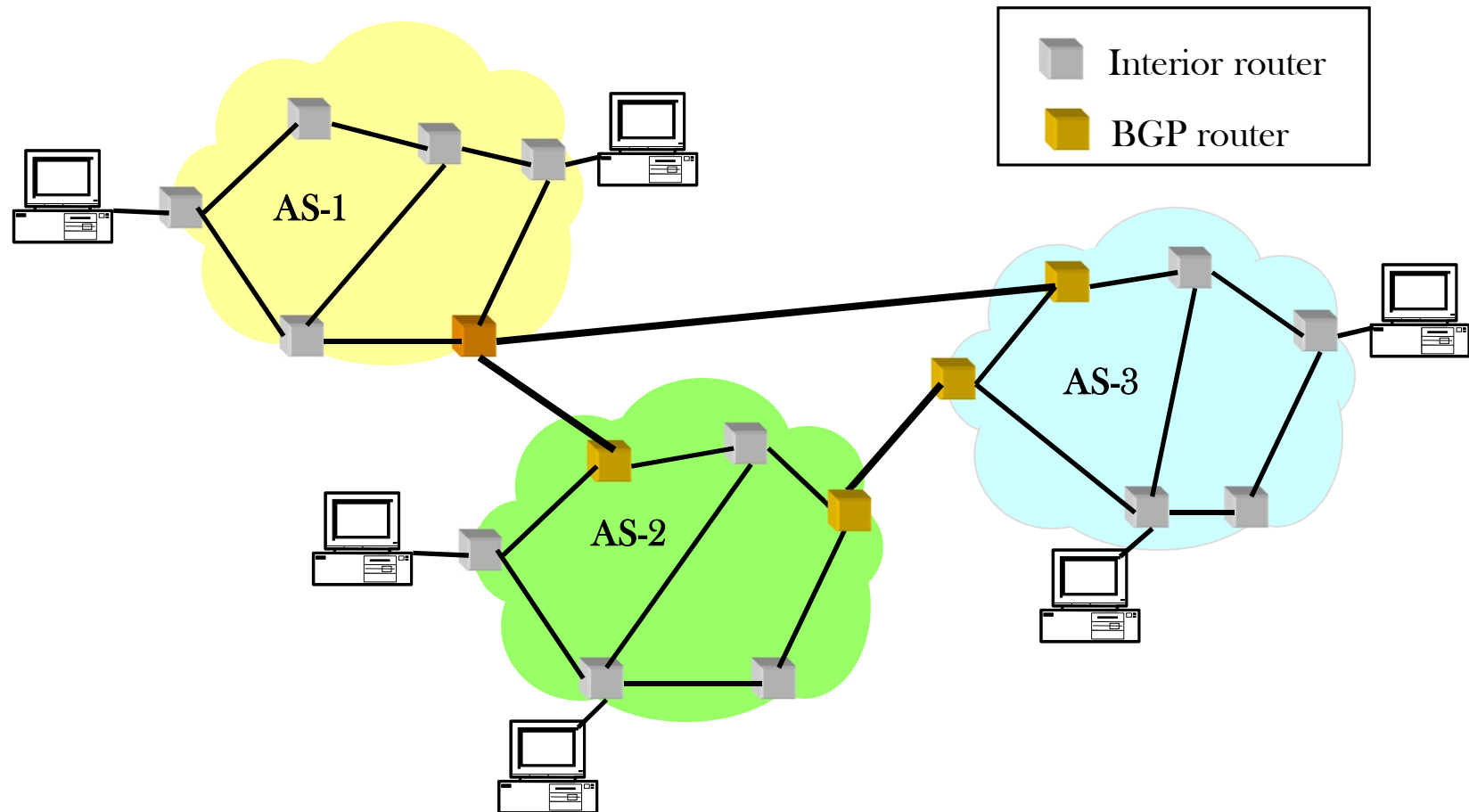


DHCP Relay Agent(DHCP中继代理)

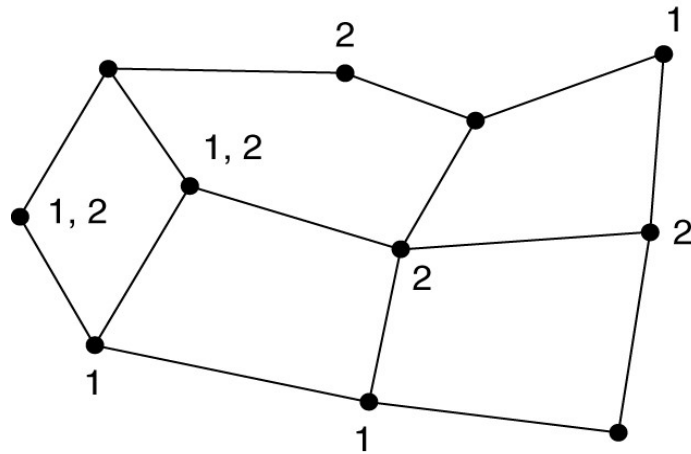
- When this is no DHCP server in the LAN, DHCP relay agent relays DHCP messages between DHCP clients and DHCP server



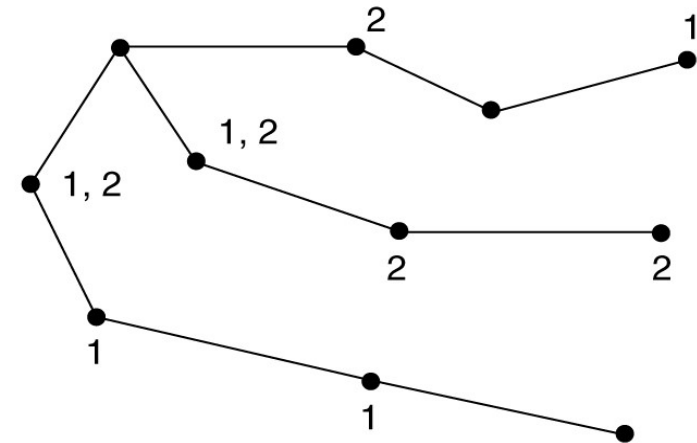
Hierarchical routing in Internet



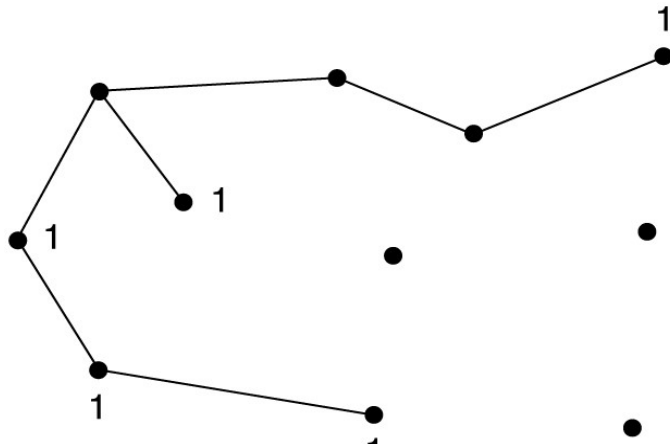
Multicast Routing: Using Group Spanning Trees (Figure 5-16)



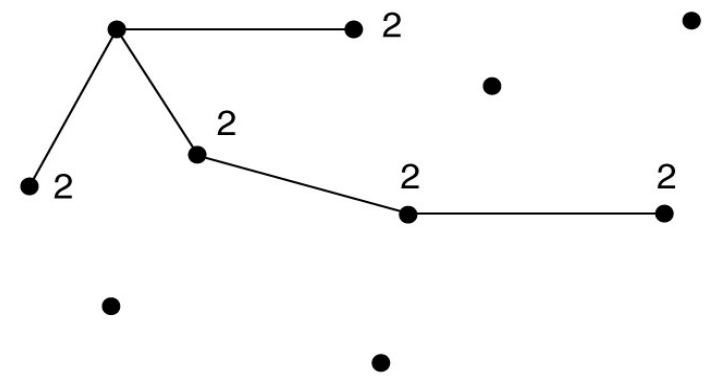
A Network



A Spanning Tree of
Leftmost Router



Multicast Tree for Group 1



Multicast Tree for Group 2